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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis



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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis

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## ABSTRACT

**Objective:** To compare the efficacy of Adjuvant treatments and identify the best scheme of adjuvant treatment for resected BTC patients.

**Design:** Systematic review and network meta-analysis.

**Data sources:** We searched studies comparing different adjuvant therapies for patients with BTC in PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials (CENTRAL), and clinicaltrials.gov from database inception to February 2020. Additionally, we manually searched references for related literatures.

**Materials and methods:** We searched studies comparing different adjuvant therapies for patients with BTC. The eligible studies were identified, and the data were extracted independently by two authors. A random-effect network meta-analysis was performed using R software. The pooled outcomes of overall survival (OS) and disease-free survival (DFS) were measured on the hazard ratio-scale and calculated as combined hazard ratios (HRs) and 95% confidence intervals (95% CIs).

**Results:** Thirteen eligible studies reporting three types of adjuvant therapies were included in our network meta-analysis. ART (HR: 0.37, 95% CI: 0.18 to 0.76) and ACRT (HR: 0.74, 95% CI: 0.55 to 0.96) were significantly more effective in prolonging OS than observation. And ACT was associated with better DFS outcome. ART showed the highest probability for ranking best to other adjuvant therapies in terms of OS and DFS.

**Conclusions:** Our primary results demonstrated that, compared to resection alone, adjuvant therapy after curative-intent resection could provide a better survival benefit for BTC patients. The optimal adjuvant therapy, the best combination of adjuvant therapies, and the optimal regimens/ dosing schedules need to be further explored.


**Keywords: Keywords:** Biliary tract cancer (BTC), adjuvant chemotherapy (ACT), adjuvant radiotherapy (ART), adjuvant chemoradiotherapy (ACRT), systematic review, network meta-analysis

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## Strengths and limitations of this study

- This is a network meta-analysis that compared the efficacy of adjuvant treatments for BTC patients after resection, and we first compared adjuvant radiotherapy with other treatments.
- This review was conducted with a strict selection criterion to lower the risk of bias and increase the homogeneity, such as only patients with negative resection margins (R0) or microscopic positive resection margins (R1), but not macroscopic involvement resections (R2) were included, and studies with unbalanced baseline profiles in age, sex, disease severity or residual tumor status were excluded.
- We did a comprehensive analysis including the transitivity assumption, local inconsistency of the model, global and local heterogeneity, sensitivity analysis, subgroup analysis, rank probability and publication bias.
- All studies regarding adjuvant radiotherapy and chemoradiotherapy were retrospective studies, which were inherent to selection bias.
- Because of the small sample sizes of adjuvant radiotherapy, the claim of optimal treatment for biliary tract cancer patients needed to be cautious.

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## INTRODUCTION

Biliary tract cancer (BTC) is divided into gallbladder cancer (GBC), intrahepatic cholangiocarcinoma (iCCA), and extrahepatic cholangiocarcinoma (eCCA), while the latter is further subclassified into perihilar cholangiocarcinoma (pCCA or Klatskin tumor), and distal cholangiocarcinoma (dCCA) [1]. BTC accounts for about 3% of digestive system cancers and 10%~15% of primary liver cancers [2]. The incidence of BTC is higher in Asian countries and lower in European countries, the United States and Australia, and the global incidence is on the rise [2, 3]. The vast majority (>90% of cases) of BTC cases are adenocarcinoma. Moreover, most cases usually are in the advanced or metastasized stage at the time of the original diagnosis. Only approximately 20% of BTC cases are considered resectable [4].

Surgical resection remains the primary curative treatment for patients with resected BTC. However, the high recurrence rates (included locoregional or distant recurrence) and low survival rates (5-year survival rates of patients were ranging from 5%～15%) are the prominent problems, even with a complete (R0) resection [4-8]. Previously published studies showed that the histologic margin status, the lymph node involvement and the intrahepatic metastasis are main prognostic factors [9-11]. Adjuvant chemotherapy (ACT), adjuvant radiotherapy (ART) and adjuvant chemoradiotherapy (ACRT) are the mainstream options following resection for adjuvant therapy.

A variety of guidelines mentioned that the postoperative adjuvant therapy could be considered an option for BTC patients[12-14]. But experts have not reached any agreement on the optimal scheme of adjuvant therapy till now since different considerations should be given based on the BTC subtypes and various prognostic factors. Moreover, it is still controversial regarding the efficacy of adjuvant therapy in the treatment of resectable BTC.

Previously, a study reported by Rangarajan K, et al. showed a significant improvement in overall survival (OS) with any adjuvant therapy after resection compared with resection only [15]. On the other hand, Horgan et al. reported a nonsignificant improvement in OS with adjuvant therapy compared with observation. But a more significant benefit from adjuvant therapy was found in patients with positive lymph node involvement and R1 resection [16]. Two network meta-analyses of adjuvant therapies in resected BTC were published previously. One made comparisons among ACT, ACRT, and resection only [17] and did not bring ART into consideration. Another one only summarized three randomized clinical trials (RCTs) covering three different ACT regimens [18], and the robustness of their results was



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limited due to the very small sample sizes. Therefore, it is essential to perform a new network meta-analysis to elucidate the efficacy of adjuvant therapy and identify the best scheme of adjuvant therapy for resected BTC patients.

## METHODS

The study design was built in accordance to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension statement for network meta-analysis for healthcare (**Supplementary Table 1**) [19].

### Search Strategy

We searched PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials (CENTRAL), and Clinicaltrials.gov from database inception to November 2020 to find relevant literatures using the main key words of "biliary tract cancer" and "adjuvant therapy". In addition, the references of the relevant literatures were manually checked for potentially relevant papers. The detailed search strategy is in **Supplementary Table 2**.

### Selection Criteria

We included studies that met the following criteria: 1) patients with histologically or cytologically confirmed resected BTC and treated with adjuvant therapy or observation after curative-intent resections (defined as negative resection margins (R0) or microscopic positive resection margins (R1), but not macroscopic involvement resections (R2)); 2) reported on at least one of the following clinical outcomes: OS or disease-free survival (DFS); 3) studies with treatment and control arms.

We excluded studies that involved: 1) ampullary carcinomas or patients with other primary cancers or neoadjuvant, palliative therapy or therapy after postoperative recurrence; 2) comparisons of the same type of adjuvant therapies; 3) unbalanced baseline profiles in age, sex, disease severity (the American Joint Committee on Cancer (AJCC) Staging ) or residual tumor status; 4) reviews, conference abstracts, posters or case reports; 5) studies not written in English or without full-text, or small sample size (less than 10 in any group).

All study titles and abstracts were screened, and then the full texts of potentially eligible articles were sequentially assessed for final inclusion.

### Data Extraction and Quality Assessment



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The following details were extracted from each study: author, study period, country, study type, sample size, tumor site (BTC type), disease severity, age, female (%), residual margin (RM) status, lymph node (LN) status, OS, and DFS. The OS was defined as the period between surgery date (or randomization in RCTs) to the date of death (or last follow-up). The DFS was defined as the time from surgery date (or randomization in RCTs) to recurrence of tumors (locoregional or distant). Moreover, we contacted the authors when there were confusions or missing data in any articles. It was excluded or considered no these data if no response was received.

The studies' quality was assessed on the Newcastle-Ottawa Scale (NOS) [20] for observational studies, based on the following domains: selection, comparability, outcome. The detailed information can be found in **Supplementary Table 3**. A study that scored 8-9 points, of which 2 points for comparability, is considered high quality; the score of 4-7 points has moderate quality, and 0-3 means low quality. The Cochrane Risk of Bias Tool was employed to grade the quality of RCTs [21].

Two investigators (Ye Chen and Baoxia Zhang) independently conducted study selection and data extraction, and four authors (Ye Chen, Baoxia Zhang, Chang Liu and Ye Cao) assessed the risk of bias of each eligible study. Any discrepancies were resolved by consensus and arbitration by the others investigators (Cheng Lyu and Meng Qiu).

**Data Synthesis and Statistical Analysis**

The primary outcomes were OS and DFS, and the outcome was measured by the hazard ratio (HR) with its 95% confidence interval (95% CI). When HRs and 95% CIs for OS and DFS were not reported in the original article, they were extracted from survival curves using Engauge Digitizer 10.9 (2014 Mark Mitchell) and estimated by the method suggested by Tierney [22].

We did the Bayesian network meta-analysis using R software (version 4.0.0, https://www.r-project.org/). The network plots were generated for different outcomes to clarify which treatments were compared directly or indirectly. A random-effects model (including subgroup analyses based on resection margin status, tumor sites, regions, and patients without distal metastasis) was used to compare all direct and indirect evidence using “Rjags” and “gemtc” packages in R software. To fit the noninformative uniform and normal prior distributions, we set the parameters with four chains, 50,000 sample iterations (n.inter), and 20,000 burn-ins (n.adapt) with a thinning interval of 1. The convergence of the chains was assessed by the Gelman-Rubin statistic and inspection of the trace plots (**Supplementary Figure 1**)



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[23, 24]. The deviance information criterion (DIC) was used to test the goodness-of-fit of consistent and inconsistent models [25].

The transitivity assumption was evaluated by comparing the distribution of clinical variables (age, the percentage female, sample size, publication year, RM status, LN status), which could be effect modifiers. We evaluated the local inconsistency of the model by the node-splitting approach [26, 27]. We assessed the global and local heterogeneity via between-study variance $\tau^2$ and $I^2$ inconsistency statistic [28-30]. Heterogeneity was considered low, moderate, and high for estimated $\tau^2$ or $I^2$ values less than 25%, between 25% and 50%, and larger than 50%, respectively. Within the Bayesian framework, the network meta-analysis provided a ranking probability of each treatment and estimated the overall rankings by calculating the surface under the cumulative ranking (SUCRA) [31, 32]. To assess the reliability of results, we performed sensitivity analyses for OS and DFS by excluding RCT studies, only including high-quality observational studies, and only including the studies for which HRs were reported in the original articles. The "netmeta" package in R software was used to generate the comparison-adjusted funnel plots to visualize publication bias. Egger's test was used to assess funnel plot asymmetry [33, 34]. The value of $P<0.05$ was considered statistically significant. All statistical tests were two-sided.

## Patient and Public Involvement

Patients or the public were not involved in the design, or conduct, or reporting, or dissemination plans of our research.

# RESULTS

## Characteristics of the Included Studies

We identified 1673 records, including 18 records manually searched by reviewing the references of relevant publications. We retrieved and reviewed 220 full-text articles after excluding 485 duplicates and 986 ineligible records by reviewing abstracts (**Figure 1**). Thirteen studies consisted of a total of 2505 patients who received one of the following four different treatments: ACT, ART, ACRT, and surgical resection alone (observation group) were included in our analysis. It is worth noting that 22 studies [11, 35-55] were excluded because of unbanlanced baseline. Of the included patients, 1567 patients received adjuvant treatments following curative-intent resection, and 938 patients had curative-intent resection without adjuvant treatments.

The main characteristics of the studies included in network meta-analysis are presented in Table 1. These 13 studies consisted of three RCTs [56-58] and 10



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retrospective studies (RS) [59-68]. Six studies[57, 58, 62, 63, 66, 67] were carried out in Western countries (USA, France, and UK) and seven studies[56, 59-61, 64, 65, 68] were carried out in Asia (Japan, Korea and China). One study [63] assessed patients with GBC, and nine studies [56, 59-62, 65-68] assessed patients with CCA. Of the nine CCA studies, one study [68] particularly examined the outcomes of iCCA, and five studies [56, 59, 60, 66, 67] examined the outcomes of eCCA , including pCCA [59] and dCCA [66, 67]. In the remaining three studies [57, 58, 64], the involved BTC subtypes were mixed.

The quality assessment of 10 retrospective studies is listed in **Supplement Table 4**, which showed eight studies [59-61, 64-68] were judged to be high quality, and two studies [62, 63] were judged to be moderate quality. Of the two moderate quality studies, they were assigned only one star in "comparability" due to the unreported resection margin status [49] or lymph node status [63]. **Supplementary Figure 2** summarises the risk of bias assessments of three RCTs. All three RCTs were conducted in an open-label fashion, one [57] did not reveal the details of allocation concealment, and all were at low risk of bias in the domains of "random sequence generation," "incomplete outcome data" and "other bias."

**Transitivity, Heterogeneity, and Inconsistency Assessment**

The transitivity assumption was assessed cross treatments in our network. Following patients' characteristics that are known modifiers of the treatment efficacy, such as age, percentage of female, sample size, publication year, RM status and LN status were evaluated and visualized by boxplots (**Supplementary Figure 3**). No difference was seen in these characteristics between different therapies (**Supplementary Table 5**). Therefore, the transitivity assumption was ensured in this network.

The heterogeneity was also evaluated between eligible studies. Low heterogeneity was detected in OS, with the global $I^2 = 10.1\%$ for a pairwise effect and $I^2 = 12.7\%$ for a consistency effect (**Supplementary Table 6**).

There was only one eligible study providing DFS data for the direct comparisons of ACT vs. ACRT and ART vs. observation; hence the heterogeneities were not estimable. Minimal heterogeneity ($I^2 = 0\%$ for a pairwise pooled effect and a consistency effect) was seen in the comparisons of ACT vs. observation and ACRT vs. observation in DFS (**Supplementary Table 6**).

The local inconsistency was evaluated by contrasting the corresponding pooled hazard ratios of OS and DFS after comparing results from pairwise meta-analysis and network meta-analysis. There was no statistical inconsistency between direct and indirect evidence after node splitting (**Supplementary Table 7**).



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To further checking the consistency at treatment level, we compared the goodness of fit of the inconsistency model (unrelated mean effects model, UME) with the consistency model. The DICs of the consistency model were 19.20 for OS and 13.95 for DFS which were slightly lower than the DICs of the inconsistency model (19.90 and 14.03 respectively), suggesting there was no evidence of inconsistency in the network. And the more parsimonious model, the consistency model, should be used for our analyses (**Supplementary Table 8**).

**Network Meta-analysis of Treatments in Biliary Tract Cancer**

A network meta-analysis was conducted to assess the efficacy (OS and DFS) of the following treatments in a Bayesian framework: observation alone after surgery, surgery with ACT, surgery with ART, and surgery with ACRT.

The network plot was presented in **Figure 2**. Data for OS were available from 12 studies that incorporated 2407 patients of whom 902 (37.5%) in the observation group, 972 (40.4%) received ACT, 43 (1.8%) received ART, and 490 (20.3%) received ACRT. The pooled OS data indicated that ART (HR: 0.37, 95% CI: 0.18 to 0.76) and ACT (HR: 0.74, 95% CI: 0.55 to 0.96) were more beneficial for BTC patients compared to observation (**Figure 3A-B**). And such benefit was not significant in ACT group (HR: 0.87, 95% CI: 0.68 to 1.06). In the comparisons between different adjuvant therapies, ART had a significant benefit in OS over ACT (HR: 0.43, 95% CI: 0.20 to 0.91) but not ACRT (HR:0.50, 95% CI: 0.23 to 1.09) (**Figure 3A**).

In terms of DFS, we included 10 studies with 1668 patients in the network meta-analysis (**Figure 2**). The pooled DFS data demonstrated a greater efficacy for ACT compared with observation (HR: 0.84, 95% CI: 0.70 to 0.99). There were no significant differences in DFS between other pairwise comparisons (**Figure 3A and Figure 3C**).

The ranking analysis was done by SUCRA. Based on both pooled data of OS and DFS, the ranking order of efficacy of treatments was consistent. The best therapy to the worse was ranked as follows: ART, ACRT, ACT, and observation (**Figure 3D**). The probability of ART being the best was approximately 95.8% for OS and 66.7% for DFS. However, compared to the second-best treatment ACRT, the ART did not show significant difference in OS and DFS.

We performed subgroup analyses for OS by patients of different residual tumor status, tumor sites, regions, and absence of distant metastasis. Five studies reported the outcomes in patients after R0 resection and six studies with the outcomes in



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patients after R1 resection (**Supplementary Figure 4A**). In both R0 and R1 groups, no survival advantage was seen in patients who underwent adjuvant therapies than surgery alone (**Supplementary Figure 5A and Supplementary Figure 6A**). In the subgroup analysis by primary tumor sites, three studies enrolled patients with GBC, and nine studies recruited patients with CCA (**Supplementary Figure 4B**). The benefit in the OS of ART was clear compared with observation in the CCA group (HR: 0.37, 95% CI: 0.18 to 0.76), but it remains unclear in the GBC group since lack of eligible ART studies. Among the comparable treatments in the GBC group, no significant differences in OS were found between ACT, ACRT and observation (**Supplementary Figure 5B and Supplementary Figure 6B**). When studies grouped as per regions, six included studies were carried out in Asian countries, and six were from the Western countries (**Supplementary Figure 4C**). There was no apparent association between adjuvant therapies and improved OS in the Asian group (**Supplementary Figure 5C and Supplementary Figure 6C**). In the Western group, the pooled results favoured ACRT (ACRT vs. Observation, HR: 0.66, 95% CI: 0.48 to 0.93). Subgroup analysis was also conducted for the ten studies investigating the OS in non-distant metastasis patients (**Supplementary Figure 4D**). The pooled results showed superior efficacy of ART compared with observation (HR: 0.37, 95% CI:0.18 to 0.74), and ART also presented a better efficacy compared to ACT (HR: 0.41, 95% CI: 0.20 to 0.84) (**Supplementary Figure 5D and Supplementary Figure 6D**). ACRT was also associated with a survival advantage over observation (HR: 0.76, 95% CI: 0.59 to 0.97).

Subgroup analyses were performed for DFS by patients for different tumor sites, regions and absence of distant metastasis (**Supplementary Figure 7**). Due to the small number of studies, the subgroup analyses for DFS by different residual tumor status cannot be conducted. When split studies by primary tumor sites, no apparent advantage in DFS was found in adjuvant therapies in either GBC or CCA groups (**Supplementary Figure 5E and Supplementary Figure 6E**). Furthermore, the stratified meta-analysis by the regions did not give a clear indication of a favourable treatment since because the confidence intervals of HRs of DFS were wide (**Supplementary Figure 5F and Supplementary Figure 6F**). In patients without distant metastasis, the results from this subgroup were similar with the primary analysis (**Supplementary Figure 5D and Supplementary Figure 6D**).

To assess the robustness of the primary results, sensitivity analyses were performed for OS and DFS by excluding RCT studies, removing moderate-quality observational studies, and only including the studies for which HRs were reported in the original articles. Eight retrospective studies for OS and seven retrospective studies



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for DFS were included in the first sensitivity analysis. The pooled results (**Figure 4A**) and the ranking profiles of comparable treatments from retrospective studies confirmed the reliability of the primary results of OS (**Supplementary Figure 8A**). In terms of DFS, ART still ranked as the best treatment, but the pooled HR of ACT vs. observation did not show statistical significance (HR: 0.87, 95% CI: 0.64 to 1.1) (**Figure 4B and Supplementary Figure 8B**). After removing moderate-quality studies, ten studies for OS and eight studies for DFS, which scoring a NOS 8～9 points, were included in the second sensitivity analysis. The exclusion of moderate-quality studies resulted in a marginal change of the pooled HRs of OS and DFS (**Supplementary Figure 8**) but did not substantially affect the treatments' ranking (**Figure 4**). Furthermore, a sensitivity analysis was performed by combining HRs reported in the original articles (7 studies for OS and six studies for DFS) and the results did not suggest a material change in the efficacy estimation (**Supplementary Figure 8**).

The visual examination of the comparison-adjusted funnel plots did not suggest the publication bias presented in our network meta-analysis (**Supplementary Figure 9**). The results of Egger's test for OS (p=0.11) and DFS (p=0.91) also rejected the presence of small-study effects.

## DISCUSSION

Biliary tract cancers (BTC) are the uncommon and heterogeneous type of cancers with a higher prevalence in Asian countries [69]. In general, BTC included the cancers raised from the intrahepatic, hilar, and distal bile duct, as well as gallbladder [7]. Surgical resection provides the only chance for cure of BTC patients at an early stage, but the survival outcomes are poor. The 5-year survival rate was as low as 10% [70]. The most recent guidelines from the National Comprehensive Cancer Network (NCCN) and the American Society of Clinical Oncology (ASCO) all recommend the use of adjuvant therapy for BTC patients after resection [12, 71]. However, experts did mention that the suggestions of using adjuvant therapy are based on a very limited number of studies, and the benefit of adjuvant therapy remains unclear in many BTC pivotal trials [56-58, 72, 73]. Therefore, it is necessary to compile up-to-date studies to validate the efficacy of adjuvant therapy in BTC patients.

In this network meta-analysis, we included 13 studies to evaluate the comparative benefits (involving 2407 patients for OS and 1668 patients for DFS) between adjuvant therapy (ACT, ART, and ACRT) following surgical resection and the curative-intent resection (observation group). Our primary results demonstrated that adjuvant therapy, especially ART and ACRT, was more effective than



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observation in OS. In line with the favourable impact of ACT on OS, DFS's pooled results suggested a statistically significant benefit for the ACT over observation. Although the ART ranked as the best treatment in terms of both OS and DFS, the small sample sizes of ART evidence should be considered, and the claim of optimal treatment for BTC patients need to be cautious. Overall, our primary analysis showed that there were survival benefits of adjuvant therapy following curative-intent resection in BTC and further support the use of adjuvant therapy in patients with resected BTC.

In 2012, Horgan et al. published a comprehensive meta-analysis of adjuvant therapy in the treatment of BTC [16]. Consistent with Horgan's analysis, we also observed a trend towards better survival outcomes with adjuvant therapy. However, the significant benefit of adjuvant therapy in patients with R1 disease was not found in our analysis. For the impact of ART on the prognosis, our result was not in line with the previously published meta-analysis. Horgan suggested the ART did not provide a greater advantage than observation while our analysis showed ART ranked first in adjuvant therapy in terms of both OS and DFS. This was possibly due to the more stringent selection of studies and the effects of small sample sizes. Studies with unreported or mixed baseline information and studies involving palliative treatment were all excluded from our analysis. In the most recent review of BTC [74], Horgan and colleagues revised their analysis and suggested that patients, especially high-risk patients, benefit from ART and ACRT after reviewing retrospective data published recently, which was in agreement with our results. In 2017, Ghidini et al. published a meta-analysis that mainly focused on the ACT in the treatment of BTC [75]. Although a few studies reported on palliative or R2 resection were included in Ghidini's analysis, a similar survival benefit was obtained in the ACT (HR: 0.59, 95% CI: 0.49–0.71).

This study presents several limitations. Firstly, due to the nature of retrospective studies, confounding factors may remain in our network meta-analysis, although we employed strict inclusion criteria on the baseline profiles. Our results derived from data of patients who underwent different operations, different adjuvant chemotherapeutic drugs radiation and chemoradiation methods, such as concurrent chemoradiation and sequential chemoradiation. The interval between operation and adjuvant therapy, length of follow-up time, and treatments after recurrence were not fully recorded. Moreover, two retrospective studies did not provide an entirely balanced baseline, and the impact on the outcome was unclear. Secondly, because of BTC's heterogeneities, it was noted that the characteristics and prognosis of BTC were not the same. The survival benefit may be mainly driven by a particular subtype



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of cancers and might change the overall treatment effects. Thirdly, most comparisons were indirectly, and direct evidence of the ART versus observation, ACRT versus ACT, and ACRT versus observation was from two or three studies. The number of patients treated with ART was small, which would result in a considerable risk of bias of its treatment benefit. Therefore, the estimates should be interpreted with caution, even though the transitivity and consistency assumptions were checked. Fourthly, data on AEs were reported only in 8 studies, of which three were compared between ACT and observation all from RCTs. Five only offered the adjuvant treatment-related adverse events but not the control group. Thus, the comparison of safety outcomes cannot be executed. Finally, the definition of OS and DFS in RCTs and retrospective studies was a bit different. In RCT, it started from the date of randomization and the date of surgery in retrospective studies.

## CONCLUSION

Our analysis supports the use of adjuvant therapy in BTC patients who undergo curative-intent resection. Combined with the results from previous meta-analysis, high-risk patients (node-positive and/or margin-positive) may obtain a more significant benefit, but further validation is needed. For patients with node-negative and R0 resection, the use of adjuvant therapy and the regimen of adjuvant therapy should be discussed case-by-case. However, the optimal adjuvant therapy, the best combination of adjuvant therapies, and the optimal regimens/ dosing schedules are still uncertain. Thorough and high-quality RCTs are warranted.

## ABBREVIATIONS

BTC: biliary tract carcinoma

OS: overall survival

DFS: disease-free survival

HR (95% CI): hazard ratio (95% confidence interval)

GBC: gallbladder cancer

CCA: cholangiocarcinoma

iCCA: intrahepatic cholangiocarcinoma

eCCA: extrahepatic cholangiocarcinoma

pCCA: perihilar cholangiocarcinoma

HDC: hepatic duct carcinoma



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dCCA: distal cholangiocarcinoma

ACT: adjuvant chemotherapy

ART: adjuvant radiotherapy

ACRT: adjuvant chemoradiotherapy

RCT: randomized clinical trial

PRIMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyse

AJCC: American Joint Committee on Cancer

RM: residual margin

R0: negative resection margin

R1: microscopic positive resection margin

R2: macroscopic involvement resection

LN: lymph node

NOS: Newcastle-Ottawa Scale

DIC: deviance information criterion

SUCRA: surface under the cumulative ranking

UME: unrelated mean effect

NCCN: National Comprehensive Cancer Network

ASCO: American Society of Clinical Oncology

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## Author Contributions

This study was designed, coordinated and supervised by Meng Qiu as the corresponding author. Ye Chen and Baoxia Zhang screened the articles, extracted the data and assessed the quality. Ye Cao and Chang Liu contacted study authors for additional information. Ye Chen, Baoxia Zhang and Ye Cao analysed, interpreted the data and prepared the material. Ye Chen, Baoxia Zhang, Chang Liu and Cheng Lyu wrote the draft of the report. All authors had reviewed and approved the final submitted version of the report. Meng Qiu was responsible for the integrity and accuracy of the data.

## Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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## References


1. Nakeeb A, Pitt HA, Sohn TA, *et al.* Cholangiocarcinoma. A spectrum of intrahepatic, perihilar, and distal tumors. Ann Surg. 1996;224(4):463-473; discussion 473-465. doi:10.1097/00000658-199610000-00005

2. Benavides M, Anton A, Gallego J, *et al.* Biliary tract cancers: SEOM clinical guidelines. Clin Transl Oncol. 2015;17(12):982-987. doi:10.1007/s12094-015-1436-2

3. Patel T. Increasing incidence and mortality of primary intrahepatic cholangiocarcinoma in the United States. Hepatology. 2001;33(6):1353-1357. doi:10.1053/jhep.2001.25087

4. Valle JW, Lamarca A, Goyal L, *et al.* New Horizons for Precision Medicine in Biliary Tract Cancers. Cancer Discov. 2017;7(9):943-962. doi:10.1158/2159-8290.CD-17-0245

5. Anderson CD, Pinson CW, Berlin J, *et al.* Diagnosis and treatment of cholangiocarcinoma. Oncologist. 2004;9(1):43-57. doi:10.1634/theoncologist.9-1-43

6. DeOliveira ML, Cunningham SC, Cameron JL, *et al.* Cholangiocarcinoma: thirty-one-year experience with 564 patients at a single institution. Ann Surg. 2007;245(5):755-762. doi:10.1097/01.sla.0000251366.62632.d3

7. Bridgewater JA, Goodman KA, Kalyan A, *et al.* Biliary Tract Cancer: Epidemiology, Radiotherapy, and Molecular Profiling. Am Soc Clin Oncol Educ Book. 2016;35:e194-203. doi:10.14694/EDBK_160831

10.1200/EDBK_160831

8. Doherty MK, Knox JJ. Adjuvant therapy for resected biliary tract cancer: a review. Chin




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

Clin Oncol. 2016;5(5):64. doi:10.21037/cco.2016.08.05

9. Mavros MN, Economopoulos KP, Alexiou VG, *et al.* Treatment and Prognosis for Patients With Intrahepatic Cholangiocarcinoma: Systematic Review and Meta-analysis. JAMA Surg. 2014;149(6):565-574. doi:10.1001/jamasurg.2013.5137

10. Murakami Y, Uemura K, Sudo T, *et al.* Prognostic factors after surgical resection for intrahepatic, hilar, and distal cholangiocarcinoma. Ann Surg Oncol. 2011;18(3):651-658. doi:10.1245/s10434-010-1325-4

11. Wirasorn K, Ngamprasertchai T, Khuntikeo N, *et al.* Adjuvant chemotherapy in resectable cholangiocarcinoma patients. J Gastroenterol Hepatol. 2013;28(12):1885-1891. doi:10.1111/jgh.12321

12. Al B. Benson I, Michael I. D'Angelica, Daniel E. Abbott, *et al.* NCCN clinical practice guidelines in oncology: hepatobiliary cancer. In. *Journal of the National Comprehensive Cancer*; 2020.

13. Valle JW, Borbath I, Khan SA, *et al.* Biliary cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. Ann Oncol. 2016;27(suppl 5):v28-v37. doi:10.1093/annonc/mdw324

14. HJ L, SK Q, F S, *et al.* Biliary tract caner: CSCO consensus for diagnosis and treatment. Chinese Clinical Oncology. 2019;24(9):828-838.

15. Rangarajan K, Simmons G, Manas D, *et al.* Systemic adjuvant chemotherapy for cholangiocarcinoma surgery: A systematic review and meta-analysis. Eur J Surg Oncol.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

2020;46(4 Pt A):684-693. doi:10.1016/j.ejso.2019.11.499

16. Horgan AM, Amir E, Walter T*, et al.* Adjuvant therapy in the treatment of biliary tract cancer: a systematic review and meta-analysis. J Clin Oncol. 2012;30(16):1934-1940. doi:10.1200/JCO.2011.40.5381

17. Zhu GQ, Shi KQ, You J*, et al.* Systematic review with network meta-analysis: adjuvant therapy for resected biliary tract cancer. Aliment Pharmacol Ther. 2014;40(7):759-770. doi:10.1111/apt.12900

18. Kish M, Chan K, Perry K*, et al.* A systematic review and network meta-analysis of adjuvant therapy for curatively resected biliary tract cancers. Curr Oncol. 2020;27(1):e20-e26. doi:10.3747/co.27.5465

19. Hutton B, Salanti G, Caldwell DM*, et al.* The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. Ann Intern Med. 2015;162(11):777-784. doi:10.7326/M14-2385

20. Wells G, Shea B, O'Connell D*, et al.* The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. In. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp: Ottawa Hospital Research Institute; 2014.

21. Julian Higgins, Douglas Altman, Sterne J. Assessing risk of bias in included studies. In: Julian Higgins, Green S, (eds). *Cochrane handbook of systematic reviews of interventions*. 5.1.0 ed. England: The Cochrane Collaboration, Wiley; 2011, 187-235.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

22. Tierney JF, Stewart LA, Ghersi D, *et al.* Practical methods for incorporating summary time-to-event data into meta-analysis. Trials. 2007;8:16. doi:10.1186/1745-6215-8-16

23. Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. Statistical Science. 1992;7:457-511.

24. Lynch SM. *Introduction to Applied Bayesian Statistics and Estimation for Social Scientists*. 1st edition ed. New York, NY: Springer-Verlag; 2007.

25. Higgins JPT, Jackson D, Barrett JK, *et al.* Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. Research Synthesis Methods. 2012;3(2):98-110. doi:10.1002/jrsm.1044

26. Dias S, Welton NJ, Caldwell DM, *et al.* Checking consistency in mixed treatment comparison meta-analysis. Stat Med. 2010;29(7-8):932-944. doi:10.1002/sim.3767

27. Van Valkenhoef G, Dias S, Ades AE, *et al.* Automated generation of node-splitting models for assessment of inconsistency in network meta-analysis. Res Synth Methods. 2016;7(1):80-93. doi:10.1002/jrsm.1167

28. Higgins JP, Thompson SG, Deeks JJ, *et al.* Measuring inconsistency in meta-analyses. BMJ. 2003;327(7414):557-560. doi:10.1136/bmj.327.7414.557

29. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. Stat Med. 2002;21(11):1539-1558. doi:10.1002/sim.1186

30. Debray TP, Damen JA, Riley RD, *et al.* A framework for meta-analysis of prediction



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

model studies with binary and time-to-event outcomes. Stat Methods Med Res. 2019;28(9):2768-2786. doi:10.1177/0962280218785504

31. Salanti G, Ades AE, Ioannidis JP. Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial. J Clin Epidemiol. 2011;64(2):163-171. doi:10.1016/j.jclinepi.2010.03.016

32. Veroniki AA, Vasiliadis HS, Higgins JP*, et al.* Evaluation of inconsistency in networks of interventions. Int J Epidemiol. 2013;42(1):332-345. doi:10.1093/ije/dys222

33. Egger M, Davey Smith G, Schneider M*, et al.* Bias in meta-analysis detected by a simple, graphical test. BMJ. 1997;315(7109):629-634. doi:10.1136/bmj.315.7109.629

34. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. Biometrics. 1994;50(4):1088-1101. https://www.ncbi.nlm.nih.gov/pubmed/7786990

35. Sagawa N, Kondo S, Morikawa T*, et al.* Effectiveness of radiation therapy after surgery for hilar cholangiocarcinoma. Surgery Today. 2005;35(7):548-552. doi:10.1007/s00595-005-2989-4

36. Borghero Y, Crane CH, Szklaruk J*, et al.* Extrahepatic bile duct adenocarcinoma: patients at high-risk for local recurrence treated with surgery and adjuvant chemoradiation have an equivalent overall survival to patients with standard-risk treated with surgery alone. Ann Surg Oncol. 2008;15(11):3147-3156. doi:10.1245/s10434-008-9998-7

37. Gold DG, Miller RC, Haddock MG*, et al.* Adjuvant therapy for gallbladder carcinoma: the



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

552 Mayo Clinic Experience. Int J Radiat Oncol Biol Phys. 2009;75(1):150-155.
553 doi:10.1016/j.ijrobp.2008.10.052

554 38. Ercolani G, Vetrone G, Grazi GL*, et al.* Intrahepatic cholangiocarcinoma: primary liver
555 resection and aggressive multimodal treatment of recurrence significantly prolong survival.
556 Ann Surg. 2010;252(1):107-114. doi:10.1097/SLA.0b013e3181e462e6

557 39. Kim K, Chie EK, Jang JY*, et al.* Postoperative chemoradiotherapy for gallbladder cancer.
558 Strahlenther Onkol. 2012;188(5):388-392. doi:10.1007/s00066-012-0074-7

559 40. Kondo N, Murakami Y, Uemura K*, et al.* Elevated perioperative serum CA 19-9 levels are
560 independent predictors of poor survival in patients with resectable cholangiocarcinoma. J
561 Surg Oncol. 2014;110(4):422-429. doi:10.1002/jso.23666

562 41. Bektas H, Yeyrek C, Kleine M*, et al.* Surgical treatment for intrahepatic
563 cholangiocarcinoma in Europe: a single center experience. J Hepatobiliary Pancreat Sci.
564 2015;22(2):131-137. doi:10.1002/jhbp.158

565 42. Hoehn RS, Wima K, Ertel AE*, et al.* Adjuvant Chemotherapy and Radiation Therapy is
566 Associated with Improved Survival for Patients with Extrahepatic Cholangiocarcinoma. Annals
567 of Surgical Oncology. 2015;22:1133-1139. doi:10.1245/s10434-015-4599-8

568 43. Sur MD, In H, Sharpe SM*, et al.* Defining the Benefit of Adjuvant Therapy Following
569 Resection for Intrahepatic Cholangiocarcinoma. Annals of Surgical Oncology.
570 2015;22(7):2209-2217. doi:10.1245/s10434-014-4275-4

571 44. Im JH, Seong J, Lee IJ*, et al.* Surgery alone versus surgery followed by chemotherapy



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

572 and radiotherapy in resected extrahepatic bile duct cancer: Treatment outcome analysis of
573 336 patients. Cancer Research and Treatment. 2016;48(2):583-595. doi:10.4143/crt.2015.091

574 45. Kim Y, Hwang I, Park S-E*, et al.* Role of adjuvant therapy after R0 resection for patients
575 with distal cholangiocarcinoma. Cancer Chemotherapy & Pharmacology. 2016;77(5):979-985.
576 doi:10.1007/s00280-016-3014-x

577 46. Kim TG. Patterns of initial failure after resection for gallbladder cancer: Implications for
578 adjuvant radiotherapy. Radiation Oncology Journal. 2017;35(4):359-367.
579 doi:10.3857/roj.2017.00388

580 47. Kim Y, Oh S, Go S-I*, et al.* The role of adjuvant therapy after R0 resection for patients
581 with intrahepatic and perihilar cholangiocarcinomas. Cancer Chemotherapy & Pharmacology.
582 2017;79(1):99-106. doi:10.1007/s00280-016-3206-4

583 48. Kim YS, Jeong CY, Song HN*, et al.* The efficacy of fluoropyrimidine-based adjuvant
584 chemotherapy on biliary tract cancer after R0 resection. Chin J Cancer. 2017;36(1):9.
585 doi:10.1186/s40880-017-0182-y

586 49. Lee J, Kang SH, Noh OK*, et al.* Adjuvant concurrent chemoradiation therapy in patients
587 with microscopic residual tumor after curative resection for extrahepatic cholangiocarcinoma.
588 Clin Transl Oncol. 2017;20(8):1011-1017. doi:10.1007/s12094-017-1815-y

589 50. Mitin T, Kristian Enestvedt C, Jemal A*, et al.* Limited Use of Adjuvant Therapy in Patients
590 with Resected Gallbladder Cancer Despite a Strong Association with Survival. Journal of the
591 National Cancer Institute. 2017;109(7). doi:10.1093/jnci/djw324



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

51. Tran Cao HS, Zhang Q, Sada YH, *et al.* The role of surgery and adjuvant therapy in lymph node-positive cancers of the gallbladder and intrahepatic bile ducts. Cancer. 2017;124(1):74-83. doi:10.1002/cncr.30968

52. Lin YK, Hsieh MC, Wang WW, *et al.* Outcomes of adjuvant treatments for resectable intrahepatic cholangiocarcinoma: Chemotherapy alone, sequential chemoradiotherapy, or concurrent chemoradiotherapy. Radiotherapy and Oncology. 2018;128(3):575-583. doi:10.1016/j.radonc.2018.05.011

53. Geng ZM, Cai ZQ, Zhang Z, *et al.* Estimating survival benefit of adjuvant therapy based on a Bayesian network prediction model in curatively resected advanced gallbladder adenocarcinoma. World Journal of Gastroenterology. 2019;25(37):5655-5666. doi:10.3748/wjg.v25.i37.5655

54. Kattepur AK, Patkar S, Goel M, *et al.* Role of Adjuvant Chemotherapy in Resected T2N0 Gall Bladder Cancer. Journal of gastrointestinal surgery : official journal of the Society for Surgery of the Alimentary Tract. 2019;23(11):2232-2238. doi:10.1007/s11605-019-04104-4

55. Navarro J, Kang I, Hwang HK, *et al.* Glucose to Lymphocyte Ratio as a Prognostic Marker in Patients With Resected pT2 Gallbladder Cancer. Journal of Surgical Research. 2019;240:17-29. doi:10.1016/j.jss.2019.02.043

56. Ebata T, Hirano S, Konishi M, *et al.* Randomized clinical trial of adjuvant gemcitabine chemotherapy versus observation in resected bile duct cancer. British journal of surgery. 2018;105(3):192-202. doi:10.1002/bjs.10776



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

612 57. Edeline J, Benabdelghani M, Bertaut A*, et al.* Gemcitabine and oxaliplatin chemotherapy
613 or surveillance in resected biliary tract cancer (Prodige 12-accord 18-Unicancer GI): a
614 randomized phase III study. Journal of clinical oncology. 2019;37(8):658 - 667.
615 doi:10.1200/JCO.18.00050

616 58. Primrose JN, Fox RP, Palmer DH*, et al.* Capecitabine compared with observation in
617 resected biliary tract cancer (BILCAP): a randomised, controlled, multicentre, phase 3 study.
618 Lancet oncology. 2019;20(5):663-673. doi:10.1016/S1470-2045(18)30915-X

619 59. Todoroki T, Ohara K, Kawamoto T*, et al.* Benefits of adjuvant radiotherapy after radical
620 resection of locally advanced main hepatic duct carcinoma. Int J Radiat Oncol Biol Phys.
621 2000;46(3):581-587. doi:10.1016/s0360-3016(99)00472-1

622 60. Kim TH, Han SS, Park SJ*, et al.* Role of adjuvant chemoradiotherapy for resected
623 extrahepatic biliary tract cancer. Int J Radiat Oncol Biol Phys. 2011;81(5):e853-859.
624 doi:10.1016/j.ijrobp.2010.12.019

625 61. Kobayashi H, Murakami Y, Uemura K*, et al.* Human equilibrative nucleoside transporter
626 1 expression predicts survival of advanced cholangiocarcinoma patients treated with
627 gemcitabine-based adjuvant chemotherapy after surgical resection. Annals of Surgery.
628 2012;256(2):288-296. doi:10.1097/SLA.0b013e3182536a42

629 62. Dover LL, Oster RA, McDonald AM*, et al.* Impact of adjuvant chemoradiation on survival
630 in patients with resectable cholangiocarcinoma. HPB (Oxford). 2016;18(10):843-850.
631 doi:10.1016/j.hpb.2016.07.008



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

63. Kim Y, Amini N, Wilson A, *et al.* Impact of Chemotherapy and External-Beam Radiation Therapy on Outcomes among Patients with Resected Gallbladder Cancer: A Multi-institutional Analysis. Annals of Surgical Oncology. 2016;23(9):2998-3008. doi:10.1245/s10434-016-5262-8

64. Morine Y, Shimada M, Ikemoto T, *et al.* Effect of adjuvant gemcitabine combined with low-dose 5-fluorouracil and cisplatin chemotherapy for advanced biliary carcinoma. Anticancer Research. 2017;37(11):6421-6428. doi:10.21873/anticanres.12096

65. Akahoshi K, Ban D, Kuboki R, *et al.* Orotate phosphoribosyltransferase as a predictor of benefit from S-1 adjuvant chemotherapy for cholangiocarcinoma patients. J Gastroenterol Hepatol. 2018;34(6):1108-1115. doi:10.1111/jgh.14477

66. Bergeat D, Turrini O, Courtin-Tanguy L, *et al.* Impact of adjuvant chemotherapy after pancreaticoduodenectomy for distal cholangiocarcinoma: a propensity score analysis from a French multicentric cohort. Langenbecks Arch Surg. 2018;403(6):701-709. doi:10.1007/s00423-018-1702-1

67. Hester C, Nassour I, Adams-Huet B, *et al.* Improved Survival in Surgically Resected Distal Cholangiocarcinoma Treated with Adjuvant Therapy: a Propensity Score Matched Analysis. J Gastrointest Surg. 2018;22(12):2080-2087. doi:10.1007/s11605-018-3875-x

68. Zheng X, Chen B, Wu JX, *et al.* Benefit of adjuvant radiotherapy following narrow-margin hepatectomy in patients with intrahepatic cholangiocarcinoma that adhere to major vessels. Cancer Management and Research. 2018;10:3973-3981. doi:10.2147/CMAR.S172940



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

69. Tariq NU, McNamara MG, Valle JW. Biliary tract cancers: current knowledge, clinical candidates and future challenges. Cancer Manag Res. 2019;11:2623-2642. doi:10.2147/CMAR.S157092

70. Khan SA, Davidson BR, Goldin RD, *et al.* Guidelines for the diagnosis and treatment of cholangiocarcinoma: an update. Gut. 2012;61(12):1657-1669. doi:10.1136/gutjnl-2011-301748

71. Shroff RT, Kennedy EB, Bachini M, *et al.* Adjuvant Therapy for Resected Biliary Tract Cancer: ASCO Clinical Practice Guideline. J Clin Oncol. 2019;37(12):1015-1027. doi:10.1200/JCO.18.02178

72. Takada T, Amano H, Yasuda H, *et al.* Is postoperative adjuvant chemotherapy useful for gallbladder carcinoma? A phase III multicenter prospective randomized controlled trial in patients with resected pancreaticobiliary carcinoma. Cancer. 2002;95(8):1685-1695. doi:10.1002/cncr.10831

73. Neoptolemos JP, Moore MJ, Cox TF, *et al.* Effect of adjuvant chemotherapy with fluorouracil plus folinic acid or gemcitabine vs observation on survival in patients with resected periampullary adenocarcinoma: the ESPAC-3 periampullary cancer randomized trial. JAMA. 2012;308(2):147-156. doi:10.1001/jama.2012.7352

74. Horgan AM, Knox JJ. Adjuvant Therapy for Biliary Tract Cancers. J Oncol Pract. 2018;14(12):701-708. doi:10.1200/JOP.18.00558

75. Ghidini M, Tomasello G, Botticelli A, *et al.* Adjuvant chemotherapy for resected biliary



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

672 tract cancers: a systematic review and meta-analysis. HPB (Oxford). 2017;19(9):741-748.

673 doi:10.1016/j.hpb.2017.05.010

674

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675 **Table 1. Characteristics of included studies.**

| Studies | Country | Study period | BTC type | Disease severity | Therapy | Regimen of CT/ type of RT | Sample size | Margin status (R0/R1) | Nodal status (negative/positive) | Female No. (%) | Age, Mean (SD/range), years | 5-Year OS (%); MOST (months) | 5-Year DFS (%); MDFST (months) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **ACT vs Observation** | | | | | | | | | | | | | |
| Kobayashi 2012 [61] | Japan | 1989-2010 | CCA | I-IV | ACT | GEM/S-1 | 51 | 39/12 | N0: 26; N+: 25 | 28 (55) | - [a] | 46.0; - | - |
| | | | | | Observation | - | 54 | 37/17 | N0: 31; N+: 23 | 14 (26) | | 23.0; - | |
| Morine 2017 [64] | Japan | 1995-2012 | CCA/ GBC | II-IVb T1-T4 | ACT | GEM+5-FU+Cis | 28 | 18/10 | 9/19 | 16 (57) | 70.9 (± 6.8) | 44.2; - | - |
| | | | | | Observation | - | 34 | 27/7 | 9/25 | 11 (32) | 66.9 (± 9.0) | 5.9; - | |
| Akahoshi 2018 [65] | Japan | 2004-2015 | CCA | 0-IVa T1-T4 | ACT | GEM; | 26 | 15/11; | 15/11; | 9 (35) | - [a] | - | 25.6; - |
| | | | | | ACT | S-1 | 36 | 23/13 | 23/13 | 6 (17) | | | 34.0; - |
| | | | | | Observation | - | 36 | 23/13 | 25/11 | 13 (36) | | | 26.8; - |
| Bergeat 2018 [66] | France | 2000-2015 | dCCA | Ia-III | ACT | GEM | 49 | 12/37 | 7/42 | 18 (37) | 62.0 (36-77) [b] | - ; 26.3 | - ; 15.5 |
| | | | | | Observation | - | 49 | 6/43 | 14/35 | 20 (41) | 65.0 (35-82) [b] | - ; 43.3 | - ; 14.7 |



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| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **ART vs Observation** | | | | | | | | | | | | | |
| Todorokia 2000 [59] | Japan | 1976-1999 | pCCA (HDC) | IVa | ART | IORT/PORT | 17 | All R1 | N0:11; N1:2; N2:4 | 7 (41) | 59.4 (34-76) | 33.9 [d]; 32.0 | - |
| | | | | | Observation | - | 19 | | N0:7; N1:4; N2:8 | 5 (26) | 60.6 (19-82) | 13.5 [d]; 10.0 | |
| Zheng 2018 [68] | China | 2007-2016 | iCCA | I-IVa | ART | IMRT | 26 | 14/12 | 22/4 | 10 (38) | 57.2 (45-76) | 55.0 [de]; 19.1 [d] | 44.0[e]; - |
| | | | | | Observation | - | 23 | 9/14 | 17/6 | 8 (35) | 55.6 (36–72) | 20.0 [de]; 9.5 [d] | 10.0[e]; - |
| **ACRT vs Observation** | | | | | | | | | | | | | |
| Kim 2011 [60] | Korea | 2001-2009 | eCCA | Ib, IIa, IIb | ACRT | 5-FU/Leu/Cap EBRT | 75 | 58/17 | N0: 39; N+: 36 | 17 (23) | 60.6 (37-74) | 41.1 [e]; - | 37.4 [e]; - |
| | | | | | Observation | - | 34 | 28/6 | N0: 26; N+: 8 | 30 (9) | 65.0 (44-85) | 35.3 [e]; - | 29.8 [e]; - |
| Dover 2016 [62] | USA | 2000-2013 | CCA | I-IV | ACRT | GEM/5-FU | 23 | 11/12 | N0: 13; N+: 10 | 10 (43) | 62.0 (38-80) [b] | - ; 30.2 | - ; 21.6 |
| | | | | | Observation | - | 72 | 56/16 | N0: 51; N+: 21 | 28 (39) | 65.0 (39-86) [b] | - ; 26.3 | - ; 17.3 |
| **ACRT vs ACT** | | | | | | | | | | | | | |
| Hester 2018 [67] | USA | 2004-2013 | dCCA | I-IV | ACRT | NG | 348 | 258/76 | N0:126; N+:211 | 135 (39) | 64.3 (±10.0) | 31.3 [e]; 32.1 | - |
| | | | | | ACT | | 348 | 256/78 | N0:118; N+:206 | 136 (39) | 64.4 (±10.8) | 29.4 [e]; 29.5 | |
| **Multiple comparison** | | | | | | | | | | | | | |
| Kim1 2016[63] | Korea | 2000-2015 | GBC | I-IV | ACT | GEM/Cis/OTHs | 61 | 54/7 | N0:21; N1/Nx:36/4 | 42 (69) | 64.5 (56-73) | Total: | Total: |



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| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | ACRT | | 44 | 36/8 | N0:15; N1/NX:24/5 | 28 (64) | 63.9 (55-70) | 33.0 [e]; 28.3 | - ; 19.8 |
| | | | | | Observation | - | 186 | 159/25 | N0:105; N1/Nx:50/31 | 122 (66) | 67.7 (56-74) | | |
| **Characteristics of included RCT studies** | | | | | | | | | | | | | |
| Ebata 2018 [56] | Japan | 2007-2011 | pCCA /dCCA | I-III T1-T4 | ACT | GEM | 117 | 106/11 | N0: 75; N1: 42 | 40 (34) | - [a] | 51.7; 62.3 | 45.0; 36.0 |
| | | | | | Observation | - | 108 | 94/14 | N0: 72; N1: 36 | 26 (24) | | 51.6; 63.8 | 44.0; 39.9 |
| Edeline 2019 [57] | France | 2009-2014 | CCA/ GBC | I-IV T1-T4 | ACT | GEMOX | 95 | 82/13 | N0:49; N+:35; Nx:11 | 38 (40) | 63.0 (33-83) [b] | 60.0 [d]; 75.8 | - ; 30.4 |
| | | | | | Observation | - | 99 | 87/12 | N0:48; N+:36; Nx:15 | 49 (50) | 63.0 (40-80) [b] | 65.0 [d]; 50.8 | - ; 18.5 |
| Primrose 2019 [58] | UK | 2006-2014 | BTC | I-IV | ACT | Cap | 223 | 139/84 | N0: 121; N1: 102 | 112 (50) | 62.0 (55-68) | 25.1 [c]; 51.1 | - ; 24.4 |
| | | | | | Observation | - | 224 | 140/84 | N0: 115; N1:108 | 112 (50) | 64.0 (55-69) | 20.5 [c]; 36.4 | - ; 17.5 |

Abbreviations: 5-Year OS (%)=5-Year overall survival probability; MOST=median OS time; 5-Year DFS (%)=5-Year disease-free survival probability; MDFST=median DFS time; N+= positive lymph node; Nx= unclear; R0=negative resection margins; R1=microscopic positive resection margins; Total= means the value of all patients in the study; BTC=biliary tract cancer; eCCA=extrahepatic cholangiocarcinoma; GBC=gallbladder cancer; CCA=cholangiocarcinoma; pCCA=Perihilar cholangiocarcinoma (or hepatic duct carcinoma=HDC); iCCA=Intrahepatic cholangiocarcinoma; dCCA=distal cholangiocarcinoma; CT= chemotherapy; RT= radiotherapy; ACT=adjuvant chemotherapy; ART=adjuvant radiotherapy; EBRT=external beam radiotherapy; ILRT, intraluminal beam radiotherapy; IMRT=intensity-modulated radiotherapy; ACRT=adjuvant chemoradiation therapy; GEM=Gemcitabine; S-1=tegafur gimeracil oteracil potassium; 5-FU=5-fluorouracil; Cap=capecitabine; Cis=cisplatin; Leu=Leucovorin; GEMOX=gemcitabine and oxaliplatin.

- indicates the data was not given originally;

[a] indicates studies were presented as younger or older than a specific age;



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[b] indicates median age was given instead of mean age;

[c] indicates overall survival rates but not survival probability;

[d] indicates a 3-Year overall survival of disease-free survival;

[e] indicates the value calculated by which way is unclear.

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## Figure Legends

**Figure 1. A PRISMA flow diagram of study selection.** BTC: biliary tract cancer; OS: overall survival; DFS: disease-free survival.

**Figure 2. Network plot of comparisons on overall survival and disease-free survival of treatments in patients with biliary tract cancer.** Each circular node represents a type of treatment. The node size corresponds to the total number of participants assigned to each treatment. Each line represents a type of direct comparison and its width corresponds to the number of studies evaluating the connected treatments. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant Chemoradiation therapy.

**Figure 3. Network meta-analysis results for efficacy outcomes. (A)** League table of the network meta-analysis. Pooled HRs and 95% CIs for OS are listed in the upper triangle and DFS are listed in the lower triangle. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. Significant results are in bold; **(B)** Forest plot of the network meta-analysis for overall survival; **(C)** Forest plot of the network meta-analysis for disease-free survival. HRs and 95% CIs are given and visually represented by the squares and error bars; **(D)** Bayesian ranking curves of comparable treatments on efficacy for patients with biliary tract cancer. Ranking curves indicate the probability of each treatment ranked first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the curves are presented in **Supplementary Table 8**. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant Chemoradiation therapy; HRs: hazard ratios; 95% CIs: 95% credible intervals; AT: adjuvant therapy.

**Figure 4. Bayesian ranking plots of comparable treatments on efficacy for patients with biliary tract cancer in sensitivity analyses.** Ranking curves indicate the probability of each comparable treatment being ranked from first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the curves are presented in Supplementary Table 9. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant



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723 Chemoradiation therapy.

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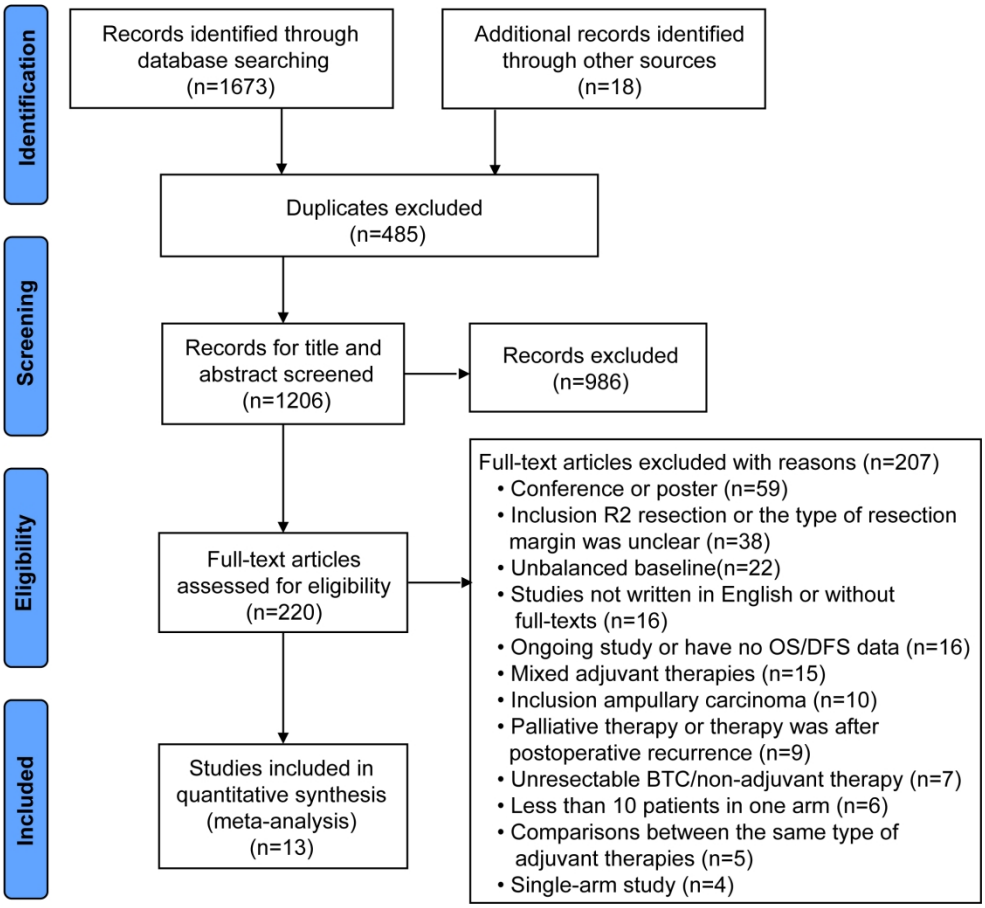



Figure 1. A PRISMA flow diagram of study selection. BTC: biliary tract cancer; OS: overall survival; DFS: disease-free survival.

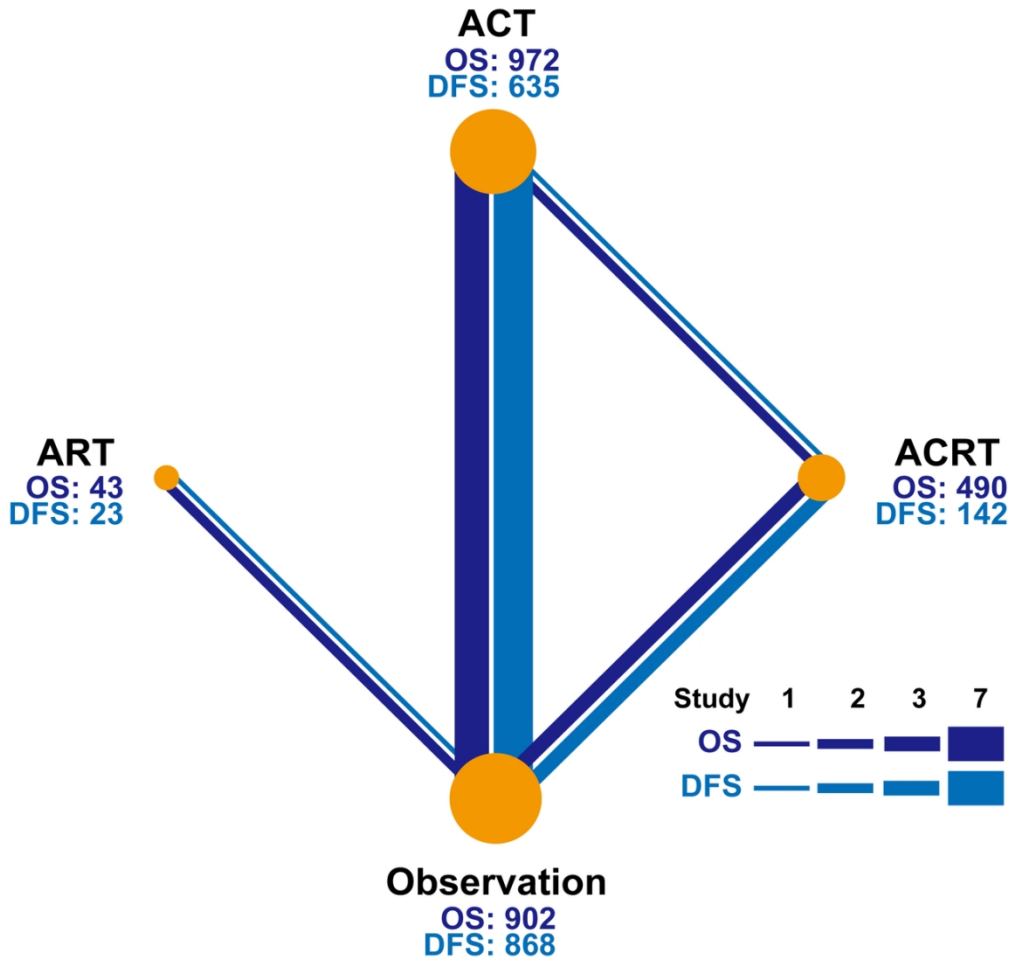



Figure 2. Network plot of comparisons on overall survival and disease-free survival of treatments in patients with biliary tract cancer. Each circular node represents a type of treatment. The node size corresponds to the total number of participants assigned to each treatment. Each line represents a type of direct comparison and its width corresponds to the number of studies evaluating the connected treatments. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant Chemoradiation therapy.

107x103mm (300 x 300 DPI)

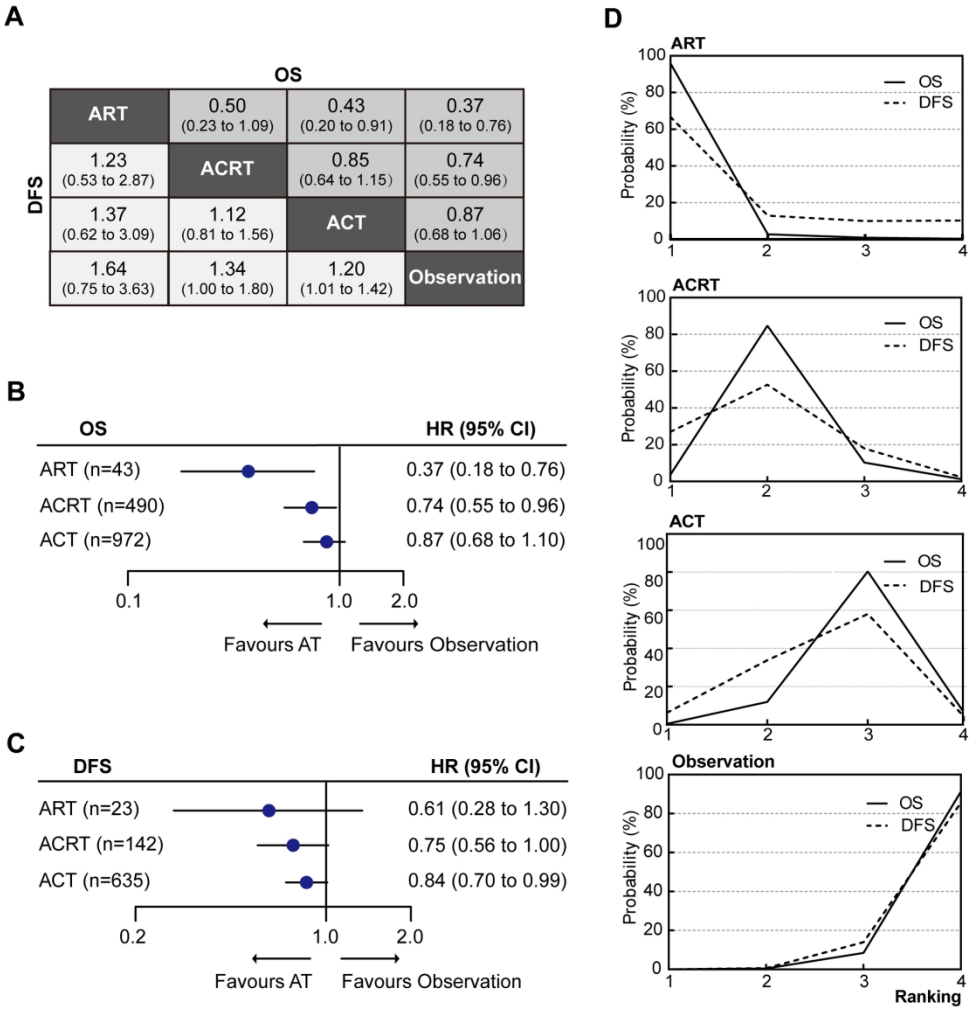



Figure 3. Network meta-analysis results for efficacy outcomes. (A) League table of the network meta-analysis. Pooled HRs and 95% CIs for OS are listed in the upper triangle and DFS are listed in the lower triangle. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. Significant results are in bold; (B) Forest plot of the network meta-analysis for overall survival; (C) Forest plot of the network meta-analysis for disease-free survival. HRs and 95% CIs are given and visually represented by the squares and error bars; (D) Bayesian ranking curves of comparable treatments on efficacy for patients with biliary tract cancer. Ranking curves indicate the probability of each treatment ranked first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the curves are presented in Supplementary Table 8. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant Chemoradiation therapy; HRs: hazard ratios; 95% CIs: 95% credible intervals; AT: adjuvant therapy.

181x182mm (300 x 300 DPI)

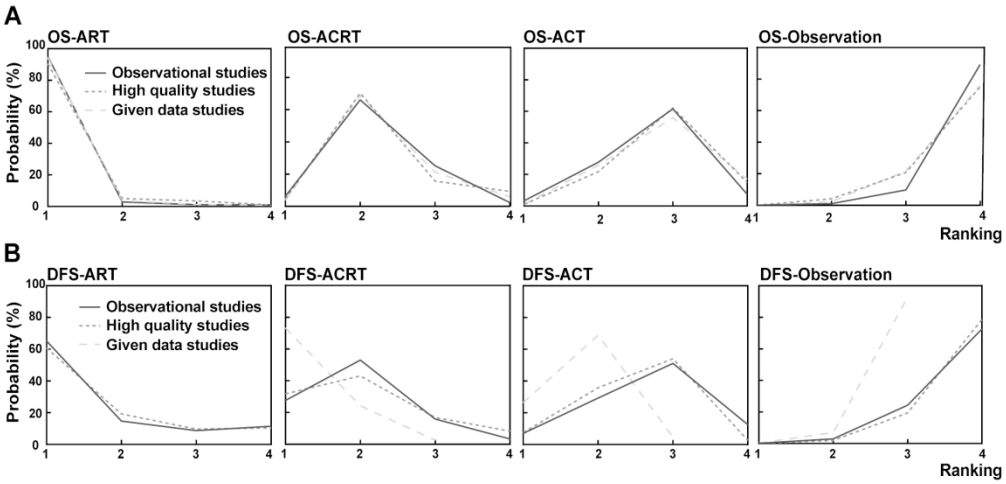



Figure 4. Bayesian ranking plots of comparable treatments on efficacy for patients with biliary tract cancer in sensitivity analyses. Ranking curves indicate the probability of each comparable treatment being ranked from first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the curves are presented in Supplementary Table 9. OS: overall survival; DFS: disease-free survival; ACT: adjuvant Chemotherapy; ART: adjuvant Radiotherapy; ACRT: adjuvant Chemoradiation therapy.

# SUPPLEMENTARYAL MATERIAL

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**Supplementary Table 1. Checklist of the PRISMA extension for network meta-analysis.**

| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review *incorporating a network meta-analysis (or related form of meta-analysis).* | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable:<br>• Background: main objectives;<br>• Methods: data sources; study eligibility criteria, participants, and interventions; study appraisal and *synthesis methods, such as network meta-analysis.*<br>• Results: number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; *treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.*<br>• Discussion/Conclusions: limitations; conclusions and implications of findings.<br>• Other: primary source of funding; systematic review registration number with registry name. | 2 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known, *including mention of why a network meta-analysis has been conducted.* | 3-4 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 4 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | / |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. *Clearly describe eligible treatments included* | 5 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | *in the treatment network, and note whether any have been clustered or merged into the same node (with justification).* | |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 5 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | Supplementaryal Materials page 6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 5 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 6 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 5 |
| Geometry of the network | S1 | Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers. | 6 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 6 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). *Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses.* | 6 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to:<br>• *Handling of multi-arm trials;*<br>• *Selection of variance structure;*<br>• *Selection of prior distributions in Bayesian analyses; and*<br>• *Assessment of model fit.* | 6 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| Assessment of Inconsistency | S2 | Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found. | 6 |
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 6 |
| Additional analyses | 16 | Describe methods of additional analyses, if done, indicating which were pre-specified. This may include, but not be limited to the following:<br>• Sensitivity or subgroup analyses;<br>• Meta-regression analyses;<br>• *Alternative formulations of the treatment network; and*<br>• *Use of alternative prior distributions for Bayesian analyses (if applicable).* | 6 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 7 |
| Presentation of network structure | S3 | Provide a network graph of the included studies to enable visualization of the geometry of the treatment network | 8 & Figure 2 |
| Summary of network geometry | S4 | Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure. | 8 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 7-8 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 7 & Supplementaryal Materials page 7 and page17 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study:1) simple summary data for each intervention group, and 2) effect estimates and confidence/credible intervals. *Modified approaches may be needed to deal with* | 7-8 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | *information from larger networks.* | |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence/credible intervals. *In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise comparisons.* If additional summary measures were explored (such as treatment rankings), these should also be presented. | 8-9 |
| Exploration for inconsistency | S5 | Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, P values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network. | 7 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 9 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression, *alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses,* and so forth [see Item 16]). | 9 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 9-10 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). *Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons).* | 10-11 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 12 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 13 |

PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analysis; PICOS = population, intervention, comparators, outcomes, study design.

*Text in italics indicates wording specific to reporting of network meta-analyses that has been added to guidance from the PRISMA statement.



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**Supplementary Table 2. Search strategies**

| Database | Query | Results |
|---|---|---|
| Pubmed | (((((((((""Biliary Tract Neoplasms""[Mesh]) OR ""Bile Duct Neoplasms""[Mesh]) OR ""Gallbladder Neoplasms""[Mesh]) OR ""Cholangiocarcinoma""[Mesh]) OR Biliary Tract Cancer) OR ""Biliary Tract Neoplasms"") OR ""Bile duct cancer"")) AND ((((((""Chemoradiotherapy, Adjuvant""[Mesh]) OR Radiotherapy, Adjuvant[Mesh]) OR ""Chemotherapy, Adjuvant""[Mesh]) OR ""Adjuvants, Immunologic""[Mesh]) OR ""adjuvant therapy"") OR ""adjuvant treatment"")) AND ((((((""Clinical Study"" [Publication Type]) OR ""Observational Study"" [Publication Type]) OR ""Clinical Trial"" [Publication Type]) OR case control study) OR ""Case-Control Studies""[Mesh]) OR (((random) OR control) OR randomized control trial)) | 349 |
| Embase | (('biliary tract cancer'/exp OR 'biliary tract tumor'/exp OR 'bile duct tumor'/exp OR 'gallbladder tumor'/exp OR 'bile duct carcinoma'/exp OR 'biliary tract cancer' OR 'biliary tract neoplasms' OR 'bile duct neoplasms' OR 'bile duct cancer' OR 'gallbladder cancer' OR cholangiocarcinoma) AND ('adjuvant therapy'/exp OR 'adjuvant chemotherapy'/exp OR 'adjuvant radiotherapy'/exp OR 'adjuvant chemoradiotherapy'/exp OR 'adjuvant treatment') AND ('clinical article'/de OR 'clinical study'/de OR 'clinical trial'/de OR 'clinical trial (topic)'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'methodology'/de OR 'multicenter study'/de OR 'observational study'/de OR 'phase 2 clinical trial'/de OR 'phase 2 clinical trial (topic)'/de OR 'phase 3 clinical trial'/de OR 'phase 3 clinical trial (topic)'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial (topic)'/de OR 'retrospective study'/de) AND ([adult]/lim OR [aged]/lim OR [middle aged]/lim OR [very elderly]/lim OR [young adult]/lim)) | 771 |
| CINAHL | ((Biliary Tract Cancer OR "bile duct cancer" OR "Gallbladder" OR "biliary tract neoplasm" OR "Cholangiocarcinoma") AND ("adjuvant therapy" OR "adjuvant treatment" OR "adjuvant chemotherapy" OR "adjuvant radiotherapy" OR "adjuvant Chemoradiotherapy" OR "adjuvant radiochemotherapy")) | 134 |
| Cochrane | ("MeSH: [Biliary Tract Neoplasms]"/exp OR "MeSH: [cholangiocarcinoma]"/exp OR ("Biliary Tract Cancer"):ti,ab,kw) AND ("MeSH: [Radiotherapy, Adjuvant]"/exp OR "MeSH: [Chemotherapy, Adjuvant]"/exp) OR "MeSH: [Chemoradiotherapy, Adjuvant]"/exp OR "Adjuvant therapy" OR "Adjuvant treatment") | 49 |
| clinicaltrials.gov | (("Biliary Tract Cancer" OR "bile duct cancer" OR "Gallbladder" OR "Cholangiocarcinoma") AND "Adjuvant") | 46 |



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**Supplementary Table 3. Checklist for quality assessment and scoring using Newcastle-Ottawa Scale.**

| **Checklist items** |
|---|
| **Selection** |
| 1. How representative was the treatment group in comparison for BTC?<br>• If yes, one star;<br>• No star if the patients were selected or selection of group was not described. |
| 2. How representative was the observation group in comparison for BTC?<br>• If drawn from the same community as the treatment group, one star;<br>• No star if drawn from a different source or selection of group was not described. |
| 3. Ascertainment of treatment: secure record or structured interview?<br>• If the treatment were confirmed from surgical records or databases (eg NCDB or SEER), one star;<br>• No star if it was not described. |
| 4. Demonstration that outcome of interest was not present at start of study.<br>• Yes, one star;<br>• No, no star. |
| **Comparability** |
| 5. Group comparable for age, sex, primary tumour site, resection margin status, tumour stage, lymph node status, distant metastasis.<br>• If yes, two stars;<br>• One star was assigned if one of these eight characteristics was differed in groups or not reported.<br>• No star was assigned if there are at least 3 characteristics in the groups differed. |
| **Outcome** |
| 6. Assessment of outcome?<br>• One star if for information ascertained by record lincage or independent blind assessment;<br>• No star if this information was not reported. |
| 7. Was follow-up long enough for outcomes to occur?<br>• Yes, one star if the follow-up period was 5 years or more;<br>• No, no star if the follow-up period was less than 5 years or this information was not reported. |
| 8. Adequacy of follow-up (one star if follow-up 90%).<br>• One star if follow-up 90%;<br>• No star if this information was not reported. |

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**Supplementary Table 4. Quality assessment of included retrospective studies.**

| Study | Selection | | | | Comparability | Outcome assessment | | | Score |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | |
| Todorokia 2000 | * | * | * | * | ** | * | * | * | 9 |
| Kim 2011 | * | * | * | * | ** | * | * | - | 8 |
| Kobayashi 2012 | * | * | * | * | ** | * | * | * | 9 |
| Dover 2016 | * | * | * | * | * | * | * | - | 7 |
| Kim1 2016 | * | * | * | * | * | * | * | - | 7 |
| Morine 2017 | * | * | * | * | ** | * | * | - | 8 |
| Akahoshi 2018 | * | * | * | * | ** | * | * | - | 8 |
| Bergeat 2018 | * | * | * | * | ** | * | * | - | 8 |
| Hester 2018 | * | * | * | * | ** | * | * | - | 8 |
| Zheng 2018 | * | * | * | * | ** | * | * | * | 9 |

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**Supplementary Table 5. Assessment of transitivity.**

| Items | Comparisons | Mean Diff. | Std.Error | Sig. | 95% CI |
|---|---|---|---|---|---|
| Age* | ART vs. Observation | -2.62 | 2.31 | 0.67 | -9.18 to 3.93 |
| | ACRT vs. Observation | 0.24 | 2.31 | 0.99 | -6.31 to 6.80 |
| | ACT vs. Observation | 1.31 | 1.82 | 0.89 | -3.87 to 6.49 |
| | ACRT vs. ART | 2.87 | 2.83 | 0.74 | -5.16 to 10.90 |
| | ACT vs. ART | 3.93 | 2.45 | 0.40 | -3.02 to 10.89 |
| | ACT vs. ACRT | 1.07 | 2.45 | 0.97 | -5.89 to 8.02 |
| Percentage female | ART vs. Observation | 2.28 | 9.82 | 0.99 | -24.90 to 29.45 |
| | ACRT vs. Observation | 6.58 | 7.42 | 0.81 | -13.96 to 27.11 |
| | ACT vs. Observation | 7.39 | 5.67 | 0.57 | -8.30 to 23.08 |
| | ACRT vs. ART | 4.30 | 11.13 | 0.98 | -26.51 to 35.11 |
| | ACT vs. ART | 5.12 | 10.05 | 0.96 | -22.69 to 32.93 |
| | ACT vs. ACRT | 0.82 | 7.73 | 0.99 | -20.56 to 22.19 |
| Sample size | ART vs. Observation | -56.25 | 71.79 | 0.86 | -254.90 to 142.40 |
| | ACRT vs. Observation | 48.50 | 54.27 | 0.81 | -101.70 to 198.70 |
| | ACT vs. Observation | 36.14 | 41.45 | 0.82 | -78.56 to 150.80 |
| | ACRT vs. ART | 104.80 | 81.40 | 0.58 | -120.50 to 330.00 |
| | ACT vs. ART | 92.39 | 73.48 | 0.60 | -110.90 to 295.70 |
| | ACT vs. ACRT | -12.36 | 56.48 | 0.99 | -168.70 to 143.90 |
| Publication year | ART vs. Observation | -6.17 | 3.74 | 0.37 | -16.50 to 4.17 |
| | ACRT vs. Observation | 0.08 | 2.82 | > 0.99 | -7.73 to 7.90 |
| | ACT vs. Observation | 2.06 | 2.16 | 0.78 | -3.91 to 8.02 |
| | ACRT vs. ART | 6.25 | 4.24 | 0.47 | -5.47 to 17.97 |
| | ACT vs. ART | 8.22 | 3.82 | 0.17 | -2.36 to 18.80 |
| | ACT vs. ACRT | 1.97 | 2.94 | 0.91 | -6.16 to 10.10 |
| Percentage R0 | ART vs. Observation | -34.85 | 19.45 | 0.30 | -88.66 to 18.97 |
| | ACRT vs. Observation | 4.60 | 14.70 | 0.99 | -36.08 to 45.29 |
| | ACT vs. Observation | 6.63 | 11.23 | 0.94 | -24.44 to 37.70 |
| | ACRT vs. ART | 39.45 | 22.05 | 0.30 | -21.58 to 100.50 |
| | ACT vs. ART | 41.48 | 19.91 | 0.19 | -13.61 to 96.56 |
| | ACT vs. ACRT | 2.03 | 15.30 | 0.99 | -40.31 to 44.37 |
| Percentage N0 | ART vs. Observation | 19.64 | 12.12 | 0.39 | -13.90 to 53.18 |
| | ACRT vs. Observation | -10.56 | 9.16 | 0.67 | -35.91 to 14.80 |
| | ACT vs. Observation | -10.92 | 7.00 | 0.42 | -30.28 to 8.45 |
| | ACRT vs. ART | -30.20 | 13.74 | 0.15 | -68.22 to 7.83 |
| | ACT vs. ART | -30.56 | 12.40 | 0.09 | -64.88 to 3.77 |
| | ACT vs. ACRT | -0.36 | 9.54 | > 0.99 | -26.75 to 26.02 |

The characteristics included age, percentage female, sample size, publication year, percentage R0 and N0 have been evaluated in the network. All the comparisons had similar mean age and other main characteristics with P value over 0.05.

* Median age was given instead of mean age in the Bergeat 2018, Dover 2016 and Edeline 2019 studies. Information of age in the Kobayashi 2012, Akahoshi 2018 and Ebata 2018 studies were presented as younger or older than a specific age that couldn't be integrated in the figure.



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**Supplementary Table 6. Results of global heterogeneity and local heterogeneity.**

| Comparisons | Local | | Global | | | Heterogeneity assessment |
|---|---|---|---|---|---|---|
| | Pair-wise ($I^2$) | Network ($I^2$) | Between study variance ($\tau 2$) | Pair-wise ($I^2$) | Consistency effect ($I^2$) | |
| **Overall survival for BTC patients** | | | | | | |
| ACT vs Observation | 24.9% | 18.3% | 0.025 | 10.1% | 12.7% | low to moderate |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 29.4% | 20.9% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients with R0 resection margin** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.039 | 0.0% | 0.0% | low |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients with R1 resection margin** | | | | | | |
| ACT vs Observation | 41.4% | 41.3% | 0.176 | 46.7% | 46.7% | low to high |
| ART vs Observation | NA | NA | | | | |
| ACRT vs Observation | **54.9%** | **54.9%** | | | | |
| **Overall survival for GBC patients** | | | | | | |
| ACT vs Observation | **72.5%** | **72.1%** | **0.425** | **72.5%** | **61.4%** | moderate to high |
| ACRT vs Observation | NA | NA | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Overall survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.022 | 0.0% | 0.0% | low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | **55.0%** | 32.6% | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Overall survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | **68.9%** | **68.9%** | 0.221 | **54.2%** | **54.3%** | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | NA | NA | | | | |
| **Overall survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.018 | 0.0% | 0.0% | low |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients without distal-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.016 | 0.0% | 0.0% | low |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 28.5% | 9.8% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.011 | 0.0% | 0.0% | low |
| ART vs Observation | NA | NA | | | | |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | NA | NA | | | | |

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| | | | | | | |
|---|---|---|---|---|---|---|
| **Disease-free survival for GBC patients** | | | | | | |
| ACT vs Observation | **76.0%** | **75.6%** | **0.271** | **76.0%** | **57.0%** | Moderate to high |
| ACRT vs Observation | NA | NA | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Disease-free survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.3% | 0.3% | 0.023 | 0.0% | 0.0% | low |
| ART vs Observation | NA | NA | | | | |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | 14.4% | 14.4% | 0.023 | 14.3% | 14.3% | low |
| ART vs Observation | NA | NA | | | | |
| ACRT vs Observation | NA | NA | | | | |
| **Disease-free survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.023 | 0.0% | 0.0% | low |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Disease-free survival for BTC patients for patients without distal-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.019 | 0.0% | 0.0% | low |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | NA | NA | | | | |

The numbers with high heterogeneity are in bold. NA=Not available, because only one study was included in this type of comparison. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma.

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**Supplementary Table 7. Node-splitting analysis of inconsistency.**

| Comparison | Direct effect | Indirect effect | Overall | P |
|---|---|---|---|---|
| | lnHR (95% CI) | | | |
| **Overall survival for BTC patients** | | | | |
| ACRT, ACT | 0.21 (-0.25 to 0.61) | 0.04 (-0.48 to 0.62) | 0.16 (-0.14 to 0.44) | 0.56 |
| ACRT, Observation | 0.25 (-0.10 to 0.69) | 0.42 (-0.20 to 1.10) | 0.30 (0.04 to 0.60) | 0.55 |
| ACT, Observation | 0.18 (-0.07 to 0.48) | -0.06 (-0.76 to 0.76) | 0.14 (-0.06 to 0.39) | 0.44 |
| **Overall survival for GBC patients** | | | | |
| ACRT, ACT | 0.07 (-0.80 to 0.93) | 0.13 (-0.54 to 1.0) | 0.11 (-0.28 to 0.58) | 0.87 |
| **Overall survival for CCA patients** | | | | |
| ACRT, ACT | 0.26 (-0.23 to 0.76) | 0.070 (-0.40 to 0.60) | 0.18 (-0.14 to 0.51) | 0.51 |
| ACRT, Observation | 0.19 (-0.19 to 0.67) | 0.38 (-0.17 to 0.96) | 0.26 (-0.02 to 0.61) | 0.52 |
| ACT, Observation | 0.13 (-0.13 to 0.40) | -0.07 (-0.66 to 0.64) | 0.083 (-0.14 to 0.34) | 0.51 |
| **Overall survival for BTC patients in Asia** | | | | |
| NA | NA | NA | NA | NA |
| **Overall survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.21 (-0.24 to 0.60) | 0.50 (-0.34 to 1.4) | 0.27 (-0.04 to 0.58) | 0.50 |
| ACRT, Observation | 0.45 (-0.05 to 0.97) | 0.34 (-0.40 to 0.99) | 0.41 (0.07 to 0.73) | 0.75 |
| ACT, Observation | 0.13 (-0.22 to 0.44) | 0.32 (-0.64 to 1.3) | 0.14 (-0.12 to 0.39) | 0.67 |
| **Overall survival for BTC patients without distal-metastasis** | | | | |
| ACRT, ACT | 0.21 (-0.17 to 0.55) | 0.10 (-0.34 to 0.63) | 0.19 (-0.07 to 0.45) | 0.67 |
| ACRT, Observation | 0.24 (-0.07 to 0.62) | 0.35 (-0.21 to 0.90) | 0.28 (0.03 to 0.53) | 0.67 |
| ACT, Observation | 0.12 (-0.13 to 0.37) | -0.07 (-0.65 to 0.62) | 0.09 (-0.11 to 0.30) | 0.50 |
| **Disease-free survival for BTC patients** | | | | |
| ACRT, ACT | 0.09 (-0.44 to 0.62) | 0.18 (-0.26 to 0.63) | 0.11 (-0.21 to 0.44) | 0.79 |
| **Disease-free survival for GBC patients** | | | | |
| — | — | — | — | — |
| **Disease-free survival for CCA patients** | | | | |
| — | — | — | — | — |
| **Disease-free survival for BTC patients in Asia** | | | | |
| — | — | — | — | — |
| **Disease-free survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.09 (-0.51 to 0.69) | 0.23 (-0.51 to 0.99) | 0.09 (-0.33 to 0.55) | 0.76 |
| **Disease-free survival for BTC patients without distal-metastasis** | | | | |
| ACRT, ACT | 0.091 (-0.43 to 0.61) | 0.21 (-0.22 to 0.67) | 0.14 (-0.18 to 0.48) | 0.71 |

Nonsignificant values (P >0.05) indicating no inconsistency between the direct effect and indirect effect. "—"=Not available; CI=confidence interval. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 8. Comparisons of the fitness of consistency and inconsistency models using deviance information criteria**

| Item | Model | Overall | Resection margin status | | Tumor Site | | The regions | | Non-distal metastasis |
|---|---|---|---|---|---|---|---|---|---|
| | | | R0 | R1 | GBC | CCA | Asia | Western | |
| OS | Consistency | 19.20 | 7.21 | 11.18 | 8.60 | 14.79 | 10.72 | 8.31 | 14.69 |
| | Inconsistency | 19.90 | 7.16 | 11.19 | 8.57 | 15.70 | 10.73 | 10.01 | 15.77 |
| DFS | Consistency | 13.95 | NA | NA | 6.30 | 13.54 | 9.86 | 7.88 | 11.71 |
| | Inconsistency | 14.03 | NA | NA | 6.29 | 13.60 | 10.16 | 7.89 | 11.82 |

The fitness of the Bayesian model was evaluated by deviance information criteria (DIC), which is adjusted with complexity of the model.

NA=Not applicable; OS=overall survival; DFS=disease-free survival; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma

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**Supplementary Table 9. Ranking results for overall survival (OS) and disease-free survival (DFS).**

| Treatment | Rank of probability for OS (%) | | | | | Rank of probability for DFS (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| ACT | 0.4 | 12.0 | 80.4 | 7.2 | 35.2 | 6.3 | 33.8 | 58.0 | 1.9 | 48.1 |
| ART | **95.8** | 2.8 | 1.0 | 0.4 | 98.0 | **66.7** | 13.0 | 10.0 | 10.3 | 78.7 |
| ACRT | 3.7 | 84.7 | 10.3 | 1.3 | 63.6 | 27.1 | 52.6 | 18.0 | 2.4 | 68.1 |
| Observation | 0.0 | 0.5 | 8.4 | **91.1** | 3.1 | 0.0 | 0.6 | 14.0 | **85.4** | 5.1 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 10. Rank results for OS and DFS in the sensitivity analysis.**

| Treatment | Rank of probability (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| | OS for studies including observational studies | | | | | DFS for studies including observational studies | | | | |
| ACT | 3.3 | 27.8 | 61.5 | 7.5 | 42.3 | 7.1 | 29.3 | 51.1 | 12.6 | 43.6 |
| ART | **90.5** | 5.1 | 3.6 | 1.0 | 95.0 | **65.2** | 14.7 | 8.7 | 11.4 | 77.9 |
| ACRT | 6.3 | 66.1 | 25.2 | 2.4 | 58.8 | 27.6 | 53.1 | 15.9 | 3.4 | 68.3 |
| Observation | 0.0 | 1.1 | 9.7 | **89.2** | 4.0 | 0.2 | 3.0 | 24.3 | **72.6** | 10.2 |
| **Treatment** | **OS for studies with high quality** | | | | | **DFS for studies with high quality** | | | | |
| ACT | 1.0 | 21.8 | 62.2 | 15.0 | 36.3 | 7.5 | 35.8 | 54.1 | 2.7 | 49.4 |
| ART | **94.3** | 3.7 | 1.3 | 0.7 | 97.2 | **60.7** | 19.2 | 9.7 | 10.4 | 76.7 |
| ACRT | 4.6 | 70.4 | 15.7 | 9.3 | 56.8 | 31.8 | 43.0 | 16.9 | 8.3 | 66.1 |
| Observation | 0.1 | 4.1 | 20.8 | **75.0** | 9.8 | 0.1 | 2.0 | 19.4 | **78.6** | 7.8 |
| **Treatment** | **OS for original reported HRs (95%CI) in the studies** | | | | | **DFS for original reported HRs (95%CI) in the studies** | | | | |
| ACT | 1.0 | 26.0 | 56.0 | 17.0 | 37.0 | 26.4 | 68.9 | 4.7 | NA | 60.9 |
| ART | **95.0** | 2.9 | 1.1 | 1.0 | 97.3 | NA | NA | NA | NA | NA |
| ACRT | 3.9 | 68.9 | 21.2 | 6.0 | 56.9 | **73.3** | 24.2 | 2.6 | NA | 85.4 |
| Observation | 0.1 | 2.3 | 21.7 | **76.0** | 8.8 | 0.3 | 6.9 | **92.8** | NA | 37.9 |

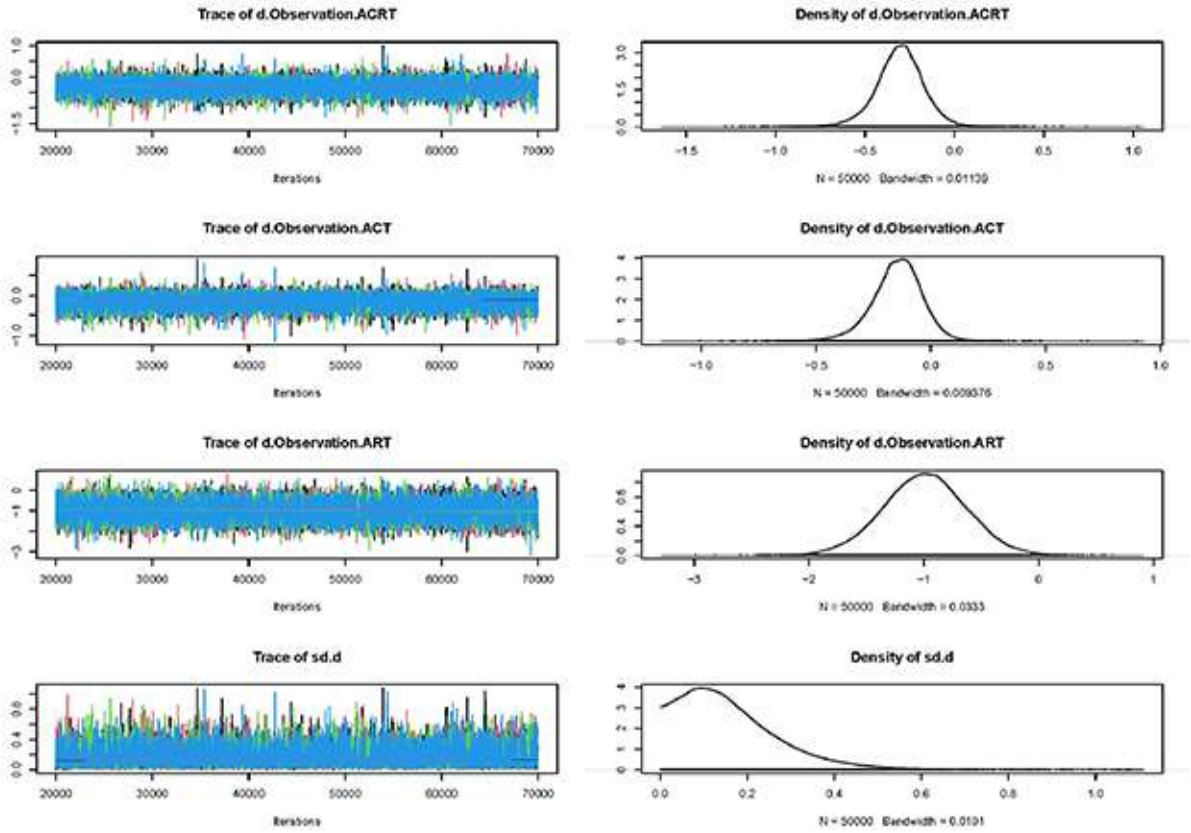
The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy. CI=confidence interval; OS=overall survival; DFS=disease-free survival.

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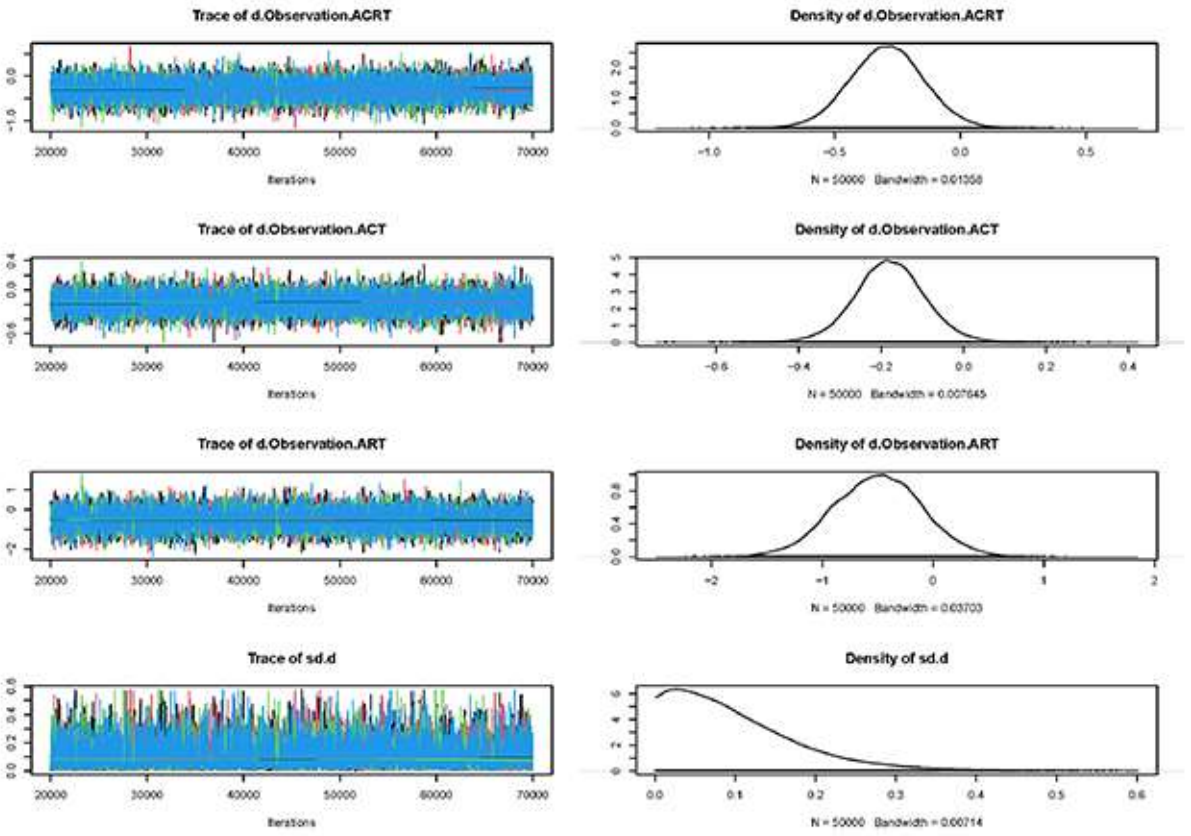


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## A. The trace plot and density plot for overall survival

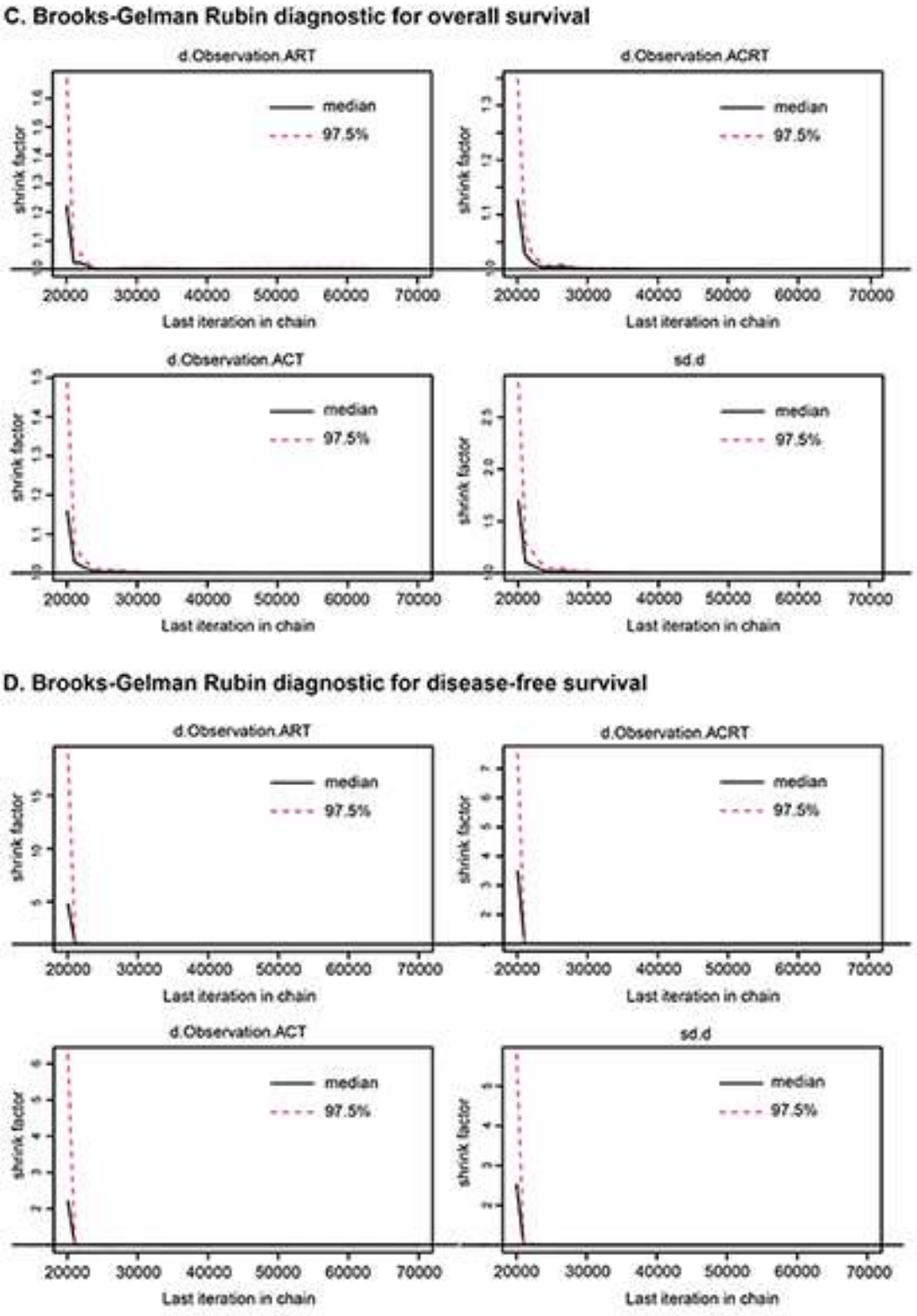



## B. The trace plot and density plot for disease-free survival





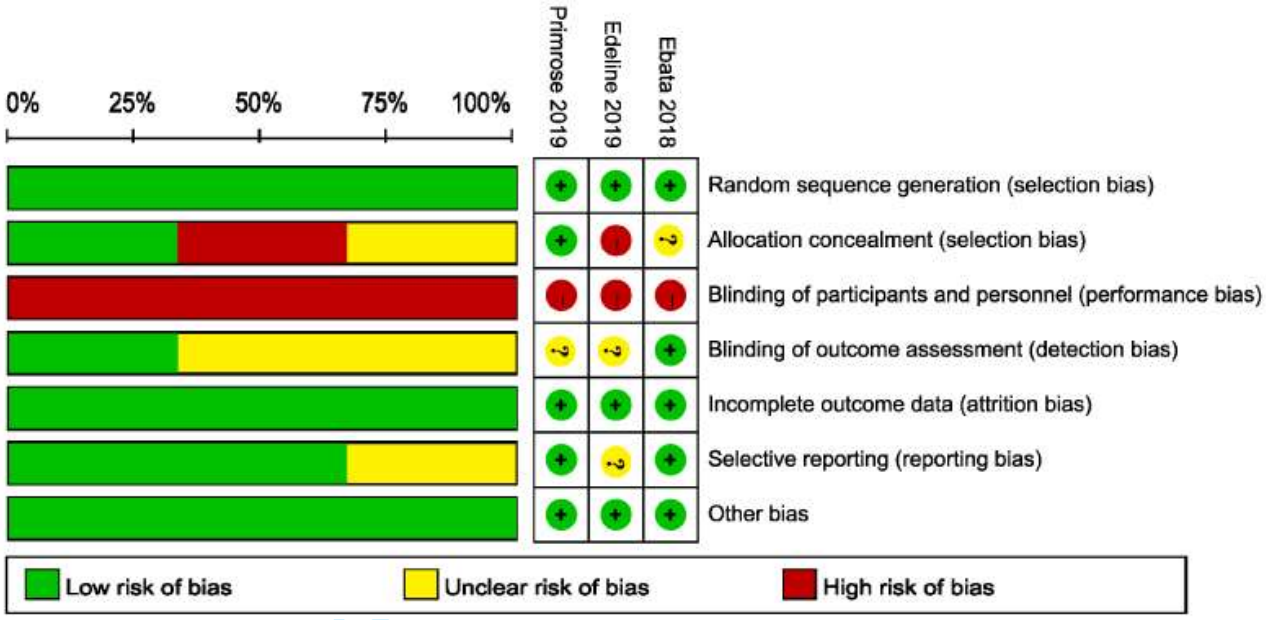

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**Supplementary Figure 1. Convergence of the consistency model by inspection of the history feature.** The trace plots and density plots for overall survival **(A)** and disease-free survival **(B)**. The Brooks-Gelman Rubin diagnostic for overall survival **(C)** and disease-free survival **(D)**.

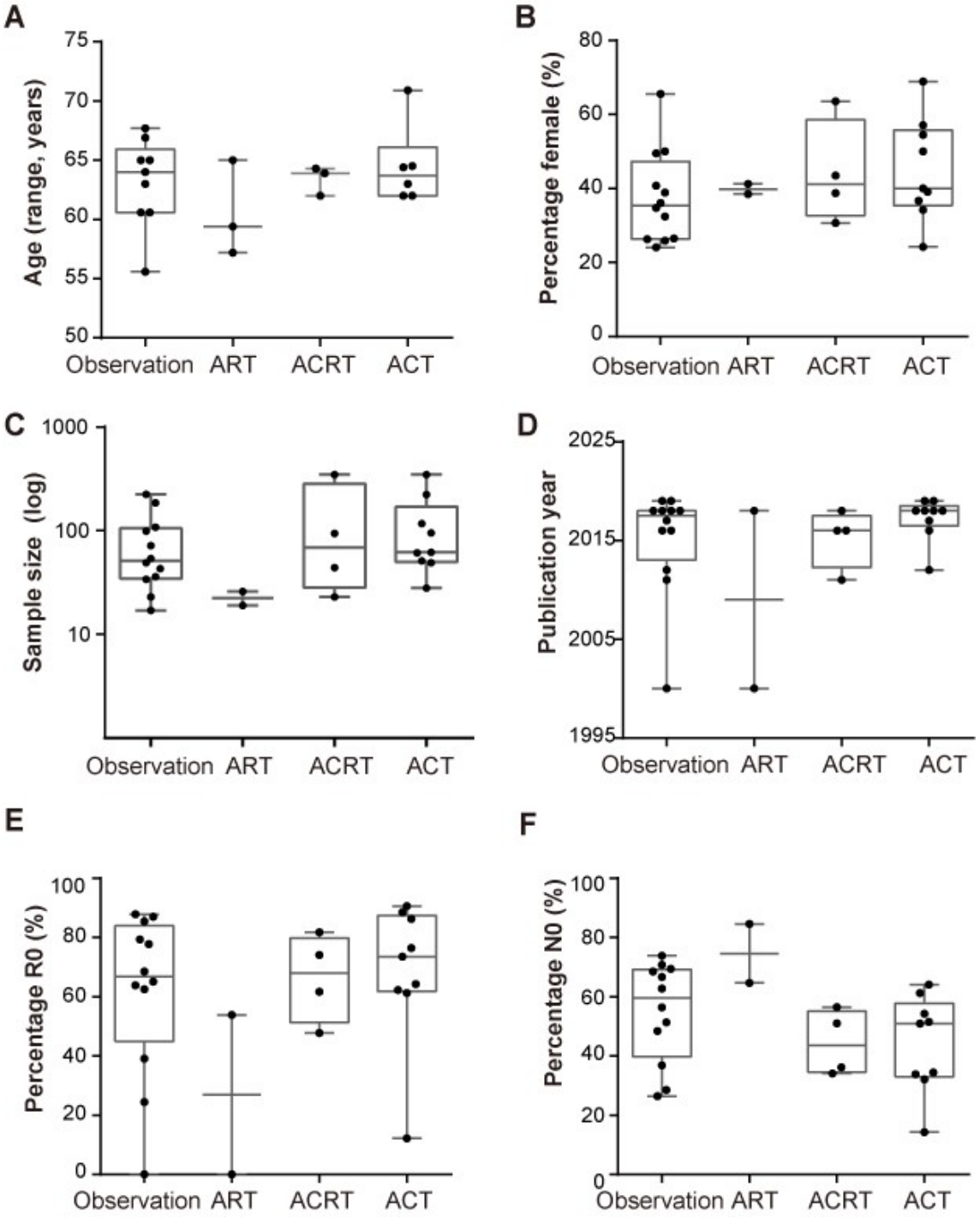


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**Supplementary Figure 2. Summary of results from quality assessment of three randomized controlled studies using the Cochrane risk of bias tool.**

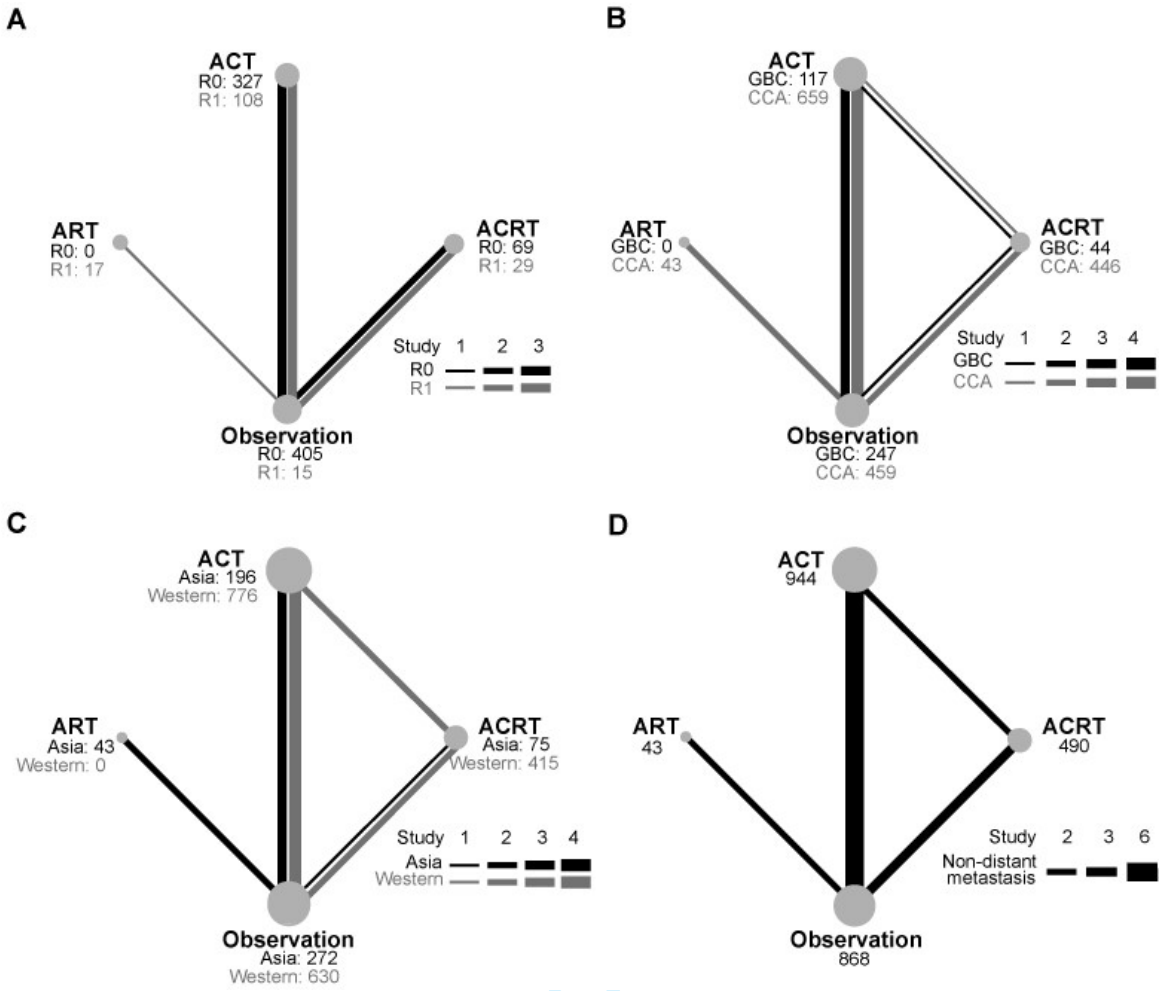
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**Supplementary Figure 3. Boxplots for distribution of mean age (A), percentage female (B), sample size (C), publication year (D), percentage R0 (E) and N0 (F) across comparisons.**



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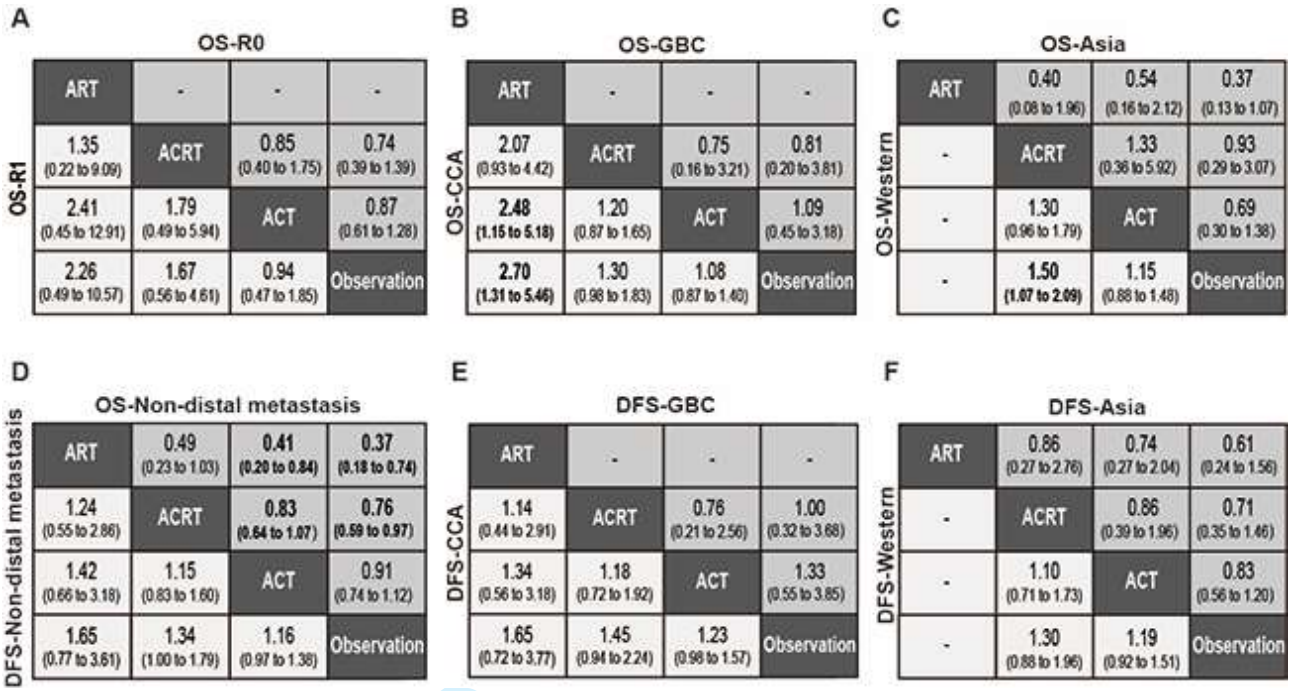



**Supplementary Figure 4. Network plots of comparisons on overall survival of treatments in subgroups analyses (patients stratified into R0, R1, GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** Comparisons on overall survival in patients with GBC and CCA; **(C)** Comparisons on overall survival in subgroups of study patients in Asia and Western; **(D)** Comparisons on overall survival in patients without distal metastasis.

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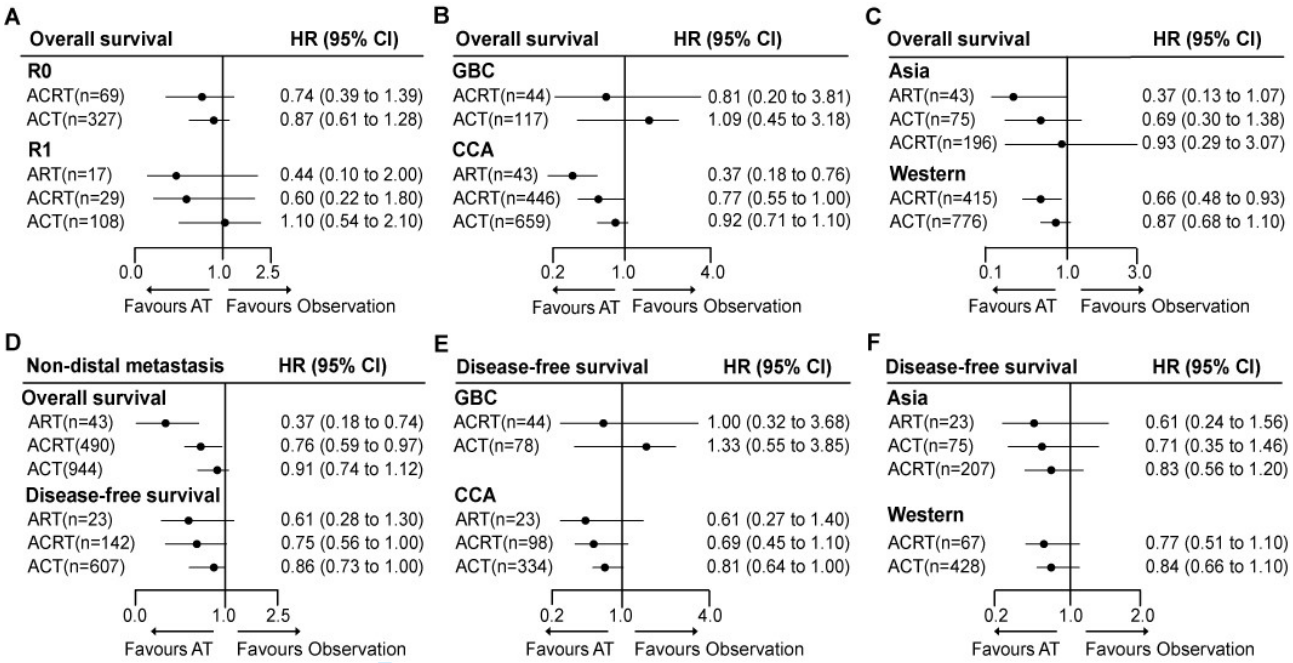



**Supplementary Figure 5. League tables of the network meta-analysis in subgroup analyses. (A)** League table of overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** and **(E)** League tables of overall survival and disease-free survival in subgroups of study patients with GBC and CCA; **(C)** and **(F)** League tables of overall survival and disease-free survival in subgroups of study patients in Asia and Western; **(D)** League table of overall survival and disease-free survival in subgroups of study patients without distal metastasis. Pooled hazard ratios (HRs) and 95% credible intervals (95% CIs) are listed in each cell.

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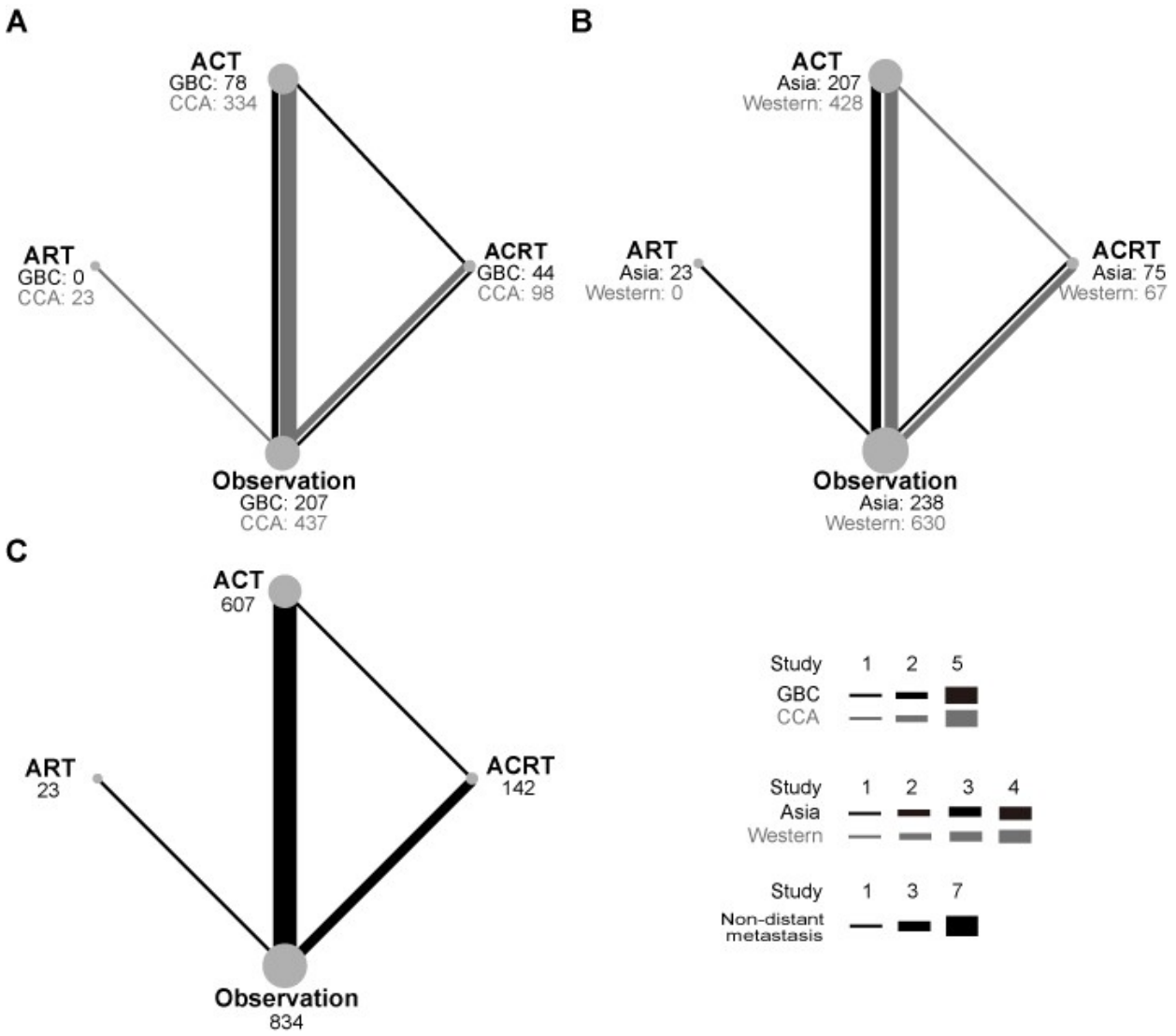



**Supplementary Figure 6. Forest plots of the network meta-analysis for OS and DFS in subgroups analyses**. **(A)** Forest plots of overall survival in the patients stratified into R0 and R1; **(B)** and **(E)** Forest plots of overall survival and disease-free survival in the patients stratified into GBC and CCA; **(C)** and **(F)** Forest plots of overall survival and disease-free survival in the patients stratified into Asia and Western; **(D)** Forest plots of overall survival and disease-free survival in non-distal metastasis patients. Effect sizes are presented as hazard ratios (HRs) with 95% confidence intervals (95% CIs).

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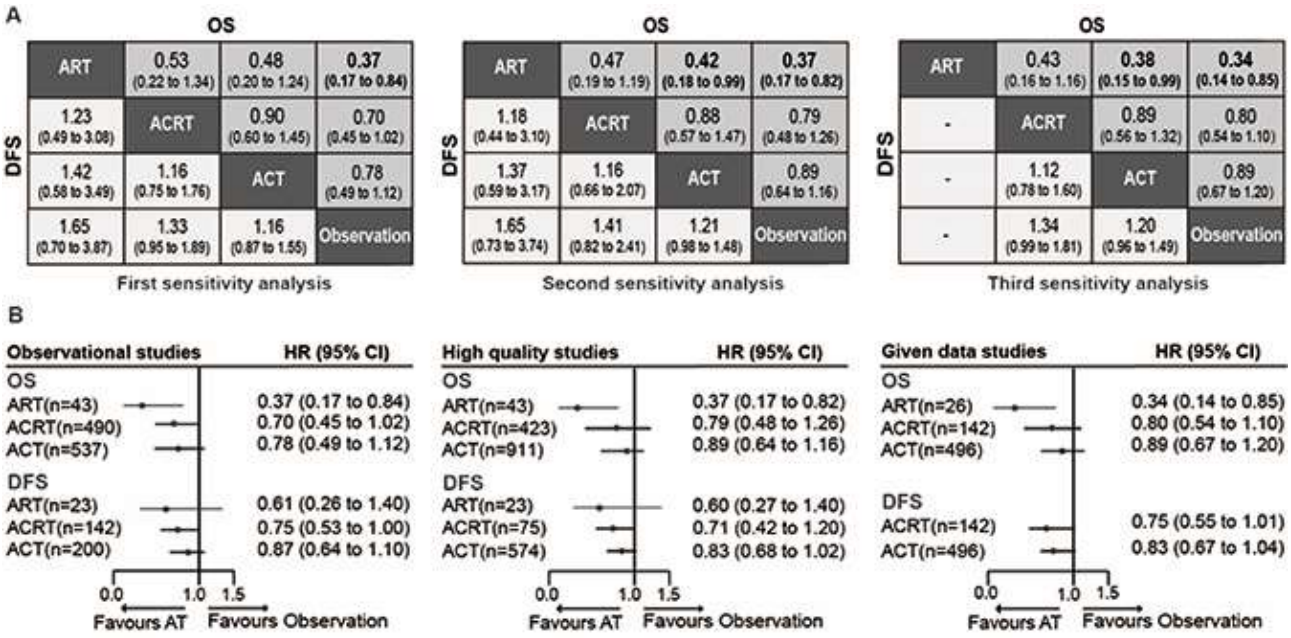
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**Supplementary Figure 7. Network plots of comparisons on disease-free survival of treatments in subgroups analyses (patients stratified into GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on disease-free survival in subgroups of study patients with GBC and CCA; **(B)** Comparisons on disease-free survival in subgroups of study patients in Asia and Western; **(C)** Comparisons on disease-free survival in subgroup of study patients without distal metastasis.



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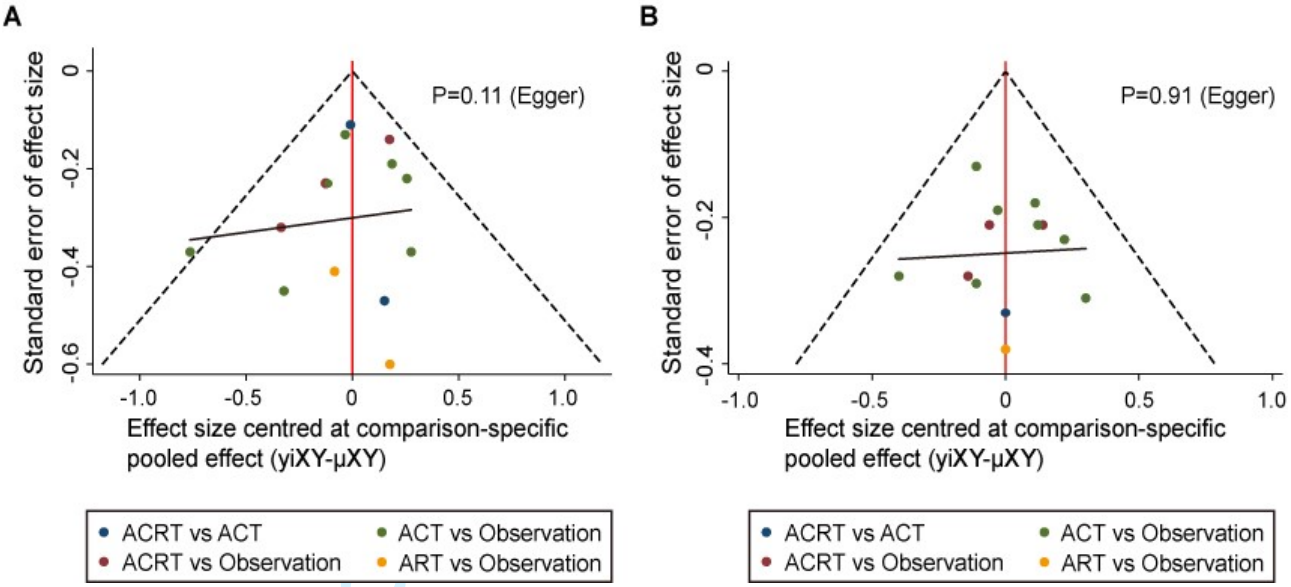



**Supplementary Figure 8. Pooled estimates of the sensitivity analysis. (A)** League tables of the network meta-analysis in sensitivity analyses; **(B)** Forest plots of the network meta-analysis in sensitivity analyses. Adjuvant treatments are ranked according to their surface under the curve cumulative ranking and compared with observation. Effect sizes are presented as Hazard ratios (HRs) with 95% confidence intervals (95% CI). OS: overall survival; DFS: disease-free survival. First sensitivity analysis: excluding RCT studies; Second sensitivity analysis: removing low-and moderate-quality observational studies; Third sensitivity analysis: only including the studies for which HRs were reported in the original articles.

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**Supplementary Figure 9. The 'comparison-adjusted' funnel plot in network meta-analysis to assess funnel plot asymmetry on the efficacy outcomes. (A)** Funnel plot asymmetry on the overall survival; **(B)** Funnel plot asymmetry on the disease-free survival. Funnel-plot asymmetry was tested b Egger's regression tests. No publication bias is detected when the P value is larger than 0.05.

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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis



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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis

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## ABSTRACT

**Objective:** This study aimed to determine the benefits of adjuvant therapy in patients with resected biliary tract cancer (BTC) and identify the optimal adjuvant treatment scheme.

**Design:** Systematic review and network meta-analysis.

**Data sources:** Studies comparing different adjuvant therapies in patients with BTC were searched in PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials, and clinicaltrials.gov databases from inception to December 2021. Additionally, the references were manually searched for the related literature.

**Materials and methods:** Eligible studies were identified, and data were extracted independently by two authors. A random-effects network meta-analysis was performed using R software. The pooled outcomes of overall survival (OS) and disease-free survival (DFS) were measured using the combined hazard ratios (HRs) with 95% confidence intervals (95% CIs).

**Results:** Nineteen eligible studies reporting three types of adjuvant therapies were included in our network meta-analysis. Adjuvant radiotherapy (ART, HR, 0.62; 95% CI, 0.42 to 0.93), adjuvant chemoradiotherapy (ACRT; HR, 0.71; 95% CI, 0.54 to 0.83), and adjuvant chemotherapy (ACT; HR, 0.84; 95% CI, 0.68 to 0.98) were more effective in prolonging OS than that of observation, with no significant difference between the three adjuvant therapies. Moreover, the improvement in DFS was also




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found in ACRT and ACT compared with that of observation (HR, 0.60; 95% CI, 0.45 to 0.75; HR, 0.82; 95% CI, 0.68 to 0.97, respectively). The DFS benefit of ACRT was greater than that of ACT (HR, 0.73; 95% CI, 0.53 to 0.95).

**Conclusions:** Our primary results demonstrated that, compared with that of observation, ACRT and ACT after radical resection could provide better OS and DFS benefits in patients with BTC. Although ACRT shows superior DFS outcomes compared to that of ACT, prospective randomized controlled trials are still needed.

**Keywords:** Biliary tract cancer, adjuvant chemotherapy, adjuvant radiotherapy, adjuvant chemoradiotherapy, systematic review, network meta-analysis

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## Strengths and limitations of this study

- This network meta-analysis compared the efficacy of three major types of adjuvant treatments for BTC patients after radical resection.
- This review was conducted with a strict selection criterion to lower the risk of bias and increase the homogeneity, including only patients with negative resection margins (R0) or microscopic positive resection margins (R1), but not macroscopic involvement resections (R2), and excluding studies with unbalanced or unclear baseline profiles in age, sex, disease severity, or residual tumour status.
- A comprehensive analysis was performed, including the transitivity assumption, local inconsistency of the model, global and local heterogeneity, sensitivity analysis, subgroup analysis, rank probability, and publication bias.
- All studies regarding ART and ACRT were retrospective studies, which were inherent to selection bias.
- Because of the small sample sizes of ART, the claim of optimal treatment for patients with BTC should be interpreted with caution. The optimal regimens and dosing schedules for ACT and ACRT need to be further explored.

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## INTRODUCTION

Biliary tract cancer (BTC) is classified into gallbladder cancer (GBC), intrahepatic cholangiocarcinoma (iCCA), and extrahepatic cholangiocarcinoma (eCCA), and eCCA is further subdivided into perihilar cholangiocarcinoma (pCCA or Klatskin tumour) and distal cholangiocarcinoma (dCCA).[1] BTC accounts for approximately 3% of digestive system cancers and 10%–15% of primary liver cancers.[2] The incidence of BTC is higher in Asian countries and lower in European countries, the United States, and Australia; however, its incidence is increasing globally.[2, 3] The vast majority (>90%) of BTC cases were adenocarcinoma. Moreover, most cases are usually in the advanced or metastatic stage at the initial diagnosis. Only approximately 20% of BTC cases are considered resectable.[4]

Surgical resection remains the primary curative treatment in patients with resected BTC. However, the high recurrence rates (including locoregional or distant recurrence) and low survival rates (5-year survival rates of patients ranging from 5% to 15%) are prominent problems, even with complete (R0) resection.[4-8] Previously published studies have shown that histologic margin status, lymph node (LN) involvement, and intrahepatic metastasis are the main prognostic factors.[9-11] Adjuvant chemotherapy (ACT), adjuvant radiotherapy (ART), and adjuvant chemoradiotherapy (ACRT) are the main options following resection for adjuvant therapy.

A variety of guidelines have suggested that postoperative adjuvant therapy could



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be considered an option for BTC patients.[12-15] The use of ACT and ACRT for patients with GBC and eCCA was supported by the National Comprehensive Cancer Network (NCCN) guidelines. Adjuvant capecitabine is recommended by the American Society of Clinical Oncology (ASCO) guidelines for patients with resected BTC for six months, based on the results of the BILCAP study, which showed a protocol-specified adjusted overall survival (OS) hazard ratio (HR) of 0.71 (95% confidence interval [CI] 0.55–0.92).[16] However, some experts still hold reservations regarding the BILCAP study due to the underpowered statistical design and concerns over data maturity.[17] The discrepancy between the BILCAP study and the PRODIGE-12/ACCORD-18 studies also results in more discussions on the optimal scheme of adjuvant therapy.[18] Despite the tremendous effort, universal agreement on the optimal scheme of adjuvant therapy has not been established due to the lack of high-quality evidence.[19]

Moreover, conflicting results were also observed in the meta-analyses on this topic. A previous study by Rangarajan et al. showed a significant improvement in OS with adjuvant therapy after resection compared with that of resection only.[20] In contrast, Horgan et al. reported a non-significant improvement in OS with adjuvant therapy compared with that of observation in the overall population. However, a more significant benefit from adjuvant therapy was found in patients with positive LN involvement and R1 resection.[21] Three network meta-analyses of adjuvant therapies in patients with resected BTC have been published previously. One made comparisons among ACT, ACRT, and resection only and did not consider ART.[22]



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One only summarized three randomized clinical trials (RCTs) covering three different ACT regimens, and the robustness of their results was limited due to the very small sample sizes.[23] A recent study indicated that ACRT could provide a survival benefit in patients with positive margins or nodal involvement, but the inclusion of patients with R2 margins is still debatable.[24]

Therefore, it is essential to perform a new network meta-analysis to elucidate the efficacy of adjuvant therapy and identify the optimal scheme of adjuvant therapy for patients with resected BTC.

## METHODS

The study design was built in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension statement for network meta-analysis for healthcare (**Supplementary Table 1**).[25]

### Search Strategy

PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials (CENTRAL), and ClinicalTrials. gov databases were searched from inception to December 2021 to find relevant literature using the main keywords “biliary tract cancer” and “adjuvant therapy.” In addition, references in the relevant literature were manually checked for potentially relevant papers. The detailed search strategy is presented in **Supplementary Table 2**.

### Selection Criteria



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Studies that met the following criteria were included: (1) those including patients with histologically or cytologically confirmed resected BTC and treated with adjuvant therapy or observation after curative-intent resection (defined as negative resection margins [R0] or microscopic positive resection margins [R1], but not macroscopic involvement resections [R2]); (2) studies reporting at least one of the following clinical outcomes, OS or disease-free survival (DFS); and (3) studies with treatment and control arms.

Studies that involved the following were excluded: (1) studies including patients with ampullary carcinomas or other primary cancers or neoadjuvant therapy, palliative therapy, or therapy after postoperative recurrence; (2) studies comparing the same type of adjuvant therapies; (3) those with unbalanced baseline profiles in age, sex, disease severity (American Joint Committee on Cancer staging), or residual tumour status; (4) reviews, conference abstracts, posters, or case reports; and (5) studies not written in English or without a full text, or with a small sample size (<10 in any group).

All study titles and abstracts were screened, and then the full texts of potentially eligible articles were sequentially assessed for final inclusion.

**Data Extraction and Quality Assessment**

The following details were extracted from each study: author, study period, country, study type, sample size, tumour site (BTC type), disease severity, age, female sex (%), resection margin (RM) status, LN status, OS, and DFS. OS was defined as the



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period from the date of surgery (or randomization in RCTs) to the date of death (or last follow-up). DFS was defined as the time from the surgery date (or randomization in RCTs) to recurrence of tumours (locoregional or distant). Moreover, the authors were contacted when there was confusion or missing data in any article. Data were excluded or not considered if no response was received.

The quality of the studies was assessed using the Newcastle–Ottawa Scale (NOS) for observational studies, based on the following domains: selection, comparability, and outcome.[26] The detailed information can be found in **Supplementary Table 3**. A study that scored 8–9 points, of which 2 points for comparability, is considered high quality; a score of 4–7 points indicates moderate quality, and 0–3 means low quality. The Cochrane Risk of Bias Tool was used to grade the RCT quality.[27]

Two investigators (Ye Chen and Baoxia Zhang) independently conducted the study selection and data extraction, and four authors (Ye Chen, Baoxia Zhang, Chang Liu, and Ye Cao) assessed the risk of bias of each eligible study. Discrepancies were resolved by consensus and arbitration by other investigators (Cheng Lyu and Meng Qiu).

**Data Synthesis and Statistical Analysis**

The primary outcomes were OS and DFS and measured using the HR with a 95% CI. When HRs and 95% CIs for OS and DFS were not reported in the original article, they were extracted from survival curves using Engauge Digitizer 10.9 (2014 Mark Mitchell) and estimated by the method suggested by Tierney.[28]



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A Bayesian network meta-analysis was performed using the R software (version 4.0.0, https://www.r-project.org/). Network plots were generated for different outcomes to clarify which treatments were compared directly or indirectly. A random-effects model (including subgroup analyses based on the RM status, tumour sites, regions, and patients without distal metastasis) was used to compare all direct and indirect evidence using "Rjags" and "gemtc" packages in the R software. To fit the non-informative uniform and normal prior distributions, the parameters were set with four chains, 50,000 sample iterations (n.inter), and 20,000 burn-ins (n.adapt) with a thinning interval of 1. The convergence of the chains was assessed using the Gelman–Rubin statistics and inspection of the trace plots (**Supplementary Figure 1**).[29, 30] The deviance information criterion (DIC) was used to test the goodness of fit of consistent and inconsistent models.[31]

The transitivity assumption was evaluated by comparing the distribution of clinical variables (age, percentage of females, sample size, publication year, RM status, LN status), which could be effect modifiers. The local inconsistency of the model was evaluated using the node-splitting approach.[32, 33] The global and local heterogeneity was assessed via between-study variance $\tau^2$ and $I^2$ inconsistency statistic.[34-36] Heterogeneity was considered low, moderate, and high for estimated $\tau^2$ or $I^2$ values < 25%, between 25% and 50%, and larger than 50%, respectively. Within the Bayesian framework, the network meta-analysis provided a ranking probability of each treatment and estimated the overall rankings by calculating the surface under the cumulative ranking (SUCRA).[37, 38] To assess the reliability of



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results, sensitivity analyses for OS and DFS were performed by excluding RCT studies, only including high-quality studies, and only including the studies for which HRs were reported in the original articles. The "netmeta" package in the R software was used to generate the comparison-adjusted funnel plots to visualize publication bias. Egger's test was used to assess funnel plot asymmetry.[39, 40] *P*-values < 0.05 were considered statistically significant. All statistical tests were two-sided.

## Patient and Public Involvement

Patients or the public were not involved in our research design, conduct, reporting, or dissemination plans.

## RESULTS

### Characteristics of the Included Studies

A total of 2,146 records were identified, including 49 records that were manually searched by reviewing the references of relevant publications. After reviewing the abstracts, 264 full-text articles were retrieved and reviewed after excluding 688 duplicates and 1,194 ineligible records (**Figure 1**). Nineteen studies comprising a total of 5,595 patients who received one of the following four different treatments after radical resection—ACT, ART, ACRT, and surgical resection alone (observation group)—were included in our analysis. It is worth noting that 34 studies[11, 41-73] were excluded because of unbalanced or unclear baselines. Of the included patients, 2,774 patients received adjuvant treatments following curative-intent resection, and



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2,821 patients underwent curative-intent resection without adjuvant treatments.

The main characteristics of the studies included in the network meta-analysis are presented in **Table 1**. These 19 studies consisted of 3 RCTs[16, 74, 75] and 16 retrospective studies[76-91]. The patients of 6 studies[16, 75, 79, 83, 84, 91] were Westerners (Americans, French, and British) and 13 studies[74, 76-78, 80-82, 85-90] were Asian (Japanese, Korean, Indian, and Chinese). Four studies[80, 87, 88, 91] assessed patients with GBC, and 11 studies[74, 76-79, 82-86, 89] assessed patients with CCA. Of the 11 studies, 2 studies[85, 86] particularly examined the outcomes of iCCA, and 6 studies[74, 76, 77, 83, 84, 89] examined the outcomes of eCCA, including pCCA[76, 89] and dCCA[83, 84]. In the remaining 4 studies[16, 75, 81, 90], the BTC subtypes involved were varied.

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**Table 1. Characteristics of included studies.**

| Studies | Country | Study period | BTC type | Disease severity | Therapy | Regimen of CT/ type of RT | Sample size | Margin status (R0/R1) | Nodal status (negative/positive) | Female No. (%) | Age, Mean (SD/range), years | 5-Year OS (%); MOST (months) | 5-Year DFS (%); MDFST (months) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Characteristics of included observational studies** | | | | | | | | | | | | | |
| **ACT vs Observation** | | | | | | | | | | | | | |
| Kobayashi 2012[78] | Japan | 1989-2010 | CCA | I-IV | ACT<br>Observation | GEM/S-1<br>- | 51<br>54 | 39/12<br>37/17 | N0:26; N+: 25<br>N0: 31; N+: 23 | 28 (55)<br>14 (26) | - [a] | 46.0; -<br>23.0; - | - |
| Morine 2017[81] | Japan | 1995-2012 | CCA/ GBC | II-IVb T1-T4 | ACT<br>Observation | GEM+5-FU+Cis<br>- | 28<br>34 | 18/10<br>27/7 | 9/19<br>9/25 | 16 (57)<br>11 (32) | 70.9 (± 6.8)<br>66.9 (± 9.0) | 44.2; -<br>5.9; - | - |
| Akahoshi 2018[82] | Japan | 2004-2015 | CCA | 0-IVa T1-T4 | ACT<br>ACT<br>Observation | GEM; S-1<br>-<br>- | 26<br>36<br>36 | 15/11; 23/13<br>23/13 | 15/11;<br>23/13<br>25/11 | 9 (35)<br>6 (17)<br>13 (36) | - [a] | - | 25.6; -<br>34.0; -<br>26.8; - |
| Bergeat 2018[83] | France | 2000-2015 | dCCA | Ia-III | ACT<br>Observation | GEM<br>- | 49<br>49 | 12/37<br>6/43 | 7/42<br>14/35 | 18 (37)<br>20 (41) | 62.0 (36-77) [b]<br>65.0 (35-82) [b] | - ; 26.3<br>- ; 43.3 | - ; 15.5<br>- ; 14.7 |
| Miyata 2021[90] | Japan | 2007-2018 | GBC/ pCCA/ dCCA | I-IV | ACT<br>Observation | S-1<br>- | 38<br>38 | 29/9<br>28/10 | N0: 21; N1: 17<br>N0: 22; N1: 16 | 13 (34)<br>16 (42) | 72 (52-82) [b]<br>74 (42-85) [b] | 71.0; -<br>42.9; 28.2 | 51.6; 61.2<br>35.9; 13.1 |
| **ART vs Observation** | | | | | | | | | | | | | |
| Todorokia 2000[76] | Japan | 1976-1999 | pCCA (HDC) | IVa | ART<br>Observation | EBRT<br>- | 17<br>19 | All R1 | N0:11; N1:2; N2:4<br>N0:7; N1:4; N2:8 | 7 (41)<br>5 (26) | 59.4 (34-76)<br>60.6 (19-82) | 33.9 [d]; 32.0<br>13.5 [d]; 10.0 | - |
| Jiang 2010[86] | China | 1998-2008 | iCCA | - | ART<br>Observation | EBRT | 24<br>66 | - | N1: 8; Nx: 16<br>N1: 23; Nx: 43 | 7 (29)<br>31 (47) | - [a] | 11.9; 19.1<br>9.5; 1.5 | - |
| Zheng 2018[85] | China | 2007-2016 | iCCA | I-IVa | ART<br>Observation | IMRT<br>- | 26<br>23 | 14/12<br>9/14 | 22/4<br>17/6 | 10 (38)<br>8 (35) | 57.2 (45-76)<br>55.6 (36–72) | 55.0 [de]; 19.1 [d]<br>20.0 [de]; 9.5 [d] | 44.0[e]; -<br>10.0[e]; - |
| **ACRT vs Observation** | | | | | | | | | | | | | |
| Kim 2011[77] | Korea | 2001-2009 | eCCA | Ib, IIa, IIb | ACRT<br>Observation | 5-FU/Leu/Cap EBRT<br>- | 75<br>34 | 58/17<br>28/6 | N0: 39; N+: 36<br>N0: 26; N+: 8 | 17 (23)<br>30 (9) | 60.6 (37-74)<br>65.0 (44-85) | 41.1 [e]; -<br>35.3 [e]; - | 37.4 [e]; -<br>29.8 [e]; - |

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| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dover 2016[79] | USA | 2000-2013 | CCA | I-IV | ACRT<br>Observation | GEM/5-FU<br>- | 23<br>72 | 11/12<br>56/16 | N0: 13; N+: 10<br>N0: 51; N+: 21 | 10 (43)<br>28 (39) | 62.0 (38-80) [b]<br>65.0 (39-86) [b] | - ; 30.2<br>- ; 26.3 | - ; 21.6<br>- ; 17.3 |
| **ACRT vs ACT** | | | | | | | | | | | | | |
| Gu 2017[87] | China | 2003-2013 | GBC | II-IVa | ACRT<br>Observation | Cap/S-1/Ox+5-FU/Cap/GEM EBRT | 39<br>39 | All R0 | - | 29 (74)<br>26 (67) | 61 (35-77) [b]<br>63 (45-85) [b] | 42.4; 27<br>17.9; 13 | 48.7 [d]; 23<br>13.5 [d]; 7 |
| Hester 2018[84] | USA | 2004-2013 | dCCA | I-IV | ACRT<br>ACT | - | 348<br>348 | 258/76<br>256/78 | N0:126; N+:211<br>N0:118; N+:206 | 135 (39)<br>136 (39) | 64.3 (±10.0)<br>64.4 (±10.8) | 31.3 [e]; 32.1<br>29.4 [e]; 29.5 | - |
| Choudhary 2019[88] | Indian | 2008-2017 | GBC | II- III | ACRT<br>ACT | GEM+Cis EBRT/3DCRT/IMRT | 23<br>20 | 17/2 (Rx:4)<br>16/2 (Rx:2) | - | 20 (87)<br>17 (85) | 55 (24-73) [ab] | - | - |
| **Multiple comparison** | | | | | | | | | | | | | |
| Kim 2016[80] | Korea | 2000-2015 | GBC | I-IV | ACT<br>ACRT<br>Observation | GEM/Cis/OTHs<br>- | 61<br>44<br>186 | 54/7<br>36/8<br>159/25 | N0:21; N1/Nx:36/4<br>N0:15; N1/Nx:24/5<br>N0:105; N1/Nx:50/31 | 42 (69)<br>28 (64)<br>122 (66) | 64.5 (56-73)<br>63.9 (55-70)<br>67.7 (56-74) | Total:<br>33.0 [e]; 28.3 | Total:<br>- ; 19.8 |
| Im 2021[89] | Korea | 2001-2017 | pCCA | II-IVa | ACT<br>ART<br>ACRT<br>Observation | 5-FU/GEM based EBRT/IMRT<br>- | 45<br>16<br>16<br>37 | 31/14<br>2/14<br>5/11<br>25/12 | - | 22 (33)<br>10 (63)<br>4 (19)<br>35 (38.9) | 65 (32-82) [ab] | 38.9 ; -<br>8.2; -<br>49.4; -<br>27.0; - | 24.6; -<br>5.6; -<br>40.4; -<br>24.5; - |
| Wan 2021[91] | USA (SEER database) | 1973-2015 | GBC | II-IV | ACT<br>ACRT<br>Observation | - | 444<br>542<br>1703 | - | N0: 162; Nx: 282<br>N0: 156; Nx: 386<br>N0: 961; Nx: 742 | 311 (70)<br>374 (69)<br>1192 (70) | - [a] | 38.9/15.0; -<br>41.5/21.1; -<br>32.8/11.3; - | - |
| **Characteristics of included RCT studies** | | | | | | | | | | | | | |
| Ebata 2018[74] | Japan | 2007-2011 | pCCA/dCCA | I-III<br>T1-T4 | ACT<br>Observation | GEM<br>- | 117<br>108 | 106/11<br>94/14 | N0: 75; N1: 42<br>N0: 72; N1: 36 | 40 (34)<br>26 (24) | - [a] | 51.7; 62.3<br>51.6; 63.8 | 45.0; 36.0<br>44.0; 39.9 |
| Edeline 2019[75] | France | 2009-2014 | CCA/GBC | I-IV<br>T1-T4 | ACT<br>Observation | GEMOX<br>- | 95<br>99 | 82/13<br>87/12 | N0:49; N+:35; Nx:11<br>N0:48; N+:36; Nx:15 | 38 (40)<br>49 (50) | 63.0 (33-83) [b]<br>63.0 (40-80) [b] | 60.0 [d]; 75.8<br>65.0 [d]; 50.8 | - ; 30.4<br>- ; 18.5 |
| Primrose 2019[16] | UK | 2006-2014 | BTC | I-IV | ACT<br>Observation | Cap<br>- | 223<br>224 | 139/84<br>140/84 | N0: 121; N1: 102<br>N0: 115; N1:108 | 112 (50)<br>112 (50) | 62.0 (55-68)<br>64.0 (55-69) | 25.1 [c]; 51.1<br>20.5 [c]; 36.4 | - ; 24.4<br>- ; 17.5 |

220 Abbreviations: 5-Year OS (%)=5-Year overall survival probability; MOST=median OS time; 5-Year DFS (%)=5-Year disease-free survival probability; MDFST=median DFS time; N+=



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positive lymph node; Nx= unclear; R0=negative resection margins; R1=microscopic positive resection margins; Rx=unknown; Total= means the value of all patients in the study; BTC=biliary tract cancer; eCCA=extrahepatic cholangiocarcinoma; GBC=gallbladder cancer; CCA=cholangiocarcinoma; pCCA=Perihilar cholangiocarcinoma (or hepatic duct carcinoma=HDC); iCCA=Intrahepatic cholangiocarcinoma; dCCA=distal cholangiocarcinoma; CT= chemotherapy; RT= radiotherapy; ACT=adjuvant chemotherapy; ART=adjuvant radiotherapy; EBRT=external beam radiotherapy; ILRT=intraluminal beam radiotherapy; IMRT=intensity-modulated radiotherapy; 3DCRT=three-dimensional conformal radiotherapy; ACRT=adjuvant chemoradiation therapy; GEM=Gemcitabine; S-1=tegafur gimeracil oteracil potassium; 5-FU=5-fluorouracil; Cap=capecitabine; Cis=cisplatin; Leu=Leucovorin; Ox=oxaliplatin; GEMOX=gemcitabine and oxaliplatin.

- indicates that the data was not given originally;

[a] indicates that studies were presented as younger or older than a specific age;
[b] indicates that median age was given instead of mean age;
[c] indicates overall survival rates but not survival probability;
[d] indicates a 3-year overall survival of disease-free survival;
[e] indicates that the method of the value calculated is unclear.

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The quality assessment of 16 retrospective studies is listed in **Supplementary Table 4**, which shows that 12 studies[76-78, 81-85, 87, 89-91] were judged to be of high quality, and 4 studies[79, 80, 86, 88] were judged to be of moderate quality. Of these studies, four were assigned only one star in "comparability" due to the unreported RM[79, 86] or LN status[80, 88]. Ten studies obtained 0 star in item 8 in relation to "outcome" due to the inadequacy of follow-up.[77, 79-84, 86-88, 90] **Supplementary Figure 1** summarizes the risk of bias assessments of three RCTs. All three RCTs were conducted in an open-label fashion, one[75] did not reveal the details of allocation concealment, and all were at low risk of bias in the domains of "random sequence generation," "incomplete outcome data," and "other bias."

**Transitivity, Heterogeneity, and Inconsistency Assessment**

The transitivity assumption was assessed across treatments in our network. Patient characteristics that are known modifiers of treatment efficacy, such as age, percentage of female participants, sample size, publication year, RM status, and LN status, were evaluated and visualized using box plots (**Supplementary Figure 2**). No difference in these characteristics was observed between the different therapies (**Supplementary Table 5**). Therefore, the transitivity assumption was ensured in this network.

Heterogeneity was also evaluated between eligible studies. Low global heterogeneity was detected in the OS, with $I^2 = 0.0\%$ for a pairwise effect and $I^2 = 19.4\%$ for a consistency effect and $\tau^2 = 0.022$ for the between-study variance (**Supplementary Table 6**). Minimal global heterogeneity was seen in the DFS, with



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$I^2$ = 0.0% for a pairwise pooled effect and a consistency effect (**Supplementary Table 6**). High local heterogeneities were detected in the OS for the direct comparison of ART vs. observation and the DFS for the direct comparison of ACRT vs. observation, with $I^2$ = 81.9% and 82.2%, respectively, for the pooled network effect.

The local inconsistency was evaluated by comparing the corresponding pooled HRs of OS and DFS after comparing the results from pairwise and network meta-analyses. Statistical inconsistency between the direct and indirect evidence after node-splitting was not found, except for the comparison of ACRT vs. ART ($P < 0.001$) and ACT vs. ART ($P = 0.001$) in OS and ACRT vs. ART ($P = 0.009$) in DFS, owing to the fact that only one study contained OS and DFS data comparing ACRT vs. ART and ACT vs. ART (**Supplementary Table 7**).

To further check the consistency at the treatment level, the goodness of fit of the inconsistency model (unrelated mean effects model, UME) was compared with that of the consistency model. The DICs of the consistency model were 46.17 for OS and 29.93 for DFS, which were similar to the DICs of the inconsistency model (46.05 and 31.15, respectively), suggesting no evidence of inconsistency in the network. The more parsimonious model, the consistency model, was used for our analyses (**Supplementary Table 8**).

**Network Meta-analysis of Treatments in Biliary Tract Cancer**

A network meta-analysis was conducted to assess the efficacy (OS and DFS) of the



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following treatments in a Bayesian framework: observation alone after surgery, surgery with ACT, surgery with ART, and surgery with ACRT.

A network plot is shown in **Figure 2**. OS data were available from 18 studies that included 5,497 patients, of whom 2,785 (50.7%) were in the observation group, 1,519 (27.6%) received ACT, 83 (1.5%) received ART, and 1,110 (20.2%) received ACRT. The pooled OS data indicated that ART (HR, 0.62; 95% CI, 0.42 to 0.93), ACRT (HR, 0.71; 95% CI, 0.54 to 0.83), and ACT (HR, 0.84; 95% CI, 0.68 to 0.98) were more beneficial in patients with BTC compared with that of observation (**Figure 3A–B**). No significant benefits in OS were observed in the comparisons between the different adjuvant therapies (**Figure 3A**).

In terms of DFS, 14 studies with 1,979 patients were included in the network meta-analysis (**Figure 2**). The pooled DFS data demonstrated a significant improvement for ACRT and ACT compared with that of observation (HR, 0.60; 95% CI, 0.45 to 0.75, and HR, 0.82; 95% CI, 0.68 to 0.97, respectively). Furthermore, a greater efficacy for ACRT was obtained compared with that of ACT (HR, 0.73; 95% CI, 0.53 to 0.95). Significant differences in DFS were not observed between the other pairwise comparisons (**Figure 3A and 3C**).

The ranking analysis was performed using SUCRA. Based on the pooled OS and DFS data, the ranking order of OS and DFS was inconsistent. The best therapy for OS was ranked as follows: ART, ACRT, ACT, and observation (**Figure 3D**). The best SUCRA value of ART was 87.0%, which was close to that of ACRT with a SUCRA



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value of 75.3% (**Supplementary Table 9**). As for DFS, the best therapy was ranked as ACRT, ACT, ART, and observation (**Figure 3D**). The SUCRA value of ACRT was approximately 97.1%, which was far higher than that of the others (**Supplementary Table 9**).

Subgroup analyses for OS in patients with different residual tumour status, tumour sites, regions, and absence of distant metastasis were performed. Seven studies reported the outcomes of patients after R0 resection, and seven studies reported the outcomes of patients after R1 resection (**Supplementary Figure 3A**). In the R0 group, only ACRT had a survival advantage compared to that of surgery alone. In the R1 group, no survival advantage was observed in patients who underwent adjuvant therapies than in those with surgery alone (**Figure 4A, Supplementary Figure 4A**). In the subgroup analysis by primary tumour sites, six studies enrolled patients with GBC, and 11 studies recruited patients with CCA (**Supplementary Figure 3B**). The OS benefit of ACRT was clear compared with that of observation in the CCA group (HR, 0.62; 95% CI, 0.36 0.85). The benefit of ART is unclear due to the lack of eligible ART studies. Among the comparable treatments in the GBC group, no significant differences in OS were found between ACT, ACRT, and observation (**Figure 4B, Supplementary Figure 4B**). When studies were grouped according to region, 12 studies included patients in Asian countries and 7 from Western countries (**Supplementary Figure 3C**). In both the Asian and Western groups, the pooled OS results favoured ACRT (HR, 0.42; 95% CI, 0.20 to 0.74 in Asia; HR, 0.76; 95% CI, 0.63 to 0.91 in Western countries) (**Figure 4C,**



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**Supplementary Figure 4C** ). Subgroup analysis was also conducted for 15 studies investigating OS in patients with non-distant metastasis (**Supplementary Figure 3D**). The pooled results showed superior efficacy of ACRT compared with that of observation (HR, 0.74; 95% CI, 0.60 to 0.85); moreover, ACRT presented a better efficacy compared to that of ACT (HR, 0.82; 95% CI, 0.67 to 0.96) (**Figure 4D, Supplementary Figure 4D**).

Subgroup analyses were also performed for DFS in patients with different tumour sites and regions and the absence of distant metastasis (**Supplementary Figure 5**). Due to the small number of studies, subgroup analyses for DFS by different residual tumour statuses cannot be conducted. In patients without distant metastasis, the results from this subgroup were similar to that of the primary analysis; ACRT demonstrated statistically significant improvement with an HR of 0.61 (**Figure 4E, Supplementary Figure 4D**). When studies were split by primary tumour sites, only ACRT showed an apparent advantage in patients with CCA (**Figure 4F, Supplementary Figure 4E**). Furthermore, the stratified meta-analysis by region indicated that the favourable treatment was ACRT in Asia (HR, 0.47; 95% CI, 0.28 to 0.68) (**Figure 4G, Supplementary Figure 4F**).

To assess the robustness of the primary results, sensitivity analyses were performed for OS and DFS by excluding RCT studies, removing moderate-quality observational studies, and only including the studies for which HRs were reported in the original articles. The first sensitivity analysis included 15 retrospective studies of



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OS and 11 retrospective studies of DFS. The pooled results (**Supplementary Figure 6**) and the ranking profiles of comparable treatments from retrospective studies confirmed the reliability of the primary OS results (**Figure 5A**, **Supplementary Table 10**). In terms of DFS, ACRT was still ranked as the best treatment option (**Figure 5B**). Furthermore, the second and third sensitivity analyses were performed by removing moderate-quality observational studies (14 studies for OS and 11 studies for DFS) and by combining HRs reported in the original articles (11 studies for OS and 8 studies for DFS). The results did not suggest a material change in the efficacy estimation for ACRT, but ART no longer showed advantages over observation in both OS and DFS; the improvement in OS and DFS between ACT and observation was insignificant except for the pooled DFS in high-quality studies (**Supplementary Figure 6**).

The visual examination of the comparison-adjusted funnel plots did not suggest a publication bias for OS presented in our network meta-analysis (**Supplementary Figure 7A**). The result of Egger's test ($P = 0.251$) also rejected the presence of small-study effects. However, both the visual examination of the comparison-adjusted funnel plots and the results of Egger's test ($P = 0.018$) suggested a publication bias for DFS in our network meta-analysis (**Supplementary Figure 7B**).

## DISCUSSION

BTCs are an uncommon and heterogeneous type of cancer with a higher prevalence in Asian countries.[92] In general, BTCs include cancers raised from the intrahepatic,



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hilar, and distal bile ducts, as well as the gallbladder.[7] Surgical resection provides the only chance for cure in patients with BTC at an early stage, but the survival outcomes are poor. The 5-year survival rate was as low as 10%.[93] The most recent NCCN and ASCO guidelines recommend the use of adjuvant therapy for BTC patients after resection.[12, 15] However, experts mentioned that the use of adjuvant therapy is based on a very limited number of studies, and the benefit of adjuvant therapy remains unclear in many pivotal BTC trials.[16, 74, 75, 94, 95] Therefore, it is necessary to compile up-to-date studies to validate the efficacy of adjuvant therapy in BTC patients.

In this network meta-analysis, 19 studies were included to evaluate the comparative benefits (involving 5,497 patients for OS and 1,979 patients for DFS) between adjuvant therapy (ACT, ART, and ACRT) following surgical resection and curative-intent resection (observation group). Our primary results demonstrated that adjuvant therapy was more effective than that of observation in OS. However, no statistically significant difference was detected between ACT, ACRT, and ART. Moreover, the pooled DFS results suggested a statistically significant benefit for ACRT and ACT over observation. Although ART was ranked first in OS, its DFS result was inconsistent with OS, and caution should be exercised regarding the bias that arise from the small sample size effect of ART evidence (only 83 patients). In addition, a previously published meta-analysis of adjuvant therapy in the treatment of BTC suggested that ART did not provide a significant advantage over observation.[21] We hold reservations on the conclusion of the efficacy of ART, and



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further evidence is needed to elucidate this matter.

Combining the results of OS with DFS, ACRT, and ACT after radical resection could provide a meaningful survival benefit in patients with BTC. This was in line with another meta-analysis that showed that both ACT (HR, 0.61; 95% CI, 0.47–0.79) and ACRT (HR, 0.35; 95% CI, 0.14–0.83) could significantly improve the clinical survival of patients with resected BTC compared to that of surgery alone.[96] As known, the BILCAP study is a unique positive randomized trial of ACT in patients with BTC. Despite the concerns regarding the findings of the BILCAP trial, this study established adjuvant capecitabine as the new standard of care for resected BTC. However, no randomized trials of ACRT vs. ACT are unavailable. Although in our study, the ACRT ranked high in terms of OS and DFS (second place for OS, first place for DFS), whether it is genuinely superior to ACT needs further confirmation in prospective RCTs.

The types of patients who are more likely to benefit from adjuvant therapy are also the focus. A previous phase II study indicated that the risk factors were GBC, eCCA, pathologic stage T2–4, positive LN, or positive RM.[97] In our study, we were only able to conduct subgroup analyses on the effects of the RM, primary tumour site, regions, and absence of distant metastasis, but not LN and tumour size, on adjuvant therapy due to the availability of data. We observed that ACRT showed better OS or DFS benefits than that of ACT in CCA and Asian patients and ranked first in each subgroup (**Supplementary Table 11**). We noticed that approximately 95% (for OS



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analysis) and 80% (for DFS analysis) of CCA patients treated with ACRT had eCCA. The benefit of ACRT observed in CCA patients may be mainly derived from eCCA patients. Inconsistent with most studies, ACRT has an OS benefit in the R0 group, but not in the R1 group. This was possibly due to the more stringent selection of studies and the effects of small sample sizes. Subgroup analyses for DFS by R0 or R1 were not conducted because of limited data. Therefore, the subgroups of patients who could benefit more from postoperative adjuvant therapy need to be further elucidated. We look forward to seeing more RCTs in this field, especially head-to-head trials and trials designed for more specific subgroups. Using the primary tumour site, tumour size, disease severity, LN metastases, and RM as stratification factors in future studies would reduce selection bias due to the heterogeneity of the population.

This study had several limitations. First, RCT data were only available to compare ACT and observations. Unfortunately, there are neither RCTs for ART and ACRT nor head-to-head RCTs between different adjuvant treatments. The notable differences in the study design level may introduce confounding factors in our data analysis, although data transitivity and consistency could be assumed statistically. Second, the included studies spanned over a long period; operative techniques, chemotherapeutic drugs, and radiation methods have changed and improved over time. These changes in treatment methods could potentially bias our results, but the impact on the outcome was unclear and difficult to interpret. Third, most comparisons were indirect, and direct evidence was obtained from two or three studies. Furthermore, we noticed that the number of patients treated with ART was small.



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These may result in a considerable risk of bias. Fourth, the definitions of OS and DFS in RCTs and retrospective studies were calculated differently. In RCTs, it started from the date of randomization to the date of surgery in retrospective studies. The span was relatively short compared with the expected survival duration, but the slight variation in data collection should still be considered. Finally, we failed to compare the safety outcomes due to a lack of sufficient data on adverse events. Taking all these into account, our estimates should be interpreted with caution.

## CONCLUSION

Our primary results demonstrated that, compared to that of observation, ACRT and ACT after radical resection could provide better OS and DFS benefits in patients with BTC. Although ACRT shows superior DFS outcomes compared to that of ACT, thorough and high-quality RCTs between ACRT and ACT are warranted. Optimal regimens and dosing schedules still need to be explored.

## ABBREVIATIONS

BTC: biliary tract carcinoma

OS: overall survival

DFS: disease-free survival

HR (95% CI): hazard ratio (95% confidence interval)

GBC: gallbladder cancer

CCA: cholangiocarcinoma

iCCA: intrahepatic cholangiocarcinoma

eCCA: extrahepatic cholangiocarcinoma

pCCA: perihilar cholangiocarcinoma



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HDC: hepatic duct carcinoma

dCCA: distal cholangiocarcinoma

ACT: adjuvant chemotherapy

ART: adjuvant radiotherapy

ACRT: adjuvant chemoradiotherapy

RCT: randomized clinical trial

PRIMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyse

AJCC: American Joint Committee on Cancer

RM: resection margin

R0: negative resection margin

R1: microscopic positive resection margin

R2: macroscopic involvement resection

LN: lymph node

NOS: Newcastle-Ottawa Scale

DIC: deviance information criterion

SUCRA: surface under the cumulative ranking

UME: unrelated mean effect

NCCN: National Comprehensive Cancer Network

ASCO: American Society of Clinical Oncology

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## Author Contributions

As the corresponding author, this study was designed, coordinated, and supervised by Meng Qiu. Ye Chen, Baoxia Zhang, and Chang Liu screened the articles, Ye Chen and Baoxia Zhang extracted the data and assessed the quality. Ye Cao and Chang Liu contacted study authors for additional information. Ye Chen, Baoxia Zhang, and Ye Cao analyzed and prepared the material. Ye Chen, Baoxia Zhang, Chang Liu, and Cheng Lyu interpreted the data and wrote the draft of the report. All authors had reviewed and approved the final submitted version of the report. Meng Qiu was responsible for the integrity and accuracy of the data.

## Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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**Data availability statement:** All data relevant to the study are included in the article or uploaded as supplemental information. All the data extracted, analyzed, and reported in this paper were from published literature.



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## References

1. Nakeeb A, Pitt HA, Sohn TA, et al. Cholangiocarcinoma. A spectrum of intrahepatic, perihilar, and distal tumors. *Ann Surg* 1996;224(4):463-73; discussion 73-5.

2. Benavides M, Anton A, Gallego J, et al. Biliary tract cancers: SEOM clinical guidelines. *Clin Transl Oncol* 2015;17(12):982-7.

3. Patel T. Increasing incidence and mortality of primary intrahepatic cholangiocarcinoma in the United States. *Hepatology* 2001;33(6):1353-7.

4. Valle JW, Lamarca A, Goyal L, et al. New Horizons for Precision Medicine in Biliary Tract Cancers. *Cancer Discov* 2017;7(9):943-62.

5. Anderson CD, Pinson CW, Berlin J, et al. Diagnosis and treatment of cholangiocarcinoma. *Oncologist* 2004;9(1):43-57.

6. DeOliveira ML, Cunningham SC, Cameron JL, et al. Cholangiocarcinoma: thirty-one-year experience with 564 patients at a single institution. *Ann Surg* 2007;245(5):755-62.

7. Bridgewater JA, Goodman KA, Kalyan A, et al. Biliary Tract Cancer: Epidemiology, Radiotherapy, and Molecular Profiling. *Am Soc Clin Oncol Educ Book* 2016;35:e194-203.

8. Doherty MK, Knox JJ. Adjuvant therapy for resected biliary tract cancer: a review. *Chin Clin Oncol* 2016;5(5):64.

9. Mavros MN, Economopoulos KP, Alexiou VG, et al. Treatment and Prognosis for Patients With Intrahepatic Cholangiocarcinoma: Systematic Review and Meta-analysis. *JAMA Surg* 2014;149(6):565-74.

10. Murakami Y, Uemura K, Sudo T, et al. Prognostic factors after surgical resection for intrahepatic, hilar, and distal cholangiocarcinoma. *Ann Surg Oncol* 2011;18(3):651-8.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

11. Wirasorn K, Ngamprasertchai T, Khuntikeo N, et al. Adjuvant chemotherapy in resectable cholangiocarcinoma patients. *J Gastroenterol Hepatol* 2013;28(12):1885-91.

12. National Comprehensive Cancer Network. NCCN clinical practice guidelines in oncology. Hepatobiliary cancers, Version 5. 2021. https://www.nccn.org/professionals/physician_gls/pdf/hepatobiliary.pdf. (accessed 20 Jan. 2022)

13. Valle JW, Borbath I, Khan SA, et al. Biliary cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Annals of oncology : official journal of the European Society for Medical Oncology* 2016;27(suppl 5):v28-v37.

14. HJ L, SK Q, F S, et al. Biliary tract caner: CSCO consensus for diagnosis and treatment. *Chin Clin Oncol* 2019;24(9):828-38.

15. Shroff RT, Kennedy EB, Bachini M, et al. Adjuvant Therapy for Resected Biliary Tract Cancer: ASCO Clinical Practice Guideline. *J Clin Oncol* 2019;37(12):1015-27.

16. Primrose JN, Fox RP, Palmer DH, et al. Capecitabine compared with observation in resected biliary tract cancer (BILCAP): a randomised, controlled, multicentre, phase 3 study. *Lancet oncol* 2019;20(5):663-73.

17. Rizzo A, Brandi G. BILCAP trial and adjuvant capecitabine in resectable biliary tract cancer: reflections on a standard of care. *Expert Rev Gastroent* 2021;15(5):483-85.

18. Rizzo A, Brandi G. Pitfalls, challenges, and updates in adjuvant systemic treatment for resected biliary tract cancer. *Expert Rev Gastroent* 2021;15(5):547-54.

19. Lamarca A, Edeline J, McNamara MG, et al. Current standards and future perspectives in adjuvant treatment for biliary tract cancers. *Cancer treat Rev* 2020;84:101936.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Rangarajan K, Simmons G, Manas D, et al. Systemic adjuvant chemotherapy for cholangiocarcinoma surgery: A systematic review and meta-analysis. *Eur J Surg Oncol* 2020;46(4 Pt A):684-93.

21. Horgan AM, Amir E, Walter T, et al. Adjuvant therapy in the treatment of biliary tract cancer: a systematic review and meta-analysis. *J Clin Oncol* 2012;30(16):1934-40.

22. Zhu GQ, Shi KQ, You J, et al. Systematic review with network meta-analysis: adjuvant therapy for resected biliary tract cancer. *Aliment Pharmacol Ther* 2014;40(7):759-70.

23. Kish M, Chan K, Perry K, et al. A systematic review and network meta-analysis of adjuvant therapy for curatively resected biliary tract cancers. *Curr Oncol* 2020;27(1):e20-e26.

24. Chen X, Meng F, Xiong H, et al. Adjuvant Therapy for Resectable Biliary Tract Cancer: A Bayesian Network Analysis. *Front Oncol* 2021;11:600027.

25. Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Ann Intern Med* 2015;162(11):777-84.

26. Ottawa Hospital Research Institute. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. 2014. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. (accessed 29 Jun. 2020)

27. Julian Higgins, Douglas Altman, Sterne J. Assessing risk of bias in included studies. In: Julian Higgins, Green S, editors. Cochrane handbook of systematic reviews of interventions. Assessing risk of bias in included studies. 5.1.0 ed. England: The Cochrane Collaboration, Wiley; 2011. p. 187-235.

28. Tierney JF, Stewart LA, Ghersi D, et al. Practical methods for incorporating summary



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

time-to-event data into meta-analysis. *Trials* 2007;8:16.

29. Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. *Stat Sci* 1992;7:457-511.

30. Lynch SM. Introduction to Applied Bayesian Statistics and Estimation for Social Scientists. 1st edition ed. New York, NY: Springer-Verlag 2007.

31. Higgins JPT, Jackson D, Barrett JK, et al. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Res Synth Methods* 2012;3(2):98-110.

32. Dias S, Welton NJ, Caldwell DM, et al. Checking consistency in mixed treatment comparison meta-analysis. *Stat Med* 2010;29(7-8):932-44.

33. Van Valkenhoef G, Dias S, Ades AE, et al. Automated generation of node-splitting models for assessment of inconsistency in network meta-analysis. *Res Synth Methods* 2016;7(1):80-93.

34. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60.

35. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21(11):1539-58.

36. Debray TP, Damen JA, Riley RD, et al. A framework for meta-analysis of prediction model studies with binary and time-to-event outcomes. *Stat Methods Med Res* 2019;28(9):2768-86.

37. Salanti G, Ades AE, Ioannidis JP. Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial. *J Clin*



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

*Epidemiol* 2011;64(2):163-71.

38. Veroniki AA, Vasiliadis HS, Higgins JP, et al. Evaluation of inconsistency in networks of interventions. *Int J Epidemiol* 2013;42(1):332-45.

39. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;315(7109):629-34.

40. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics* 1994;50(4):1088-101.

41. Sagawa N, Kondo S, Morikawa T, et al. Effectiveness of radiation therapy after surgery for hilar cholangiocarcinoma. *Surg Today* 2005;35(7):548-52.

42. Borghero Y, Crane CH, Szklaruk J, et al. Extrahepatic bile duct adenocarcinoma: patients at high-risk for local recurrence treated with surgery and adjuvant chemoradiation have an equivalent overall survival to patients with standard-risk treated with surgery alone. *Ann Surg Oncol* 2008;15(11):3147-56.

43. Gold DG, Miller RC, Haddock MG, et al. Adjuvant therapy for gallbladder carcinoma: the Mayo Clinic Experience. *Int J Radiat Oncol* 2009;75(1):150-5.

44. Ercolani G, Vetrone G, Grazi GL, et al. Intrahepatic cholangiocarcinoma: primary liver resection and aggressive multimodal treatment of recurrence significantly prolong survival. *Ann Surg* 2010;252(1):107-14.

45. Kim K, Chie EK, Jang JY, et al. Postoperative chemoradiotherapy for gallbladder cancer. *Strahlenther Onkol* 2012;188(5):388-92.

46. Kondo N, Murakami Y, Uemura K, et al. Elevated perioperative serum CA 19-9 levels are independent predictors of poor survival in patients with resectable cholangiocarcinoma. *J*



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

*Surg Oncol* 2014;110(4):422-9.

47. Bektas H, Yeyrek C, Kleine M, et al. Surgical treatment for intrahepatic cholangiocarcinoma in Europe: a single center experience. *J Hepato-Bil-Pan Sci* 2015;22(2):131-7.

48. Hoehn RS, Wima K, Ertel AE, et al. Adjuvant Chemotherapy and Radiation Therapy is Associated with Improved Survival for Patients with Extrahepatic Cholangiocarcinoma. *Ann Surg Oncol* 2015;22:1133-39.

49. Sur MD, In H, Sharpe SM, et al. Defining the Benefit of Adjuvant Therapy Following Resection for Intrahepatic Cholangiocarcinoma. *Ann Surg Oncol* 2015;22(7):2209-17.

50. Im JH, Seong J, Lee IJ, et al. Surgery alone versus surgery followed by chemotherapy and radiotherapy in resected extrahepatic bile duct cancer: Treatment outcome analysis of 336 patients. *Cancer Res Treat* 2016;48(2):583-95.

51. Kim Y, Hwang I, Park S-E, et al. Role of adjuvant therapy after R0 resection for patients with distal cholangiocarcinoma. *Cancer Chemoth Pharm* 2016;77(5):979-85.

52. Kim TG. Patterns of initial failure after resection for gallbladder cancer: Implications for adjuvant radiotherapy. *Radiat Oncol J* 2017;35(4):359-67.

53. Kim Y, Oh S, Go S-I, et al. The role of adjuvant therapy after R0 resection for patients with intrahepatic and perihilar cholangiocarcinomas. *Cancer Chemoth Pharm* 2017;79(1):99-106.

54. Kim YS, Jeong CY, Song HN, et al. The efficacy of fluoropyrimidine-based adjuvant chemotherapy on biliary tract cancer after R0 resection. *Chin J Cancer* 2017;36(1):9.

55. Lee J, Kang SH, Noh OK, et al. Adjuvant concurrent chemoradiation therapy in patients



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

with microscopic residual tumor after curative resection for extrahepatic cholangiocarcinoma. *ClinTransl Oncol* 2017;20(8):1011-17.

56. Mitin T, Kristian Enestvedt C, Jemal A, et al. Limited Use of Adjuvant Therapy in Patients with Resected Gallbladder Cancer Despite a Strong Association with Survival. *J Natl Cancer Inst* 2017;109(7).

57. Tran Cao HS, Zhang Q, Sada YH, et al. The role of surgery and adjuvant therapy in lymph node-positive cancers of the gallbladder and intrahepatic bile ducts. *Cancer* 2017;124(1):74-83.

58. Lin YK, Hsieh MC, Wang WW, et al. Outcomes of adjuvant treatments for resectable intrahepatic cholangiocarcinoma: Chemotherapy alone, sequential chemoradiotherapy, or concurrent chemoradiotherapy. *Radiother Oncol* 2018;128(3):575-83.

59. Geng ZM, Cai ZQ, Zhang Z, et al. Estimating survival benefit of adjuvant therapy based on a Bayesian network prediction model in curatively resected advanced gallbladder adenocarcinoma. *World J Gastroentero* 2019;25(37):5655-66.

60. Kattepur AK, Patkar S, Goel M, et al. Role of Adjuvant Chemotherapy in Resected T2N0 Gall Bladder Cancer. *J Gastrointest Surg* 2019;23(11):2232-38.

61. Navarro J, Kang I, Hwang HK, et al. Glucose to Lymphocyte Ratio as a Prognostic Marker in Patients With Resected pT2 Gallbladder Cancer. *J Surg Res* 2019;240:17-29.

62. Kim TH, Woo SM, Lee WJ, et al. Benefit of Adjuvant Chemoradiotherapy in Resected Gallbladder Carcinoma. *Sci Rep* 2019;9(1):N.PAG-N.PAG.

63. Kizy S, Altman AM, Marmor S, et al. Surgical resection of lymph node positive intrahepatic cholangiocarcinoma may not improve survival. *HPB* 2019;21(2):235-41.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

64. Lu J, Li B, Li FY, et al. Prognostic significance of mucinous component in hilar cholangiocarcinoma after curative-intent resection. *J Surg Oncol* 2019;120(8):1341-49.

65. Han D, Yang J, Xu F, et al. Prognostic factors in patients with gallbladder adenocarcinoma identified using competing-risks analysis: A study of cases in the SEER database. *Medicine* 2020;99(31):e21322.

66. Kim H, Heo MH, Kim JY. Comparison of the effects of adjuvant concurrent chemoradiotherapy and chemotherapy for resected biliary tract cancer. *BMC Gastroenterol* 2020;20(1):20.

67. Mao W, Deng F, Wang D, et al. Treatment of advanced gallbladder cancer: A SEER-based study. *Cancer Med* 2020;9(1):141-50.

68. Zhou W, Qian L, Rong Y, et al. Prognostic factors and patterns of recurrence after curative resection for patients with distal cholangiocarcinoma. *Radiothe Oncol* 2020;147:111-17.

69. Zhu WH, Xie WY, Zhang ZD, et al. Postoperative Complications and Survival Analysis of Surgical Resection for Hilar Cholangiocarcinoma: A Retrospective Study of Fifty-Nine Consecutive Patients. *Chin Med Sci J* 2020;35(2):157-69.

70. Benzing C, Krenzien F, Mieg A, et al. A tailored approach in lymph node-positive perihilar cholangiocarcinoma. *Langenbeck Arch Surg* 2021;406(5):1499-509.

71. Chang WI, Kim BH, Kang HC, et al. The Role of Adjuvant Chemoradiotherapy in Nonhilar Extrahepatic Bile Duct Cancer: A Long-Term Single-Institution Analysis. *Int J Radiat Oncol* 2021;111(2):395-404.

72. Kosaka H, Kaibori M, Matsui K, et al. Investigation of a Tumor Location-Specific



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

669 Therapeutic Strategy for Intrahepatic Cholangiocarcinoma. *APJCP* 2021;22(5):1485-93.

670 73. Sumiyoshi T, Uemura K, Kondo N, et al. The prognostic impact of peritoneal washing
671 cytology for otherwise resectable extrahepatic cholangiocarcinoma patients. *Surg today*
672 2021;51(7):1227-31.

673 74. Ebata T, Hirano S, Konishi M, et al. Randomized clinical trial of adjuvant gemcitabine
674 chemotherapy versus observation in resected bile duct cancer. *Brit J Surg* 2018;105(3):192-
675 202.

676 75. Edeline J, Benabdelghani M, Bertaut A, et al. Gemcitabine and oxaliplatin chemotherapy
677 or surveillance in resected biliary tract cancer (Prodige 12-accord 18-Unicancer GI): a
678 randomized phase III study. *J Clin Oncol* 2019;37(8):658-67.

679 76. Todoroki T, Ohara K, Kawamoto T, et al. Benefits of adjuvant radiotherapy after radical
680 resection of locally advanced main hepatic duct carcinoma. *Int J Radiat Oncol*
681 2000;46(3):581-7.

682 77. Kim TH, Han SS, Park SJ, et al. Role of adjuvant chemoradiotherapy for resected
683 extrahepatic biliary tract cancer. *Int J Radiat Oncol* 2011;81(5):e853-9.

684 78. Kobayashi H, Murakami Y, Uemura K, et al. Human equilibrative nucleoside transporter
685 1 expression predicts survival of advanced cholangiocarcinoma patients treated with
686 gemcitabine-based adjuvant chemotherapy after surgical resection. *Ann Surg*
687 2012;256(2):288-96.

688 79. Dover LL, Oster RA, McDonald AM, et al. Impact of adjuvant chemoradiation on survival
689 in patients with resectable cholangiocarcinoma. *HPB* 2016;18(10):843-50.

690 80. Kim Y, Amini N, Wilson A, et al. Impact of Chemotherapy and External-Beam Radiation



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

691 Therapy on Outcomes among Patients with Resected Gallbladder Cancer: A Multi-institutional
692 Analysis. *Ann Surg Oncol* 2016;23(9):2998-3008.
693 81. Morine Y, Shimada M, Ikemoto T, et al. Effect of adjuvant gemcitabine combined with
694 low-dose 5-fluorouracil and cisplatin chemotherapy for advanced biliary carcinoma.
695 *Anticancer Res* 2017;37(11):6421-28.
696 82. Akahoshi K, Ban D, Kuboki R, et al. Orotate phosphoribosyltransferase as a predictor of
697 benefit from S-1 adjuvant chemotherapy for cholangiocarcinoma patients. *J Gastroen Hepatol*
698 2018;34(6):1108-15.
699 83. Bergeat D, Turrini O, Courtin-Tanguy L, et al. Impact of adjuvant chemotherapy after
700 pancreaticoduodenectomy for distal cholangiocarcinoma: a propensity score analysis from a
701 French multicentric cohort. *Langenbeck Arch Surg* 2018;403(6):701-09.
702 84. Hester C, Nassour I, Adams-Huet B, et al. Improved Survival in Surgically Resected
703 Distal Cholangiocarcinoma Treated with Adjuvant Therapy: a Propensity Score Matched
704 Analysis. *J Gastrointest Surg* 2018;22(12):2080-87.
705 85. Zheng X, Chen B, Wu JX, et al. Benefit of adjuvant radiotherapy following narrow-margin
706 hepatectomy in patients with intrahepatic cholangiocarcinoma that adhere to major vessels.
707 *Cancer Manag Res* 2018;10:3973-81.
708 86. Jiang W, Zeng ZC, Tang ZY, et al. Benefit of radiotherapy for 90 patients with resected
709 intrahepatic cholangiocarcinoma and concurrent lymph node metastases. *J Cancer Res Clin*
710 2010;136(9):1323-31.
711 87. Gu B, Qian L, Yu H, et al. Concurrent Chemoradiotherapy in Curatively Resected
712 Gallbladder Carcinoma: A Propensity Score-Matched Analysis. *Int J Radiat Oncol*



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2018;100(1):138-45.

88. Choudhary S, Gupta N, Verma CP, et al. Influence of adjuvant therapy on pattern of failure and survival in curatively resected gallbladder carcinoma. *J Cancer Res Ther* 2021;17(4):1064-68.

89. Im JH, Choi GH, Lee WJ, et al. Adjuvant radiotherapy and chemotherapy offer a recurrence and survival benefit in patients with resected perihilar cholangiocarcinoma. *J Cancer Res Clin Oncol* 2021;147(8):2435-45.

90. Miyata Y, Kogure R, Nakazawa A, et al. The efficacy of s-1 as adjuvant chemotherapy for resected biliary tract carcinoma: A propensity score-matching analysis. *J Clin Med* 2021;10(5):1-10.

91. Wan W, Zheng B, Sun W, et al. Adjuvant Therapy in Resected Nonmetastatic Stage II–IV Gallbladder Cancer: A Generalized Propensity Score Analysis. *Oncology Research & Treatment* 2021;44(7-8):390-99.

92. Tariq NU, McNamara MG, Valle JW. Biliary tract cancers: current knowledge, clinical candidates and future challenges. *Cancer Manag Res* 2019;11:2623-42.

93. Khan SA, Davidson BR, Goldin RD, et al. Guidelines for the diagnosis and treatment of cholangiocarcinoma: an update. *Gut* 2012;61(12):1657-69.

94. Takada T, Amano H, Yasuda H, et al. Is postoperative adjuvant chemotherapy useful for gallbladder carcinoma? A phase III multicenter prospective randomized controlled trial in patients with resected pancreaticobiliary carcinoma. *Cancer* 2002;95(8):1685-95.

95. Neoptolemos JP, Moore MJ, Cox TF, et al. Effect of adjuvant chemotherapy with fluorouracil plus folinic acid or gemcitabine vs observation on survival in patients with



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735 resected periampullary adenocarcinoma: the ESPAC-3 periampullary cancer randomized trial.
736 *JAMA* 2012;308(2):147-56.
737 96. Song S, Yang W, Tian H, et al. The efficacy and safety of 5-fluorouracil based adjuvant
738 therapy in resected biliary tract cancer: A systematic review and meta-analysis. *Clin Res*
739 *Hepatol Gas* 2022;46(2).
740 97. Ben-Josef E, Guthrie KA, El-Khoueiry AB, et al. SWOG S0809: A Phase II Intergroup
741 Trial of Adjuvant Capecitabine and Gemcitabine Followed by Radiotherapy and Concurrent
742 Capecitabine in Extrahepatic Cholangiocarcinoma and Gallbladder Carcinoma. *J Clin Oncol*
743 2015;33(24):2617-22.
744

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## Figure Legends

**Figure 1. A PRISMA flow diagram of the study selection.** BTC, biliary tract cancer; OS, overall survival; DFS, disease-free survival.

**Figure 2. Network plot of comparisons on overall survival and disease-free survival of treatments in patients with biliary tract cancer.** Each circular node represents a type of treatment. The node size corresponds to the total number of participants assigned to each treatment. Each line represents a type of direct comparison, and its width corresponds to the number of studies evaluating the connected treatments. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy.

**Figure 3. Network meta-analysis results for efficacy outcomes. (A)** League table of the network meta-analysis. Pooled HRs and 95% CIs for the overall survival (OS) and disease-free survival (DFS) are listed in the upper and lower triangle, respectively. The estimate in each cell compares the row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment. Significant results are in bold. **(B)** Forest plot of the network meta-analysis for OS. **(C)** Forest plot of the network meta-analysis for DFS. HRs and 95%



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CIs are provided and visually represented by the squares and error bars. **(D)** Bayesian ranking curves of comparable treatments on efficacy for patients with biliary tract cancer. Ranking curves indicate the probability of each treatment ranked first to last on OS (solid lines) and DFS (dotted lines). Data of the curves are presented in **Supplementary Table 9**. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy; HRs, hazard ratios; 95% CIs, 95% confidence intervals; AT, adjuvant therapy.

**Figure 4. Forest plots of the network meta-analysis for overall survival (OS) and disease-free survival (DFS) in the subgroup analyses.** (A) Forest plots of OS in patients stratified into the R0 and R1 groups. (B) and (E) Forest plots of OS and DFS in patients stratified into the GBC and CCA groups. (C) and (F) Forest plots of OS and DFS in patients stratified into the Asian and Western groups. (D) Forest plots of OS and DFS in patients with non-distant metastasis. Effect sizes are presented as hazard ratios with 95% confidence intervals.

**Figure 5. Bayesian ranking plots of comparable treatments on efficacy in patients with biliary tract cancer in the sensitivity analyses.** Ranking curves indicate the probability of each comparable treatment being ranked from first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the



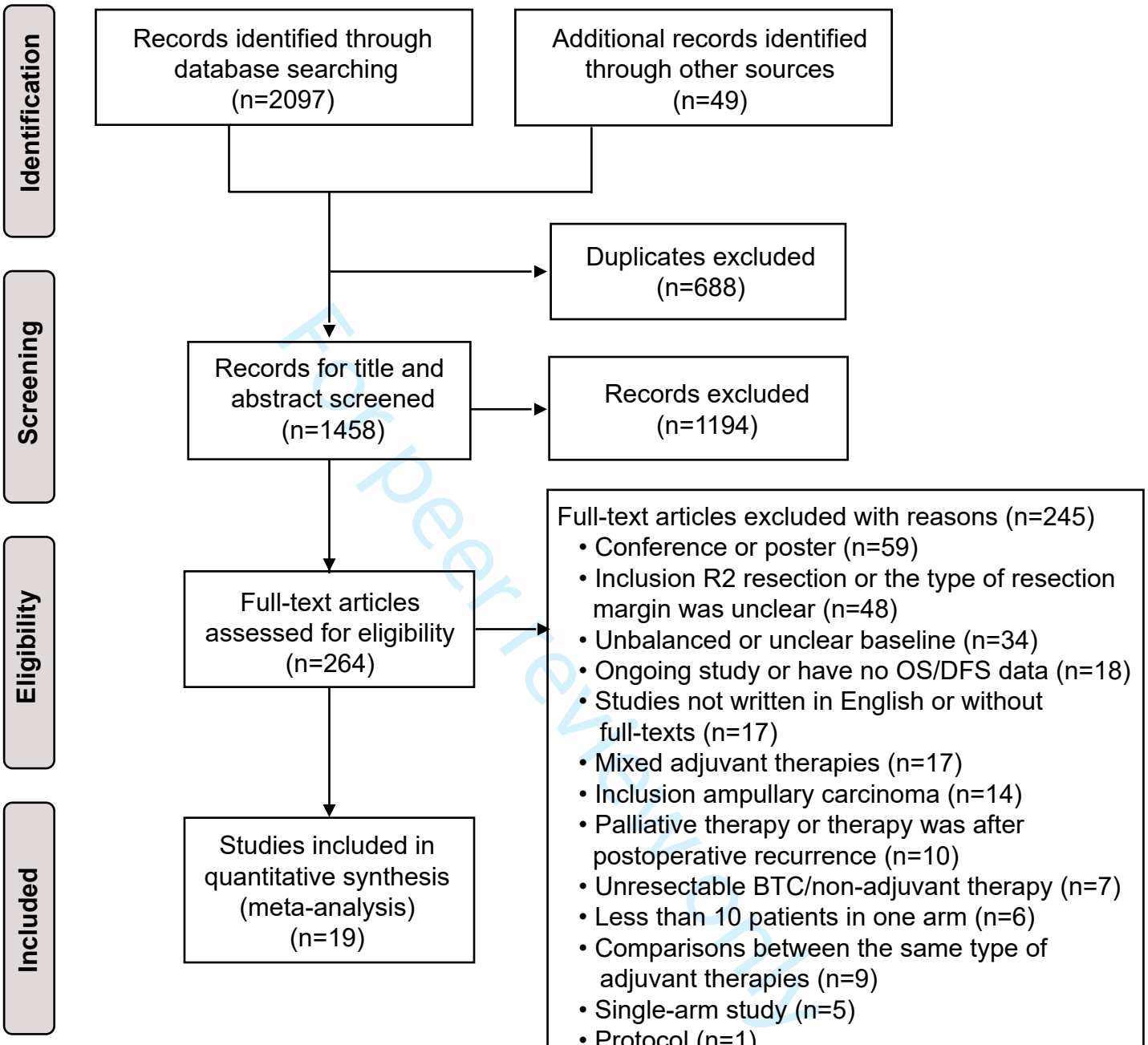
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curves are presented in **Supplementary Table 10**. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy.

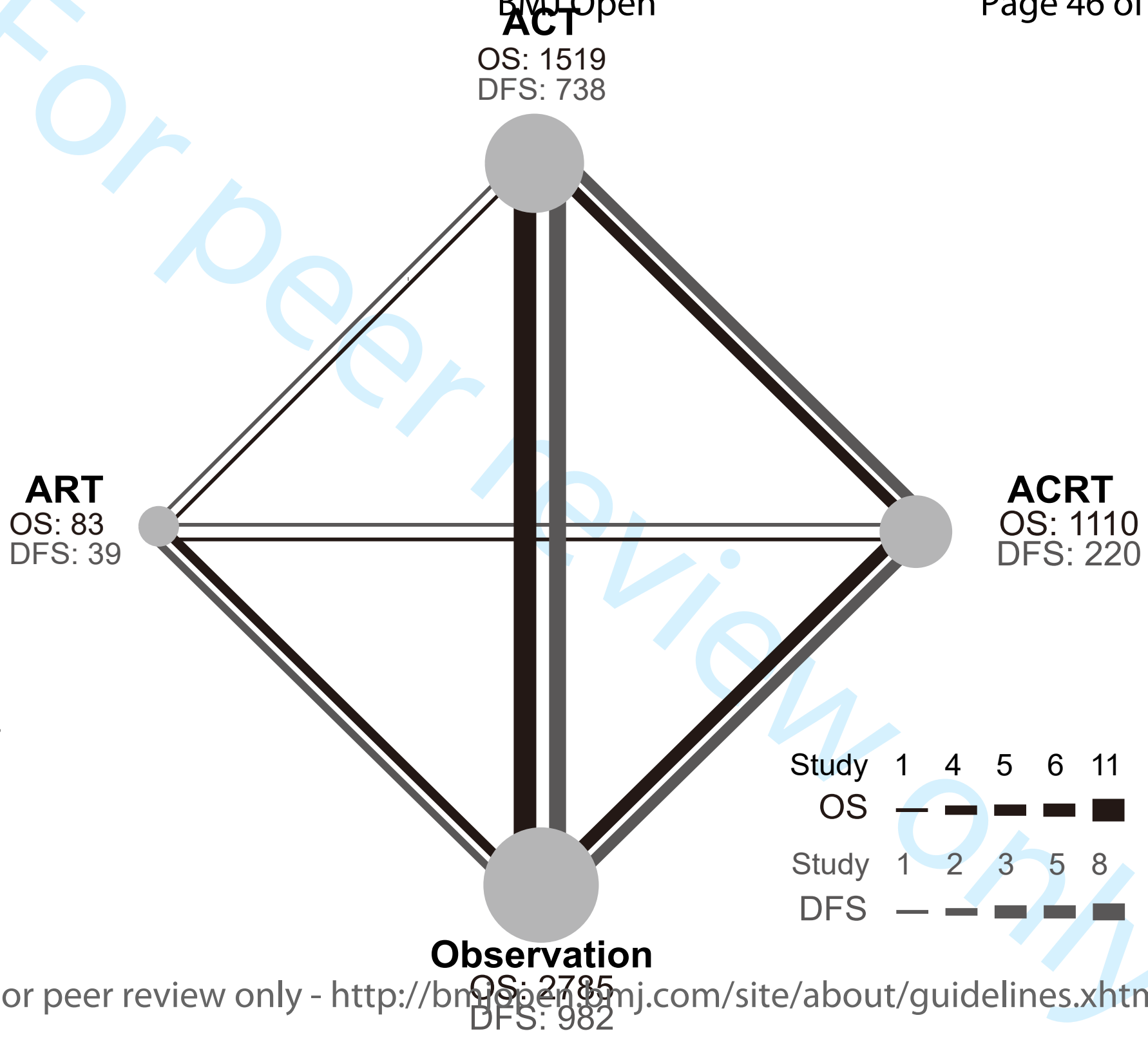
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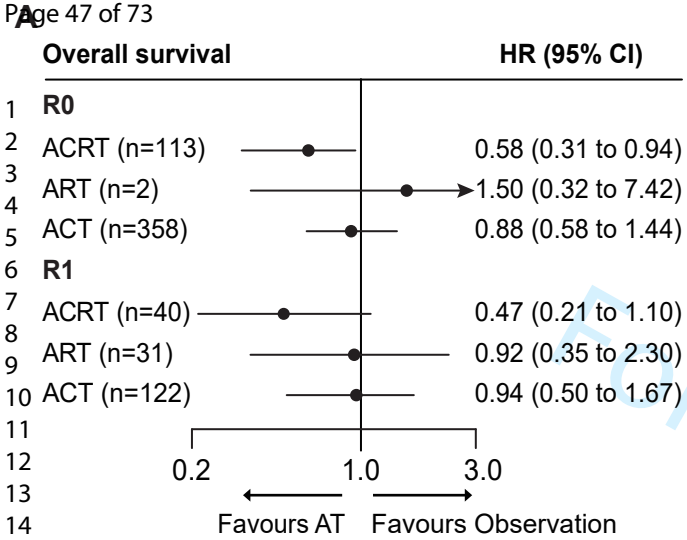
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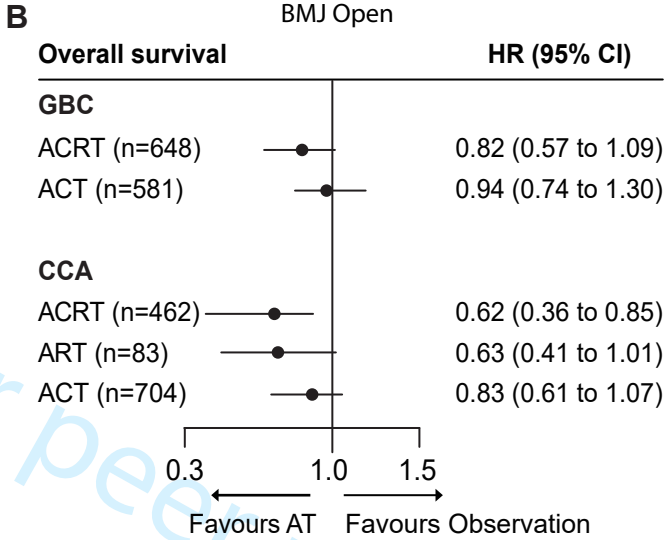


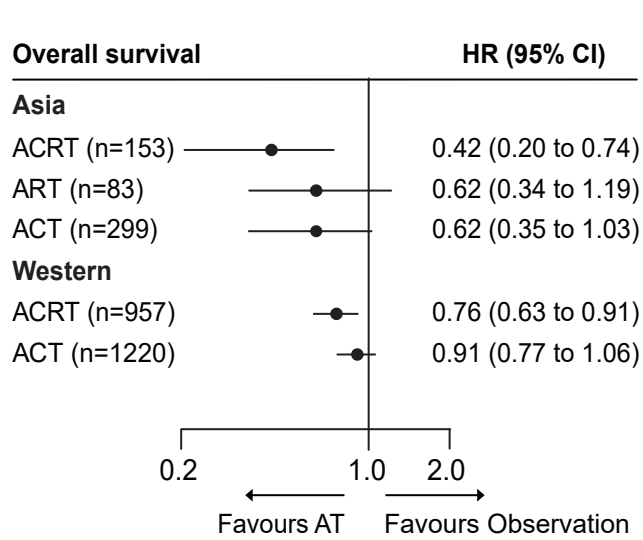

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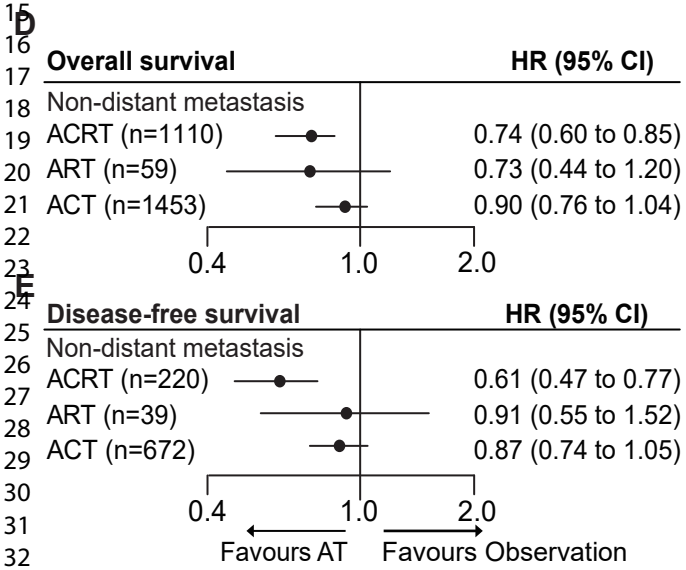


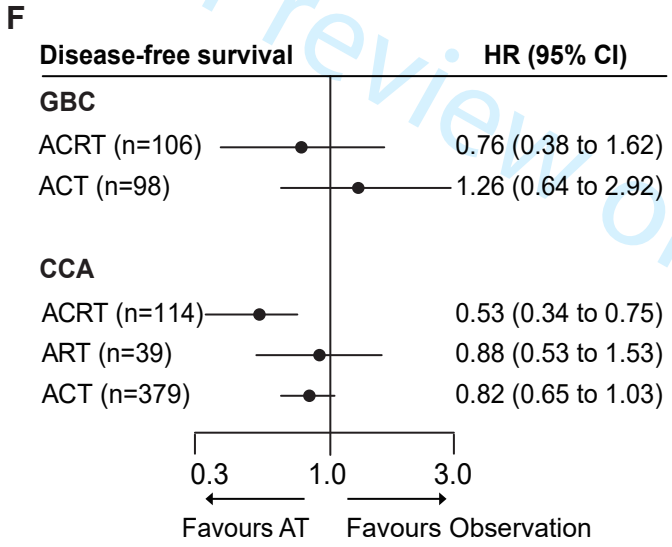

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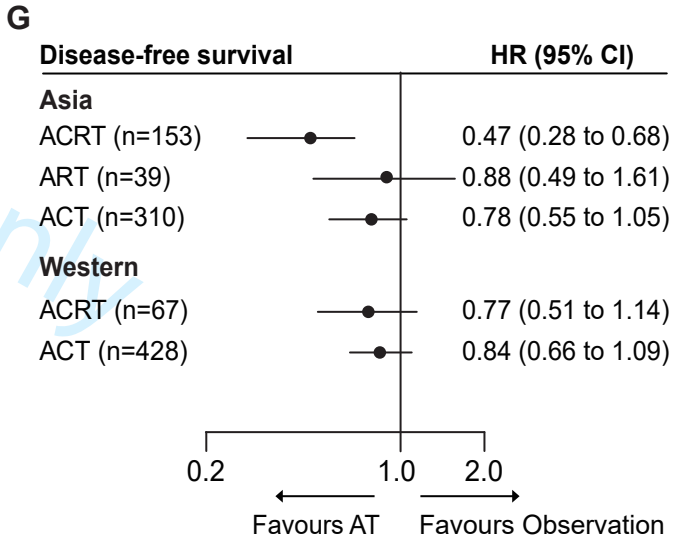


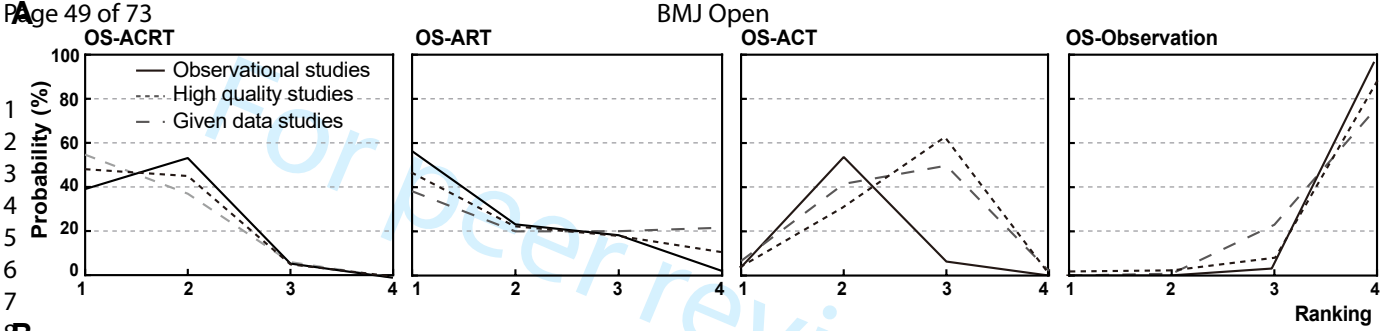




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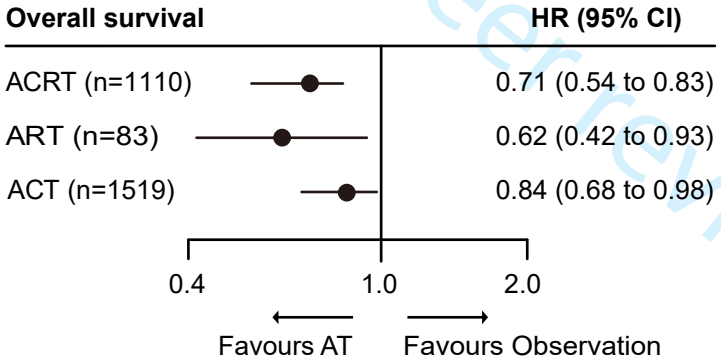



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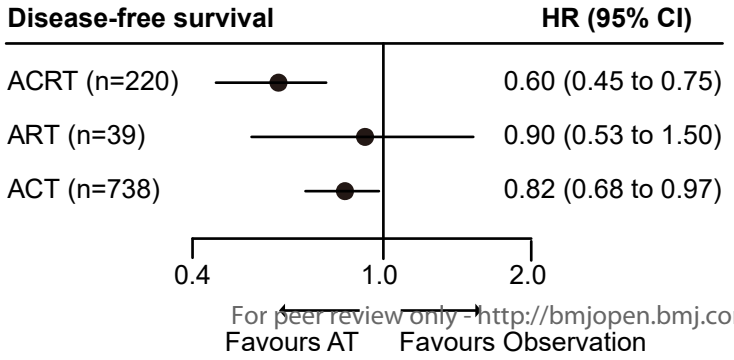
**A**

| Disease-free survival \ Overall survival | ACRT | ART | ACT | Observation |
|---|---|---|---|---|
| ACRT | | 1.14 (0.69 to 1.74) | 0.84 (0.67 to 1.01) | **0.71 (0.54 to 0.83)** |
| ART | 1.51 (0.86 to 2.73) | | 0.74 (0.49 to 1.17) | **0.62 (0.42 to 0.93)** |
| ACT | **1.37 (1.05 to 1.87)** | 0.91 (0.53 to 1.56) | | **0.84 (0.68 to 0.98)** |
| Observation | **1.68 (1.33 to 2.24)** | 1.11 (0.66 to 1.89) | **1.22 (1.03 to 1.46)** | |

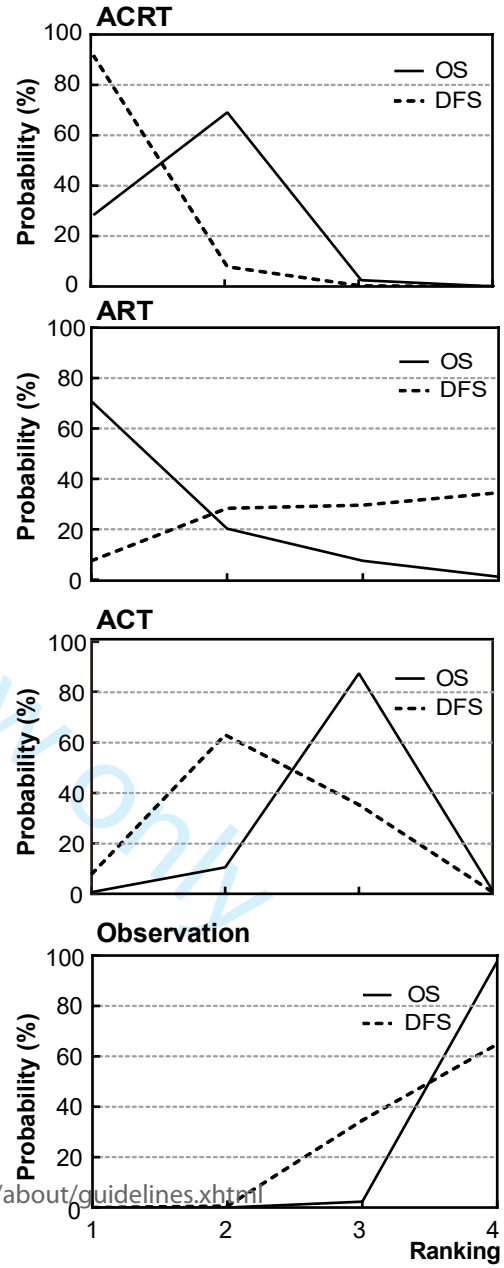
**B**




**C**




**D**




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# SUPPLEMENTAL MATERIAL

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**Supplementary Table 1. Checklist of the PRISMA extension for network meta-analysis.**

| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review *incorporating a network meta-analysis (or related form of meta-analysis).* | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable:<br>• Background: main objectives;<br>• Methods: data sources; study eligibility criteria, participants, and interventions; study appraisal and *synthesis methods, such as network meta-analysis.*<br>• Results: number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; *treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.*<br>• Discussion/Conclusions: limitations; conclusions and implications of findings.<br>• Other: primary source of funding; systematic review registration number with registry name. | 2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known, *including mention of why a network meta-analysis has been conducted.* | 4-7 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5-7 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | / |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. *Clearly describe eligible treatments included in the treatment network, and note whether any have been clustered or merged into the same node (with justification).* | 7-8 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 7 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | Supplemental Material page 6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, | 9 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | included in the meta-analysis). | |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 8-9 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 8-9 |
| Geometry of the network | S1 | Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers. | 9 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 10-11 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). *Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses.* | 9-10 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to:<br>• *Handling of multi-arm trials;*<br>• *Selection of variance structure;*<br>• *Selection of prior distributions in Bayesian analyses; and*<br>• *Assessment of model fit.* | 10 |
| Assessment of Inconsistency | S2 | Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found. | 10 |
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 11 |
| Additional analyses | 16 | Describe methods of additional analyses, if done, indicating which were pre-specified. This may include, but not be limited to the following:<br>• Sensitivity or subgroup analyses;<br>• Meta-regression analyses;<br>• *Alternative formulations of the treatment network; and*<br>• *Use of alternative prior distributions for Bayesian analyses (if applicable).* | 11 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each | 11& Figure 1 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | stage, ideally with a flow diagram. | |
| Presentation of network structure | S3 | Provide a network graph of the included studies to enable visualization of the geometry of the treatment network | 18 & Figure 2 |
| Summary of network geometry | S4 | Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure. | 18 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 13-15 & Table 1 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 16 & Supplemental Material page 7-8 and page18 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study:1) simple summary data for each intervention group, and 2) effect estimates and confidence/credible intervals. *Modified approaches may be needed to deal with information from larger networks.* | 16-17 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence/credible intervals. *In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise comparisons.* If additional summary measures were explored (such as treatment rankings), these should also be presented. | 17-18 |
| Exploration for inconsistency | S5 | Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, P values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network. | 17 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 21 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression, *alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses,* and so forth [see Item 16]). | 19-21 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 21-22 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). *Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons).* | 24-25 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 22-24 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 27 |

PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analysis; PICOS = population, intervention, comparators, outcomes, study design.

*Text in italics indicates wording specific to reporting of network meta-analyses that has been added to guidance from the PRISMA statement.

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**Supplementary Table 2. Search strategies**

| Database | Query | Results |
|---|---|---|
| Pubmed | (((((((((""Biliary Tract Neoplasms""[Mesh]) OR ""Bile Duct Neoplasms""[Mesh]) OR ""Gallbladder Neoplasms""[Mesh]) OR ""Cholangiocarcinoma""[Mesh]) OR Biliary Tract Cancer) OR ""Biliary Tract Neoplasms"") OR ""Bile duct cancer"")) AND ((((((""Chemoradiotherapy, Adjuvant""[Mesh]) OR Radiotherapy, Adjuvant[Mesh]) OR ""Chemotherapy, Adjuvant""[Mesh]) OR ""Adjuvants, Immunologic""[Mesh]) OR ""adjuvant therapy"") OR ""adjuvant treatment"")) AND ((((((""Clinical Study"" [Publication Type]) OR ""Observational Study"" [Publication Type]) OR ""Clinical Trial"" [Publication Type]) OR case control study) OR ""Case-Control Studies""[Mesh]) OR (((random) OR control) OR randomized control trial)) | 509 |
| Embase | (('biliary tract cancer'/exp OR 'biliary tract tumor'/exp OR 'bile duct tumor'/exp OR 'gallbladder tumor'/exp OR 'bile duct carcinoma'/exp OR 'biliary tract cancer' OR 'biliary tract neoplasms' OR 'bile duct neoplasms' OR 'bile duct cancer' OR 'gallbladder cancer' OR cholangiocarcinoma) AND ('adjuvant therapy'/exp OR 'adjuvant chemotherapy'/exp OR 'adjuvant radiotherapy'/exp OR 'adjuvant chemoradiotherapy'/exp OR 'adjuvant treatment') AND ('clinical article'/de OR 'clinical study'/de OR 'clinical trial'/de OR 'clinical trial (topic)'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'methodology'/de OR 'multicenter study'/de OR 'observational study'/de OR 'phase 2 clinical trial'/de OR 'phase 2 clinical trial (topic)'/de OR 'phase 3 clinical trial'/de OR 'phase 3 clinical trial (topic)'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial (topic)'/de OR 'retrospective study'/de) AND ([adult]/lim OR [aged]/lim OR [middle aged]/lim OR [very elderly]/lim OR [young adult]/lim)) | 892 |
| CINAHL | ((Biliary Tract Cancer OR "bile duct cancer" OR "Gallbladder" OR "biliary tract neoplasm" OR "Cholangiocarcinoma") AND ("adjuvant therapy" OR "adjuvant treatment" OR "adjuvant chemotherapy" OR "adjuvant radiotherapy" OR "adjuvant Chemoradiotherapy" OR "adjuvant radiochemotherapy")) | 274 |
| Cochrane | ("MeSH: [Biliary Tract Neoplasms]"/exp OR "MeSH: [cholangiocarcinoma]"/exp OR ("Biliary Tract Cancer"):ti,ab,kw) AND ("MeSH: [Radiotherapy, Adjuvant]"/exp OR "MeSH: [Chemotherapy, Adjuvant]"/exp) OR "MeSH: [Chemoradiotherapy, Adjuvant]"/exp OR "Adjuvant therapy" OR "Adjuvant treatment") | 52 |
| clinicaltrials.gov | (("Biliary Tract Cancer" OR "bile duct cancer" OR "Gallbladder" OR "Cholangiocarcinoma") AND "Adjuvant") | 46 |

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**Supplementary Table 3. Checklist for quality assessment and scoring using Newcastle-Ottawa Scale.**

| **Checklist items** |
|---|
| **Selection** |
| 1. How representative was the treatment group in comparison for BTC?<br>• If yes, one star;<br>• No star if the patients were selected or selection of group was not described. |
| 2. How representative was the observation group in comparison for BTC?<br>• If drawn from the same community as the treatment group, one star;<br>• No star if drawn from a different source or selection of group was not described. |
| 3. Ascertainment of treatment: secure record or structured interview?<br>• If the treatment were confirmed from surgical records or databases (eg NCDB or SEER), one star;<br>• No star if it was not described. |
| 4. Demonstration that outcome of interest was not present at start of study.<br>• Yes, one star;<br>• No, no star. |
| **Comparability** |
| 5. Group comparable for age, sex, primary tumour site, resection margin status, tumour stage, lymph node status, distant metastasis.<br>• If yes, two stars;<br>• One star was assigned if one of these eight characteristics was differed in groups or not reported.<br>• No star was assigned if there are at least 3 characteristics in the groups differed. |
| **Outcome** |
| 6. Assessment of outcome?<br>• One star if for information ascertained by record lincage or independent blind assessment;<br>• No star if this information was not reported. |
| 7. Was follow-up long enough for outcomes to occur?<br>• Yes, one star if the follow-up period was 5 years or more;<br>• No, no star if the follow-up period was less than 5 years or this information was not reported. |
| 8. Adequacy of follow-up.<br>• One star if follow-up 90%;<br>• No star if this information was not reported. |

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**Supplementary Table 4. Quality assessment of included retrospective studies.**

| Study | Selection | | | | Comparability | Outcome assessment | | | Score |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | |
| Todorokia 2000 | * | * | * | * | ** | * | * | * | 9 |
| Jiang 2010 | * | * | * | * | * | * | * | * | 8 |
| Kim 2011 | * | * | * | * | ** | * | * | - | 8 |
| Kobayashi 2012 | * | * | * | * | ** | * | * | * | 9 |
| Dover 2016 | * | * | * | * | * | * | * | - | 7 |
| Kim 2016 | * | * | * | * | * | * | * | - | 7 |
| Morine 2017 | * | * | * | * | ** | * | * | - | 8 |
| Gu 2017 | * | * | * | * | ** | * | * | - | 8 |
| Akahoshi 2018 | * | * | * | * | ** | * | * | - | 8 |
| Bergeat 2018 | * | * | * | * | ** | * | * | - | 8 |
| Hester 2018 | * | * | * | * | ** | * | * | - | 8 |
| Zheng 2018 | * | * | * | * | ** | * | * | * | 9 |
| Choudhary 2019 | * | * | * | * | * | * | * | - | 7 |
| Im 2021 | * | * | * | * | ** | * | * | * | 9 |
| Miyata 2021 | * | * | * | * | ** | * | * | - | 8 |
| Wan 2021 | * | * | * | * | ** | * | * | * | 9 |

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**Supplementary Table 5. Assessment of transitivity.**

| Items | Comparisons | Mean Diff. | Std.Error | Sig. | 95% CI |
|---|---|---|---|---|---|
| Age* | ART vs. Observation | -3.594 | 2.710 | 0.557 | -11.150 to 3.961 |
| | ACRT vs. Observation | -1.327 | 2.430 | 0.947 | -8.100 to 5.445 |
| | ACT vs. Observation | 1.416 | 2.012 | 0.895 | -4.192 to 7.024 |
| | ACRT vs. ART | 2.267 | 3.178 | 0.891 | -6.592 to 11.130 |
| | ACT vs. ART | 5.010 | 2.872 | 0.327 | -2.994 to 13.010 |
| | ACT vs. ACRT | 2.743 | 2.608 | 0.722 | -4.527 to 10.010 |
| Percentage female | ART vs. Observation | 0.81 | 9.782 | 0.9998 | -25.46 to 27.09 |
| | ACRT vs. Observation | 11.09 | 7.547 | 0.47 | -9.19 to 31.36 |
| | ACT vs. Observation | 6.10 | 6.485 | 0.78 | -11.33 to 23.52 |
| | ACRT vs. ART | 10.28 | 10.78 | 0.78 | -18.68 to 39.23 |
| | ACT vs. ART | 5.28 | 10.06 | 0.95 | -21.75 to 32.32 |
| | ACT vs. ACRT | -4.99 | 7.91 | 0.92 | -26.24 to 16.26 |
| Sample size | ART vs. Observation | -145.1 | 157.3 | 0.793 | -567.7 to 277.5 |
| | ACRT vs. Observation | -25.2 | 121.4 | 0.996 | -351.3 to 300.8 |
| | ACT vs. Observation | -44.7 | 104.3 | 0.973 | -324.9 to 235.5 |
| | ACRT vs. ART | 119.9 | 173.4 | 0.900 | -345.8 to 585.6 |
| | ACT vs. ART | 100.4 | 161.9 | 0.925 | -334.5 to 535.2 |
| | ACT vs. ACRT | -19.5 | 127.2 | 0.999 | -361.2 to 322.2 |
| Publication year | ART vs. Observation | -3.750 | 2.633 | 0.492 | -10.8 to 3.323 |
| | ACRT vs. Observation | 1.375 | 2.031 | 0.905 | -4.082 to 6.832 |
| | ACT vs. Observation | 2.231 | 1.746 | 0.582 | -2.459 to 6.920 |
| | ACRT vs. ART | 5.125 | 2.901 | 0.305 | -2.669 to 12.92 |
| | ACT vs. ART | 5.981 | 2.709 | 0.139 | -1.297 to 13.2 |
| | ACT vs. ACRT | 0.856 | 2.129 | 0.978 | -4.864 to 6.575 |
| Percentage R0 | ART vs. Observation | -12.66 | 17.17 | 0.881 | -58.75 to 33.42 |
| | ACRT vs. Observation | 7.61 | 13.97 | 0.947 | -29.87 to 45.09 |
| | ACT vs. Observation | 3.64 | 12.52 | 0.991 | -29.95 to 37.23 |
| | ACRT vs. ART | 20.27 | 18.70 | 0.701 | -29.90 to 70.44 |
| | ACT vs. ART | 16.30 | 17.64 | 0.792 | -31.03 to 63.64 |
| | ACT vs. ACRT | -3.97 | 14.54 | 0.993 | -42.98 to 35.03 |
| Percentage N0 | ART vs. Observation | 19.33 | 11.23 | 0.332 | -11.32 to 49.99 |
| | ACRT vs. Observation | -14.00 | 7.74 | 0.291 | -35.13 to 7.13 |
| | ACT vs. Observation | -10.90 | 5.99 | 0.285 | -27.24 to 5.44 |
| | ACRT vs. ART | -33.33 | 12.43 | 0.056 | -67.27 to 0.60 |
| | ACT vs. ART | -30.24 | 11.42 | 0.060 | -61.41 to 0.94 |
| | ACT vs. ACRT | 3.10 | 8.01 | 0.980 | -18.78 to 24.97 |

The characteristics included age, percentage female, sample size, publication year, percentage R0 and N0 have been evaluated in the network. All the comparisons had similar mean age and other main characteristics with P-value over 0.05.

* Median age was given instead of mean age in the Bergeat, Dover, Miyata, Gu, and Edeline studies. Information of age in the Jiang, Kobayashi, Akahoshi, Ebata, Choudhary, Im (2021), and Wan studies were presented as younger or older than a specific age that couldn't be integrated into the figure.



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**Supplementary Table 6. Results of global heterogeneity and local heterogeneity.**

| Comparisons | Local | | Global | | | Heterogeneity assessment |
|---|---|---|---|---|---|---|
| | Pair-wise ($I^2$) | Network ($I^2$) | Between study variance ($\tau^2$) | Pair-wise ($I^2$) | Consistency effect ($I^2$) | |
| **Overall survival for BTC patients** | | | | | | |
| ACT vs Observation | 20.3% | 20.2% | 0.022 | 0.0% | 19.4% | Low to high |
| ART vs Observation | 0.0% | **81.9%** | | | | |
| ACRT vs Observation | 33.4% | 47.5% | | | | |
| ACT vs ART | 0.0% | 49.4% | | | | |
| ACRT vs ART | 0.0% | 28.8% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients with R0 resection margin** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.069 | 0.96% | 0.0% | Low to moderate |
| ART vs Observation | NA | 0.0% | | | | |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACT vs ART | NA | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 33.5% | 21.6% | | | | |
| **Overall survival for BTC patients with R1 resection margin** | | | | | | |
| ACT vs Observation | 33.8% | 36.5% | 0.179 | 24.9% | 13.5% | Low to moderate |
| ART vs Observation | 0.0% | 34.5% | | | | |
| ACRT vs Observation | 28.1% | 28.5% | | | | |
| ACT vs ART | NA | 0.0% | | | | |
| ACRT vs ART | NA | 0.0% | | | | |
| ACRT vs ACT | NA | 0.0% | | | | |
| **Overall survival for GBC patients** | | | | | | |
| ACT vs Observation | 44.1% | 45.0% | 0.028 | 12.1% | 6.0% | Low to moderate |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.064 | 0.0% | 25.8% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | **66.7%** | **93.7%** | | | | |
| ACT vs ART | 0.0% | 47.7% | | | | |
| ACRT vs ART | 0.0% | 17.6% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | **56.5%** | **57.0%** | 0.271 | 27.5% | **64.9%** | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | **71.6%** | **91.7%** | | | | |
| ACT vs ART | 0.0% | 9.8% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.007 | 0.0% | 0.0% | Low |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients without distant-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.011 | 0.0% | 1.7% | Low to high |
| ART vs Observation | 0.0% | **63.2%** | | | | |
| ACRT vs Observation | 33.1% | 41.0% | | | | |
| ACT vs ART | 0.0% | 42.7% | | | | |
| ACRT vs ART | 0.0% | 20.6% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients** | | | | | | |



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| | | | | | | |
|---|---|---|---|---|---|---|
| ACT vs Observation | 6.1% | 6.6% | 0.017 | 0.0% | 0.0% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 20.0% | **82.2%** | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for GBC patients** | | | | | | |
| ACT vs Observation | **75.0%** | 48.1% | 0.169 | **51.2%** | 34.1% | Low to high |
| ACRT vs Observation | 9.7% | 0.0% | | | | |
| ACRT vs ACT | 3.7% | 0.0% | | | | |
| **Disease-free survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.029 | 0.0% | 0.0% | Low to moderate |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 35.0% | 47.0% | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | 32.8% | 32.4% | 0.067 | 0.0% | 0.0% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 35.8% | **67.6%** | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.023 | 0.0% | 0.0% | Low |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Disease-free survival for BTC patients for patients without distant-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.012 | 0.0% | 0.0% | Low to moderate |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 20.0% | 38.7% | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |

The numbers with high heterogeneity are in bold (We inferred the magnitude of heterogeneity by comparing the estimated $\tau 2$ to empirical distributions of heterogeneity typically found in meta-analyses. Low heterogeneity could be considered when the estimated $\tau^2$ is less than the 25% quantile of the empirical distribution, moderate heterogeneity for $\tau^2$ between 25% and 50% quantile, and high heterogeneity for $\tau^2$ larger than the 50% quantile.). NA=Not available, because only one study was included in this type of comparison; ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma.



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**Supplementary Table 7. Node-splitting analysis of inconsistency.**

| Comparison | Direct effect | Indirect effect | Overall | *P* |
|---|---|---|---|---|
| | lnHR (95% Cl) | | | |
| **Overall survival for BTC patients** | | | | |
| ACRT, ACT | 0.24 (0.02, 0.51) | 0.10 (-0.30, 0.58) | 0.18 (-0.01, 0.40) | 0.52 |
| ACRT, ART | 1.3 (0.63, 1.90) | -0.56 (-1.10, -0.08) | -0.13 (-0.53, 0.37) | **0.0000** |
| ACT, ART | 0.60 (0.05, 1.10) | -0.66 (-1.10, -0.18) | -0.31 (-0.71, 0.16) | **0.001** |
| ACRT, Observation | 0.35 (0.07, 0.73) | 0.83 (0.25, 1.60) | 0.35 (0.18, 0.61) | 0.15 |
| ACT, Observation | 0.24 (-0.03, 0.54) | 0.19 (-0.52, 0.94) | 0.17 (0.02, 0.38) | 0.89 |
| ART, Observation | 0.70 (0.29, 1.10) | -0.92 (-2.90, 1.10) | 0.48 (0.07, 0.86) | 0.13 |
| **Overall survival for BTC patients with R0 resection margin** | | | | |
| ACRT, ACT | 1.2 (-0.82, 3.3) | 0.35 (-0.31, 1.2) | 0.41 (-0.16, 1.20) | 0.43 |
| ACRT, ART | NA | NA | NA | NA |
| ACT, ART | 0.92 (-1.30, 3.20) | 0.05 (-2.20, 2.30) | 0.53 (-1.00, 2.10) | 0.58 |
| ACRT, Observation | 0.49 (-0.03, 1.20) | 1.4 (-0.69, 3.40) | 0.53 (0.07, 1.20) | 0.41 |
| ACT, Observation | 0.14 (-0.40, 0.65) | -0.24 (-2.00, 1.60) | 0.12 (-0.35, 0.54) | 0.69 |
| ART, Observation | 0.37 (-1.80, 2.40) | -0.90 (-3.00, 1.20) | -0.41 (-2.00, 1.10) | 0.41 |
| **Overall survival for BTC patients with R1 resection margin** | | | | |
| ACRT, ACT | 0.60 (-0.69, 1.90) | 0.57 (-0.80, 1.80) | 0.68 (-0.32, 1.60) | 0.97 |
| ACRT, ART | 1.20 (-0.17, 2.5) | -0.31 (-2.40, 1.60) | 0.66 (-0.52, 1.80) | 0.18 |
| ACT, ART | 0.57 (-0.67, 1.80) | -0.88 (-2.60, 0.86) | -0.017 (-1.10, 1.00) | 0.15 |
| **Overall survival for GBC patients** | | | | |
| ACRT, ACT | 0.17 (-0.37, 0.82) | 0.81 (-0.63, 2.50) | 0.20 (-0.09, 0.55) | 0.35 |
| ACRT, Observation | 0.25 (-0.20, 0.76) | 0.55 (-0.98, 1.90) | 0.25 (-0.02, 0.54) | 0.67 |
| ACT, Observation | 0.061 (-0.51, 0.45) | -0.35 (-2.00, 1.30) | 0.06 (-0.26, 0.30) | 0.58 |
| **Overall survival for CCA patients** | | | | |
| ACRT, ACT | 0.35 (0.03, 0.81) | 0.15 (-0.29, 0.72) | 0.30 (-0.03, 0.81) | 0.45 |
| ACRT, ART | 1.3 (0.60, 2.00) | -0.57 (-1.10, 0.01) | 0.04 (-0.51, 0.77) | **0.00035** |
| ACT, ART | 0.61 (-0.02, 1.20) | -0.70 (-1.20, -0.13) | -0.27 (-0.77, 0.29) | **0.0056** |
| ACRT, Observation | 0.56 (0.03, 1.30) | 0.69 (-0.13, 1.70) | 0.49 (0.16, 1.00) | 0.77 |
| ART, Observation | 0.69 (0.27, 1.10) | -0.98 (-3.0, 1.10) | 0.46 (-0.02, 0.91) | 0.12 |
| ACT, Observation | 0.19 (-0.20, 0.59) | 0.51 (-0.41, 1.60) | 0.18 (-0.07, 0.50) | 0.50 |
| **Overall survival for BTC patients in Asia** | | | | |
| ACRT, ACT | 0.71 (-0.18, 1.70) | -0.02 (-0.74, 0.95) | 0.33 (-0.37, 1.20) | 0.23 |
| ACRT, ART | 1.3 (0.53, 2.20) | -0.75 (-1.50, 0.05) | 0.18 (-0.56, 1.20) | **0.0018** |
| ACT, ART | 0.11 (-0.78, 1.00) | -0.27 (-1.30, 0.85) | -0.15 (-0.78, 0.56) | 0.56 |
| ACRT, Observation | 0.63 (-0.040, 1.50) | 2.00 (-0.17, 4.30) | 0.77 (0.17, 1.60) | 0.24 |
| ACT, Observation | 0.55 (-0.11, 1.20) | -0.057 (-1.4, 1.4) | 0.44 (-0.06, 0.99) | 0.41 |
| ART, Observation | 0.67 (0.07, 1.30) | 0.71 (-0.54, 1.80) | 0.59 (-0.01, 1.10) | 0.96 |
| **Overall survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.16 (-0.11, 0.39) | 0.49 (-0.28, 1.30) | 0.18 (-0.02, 0.37) | 0.40 |
| ACRT, Observation | 0.25 (-0.01, 0.55) | 0.35 (-0.26, 0.89) | 0.28 (0.09, 0.46) | 0.70 |
| ACT, Observation | 0.12 (-0.09, 0.31) | 0.32 (-0.48, 1.1) | 0.10 (-0.06, 0.26) | 0.61 |
| **Overall survival for BTC patients without distant-metastasis** | | | | |
| ACRT, ACT | 0.24 (0.03, 0.48) | 0.20 (-0.17, 0.69) | 0.20 (0.039, 0.40) | 0.86 |
| ACRT, ART | 1.30 (0.66, 1.90) | -0.74 (-1.40, -0.04) | -0.01 (-0.52, 0.53) | **0.0000** |
| ACT, ART | 0.59 (0.06, 1.10) | -0.82 (-1.50, -0.15) | -0.21 (-0.72, 0.31) | **0.001** |
| ART, Observation | 0.66 (0.13, 1.20) | -1.00 (-3.00, 0.98) | 0.32 (-0.18, 0.81) | 0.12 |
| ACRT, Observation | 0.31 (0.08, 0.69) | 0.61 (0.11, 1.40) | 0.31 (0.15, 0.52) | 0.28 |
| ACT, Observation | 0.14 (-0.12, 0.42) | 0.20 (-0.43, 0.94) | 0.10 (-0.04, 0.27) | 0.85 |
| **Disease-free survival for BTC patients** | | | | |
| ACRT, ACT | 0.49 (0.10, 0.93) | 0.26 (-0.14, 0.70) | 0.32 (0.05, 0.63) | 0.41 |
| ACRT, ART | 1.30 (0.60, 2.10) | -0.10 (-0.95, 0.72) | 0.41 (-0.14, 1.00) | **0.009** |
| ACT, ART | 0.33 (-0.27, 0.94) | -0.20 (-0.92, 0.54) | 0.09 (-0.44, 0.63) | 0.27 |
| ACRT, Observation | 0.43 (0.17, 0.75) | 1.10 (0.34, 1.80) | 0.52 (0.28, 0.81) | 0.11 |
| ACT, Observation | 0.22 (0.04, 0.41) | -0.14 (-0.87, 0.60) | 0.20 (0.03, 0.38) | 0.35 |



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| | | | | |
|---|---|---|---|---|
| ART, Observation | 0.29 (-0.30, 0.86) | -0.30 (-1.90, 1.30) | 0.11 (-0.41, 0.63) | 0.50 |
| **Disease-free survival for GBC patients** | | | | |
| ACRT, ACT | 0.30 (-0.52, 1.20) | 1.50 (-0.38, 3.30) | 0.50 (-0.20, 1.30) | 0.22 |
| ACRT, Observation | 0.30 (-0.47, 1.10) | -0.25 (-2.10, 1.60) | 0.27 (-0.48, 0.97) | 0.52 |
| ACT, Observation | -0.28 (-1.30, 0.60) | -0.15 (-2.10, 1.80) | -0.24 (-1.10, 0.44) | 0.89 |
| **Disease-free survival for CCA patients** | | | | |
| ACRT, ACT | 1.00 (0.33, 1.70) | 0.27 (-0.18, 0.78) | 0.45 (0.05, 0.95) | 0.09 |
| ACRT, ART | 1.30 (0.62, 2.10) | -0.14 (-1.10, 0.79) | 0.53 (-0.08, 1.30) | **0.017** |
| ACT, ART | 0.33 (-0.32, 0.98) | -0.19 (-0.98, 0.63) | 0.09 (-0.49, 0.67) | 0.32 |
| ACRT, Observation | 0.58 (0.20, 1.10) | 1.40 (0.02, 2.70) | 0.64 (0.29, 1.10) | 0.28 |
| ACT, Observation | 0.21 (-0.04, 0.49) | -0.15 (-1.40, 1.10) | 0.20 (-0.03, 0.44) | 0.55 |
| ART, Observation | 0.28 (-0.33, 0.89) | -0.33 (-2.00, 1.30) | 0.11 (-0.45, 0.66) | 0.50 |
| **Disease-free survival for BTC patients in Asia** | | | | |
| ACRT, ACT | 0.88 (0.29, 1.50) | 0.25 (-0.32, 0.92) | 0.50 (0.09, 1.00) | 0.14 |
| ACRT, ART | 1.30 (0.56, 2.20) | 0.02 (-0.99, 1.10) | 0.64 (-0.01, 1.40) | 0.052 |
| ACT, ART | 0.33 (-0.43, 1.10) | -0.13 (-1.00, 0.85) | 0.13 (-0.49, 0.78) | 0.44 |
| ACRT, Observation | 0.65 (0.21, 1.30) | 1.2 (0.30, 2.10) | 0.76 (0.39, 1.30) | 0.31 |
| ACT, Observation | 0.30 (-0.06, 0.70) | -0.01 (-0.90, 0.99) | 0.25 (-0.05, 0.59) | 0.52 |
| ART, Observation | 0.28 (-0.39, 0.95) | -0.30 (-2.00, 1.40) | 0.12 (-0.48, 0.72) | 0.54 |
| **Disease-free survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.09 (-0.51 to 0.69) | 0.23 (-0.51 to 0.99) | 0.09 (-0.33 to 0.55) | 0.76 |
| **Disease-free survival for BTC patients without distant-metastasis** | | | | |
| ACRT, ACT | 0.48 (0.12, 0.89) | 0.34 (-0.04, 0.77) | 0.35 (0.09, 0.65) | 0.61 |
| ACRT, ART | 1.30 (0.62, 2.00) | -0.12 (-0.94, 0.69) | 0.39 (-0.17, 0.96) | **0.01** |
| ACT, ART | 0.33 (-0.26, 0.92) | -0.26 (-0.98, 0.48) | 0.05 (-0.46, 0.57) | 0.19 |
| ACRT, Observation | 0.43 (0.18, 0.72) | 0.98 (0.27, 1.70) | 0.49 (0.25, 0.75) | 0.16 |
| ACT, Observation | 0.15 (-0.05, 0.33) | -0.13 (-0.85, 0.62) | 0.14 (-0.04, 0.30) | 0.46 |
| ART, Observation | 0.29 (-0.29, 0.86) | -0.33 (-2.00, 1.30) | 0.10 (-0.42, 0.62) | 0.46 |

Nonsignificant values ($P > 0.05$) indicate no inconsistency between direct and indirect effects. NA=Not available; CI=confidence interval. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 8. Comparisons of the fitness of consistency and inconsistency models using deviance information criteria**

| Item | Model | Overall | Resection margin status | | Tumor Site | | The regions | | Non-distant metastasis |
|---|---|---|---|---|---|---|---|---|---|
| | | | R0 | R1 | GBC | CCA | Asia | Western | |
| OS | Consistency | 46.17 | 15.37 | 16.03 | 16.95 | 31.39 | 226.5 | 11.88 | 37.79 |
| | Inconsistency | 46.05 | 17.96 | 15.99 | 16.70 | 32.54 | 228.1 | 12.93 | 38.31 |
| DFS | Consistency | 29.93 | NA | NA | 9.50 | 22.21 | 22.76 | 7.88 | 23.61 |
| | Inconsistency | 31.15 | NA | NA | 10.26 | 24.96 | 25.12 | 7.89 | 25.28 |

The fitness of the Bayesian model was evaluated by deviance information criteria (DIC), which is adjusted with the complexity of the model.

NA=Not applicable; OS=overall survival; DFS=disease-free survival; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma.

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**Supplementary Table 9. Ranking results of network meta-analysis for overall survival (OS) and disease-free survival (DFS).**

| Treatment | Rank of probability for OS (%) | | | | | Rank of probability for DFS (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| ACT | 0.7 | 10.5 | **87.5** | 1.3 | 36.9 | 7.9 | 63.1 | 35.4 | 0.7 | 54.7 |
| ART | **70.9** | 20.3 | 7.5 | 1.2 | **87.0** | 7.5 | 28.4 | 29.6 | 34.5 | 36.3 |
| ACRT | 28.4 | **69.1** | 2.5 | 0.0 | 75.3 | **91.7** | 7.9 | 0.3 | 0.0 | **97.1** |
| Observation | 0.0 | 0.0 | 2.4 | 97.5 | 0.8 | 0.0 | 0.6 | 34.6 | 64.8 | 11.9 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 10. Rank results for OS and DFS in the sensitivity analysis.**

| Treatment | Rank of probability (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| | OS for studies including observational studies | | | | | DFS for studies including observational studies | | | | |
| ACT | 3.8 | 53.7 | 6.4 | 0.0 | 43.2 | 1.6 | 60.3 | 34.6 | 3.4 | 53.4 |
| ART | **56.4** | 23.2 | 18.3 | 2.1 | **77.9** | 6.9 | 28.9 | 28.1 | 36.0 | 35.6 |
| ACRT | 39.8 | 53.7 | 6.4 | 0.0 | **77.8** | **91.4** | 8.0 | 0.5 | 0.0 | **97.0** |
| Observation | 0.0 | 0.0 | 3.2 | 96.8 | 1.1 | 0.0 | 2.7 | 36.7 | 60.6 | 14.1 |
| **Treatment** | **OS for studies with high quality** | | | | | **DFS for studies with high quality** | | | | |
| ACT | 4.0 | 31.0 | 63.0 | 2.0 | 45.7 | 0.9 | 63.3 | 34.9 | 0.9 | 54.7 |
| ART | 47.3 | 22.9 | 18.5 | 11.2 | 68.8 | 3.8 | 31.6 | 31.6 | 32.9 | 35.4 |
| ACRT | **48.7** | 45.6 | 5.6 | 0.1 | **90.0** | **95.3** | 4.4 | 0.3 | 0.0 | **98.3** |
| Observation | 0.0 | 0.5 | 6.3 | 86.7 | 1.6 | 0.0 | 0.7 | 33.2 | 66.1 | 11.5 |
| **Treatment** | **OS for original reported HRs (95%CI) in the studies** | | | | | **DFS for original reported HRs (95%CI) in the studies** | | | | |
| ACT | 6.6 | 41.5 | 49.8 | 2.1 | 50.8 | 2.2 | 77.8 | 18.3 | 1.6 | 60.2 |
| ART | 38.2 | 20.0 | 20.1 | 21.7 | 58.2 | 3.7 | 11.7 | 14.7 | 69.9 | 16.4 |
| ACRT | **55.2** | 37.7 | 7.0 | 0.1 | **82.7** | **94.0** | 5.6 | 0.3 | 0.0 | **97.9** |
| Observation | 0.0 | 0.9 | 23.0 | 76.1 | 8.3 | 0.0 | 4.9 | 66.6 | 28.5 | 25.5 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

CI=confidence interval; OS=overall survival; DFS=disease-free survival.

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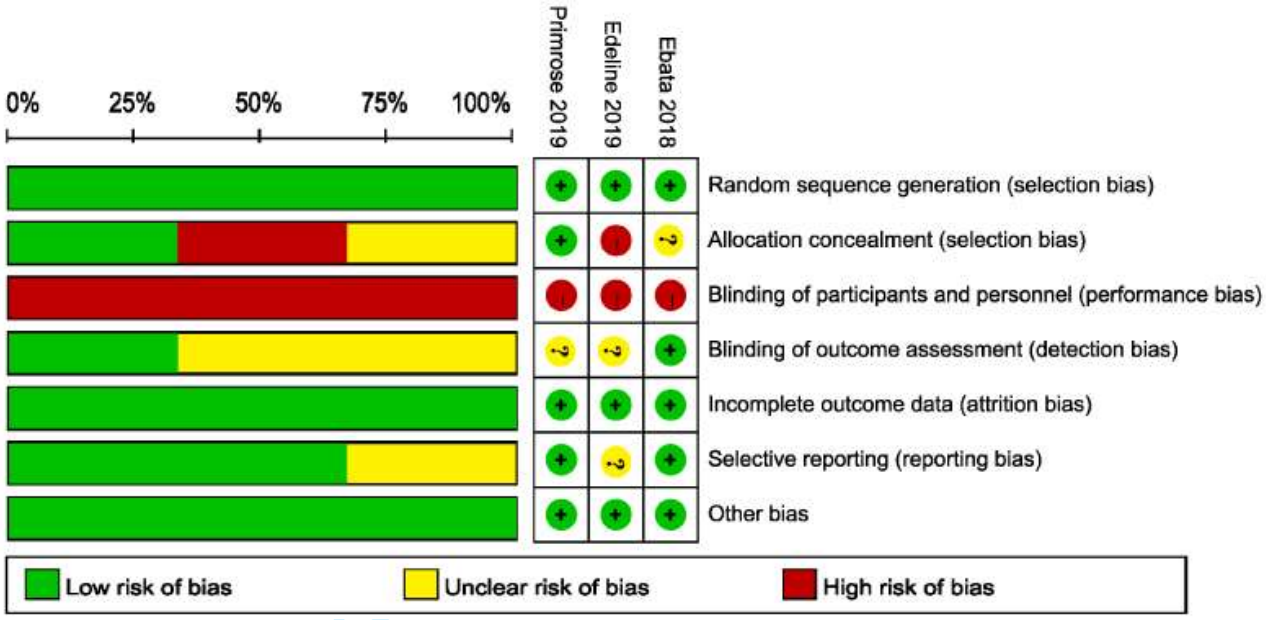
**Supplementary Table 11. Ranking results for overall survival (OS) and disease-free survival (DFS) in the subgroup analysis.**

| Treatment | Rank of probability (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| | OS for BTC patients with R0 resection margin | | | | | OS for BTC patients with R1 resection margin | | | | |
| ACT | 6.0 | 54.0 | 32.3 | 7.6 | 52.8 | 5.0 | 31.9 | 36.3 | 26.5 | 38.5 |
| ART | 11.6 | 12.0 | 7.6 | 68.8 | 22.1 | 9.9 | 36.8 | 17.5 | 35.8 | 40.3 |
| ACRT | **81.9** | 15.6 | 2.2 | 0.3 | **93.1** | **83.9** | 11.2 | 3.3 | 1.6 | **92.5** |
| Observation | 0.4 | 18.4 | 57.9 | 23.3 | 32.0 | 28.8 | 1.2 | 20.1 | 42.5 | 36.2 |
| | OS for GBC patients | | | | | OS for CCA patients | | | | |
| ACT | 7.4 | 61.3 | 31.3 | NA | 38.1 | 1.1 | 15.9 | 75.7 | 7.3 | 37.0 |
| ART | NA | NA | NA | NA | NA | 45.4 | 39.8 | 12.2 | 2.7 | 75.9 |
| ACRT | **90.4** | 8.3 | 1.3 | NA | **94.5** | 53.5 | 43.9 | 2.5 | 0.1 | **83.6** |
| Observation | 2.2 | 30.4 | 67.4 | NA | 17.4 | 0.0 | 0.4 | 9.7 | 90.0 | 3.5 |
| | OS for BTC patients in Asia | | | | | OS for BTC patients in Western | | | | |
| ACT | 8.5 | 45.1 | 43.6 | 2.8 | 53.1 | 3.2 | 87.2 | 9.6 | NA | 46.8 |
| ART | 13.5 | 37.1 | 42.7 | 6.7 | 52.5 | NA | NA | NA | NA | NA |
| ACRT | **78.0** | 17.2 | 4.7 | 0.1 | **91.0** | **96.6** | 3.1 | 0.2 | NA | **98.2** |
| Observation | 0.0 | 0.6 | 9.0 | 90.5 | 3.4 | 0.2 | 96.6 | 90.1 | NA | 5.0 |
| | OS for BTC patients without distant metastasis | | | | | DFS for BTC patients without distant metastasis | | | | |
| ACT | 0.4 | 19.9 | 73.1 | 6.7 | 38.0 | 0.3 | 53.7 | 42.0 | 4.0 | 50.1 |
| ART | **50.2** | 29.2 | 10.6 | 10.0 | 73.2 | 7.1 | 36.2 | 21.1 | 35.5 | 38.3 |
| ACRT | 49.4 | 50.0 | 0.6 | 0.0 | **82.9** | **92.6** | 7.2 | 0.2 | 0.0 | **97.5** |
| Observation | 0.0 | 0.9 | 15.7 | 83.3 | 5.8 | 0.0 | 2.9 | 36.6 | 60.5 | 14.1 |
| | DFS for GBC patients | | | | | DFS for CCA patients | | | | |
| ACT | 4.1 | 20.1 | 75.8 | NA | 14.1 | 1.2 | 57.9 | 38.2 | 2.7 | 52.6 |
| ART | NA | NA | NA | NA | NA | 4.4 | 35.0 | 28.1 | 32.5 | 37.1 |
| ACRT | **78.2** | 18.0 | 3.8 | NA | **87.2** | **94.4** | 5.2 | 0.4 | 0.0 | 37.1 |
| Observation | 17.7 | 62.0 | 20.4 | NA | 48.7 | 0.0 | 1.8 | 33.3 | 64.8 | 12.3 |
| | DFS for BTC patients in Asia | | | | | DFS for BTC patients in Western | | | | |
| ACT | 0.9 | 62.8 | 33.3 | 3.0 | 53.9 | 32.3 | 60.4 | 7.3 | NA | 62.5 |
| ART | 3.1 | 31.1 | 33.0 | 32.8 | 34.8 | NA | NA | NA | NA | NA |
| ACRT | **96.0** | 3.8 | 0.2 | 0.0 | **98.6** | **66.4** | 25.2 | 8.5 | NA | **78.9** |
| Observation | 0.0 | 2.3 | 33.4 | 64.2 | 12.7 | 1.3 | 14.5 | 84.2 | NA | 8.6 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. NA=Not available; ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.



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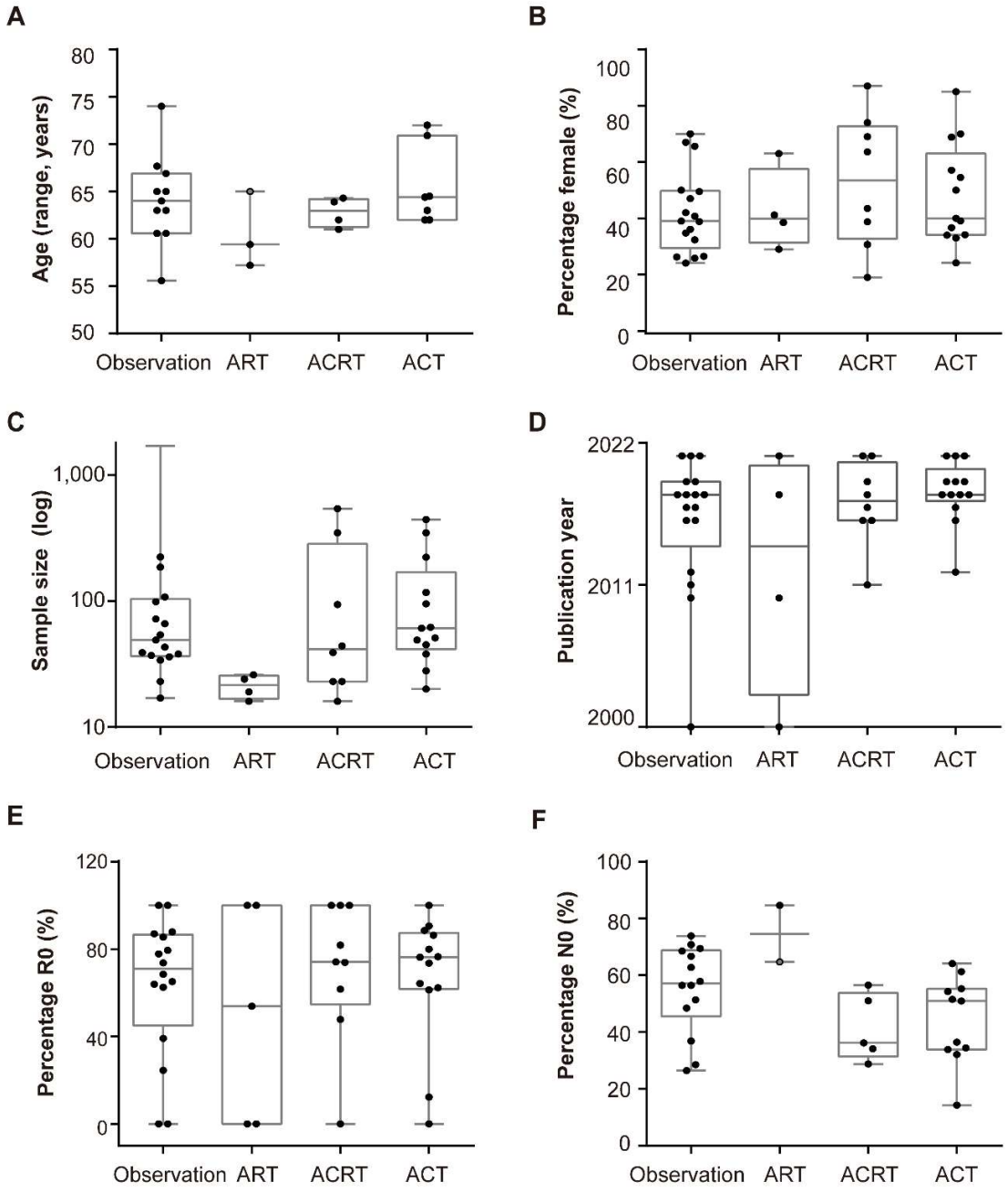



**Supplementary Figure 1. Summary of results from quality assessment of three randomized controlled studies using the Cochrane risk of bias tool.**

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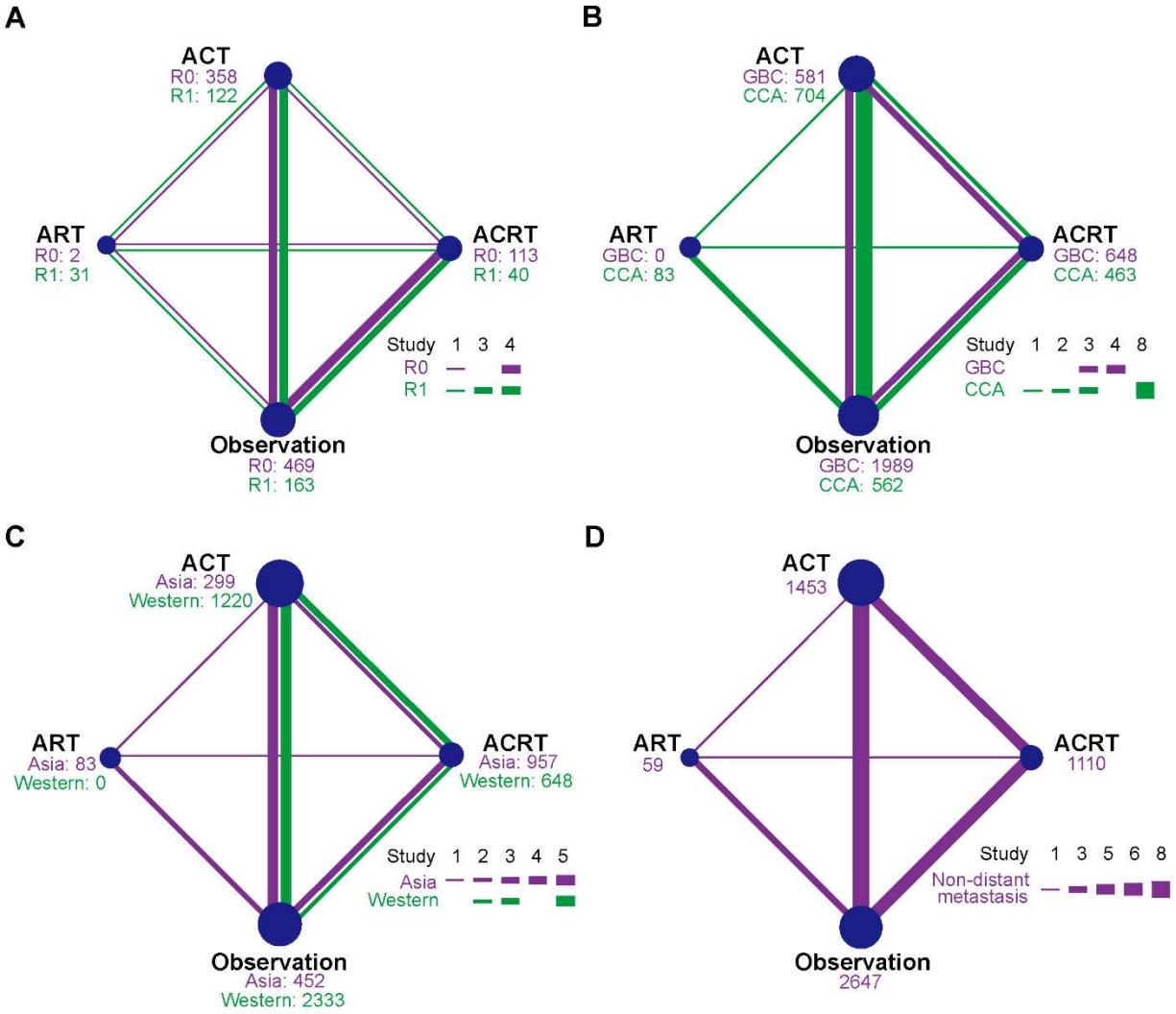
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**Supplementary Figure 2. Boxplots for distribution of mean age (A), percentage female (B), sample size (C), publication year (D), percentage R0 (E) and N0 (F) across comparisons.**



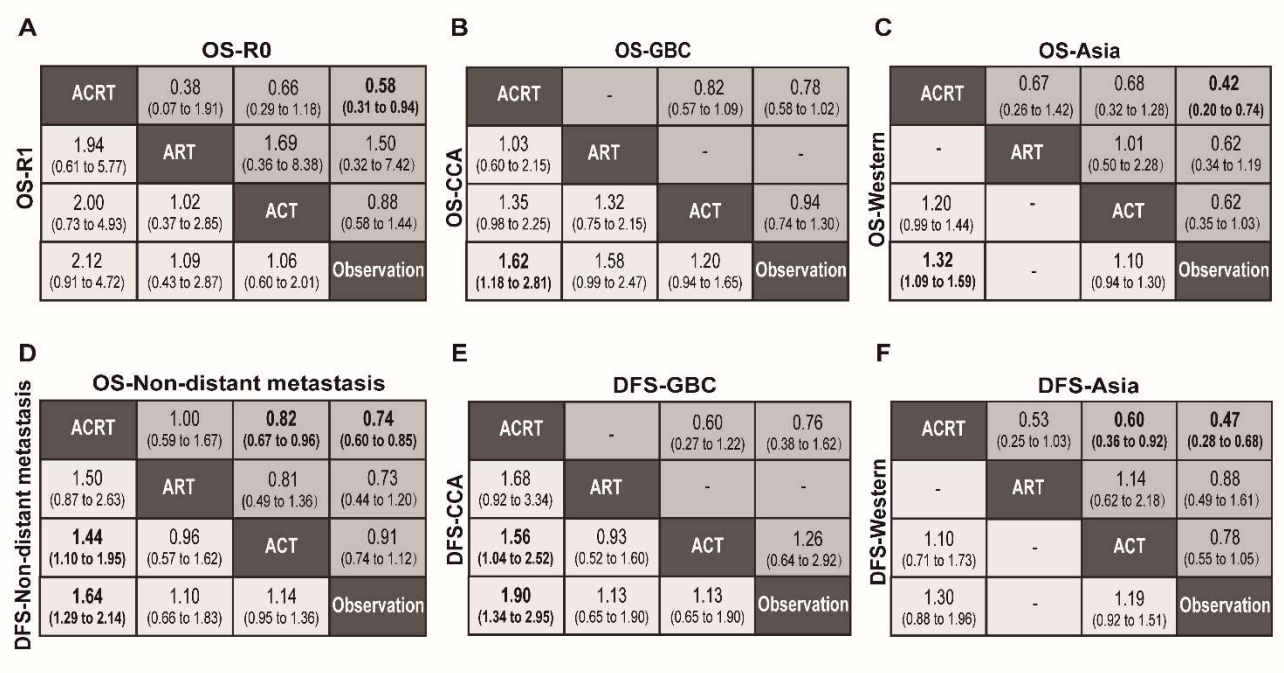
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**Supplementary Figure 3. Network plots of comparisons on overall survival of treatments in subgroups analyses (patients stratified into R0, R1, GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** Comparisons on overall survival in patients with GBC and CCA; **(C)** Comparisons on overall survival in subgroups of study patients in Asia and Western; **(D)** Comparisons on overall survival in patients without distal metastasis.



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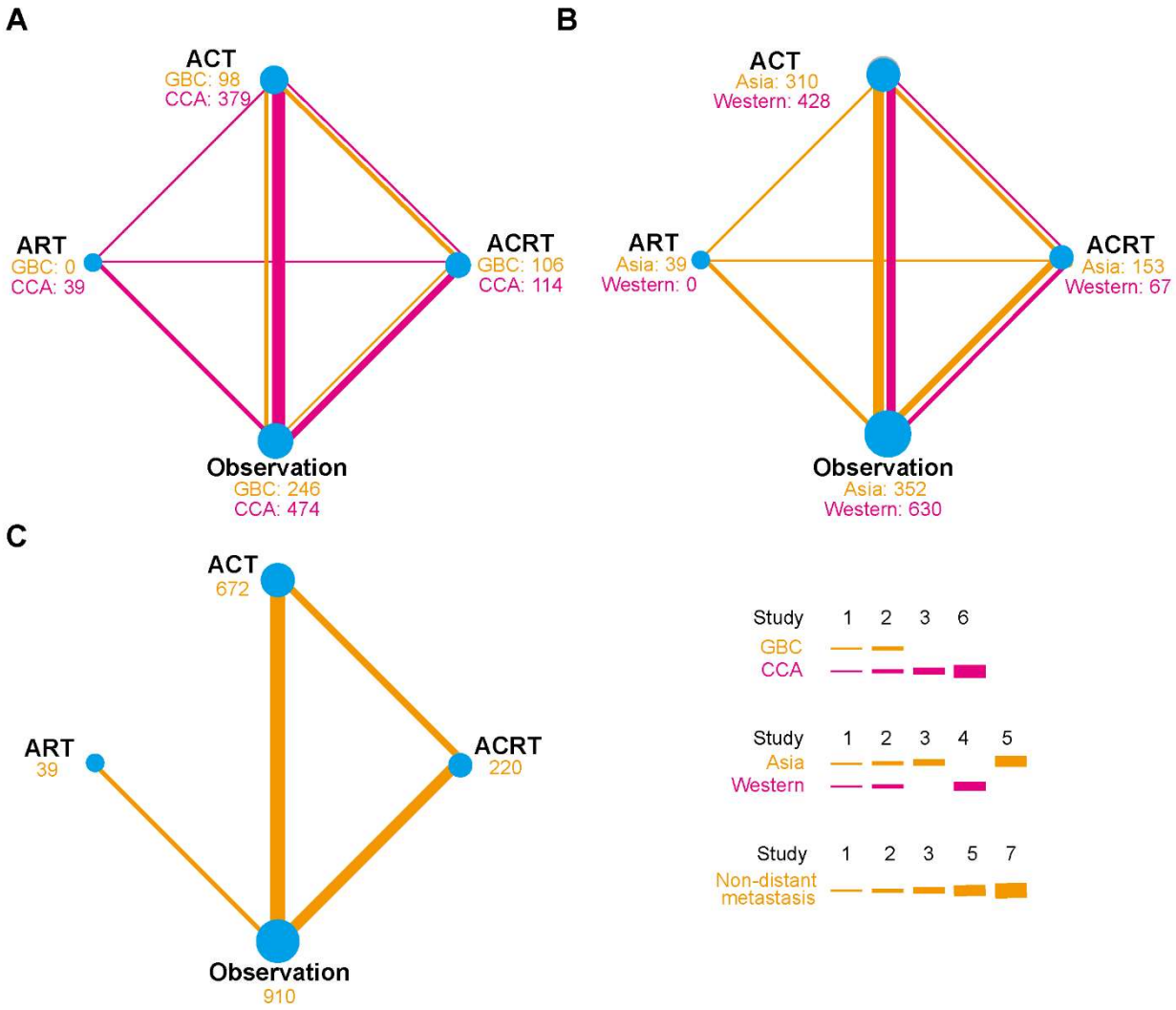



**Supplementary Figure 4. League tables of the network meta-analysis in subgroup analyses. (A)** League table of overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** and **(E)** League tables of overall survival and disease-free survival in subgroups of study patients with GBC and CCA; **(C)** and **(F)** League tables of overall survival and disease-free survival in subgroups of study patients in Asia and Western; **(D)** League table of overall survival and disease-free survival in subgroups of study patients without distal metastasis. Pooled hazard ratios (HRs) and 95% credible intervals (95% CIs) are listed in each cell. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment.

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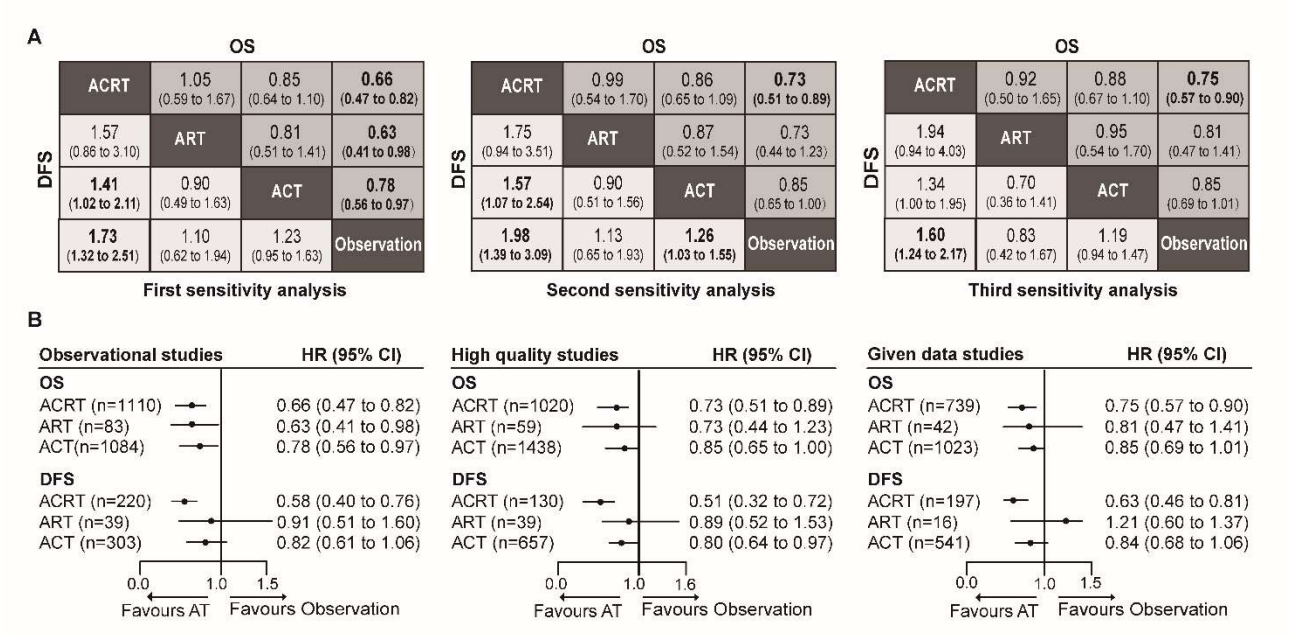
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**Supplementary Figure 5. Network plots of comparisons on disease-free survival of treatments in subgroups analyses (patients stratified into GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on disease-free survival in subgroups of study patients with GBC and CCA; **(B)** Comparisons on disease-free survival in subgroups of study patients in Asia and Western; **(C)** Comparisons on disease-free survival in the subgroup of study patients without distal metastasis.



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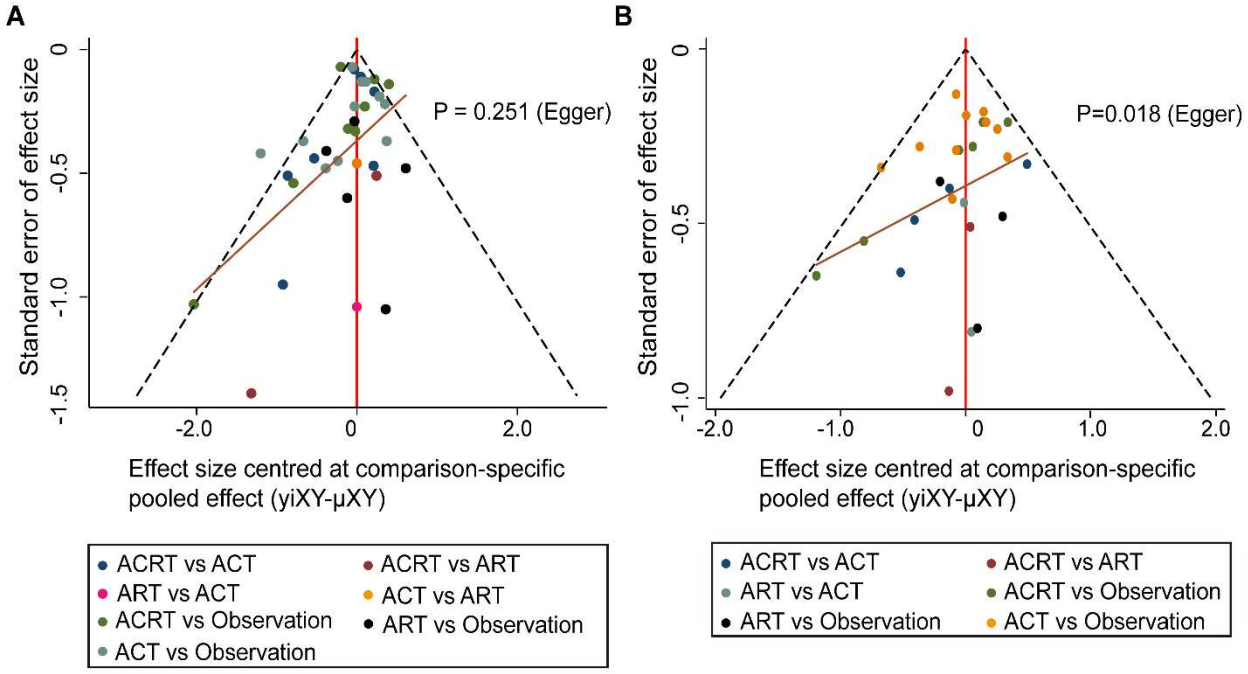



**Supplementary Figure 6. Pooled estimates of the sensitivity analysis. (A)** League tables of the network meta-analysis in sensitivity analyses. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment. **(B)** Forest plots of the network meta-analysis in sensitivity analyses. Adjuvant treatments are ranked according to their surface under the cumulative ranking curve and compared with observation. Effect sizes are presented as hazard ratios (HRs) with 95% confidence intervals (95% CI). OS: overall survival; DFS: disease-free survival. First sensitivity analysis: excluding RCT studies; Second sensitivity analysis: removing low-and moderate-quality observational studies; Third sensitivity analysis: only including the studies for which HRs were reported in the original articles.

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**Supplementary Figure 7. The 'comparison-adjusted' funnel plot in network meta-analysis to assess funnel plot asymmetry on the efficacy outcomes. (A)** Funnel plot asymmetry on the overall survival; **(B)** Funnel plot asymmetry on the disease-free survival. Funnel-plot asymmetry was tested b Egger's regression tests. No publication bias is detected when the P-value is larger than 0.05.

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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis



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# Clinical efficacy of adjuvant treatments for patients with resected biliary tract cancer: a systematic review and network meta-analysis

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## ABSTRACT

**Objective:** This study aimed to determine the benefits of adjuvant therapy in patients with resected biliary tract cancer (BTC) and identify the optimal adjuvant treatment scheme.

**Design:** Systematic review and network meta-analysis.

**Data sources:** Studies comparing different adjuvant therapies in patients with BTC were searched in PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials, and clinicaltrials.gov databases from inception to December 2021. Additionally, the references were manually searched for the related literature.

**Materials and methods:** Eligible studies were identified, and data were extracted independently by two authors. A random-effects network meta-analysis was performed using R software. The pooled outcomes of overall survival (OS) and disease-free survival (DFS) were measured using the combined hazard ratios (HRs) with 95% confidence intervals (95% CIs).

**Results:** Nineteen eligible studies reporting three types of adjuvant therapies were included in our network meta-analysis. Adjuvant radiotherapy (ART, HR, 0.62; 95% CI, 0.42 to 0.93), adjuvant chemoradiotherapy (ACRT; HR, 0.71; 95% CI, 0.54 to 0.83), and adjuvant chemotherapy (ACT; HR, 0.84; 95% CI, 0.68 to 0.98) were more effective in prolonging OS than that of observation, with no significant difference between the three adjuvant therapies. Moreover, the improvement in DFS was also




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found in ACRT and ACT compared with that of observation (HR, 0.60; 95% CI, 0.45 to 0.75; HR, 0.82; 95% CI, 0.68 to 0.97, respectively). Furthermore, ACRT obtained a slightly better DFS benefit compared with that of ACT (HR, 0.73; 95% CI, 0.53 to 0.95).

**Conclusions:** Our primary results demonstrated that, compared with that of observation, ACRT and ACT after radical resection could provide better OS and DFS benefits in patients with BTC. However, ART only showed improvement in OS, but not in DFS. Due to the lack of head-to-head studies of ACT, ACRT, and ART, the above results need to be further verified by prospective randomized controlled trials.

**Keywords:** Biliary tract cancer, adjuvant chemotherapy, adjuvant radiotherapy, adjuvant chemoradiotherapy, systematic review, network meta-analysis

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## Strengths and limitations of this study

- This network meta-analysis compared the efficacy of three major types of adjuvant treatments for BTC patients after radical resection.
- This review was conducted with a strict selection criterion to lower the risk of bias and increase the homogeneity, including only patients with negative resection margins (R0) or microscopic positive resection margins (R1), but not macroscopic involvement resections (R2), and excluding studies with unbalanced or unclear baseline profiles in age, sex, disease severity, or residual tumour status.
- A comprehensive analysis was performed, including the transitivity assumption, local inconsistency of the model, global and local heterogeneity, sensitivity analysis, subgroup analysis, rank probability, and publication bias.
- All studies regarding ART and ACRT were retrospective studies, which were inherent to selection bias.
- Because of the small sample sizes of ART, the claim of optimal treatment for patients with BTC should be interpreted with caution. The optimal regimens and dosing schedules for ACT and ACRT need to be further explored.

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## INTRODUCTION

Biliary tract cancer (BTC) is classified into gallbladder cancer (GBC), intrahepatic cholangiocarcinoma (iCCA), and extrahepatic cholangiocarcinoma (eCCA), and eCCA is further subdivided into perihilar cholangiocarcinoma (pCCA or Klatskin tumour) and distal cholangiocarcinoma (dCCA).[1] BTC accounts for approximately 3% of digestive system cancers and 10%–15% of primary liver cancers.[2] The incidence of BTC is higher in Asian countries and lower in European countries, the United States, and Australia; however, its incidence is increasing globally.[2, 3] The vast majority (>90%) of BTC cases were adenocarcinoma. Moreover, most cases are usually in the advanced or metastatic stage at the initial diagnosis. Only approximately 20% of BTC cases are considered resectable.[4]

Surgical resection remains the primary curative treatment in patients with resected BTC. However, the high recurrence rates (including locoregional or distant recurrence) and low survival rates (5-year survival rates of patients ranging from 5% to 15%) are prominent problems, even with complete (R0) resection.[4-8] Previously published studies have shown that histologic margin status, lymph node (LN) involvement, and intrahepatic metastasis are the main prognostic factors.[9-11] Adjuvant chemotherapy (ACT), adjuvant radiotherapy (ART), and adjuvant chemoradiotherapy (ACRT) are the main options following resection for adjuvant therapy.

A variety of guidelines have suggested that postoperative adjuvant therapy could



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be considered an option for BTC patients.[12-15] The use of ACT and ACRT for patients with GBC and eCCA was supported by the National Comprehensive Cancer Network (NCCN) guidelines. Adjuvant capecitabine is recommended by the American Society of Clinical Oncology (ASCO) guidelines for patients with resected BTC for six months, based on the results of the BILCAP study, which showed a protocol-specified adjusted overall survival (OS) hazard ratio (HR) of 0.71 (95% confidence interval [CI] 0.55–0.92).[16] However, some experts still hold reservations regarding the BILCAP study due to the underpowered statistical design and concerns over data maturity.[17] The discrepancy between the BILCAP study and the PRODIGE-12/ACCORD-18 studies also results in more discussions on the optimal scheme of adjuvant therapy.[18] Despite the tremendous effort, universal agreement on the optimal scheme of adjuvant therapy has not been established due to the lack of high-quality evidence.[19]

Moreover, conflicting results were also observed in the meta-analyses on this topic. A previous study by Rangarajan et al. showed a significant improvement in OS with adjuvant therapy after resection compared with that of resection only.[20] In contrast, Horgan et al. reported a non-significant improvement in OS with adjuvant therapy compared with that of observation in the overall population. However, a more significant benefit from adjuvant therapy was found in patients with positive LN involvement and R1 resection.[21] Three network meta-analyses of adjuvant therapies in patients with resected BTC have been published previously. One made comparisons among ACT, ACRT, and resection only and did not consider ART.[22]



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One only summarized three randomized clinical trials (RCTs) covering three different ACT regimens, and the robustness of their results was limited due to the very small sample sizes.[23] A recent study indicated that ACRT could provide a survival benefit in patients with positive margins or nodal involvement, but the inclusion of patients with R2 margins is still debatable.[24]

Therefore, it is essential to perform a new network meta-analysis to elucidate the efficacy of adjuvant therapy and identify the optimal scheme of adjuvant therapy for patients with resected BTC.

## METHODS

The study design was built in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) extension statement for network meta-analysis for healthcare (**Supplementary Table 1**).[25]

### Search Strategy

PubMed, Embase, CINAHL, Cochrane Central Register of Controlled Trials (CENTRAL), and ClinicalTrials. gov databases were searched from inception to December 2021 to find relevant literature using the main keywords “biliary tract cancer” and “adjuvant therapy.” In addition, references in the relevant literature were manually checked for potentially relevant papers. The detailed search strategy is presented in **Supplementary Table 2**.

### Selection Criteria



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Studies that met the following criteria were included: (1) those including patients with histologically or cytologically confirmed resected BTC and treated with adjuvant therapy or observation after curative-intent resection (defined as negative resection margins [R0] or microscopic positive resection margins [R1], but not macroscopic involvement resections [R2]); (2) studies reporting at least one of the following clinical outcomes, OS or disease-free survival (DFS); and (3) studies with treatment and control arms.

Studies that involved the following were excluded: (1) studies including patients with ampullary carcinomas or other primary cancers or neoadjuvant therapy, palliative therapy, or therapy after postoperative recurrence; (2) studies comparing the same type of adjuvant therapies; (3) those with unbalanced baseline profiles in age, sex, disease severity (American Joint Committee on Cancer staging), or residual tumour status; (4) reviews, conference abstracts, posters, or case reports; and (5) studies not written in English or without a full text, or with a small sample size (<10 in any group).

All study titles and abstracts were screened, and then the full texts of potentially eligible articles were sequentially assessed for final inclusion.

**Data Extraction and Quality Assessment**

The following details were extracted from each study: author, study period, country, study type, sample size, tumour site (BTC type), disease severity, age, female sex (%), resection margin (RM) status, LN status, OS, and DFS. OS was defined as the



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period from the date of surgery (or randomization in RCTs) to the date of death (or last follow-up). DFS was defined as the time from the surgery date (or randomization in RCTs) to recurrence of tumours (locoregional or distant). Moreover, the authors were contacted when there was confusion or missing data in any article. Data were excluded or not considered if no response was received.

The quality of the studies was assessed using the Newcastle–Ottawa Scale (NOS) for observational studies, based on the following domains: selection, comparability, and outcome.[26] The detailed information can be found in **Supplementary Table 3**. A study that scored 8–9 points, of which 2 points for comparability, is considered high quality; a score of 4–7 points indicates moderate quality, and 0–3 means low quality. The Cochrane Risk of Bias Tool was used to grade the RCT quality.[27]

Two investigators (Ye Chen and Baoxia Zhang) independently conducted the study selection and data extraction, and four authors (Ye Chen, Baoxia Zhang, Chang Liu, and Ye Cao) assessed the risk of bias of each eligible study. Discrepancies were resolved by consensus and arbitration by other investigators (Cheng Lyu and Meng Qiu).

**Data Synthesis and Statistical Analysis**

The primary outcomes were OS and DFS and measured using the HR with a 95% CI. When HRs and 95% CIs for OS and DFS were not reported in the original article, they were extracted from survival curves using Engauge Digitizer 10.9 (2014 Mark Mitchell) and estimated by the method suggested by Tierney.[28]



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A Bayesian network meta-analysis was performed using the R software (version 4.0.0, https://www.r-project.org/). Network plots were generated for different outcomes to clarify which treatments were compared directly or indirectly. A random-effects model (including subgroup analyses based on the RM status, tumour sites, regions, and patients without distal metastasis) was used to compare all direct and indirect evidence using "Rjags" and "gemtc" packages in the R software. To fit the non-informative uniform and normal prior distributions, the parameters were set with four chains, 50,000 sample iterations (n.inter), and 20,000 burn-ins (n.adapt) with a thinning interval of 1. The convergence of the chains was assessed using the Gelman–Rubin statistics and inspection of the trace plots (**Supplementary Figure 1**).[29, 30] The deviance information criterion (DIC) was used to test the goodness of fit of consistent and inconsistent models.[31]

The transitivity assumption was evaluated by comparing the distribution of clinical variables (age, percentage of females, sample size, publication year, RM status, LN status), which could be effect modifiers. The local inconsistency of the model was evaluated using the node-splitting approach.[32, 33] The global and local heterogeneity was assessed via between-study variance $\tau^2$ and $I^2$ inconsistency statistic.[34-36] Heterogeneity was considered low, moderate, and high for estimated $\tau^2$ or $I^2$ values < 25%, between 25% and 50%, and larger than 50%, respectively. Within the Bayesian framework, the network meta-analysis provided a ranking probability of each treatment and estimated the overall rankings by calculating the surface under the cumulative ranking (SUCRA).[37, 38] To assess the reliability of



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results, sensitivity analyses for OS and DFS were performed by excluding RCT studies, only including high-quality studies, and only including the studies for which HRs were reported in the original articles. The "netmeta" package in the R software was used to generate the comparison-adjusted funnel plots to visualize publication bias. Egger's test was used to assess funnel plot asymmetry.[39, 40] *P*-values < 0.05 were considered statistically significant. All statistical tests were two-sided.

## Patient and Public Involvement

Patients or the public were not involved in our research design, conduct, reporting, or dissemination plans.

## RESULTS

### Characteristics of the Included Studies

A total of 2,146 records were identified, including 49 records that were manually searched by reviewing the references of relevant publications. After reviewing the abstracts, 264 full-text articles were retrieved and reviewed after excluding 688 duplicates and 1,194 ineligible records (**Figure 1**). Nineteen studies comprising a total of 5,595 patients who received one of the following four different treatments after radical resection—ACT, ART, ACRT, and surgical resection alone (observation group)—were included in our analysis. It is worth noting that 34 studies[11, 41-73] were excluded because of unbalanced or unclear baselines. Of the included patients, 2,774 patients received adjuvant treatments following curative-intent resection, and



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2,821 patients underwent curative-intent resection without adjuvant treatments.

The main characteristics of the studies included in the network meta-analysis are presented in **Table 1**. These 19 studies consisted of 3 RCTs[16, 74, 75] and 16 retrospective studies[76-91]. The patients of 6 studies[16, 75, 79, 83, 84, 91] were Westerners (Americans, French, and British), and 13 studies[74, 76-78, 80-82, 85-90] were Asian (Japanese, Korean, Indian, and Chinese). Four studies[80, 87, 88, 91] assessed patients with GBC, and 11 studies[74, 76-79, 82-86, 89] assessed patients with CCA. Of the 11 studies, 2 studies[85, 86] particularly examined the outcomes of iCCA, and 6 studies[74, 76, 77, 83, 84, 89] examined the outcomes of eCCA, including pCCA[76, 89] and dCCA[83, 84]. In the remaining 4 studies[16, 75, 81, 90], the BTC subtypes involved were varied.

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**Table 1. Characteristics of included studies.**

| Studies | Country | Study period | BTC type | Disease severity | Therapy | Regimen of CT/ type of RT | Sample size | Margin status (R0/R1) | Nodal status (negative/positive) | Female No. (%) | Age, Mean (SD/range), years | 5-Year OS (%); MOST (months) | 5-Year DFS (%); MDFST (months) |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Characteristics of included observational studies** | | | | | | | | | | | | | |
| **ACT vs Observation** | | | | | | | | | | | | | |
| Kobayashi 2012[78] | Japan | 1989-2010 | CCA | I-IV | ACT<br>Observation | GEM/S-1<br>- | 51<br>54 | 39/12<br>37/17 | N0:26; N+: 25<br>N0: 31; N+: 23 | 28 (55)<br>14 (26) | - [a] | 46.0; -<br>23.0; - | - |
| Morine 2017[81] | Japan | 1995-2012 | CCA/ GBC | II-IVb T1-T4 | ACT<br>Observation | GEM+5-FU+Cis<br>- | 28<br>34 | 18/10<br>27/7 | 9/19<br>9/25 | 16 (57)<br>11 (32) | 70.9 (± 6.8)<br>66.9 (± 9.0) | 44.2; -<br>5.9; - | - |
| Akahoshi 2018[82] | Japan | 2004-2015 | CCA | 0-IVa T1-T4 | ACT<br>ACT<br>Observation | GEM; S-1<br>-<br>- | 26<br>36<br>36 | 15/11; 23/13<br>23/13 | 15/11;<br>23/13<br>25/11 | 9 (35)<br>6 (17)<br>13 (36) | - [a] | - | 25.6; -<br>34.0; -<br>26.8; - |
| Bergeat 2018[83] | France | 2000-2015 | dCCA | Ia-III | ACT<br>Observation | GEM<br>- | 49<br>49 | 12/37<br>6/43 | 7/42<br>14/35 | 18 (37)<br>20 (41) | 62.0 (36-77) [b]<br>65.0 (35-82) [b] | - ; 26.3<br>- ; 43.3 | - ; 15.5<br>- ; 14.7 |
| Miyata 2021[90] | Japan | 2007-2018 | GBC/ pCCA/ dCCA | I-IV | ACT<br>Observation | S-1<br>- | 38<br>38 | 29/9<br>28/10 | N0: 21; N1: 17<br>N0: 22; N1: 16 | 13 (34)<br>16 (42) | 72 (52-82) [b]<br>74 (42-85) [b] | 71.0; -<br>42.9; 28.2 | 51.6; 61.2<br>35.9; 13.1 |
| **ART vs Observation** | | | | | | | | | | | | | |
| Todorokia 2000[76] | Japan | 1976-1999 | pCCA (HDC) | IVa | ART<br>Observation | EBRT<br>- | 17<br>19 | All R1 | N0:11; N1:2; N2:4<br>N0:7; N1:4; N2:8 | 7 (41)<br>5 (26) | 59.4 (34-76)<br>60.6 (19-82) | 33.9 [d]; 32.0<br>13.5 [d]; 10.0 | - |
| Jiang 2010[86] | China | 1998-2008 | iCCA | - | ART<br>Observation | EBRT | 24<br>66 | - | N1: 8; Nx: 16<br>N1: 23; Nx: 43 | 7 (29)<br>31 (47) | - [a] | 11.9; 19.1<br>9.5; 1.5 | - |
| Zheng 2018[85] | China | 2007-2016 | iCCA | I-IVa | ART<br>Observation | IMRT<br>- | 26<br>23 | 14/12<br>9/14 | 22/4<br>17/6 | 10 (38)<br>8 (35) | 57.2 (45-76)<br>55.6 (36–72) | 55.0 [de]; 19.1 [d]<br>20.0 [de]; 9.5 [d] | 44.0[e]; -<br>10.0[e]; - |
| **ACRT vs Observation** | | | | | | | | | | | | | |
| Kim 2011[77] | Korea | 2001-2009 | eCCA | Ib, IIa, IIb | ACRT<br>Observation | 5-FU/Leu/Cap EBRT<br>- | 75<br>34 | 58/17<br>28/6 | N0: 39; N+: 36<br>N0: 26; N+: 8 | 17 (23)<br>30 (9) | 60.6 (37-74)<br>65.0 (44-85) | 41.1 [e]; -<br>35.3 [e]; - | 37.4 [e]; -<br>29.8 [e]; - |



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| | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Dover 2016[79] | USA | 2000-2013 | CCA | I-IV | ACRT<br>Observation | GEM/5-FU<br>- | 23<br>72 | 11/12<br>56/16 | N0: 13; N+: 10<br>N0: 51; N+: 21 | 10 (43)<br>28 (39) | 62.0 (38-80) [b]<br>65.0 (39-86) [b] | - ; 30.2<br>- ; 26.3 | - ; 21.6<br>- ; 17.3 |
| **ACRT vs ACT** | | | | | | | | | | | | | |
| Gu 2017[87] | China | 2003-2013 | GBC | II-IVa | ACRT<br>Observation | Cap/S-1/Ox+5-FU/Cap/GEM EBRT | 39<br>39 | All R0 | - | 29 (74)<br>26 (67) | 61 (35-77) [b]<br>63 (45-85) [b] | 42.4; 27<br>17.9; 13 | 48.7 [d]; 23<br>13.5 [d]; 7 |
| Hester 2018[84] | USA | 2004-2013 | dCCA | I-IV | ACRT<br>ACT | - | 348<br>348 | 258/76<br>256/78 | N0:126; N+:211<br>N0:118; N+:206 | 135 (39)<br>136 (39) | 64.3 (±10.0)<br>64.4 (±10.8) | 31.3 [e]; 32.1<br>29.4 [e]; 29.5 | - |
| Choudhary 2019[88] | Indian | 2008-2017 | GBC | II- III | ACRT<br>ACT | GEM+Cis EBRT/3DCRT/IMRT | 23<br>20 | 17/2 (Rx:4)<br>16/2 (Rx:2) | - | 20 (87)<br>17 (85) | 55 (24-73) [ab] | - | - |
| **Multiple comparison** | | | | | | | | | | | | | |
| Kim 2016[80] | Korea | 2000-2015 | GBC | I-IV | ACT<br>ACRT<br>Observation | GEM/Cis/OTHs<br>- | 61<br>44<br>186 | 54/7<br>36/8<br>159/25 | N0:21; N1/Nx:36/4<br>N0:15; N1/Nx:24/5<br>N0:105; N1/Nx:50/31 | 42 (69)<br>28 (64)<br>122 (66) | 64.5 (56-73)<br>63.9 (55-70)<br>67.7 (56-74) | Total:<br>33.0 [e]; 28.3 | Total:<br>- ; 19.8 |
| Im 2021[89] | Korea | 2001-2017 | pCCA | II-IVa | ACT<br>ART<br>ACRT<br>Observation | 5-FU/GEM based EBRT/IMRT<br>- | 45<br>16<br>16<br>37 | 31/14<br>2/14<br>5/11<br>25/12 | - | 22 (33)<br>10 (63)<br>4 (19)<br>35 (38.9) | 65 (32-82) [ab] | 38.9 ; -<br>8.2; -<br>49.4; -<br>27.0; - | 24.6; -<br>5.6; -<br>40.4; -<br>24.5; - |
| Wan 2021[91] | USA (SEER database) | 1973-2015 | GBC | II-IV | ACT<br>ACRT<br>Observation | - | 444<br>542<br>1703 | - | N0: 162; Nx: 282<br>N0: 156; Nx: 386<br>N0: 961; Nx: 742 | 311 (70)<br>374 (69)<br>1192 (70) | - [a] | 38.9/15.0; -<br>41.5/21.1; -<br>32.8/11.3; - | - |
| **Characteristics of included RCT studies** | | | | | | | | | | | | | |
| Ebata 2018[74] | Japan | 2007-2011 | pCCA/dCCA | I-III<br>T1-T4 | ACT<br>Observation | GEM<br>- | 117<br>108 | 106/11<br>94/14 | N0: 75; N1: 42<br>N0: 72; N1: 36 | 40 (34)<br>26 (24) | - [a] | 51.7; 62.3<br>51.6; 63.8 | 45.0; 36.0<br>44.0; 39.9 |
| Edeline 2019[75] | France | 2009-2014 | CCA/GBC | I-IV<br>T1-T4 | ACT<br>Observation | GEMOX<br>- | 95<br>99 | 82/13<br>87/12 | N0:49; N+:35; Nx:11<br>N0:48; N+:36; Nx:15 | 38 (40)<br>49 (50) | 63.0 (33-83) [b]<br>63.0 (40-80) [b] | 60.0 [d]; 75.8<br>65.0 [d]; 50.8 | - ; 30.4<br>- ; 18.5 |
| Primrose 2019[16] | UK | 2006-2014 | BTC | I-IV | ACT<br>Observation | Cap<br>- | 223<br>224 | 139/84<br>140/84 | N0: 121; N1: 102<br>N0: 115; N1:108 | 112 (50)<br>112 (50) | 62.0 (55-68)<br>64.0 (55-69) | 25.1 [c]; 51.1<br>20.5 [c]; 36.4 | - ; 24.4<br>- ; 17.5 |

222 Abbreviations: 5-Year OS (%)=5-Year overall survival probability; MOST=median OS time; 5-Year DFS (%)=5-Year disease-free survival probability; MDFST=median DFS time; N+=



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positive lymph node; Nx= unclear; R0=negative resection margins; R1=microscopic positive resection margins; Rx=unknown; Total= means the value of all patients in the study; BTC=biliary tract cancer; eCCA=extrahepatic cholangiocarcinoma; GBC=gallbladder cancer; CCA=cholangiocarcinoma; pCCA=Perihilar cholangiocarcinoma (or hepatic duct carcinoma=HDC); iCCA=Intrahepatic cholangiocarcinoma; dCCA=distal cholangiocarcinoma; CT= chemotherapy; RT= radiotherapy; ACT=adjuvant chemotherapy; ART=adjuvant radiotherapy; EBRT=external beam radiotherapy; ILRT=intraluminal beam radiotherapy; IMRT=intensity-modulated radiotherapy; 3DCRT=three-dimensional conformal radiotherapy; ACRT=adjuvant chemoradiation therapy; GEM=Gemcitabine; S-1=tegafur gimeracil oteracil potassium; 5-FU=5-fluorouracil; Cap=capecitabine; Cis=cisplatin; Leu=Leucovorin; Ox=oxaliplatin; OTHs= others; GEMOX=gemcitabine and oxaliplatin.

- indicates that the data was not given originally;

[a] indicates that studies were presented as younger or older than a specific age;
[b] indicates that median age was given instead of mean age;
[c] indicates overall survival rates but not survival probability;
[d] indicates a 3-year overall survival of disease-free survival;
[e] indicates that the method of the value calculated is unclear.

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The quality assessment of 16 retrospective studies is listed in **Supplementary Table 4**, which shows that 12 studies[76-78, 81-85, 87, 89-91] were judged to be of high quality, and 4 studies[79, 80, 86, 88] were judged to be of moderate quality. Of these studies, four were assigned only one star in "comparability" due to the unreported RM[79, 86] or LN status[80, 88]. Ten studies obtained 0 star in item 8 in relation to "outcome" due to the inadequacy of follow-up.[77, 79-84, 86-88, 90] **Supplementary Figure 1** summarizes the risk of bias assessments of three RCTs. All three RCTs were conducted in an open-label fashion, one[75] did not reveal the details of allocation concealment, and all were at low risk of bias in the domains of "random sequence generation," "incomplete outcome data," and "other bias."

**Transitivity, Heterogeneity, and Inconsistency Assessment**

The transitivity assumption was assessed across treatments in our network. Patient characteristics that are known modifiers of treatment efficacy, such as age, percentage of female participants, sample size, publication year, RM status, and LN status, were evaluated and visualized using box plots (**Supplementary Figure 2**). No difference in these characteristics was observed between the different therapies (**Supplementary Table 5**). Therefore, the transitivity assumption was ensured in this network.

Heterogeneity was also evaluated between eligible studies. Low global heterogeneity was detected in the OS, with $I^2 = 0.0\%$ for a pairwise effect and $I^2 = 19.4\%$ for a consistency effect and $\tau^2 = 0.022$ for the between-study variance (**Supplementary Table 6**). Minimal global heterogeneity was seen in the DFS, with



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$I^2$ = 0.0% for a pairwise pooled effect and a consistency effect (**Supplementary Table 6**). High local heterogeneities were detected in the OS for the direct comparison of ART vs. observation and the DFS for the direct comparison of ACRT vs. observation, with $I^2$ = 81.9% and 82.2%, respectively, for the pooled network effect.

The local inconsistency was evaluated by comparing the corresponding pooled HRs of OS and DFS after comparing the results from pairwise and network meta-analyses. Statistical inconsistency between the direct and indirect evidence after node-splitting was not found, except for the comparison of ACRT vs. ART ($P < 0.001$) and ACT vs. ART ($P = 0.001$) in OS and ACRT vs. ART ($P = 0.009$) in DFS, owing to the fact that only one study contained OS and DFS data comparing ACRT vs. ART and ACT vs. ART (**Supplementary Table 7**).

To further check the consistency at the treatment level, the goodness of fit of the inconsistency model (unrelated mean effects model, UME) was compared with that of the consistency model. The DICs of the consistency model were 46.17 for OS and 29.93 for DFS, which were similar to the DICs of the inconsistency model (46.05 and 31.15, respectively), suggesting no evidence of inconsistency in the network. The more parsimonious model, the consistency model, was used for our analyses (**Supplementary Table 8**).

**Network Meta-analysis of Treatments in Biliary Tract Cancer**

A network meta-analysis was conducted to assess the efficacy (OS and DFS) of the



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following treatments in a Bayesian framework: observation alone after surgery, surgery with ACT, surgery with ART, and surgery with ACRT.

A network plot is shown in **Figure 2**. OS data were available from 18 studies that included 5,497 patients, of whom 2,785 (50.7%) were in the observation group, 1,519 (27.6%) received ACT, 83 (1.5%) received ART, and 1,110 (20.2%) received ACRT. The pooled OS data indicated that ART (HR, 0.62; 95% CI, 0.42 to 0.93), ACRT (HR, 0.71; 95% CI, 0.54 to 0.83), and ACT (HR, 0.84; 95% CI, 0.68 to 0.98) were more beneficial in patients with BTC compared with that of observation (**Figure 3A–B**). No significant benefits in OS were observed in the comparisons between the different adjuvant therapies (**Figure 3A**).

In terms of DFS, 14 studies with 1,979 patients were included in the network meta-analysis (**Figure 2**). The pooled DFS data demonstrated a significant improvement for ACRT and ACT compared with that of observation (HR, 0.60; 95% CI, 0.45 to 0.75, and HR, 0.82; 95% CI, 0.68 to 0.97, respectively). Furthermore, a slightly better efficacy for ACRT was obtained compared with that of ACT (HR, 0.73; 95% CI, 0.53 to 0.95). Significant differences in DFS were not observed between the other pairwise comparisons (**Figure 3A and 3C**).

The ranking analysis was performed using SUCRA. Based on the pooled OS and DFS data, the ranking order of OS and DFS was inconsistent. The best therapy for OS was ranked as follows: ART, ACRT, ACT, and observation (**Figure 3D**). The best SUCRA value of ART was 87.0%, which was close to that of ACRT with a SUCRA



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value of 75.3% (**Supplementary Table 9**). As for DFS, the best therapy was ranked as ACRT, ACT, ART, and observation (**Figure 3D**). The SUCRA value of ACRT was approximately 97.1%, which was far higher than that of the others (**Supplementary Table 9**).

Subgroup analyses for OS in patients with different residual tumour status, tumour sites, regions, and absence of distant metastasis were performed. Seven studies reported the outcomes of patients after R0 resection, and seven studies reported the outcomes of patients after R1 resection (**Supplementary Figure 3A**). In the R0 group, only ACRT had a survival advantage compared to that of surgery alone. In the R1 group, no survival advantage was observed in patients who underwent adjuvant therapies than in those with surgery alone (**Figure 4A, Supplementary Figure 4A**). In the subgroup analysis by primary tumour sites, six studies enrolled patients with GBC, and 11 studies recruited patients with CCA (**Supplementary Figure 3B**). The OS benefit of ACRT was clear compared with that of observation in the CCA group (HR, 0.62; 95% CI, 0.36 0.85). The benefit of ART is unclear due to the lack of eligible ART studies. Among the comparable treatments in the GBC group, no significant differences in OS were found between ACT, ACRT, and observation (**Figure 4B, Supplementary Figure 4B**). When studies were grouped according to region, 12 studies included patients in Asian countries and 7 from Western countries (**Supplementary Figure 3C**). In both the Asian and Western groups, the pooled OS results favoured ACRT (HR, 0.42; 95% CI, 0.20 to 0.74 in Asia; HR, 0.76; 95% CI, 0.63 to 0.91 in Western countries) (**Figure 4C,**



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**Supplementary Figure 4C**). Subgroup analysis was also conducted for 15 studies investigating OS in patients with non-distant metastasis (**Supplementary Figure 3D**). The pooled results showed superior efficacy of ACRT compared with that of observation (HR, 0.74; 95% CI, 0.60 to 0.85). Moreover, ACRT tended to be more effective compared to that of ACT (HR, 0.82; 95% CI, 0.67 to 0.96) (**Figure 4D, Supplementary Figure 4D**).

Subgroup analyses were also performed for DFS in patients with different tumour sites and regions and the absence of distant metastasis (**Supplementary Figure 5**). Due to the small number of studies, subgroup analyses for DFS by different residual tumour statuses cannot be conducted. In patients without distant metastasis, the results from this subgroup were similar to that of the primary analysis; ACRT demonstrated statistically significant improvement with an HR of 0.61 (**Figure 4E, Supplementary Figure 4D**). When studies were split by primary tumour sites, only ACRT showed an apparent advantage in patients with CCA (**Figure 4F, Supplementary Figure 4E**). Furthermore, the stratified meta-analysis by region indicated that the favourable treatment was ACRT in Asia (HR, 0.47; 95% CI, 0.28 to 0.68) (**Figure 4G, Supplementary Figure 4F**).

To assess the robustness of the primary results, sensitivity analyses were performed for OS and DFS by excluding RCT studies, removing moderate-quality observational studies, and only including the studies for which HRs were reported in the original articles. The first sensitivity analysis included 15 retrospective studies of



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OS and 11 retrospective studies of DFS. The pooled results (**Supplementary Figure 6**) and the ranking profiles of comparable treatments from retrospective studies confirmed the reliability of the primary OS results (**Figure 5A**, **Supplementary Table 10**). In terms of DFS, ACRT was still ranked as the best treatment option (**Figure 5B**). Furthermore, the second and third sensitivity analyses were performed by removing moderate-quality observational studies (14 studies for OS and 11 studies for DFS) and by combining HRs reported in the original articles (11 studies for OS and 8 studies for DFS). The results did not suggest a material change in the efficacy estimation for ACRT, but ART no longer showed advantages over observation in both OS and DFS; the improvement in OS and DFS between ACT and observation was insignificant except for the pooled DFS in high-quality studies (**Supplementary Figure 6**).

The visual examination of the comparison-adjusted funnel plots did not suggest a publication bias for OS presented in our network meta-analysis (**Supplementary Figure 7A**). The result of Egger's test ($P = 0.251$) also rejected the presence of small-study effects. However, both the visual examination of the comparison-adjusted funnel plots and the results of Egger's test ($P = 0.018$) suggested a publication bias for DFS in our network meta-analysis (**Supplementary Figure 7B**).

## DISCUSSION

BTCs are an uncommon and heterogeneous type of cancer with a higher prevalence in Asian countries.[92] In general, BTCs include cancers raised from the intrahepatic,



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hilar, and distal bile ducts, as well as the gallbladder.[7] Surgical resection provides the only chance for cure in patients with BTC at an early stage, but the survival outcomes are poor. The 5-year survival rate was as low as 10%.[93] The most recent NCCN and ASCO guidelines recommend the use of adjuvant therapy for BTC patients after resection.[12, 15] However, experts mentioned that the use of adjuvant therapy is based on a very limited number of studies, and the benefit of adjuvant therapy remains unclear in many pivotal BTC trials.[16, 74, 75, 94, 95] Therefore, it is necessary to compile up-to-date studies to validate the efficacy of adjuvant therapy in BTC patients.

In this network meta-analysis, 19 studies were included to evaluate the comparative benefits (involving 5,497 patients for OS and 1,979 patients for DFS) between adjuvant therapy (ACT, ART, and ACRT) following surgical resection and curative-intent resection (observation group). Our primary results demonstrated that adjuvant therapy was more effective than that of observation in OS. However, no statistically significant difference was detected between ACT, ACRT, and ART. Moreover, the pooled DFS results suggested a statistically significant benefit for ACRT and ACT over observation. Although ART was ranked first in OS, its DFS result was inconsistent with OS, and caution should be exercised regarding the bias that arises from the small sample size effect of ART evidence (only 83 patients). In addition, a previously published meta-analysis of adjuvant therapy in the treatment of BTC suggested that ART did not provide a significant advantage over observation.[21] We hold reservations on the conclusion of the efficacy of ART, and



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further evidence is needed to elucidate this matter.

Combining the results of OS with DFS, ACRT, and ACT after radical resection could provide a survival benefit in patients with BTC. This was in line with another meta-analysis that showed that both ACT (HR, 0.61; 95% CI, 0.47–0.79) and ACRT (HR, 0.35; 95% CI, 0.14–0.83) could significantly improve the clinical survival of patients with resected BTC compared to that of surgery alone.[96] As known, the BILCAP study is a unique positive randomized trial of ACT in patients with BTC. Despite the concerns regarding the findings of the BILCAP trial, this study established adjuvant capecitabine as the new standard of care for resected BTC. However, no randomized trials of ACRT vs. ACT are available. Although in our study, the ACRT ranked high in terms of OS and DFS (second place for OS, first place for DFS), whether it is genuinely superior to ACT needs further confirmation in prospective RCTs.

The types of patients who are more likely to benefit from adjuvant therapy are also the focus. A previous phase II study indicated that the risk factors were GBC, eCCA, pathologic stage T2–4, positive LN, or positive RM.[97] In our study, we were only able to conduct subgroup analyses on the effects of the RM, primary tumour site, regions, and absence of distant metastasis, but not LN and tumour size, on adjuvant therapy due to the availability of data. We observed that ACRT showed a modest improvement in DFS than that of ACT in CCA and Asian patients and ranked first in each subgroup (**Supplementary Table 11**). We noticed that approximately 95% (for



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OS analysis) and 80% (for DFS analysis) of CCA patients treated with ACRT had eCCA. The benefit of ACRT observed in CCA patients may be mainly derived from eCCA patients. Inconsistent with most studies, ACRT has an OS benefit in the R0 group, but not in the R1 group. This was possibly due to the more stringent selection of studies and the effects of small sample sizes. Subgroup analyses for DFS by R0 or R1 were not conducted because of limited data. Therefore, the subgroups of patients who could benefit more from postoperative adjuvant therapy need to be further elucidated. We look forward to seeing more randomized controlled studies in this field, especially head-to-head trials and trials designed for more specific subgroups. Using the primary tumour site, tumour size, disease severity, LN metastases, and RM as stratification factors in future studies would reduce selection bias due to the heterogeneity of the population.

This study had several limitations. First, RCT data were only available to compare ACT and observations. Unfortunately, there are neither RCTs for ART and ACRT nor head-to-head RCTs between different adjuvant treatments. The notable differences in the study design level may introduce confounding factors in our data analysis, although data transitivity and consistency could be assumed statistically. Second, the included studies spanned over a 45-year period during which operative techniques and methods have changed and improved over time. These changes in treatment methods could potentially bias our results, but the impact on the outcome was unclear and difficult to interpret. Third, most comparisons were indirect, and direct evidence was obtained from two or three studies. The comparisons between the



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different treatment modalities and treatment regimens may substantially contribute to heterogeneity among included studies. Patients in ACT, ART or ACRT groups were treated with different modalities. Even within the same treatment modality, different regimens offered various efficacy results. Such as, the adjuvant capecitabine monotherapy (BILCAP study) appeared more effective compared with observation, whereas adjuvant gemcitabine (BCAT study) or gemcitabine plus oxaliplatin (PRODIGE 12 study) did not.[98] Furthermore, we noticed that the number of patients treated with ART was small. These may result in a considerable risk of bias. Fourth, the definitions of OS and DFS in RCTs and retrospective studies were calculated differently. In RCTs, it started from the date of randomization to the date of surgery in retrospective studies. The span was relatively short compared with the expected survival duration, but the slight variation in data collection should still be considered. Finally, we failed to compare the safety outcomes due to a lack of sufficient data on adverse events. Taking all these into account, our estimates should be interpreted with caution.

## CONCLUSIONS

Our primary results demonstrated that, compared with that of observation, ACRT and ACT after radical resection could provide better OS and DFS benefits in patients with BTC. However, ART only showed improvement in OS. ACRT had a modest DFS advantage compared to that of ACT. Due to the absence of direct evidence from head-to-head prospective studies, thorough and high-quality RCTs are warranted to



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consolidate our results further. Optimal regimens and dosing schedules still need to be explored.

## ABBREVIATIONS

BTC: biliary tract carcinoma

OS: overall survival

DFS: disease-free survival

HR (95% CI): hazard ratio (95% confidence interval)

GBC: gallbladder cancer

CCA: cholangiocarcinoma

iCCA: intrahepatic cholangiocarcinoma

eCCA: extrahepatic cholangiocarcinoma

pCCA: perihilar cholangiocarcinoma

HDC: hepatic duct carcinoma

dCCA: distal cholangiocarcinoma

ACT: adjuvant chemotherapy

ART: adjuvant radiotherapy

ACRT: adjuvant chemoradiotherapy

RCT: randomized clinical trial

PRIMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyse

AJCC: American Joint Committee on Cancer

RM: resection margin

R0: negative resection margin

R1: microscopic positive resection margin

R2: macroscopic involvement resection

LN: lymph node

NOS: Newcastle-Ottawa Scale



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DIC: deviance information criterion

SUCRA: surface under the cumulative ranking

UME: unrelated mean effect

NCCN: National Comprehensive Cancer Network

ASCO: American Society of Clinical Oncology

RCTs: randomized controlled studie

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## Author Contributions

As the corresponding author, this study was designed, coordinated, and supervised by Meng Qiu. Ye Chen, Baoxia Zhang, and Chang Liu screened the articles, Ye Chen and Baoxia Zhang extracted the data and assessed the quality. Ye Cao and Chang Liu contacted study authors for additional information. Ye Chen, Baoxia Zhang, and Ye Cao analyzed and prepared the material. Ye Chen, Baoxia Zhang, Chang Liu, and Cheng Lyu interpreted the data and wrote the draft of the report. All authors had reviewed and approved the final submitted version of the report. Meng Qiu was responsible for the integrity and accuracy of the data.

## Conflict of Interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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**Ethics approval:** Ethical approval for this study was not applicable because it reviewed existing literature and did not involve any handling of individual patient data.

**Data availability statement:** All data relevant to the study are included in the article or uploaded as supplemental information. All the data extracted, analyzed, and reported in this paper were from published literature.



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## References


1. Nakeeb A, Pitt HA, Sohn TA, et al. Cholangiocarcinoma. A spectrum of intrahepatic, perihilar, and distal tumors. *Ann Surg* 1996;224(4):463-73; discussion 73-5.

2. Benavides M, Anton A, Gallego J, et al. Biliary tract cancers: SEOM clinical guidelines. *Clin Transl Oncol* 2015;17(12):982-7.

3. Patel T. Increasing incidence and mortality of primary intrahepatic cholangiocarcinoma in the United States. *Hepatology* 2001;33(6):1353-7.

4. Valle JW, Lamarca A, Goyal L, et al. New Horizons for Precision Medicine in Biliary Tract Cancers. *Cancer Discov* 2017;7(9):943-62.

5. Anderson CD, Pinson CW, Berlin J, et al. Diagnosis and treatment of cholangiocarcinoma. *Oncologist* 2004;9(1):43-57.

6. DeOliveira ML, Cunningham SC, Cameron JL, et al. Cholangiocarcinoma: thirty-one-year experience with 564 patients at a single institution. *Ann Surg* 2007;245(5):755-62.

7. Bridgewater JA, Goodman KA, Kalyan A, et al. Biliary Tract Cancer: Epidemiology, Radiotherapy, and Molecular Profiling. *Am Soc Clin Oncol Educ Book* 2016;35:e194-203.

8. Doherty MK, Knox JJ. Adjuvant therapy for resected biliary tract cancer: a review. *Chin Clin Oncol* 2016;5(5):64.

9. Mavros MN, Economopoulos KP, Alexiou VG, et al. Treatment and Prognosis for Patients With Intrahepatic Cholangiocarcinoma: Systematic Review and Meta-analysis. *JAMA Surg* 2014;149(6):565-74.

10. Murakami Y, Uemura K, Sudo T, et al. Prognostic factors after surgical resection for intrahepatic, hilar, and distal cholangiocarcinoma. *Ann Surg Oncol* 2011;18(3):651-8.




For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

11. Wirasorn K, Ngamprasertchai T, Khuntikeo N, et al. Adjuvant chemotherapy in resectable cholangiocarcinoma patients. *J Gastroenterol Hepatol* 2013;28(12):1885-91.

12. National Comprehensive Cancer Network. NCCN clinical practice guidelines in oncology. Hepatobiliary cancers, Version 5. 2021. https://www.nccn.org/professionals/physician_gls/pdf/hepatobiliary.pdf. (accessed 20 Jan. 2022)

13. Valle JW, Borbath I, Khan SA, et al. Biliary cancer: ESMO Clinical Practice Guidelines for diagnosis, treatment and follow-up. *Annals of oncology : official journal of the European Society for Medical Oncology* 2016;27(suppl 5):v28-v37.

14. HJ L, SK Q, F S, et al. Biliary tract caner: CSCO consensus for diagnosis and treatment. *Chin Clin Oncol* 2019;24(9):828-38.

15. Shroff RT, Kennedy EB, Bachini M, et al. Adjuvant Therapy for Resected Biliary Tract Cancer: ASCO Clinical Practice Guideline. *J Clin Oncol* 2019;37(12):1015-27.

16. Primrose JN, Fox RP, Palmer DH, et al. Capecitabine compared with observation in resected biliary tract cancer (BILCAP): a randomised, controlled, multicentre, phase 3 study. *Lancet oncol* 2019;20(5):663-73.

17. Rizzo A, Brandi G. BILCAP trial and adjuvant capecitabine in resectable biliary tract cancer: reflections on a standard of care. *Expert Rev Gastroent* 2021;15(5):483-85.

18. Rizzo A, Brandi G. Pitfalls, challenges, and updates in adjuvant systemic treatment for resected biliary tract cancer. *Expert Rev Gastroent* 2021;15(5):547-54.

19. Lamarca A, Edeline J, McNamara MG, et al. Current standards and future perspectives in adjuvant treatment for biliary tract cancers. *Cancer treat Rev* 2020;84:101936.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

20. Rangarajan K, Simmons G, Manas D, et al. Systemic adjuvant chemotherapy for cholangiocarcinoma surgery: A systematic review and meta-analysis. *Eur J Surg Oncol* 2020;46(4 Pt A):684-93.

21. Horgan AM, Amir E, Walter T, et al. Adjuvant therapy in the treatment of biliary tract cancer: a systematic review and meta-analysis. *J Clin Oncol* 2012;30(16):1934-40.

22. Zhu GQ, Shi KQ, You J, et al. Systematic review with network meta-analysis: adjuvant therapy for resected biliary tract cancer. *Aliment Pharmacol Ther* 2014;40(7):759-70.

23. Kish M, Chan K, Perry K, et al. A systematic review and network meta-analysis of adjuvant therapy for curatively resected biliary tract cancers. *Curr Oncol* 2020;27(1):e20-e26.

24. Chen X, Meng F, Xiong H, et al. Adjuvant Therapy for Resectable Biliary Tract Cancer: A Bayesian Network Analysis. *Front Oncol* 2021;11:600027.

25. Hutton B, Salanti G, Caldwell DM, et al. The PRISMA extension statement for reporting of systematic reviews incorporating network meta-analyses of health care interventions: checklist and explanations. *Ann Intern Med* 2015;162(11):777-84.

26. Ottawa Hospital Research Institute. The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analyses. 2014. http://www.ohri.ca/programs/clinical_epidemiology/oxford.asp. (accessed 29 Jun. 2020)

27. Julian Higgins, Douglas Altman, Sterne J. Assessing risk of bias in included studies. In: Julian Higgins, Green S, editors. Cochrane handbook of systematic reviews of interventions. Assessing risk of bias in included studies. 5.1.0 ed. England: The Cochrane Collaboration, Wiley; 2011. p. 187-235.

28. Tierney JF, Stewart LA, Ghersi D, et al. Practical methods for incorporating summary



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

time-to-event data into meta-analysis. *Trials* 2007;8:16.

29. Gelman A, Rubin DB. Inference from iterative simulation using multiple sequences. *Stat Sci* 1992;7:457-511.

30. Lynch SM. Introduction to Applied Bayesian Statistics and Estimation for Social Scientists. 1st edition ed. New York, NY: Springer-Verlag 2007.

31. Higgins JPT, Jackson D, Barrett JK, et al. Consistency and inconsistency in network meta-analysis: concepts and models for multi-arm studies. *Res Synth Methods* 2012;3(2):98-110.

32. Dias S, Welton NJ, Caldwell DM, et al. Checking consistency in mixed treatment comparison meta-analysis. *Stat Med* 2010;29(7-8):932-44.

33. Van Valkenhoef G, Dias S, Ades AE, et al. Automated generation of node-splitting models for assessment of inconsistency in network meta-analysis. *Res Synth Methods* 2016;7(1):80-93.

34. Higgins JP, Thompson SG, Deeks JJ, et al. Measuring inconsistency in meta-analyses. *BMJ* 2003;327(7414):557-60.

35. Higgins JP, Thompson SG. Quantifying heterogeneity in a meta-analysis. *Stat Med* 2002;21(11):1539-58.

36. Debray TP, Damen JA, Riley RD, et al. A framework for meta-analysis of prediction model studies with binary and time-to-event outcomes. *Stat Methods Med Res* 2019;28(9):2768-86.

37. Salanti G, Ades AE, Ioannidis JP. Graphical methods and numerical summaries for presenting results from multiple-treatment meta-analysis: an overview and tutorial. *J Clin*



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

*Epidemiol* 2011;64(2):163-71.

38. Veroniki AA, Vasiliadis HS, Higgins JP, et al. Evaluation of inconsistency in networks of interventions. *Int J Epidemiol* 2013;42(1):332-45.

39. Egger M, Davey Smith G, Schneider M, et al. Bias in meta-analysis detected by a simple, graphical test. *BMJ* 1997;315(7109):629-34.

40. Begg CB, Mazumdar M. Operating characteristics of a rank correlation test for publication bias. *Biometrics* 1994;50(4):1088-101.

41. Sagawa N, Kondo S, Morikawa T, et al. Effectiveness of radiation therapy after surgery for hilar cholangiocarcinoma. *Surg Today* 2005;35(7):548-52.

42. Borghero Y, Crane CH, Szklaruk J, et al. Extrahepatic bile duct adenocarcinoma: patients at high-risk for local recurrence treated with surgery and adjuvant chemoradiation have an equivalent overall survival to patients with standard-risk treated with surgery alone. *Ann Surg Oncol* 2008;15(11):3147-56.

43. Gold DG, Miller RC, Haddock MG, et al. Adjuvant therapy for gallbladder carcinoma: the Mayo Clinic Experience. *Int J Radiat Oncol* 2009;75(1):150-5.

44. Ercolani G, Vetrone G, Grazi GL, et al. Intrahepatic cholangiocarcinoma: primary liver resection and aggressive multimodal treatment of recurrence significantly prolong survival. *Ann Surg* 2010;252(1):107-14.

45. Kim K, Chie EK, Jang JY, et al. Postoperative chemoradiotherapy for gallbladder cancer. *Strahlenther Onkol* 2012;188(5):388-92.

46. Kondo N, Murakami Y, Uemura K, et al. Elevated perioperative serum CA 19-9 levels are independent predictors of poor survival in patients with resectable cholangiocarcinoma. *J*



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

*Surg Oncol* 2014;110(4):422-9.

47. Bektas H, Yeyrek C, Kleine M, et al. Surgical treatment for intrahepatic cholangiocarcinoma in Europe: a single center experience. *J Hepato-Bil-Pan Sci* 2015;22(2):131-7.

48. Hoehn RS, Wima K, Ertel AE, et al. Adjuvant Chemotherapy and Radiation Therapy is Associated with Improved Survival for Patients with Extrahepatic Cholangiocarcinoma. *Ann Surg Oncol* 2015;22:1133-39.

49. Sur MD, In H, Sharpe SM, et al. Defining the Benefit of Adjuvant Therapy Following Resection for Intrahepatic Cholangiocarcinoma. *Ann Surg Oncol* 2015;22(7):2209-17.

50. Im JH, Seong J, Lee IJ, et al. Surgery alone versus surgery followed by chemotherapy and radiotherapy in resected extrahepatic bile duct cancer: Treatment outcome analysis of 336 patients. *Cancer Res Treat* 2016;48(2):583-95.

51. Kim Y, Hwang I, Park S-E, et al. Role of adjuvant therapy after R0 resection for patients with distal cholangiocarcinoma. *Cancer Chemoth Pharm* 2016;77(5):979-85.

52. Kim TG. Patterns of initial failure after resection for gallbladder cancer: Implications for adjuvant radiotherapy. *Radiat Oncol J* 2017;35(4):359-67.

53. Kim Y, Oh S, Go S-I, et al. The role of adjuvant therapy after R0 resection for patients with intrahepatic and perihilar cholangiocarcinomas. *Cancer Chemoth Pharm* 2017;79(1):99-106.

54. Kim YS, Jeong CY, Song HN, et al. The efficacy of fluoropyrimidine-based adjuvant chemotherapy on biliary tract cancer after R0 resection. *Chin J Cancer* 2017;36(1):9.

55. Lee J, Kang SH, Noh OK, et al. Adjuvant concurrent chemoradiation therapy in patients



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

with microscopic residual tumor after curative resection for extrahepatic cholangiocarcinoma. *ClinTransl Oncol* 2017;20(8):1011-17.

56. Mitin T, Kristian Enestvedt C, Jemal A, et al. Limited Use of Adjuvant Therapy in Patients with Resected Gallbladder Cancer Despite a Strong Association with Survival. *J Natl Cancer Inst* 2017;109(7).

57. Tran Cao HS, Zhang Q, Sada YH, et al. The role of surgery and adjuvant therapy in lymph node-positive cancers of the gallbladder and intrahepatic bile ducts. *Cancer* 2017;124(1):74-83.

58. Lin YK, Hsieh MC, Wang WW, et al. Outcomes of adjuvant treatments for resectable intrahepatic cholangiocarcinoma: Chemotherapy alone, sequential chemoradiotherapy, or concurrent chemoradiotherapy. *Radiother Oncol* 2018;128(3):575-83.

59. Geng ZM, Cai ZQ, Zhang Z, et al. Estimating survival benefit of adjuvant therapy based on a Bayesian network prediction model in curatively resected advanced gallbladder adenocarcinoma. *World J Gastroentero* 2019;25(37):5655-66.

60. Kattepur AK, Patkar S, Goel M, et al. Role of Adjuvant Chemotherapy in Resected T2N0 Gall Bladder Cancer. *J Gastrointest Surg* 2019;23(11):2232-38.

61. Navarro J, Kang I, Hwang HK, et al. Glucose to Lymphocyte Ratio as a Prognostic Marker in Patients With Resected pT2 Gallbladder Cancer. *J Surg Res* 2019;240:17-29.

62. Kim TH, Woo SM, Lee WJ, et al. Benefit of Adjuvant Chemoradiotherapy in Resected Gallbladder Carcinoma. *Sci Rep* 2019;9(1):N.PAG-N.PAG.

63. Kizy S, Altman AM, Marmor S, et al. Surgical resection of lymph node positive



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

intrahepatic cholangiocarcinoma may not improve survival. *HPB* 2019;21(2):235-41.

64. Lu J, Li B, Li FY, et al. Prognostic significance of mucinous component in hilar cholangiocarcinoma after curative-intent resection. *J Surg Oncol* 2019;120(8):1341-49.

65. Han D, Yang J, Xu F, et al. Prognostic factors in patients with gallbladder adenocarcinoma identified using competing-risks analysis: A study of cases in the SEER database. *Medicine* 2020;99(31):e21322.

66. Kim H, Heo MH, Kim JY. Comparison of the effects of adjuvant concurrent chemoradiotherapy and chemotherapy for resected biliary tract cancer. *BMC Gastroenterol* 2020;20(1):20.

67. Mao W, Deng F, Wang D, et al. Treatment of advanced gallbladder cancer: A SEER-based study. *Cancer Med* 2020;9(1):141-50.

68. Zhou W, Qian L, Rong Y, et al. Prognostic factors and patterns of recurrence after curative resection for patients with distal cholangiocarcinoma. *Radiothe Oncol* 2020;147:111-17.

69. Zhu WH, Xie WY, Zhang ZD, et al. Postoperative Complications and Survival Analysis of Surgical Resection for Hilar Cholangiocarcinoma: A Retrospective Study of Fifty-Nine Consecutive Patients. *Chin Med Sci J* 2020;35(2):157-69.

70. Benzing C, Krenzien F, Mieg A, et al. A tailored approach in lymph node-positive perihilar cholangiocarcinoma. *Langenbeck Arch Surg* 2021;406(5):1499-509.

71. Chang WI, Kim BH, Kang HC, et al. The Role of Adjuvant Chemoradiotherapy in Nonhilar Extrahepatic Bile Duct Cancer: A Long-Term Single-Institution Analysis. *Int J Radiat Oncol* 2021;111(2):395-404.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

72. Kosaka H, Kaibori M, Matsui K, et al. Investigation of a Tumor Location-Specific Therapeutic Strategy for Intrahepatic Cholangiocarcinoma. *APJCP* 2021;22(5):1485-93.

73. Sumiyoshi T, Uemura K, Kondo N, et al. The prognostic impact of peritoneal washing cytology for otherwise resectable extrahepatic cholangiocarcinoma patients. *Surg today* 2021;51(7):1227-31.

74. Ebata T, Hirano S, Konishi M, et al. Randomized clinical trial of adjuvant gemcitabine chemotherapy versus observation in resected bile duct cancer. *Brit J Surg* 2018;105(3):192-202.

75. Edeline J, Benabdelghani M, Bertaut A, et al. Gemcitabine and oxaliplatin chemotherapy or surveillance in resected biliary tract cancer (Prodige 12-accord 18-Unicancer GI): a randomized phase III study. *J Clin Oncol* 2019;37(8):658-67.

76. Todoroki T, Ohara K, Kawamoto T, et al. Benefits of adjuvant radiotherapy after radical resection of locally advanced main hepatic duct carcinoma. *Int J Radiat Oncol* 2000;46(3):581-7.

77. Kim TH, Han SS, Park SJ, et al. Role of adjuvant chemoradiotherapy for resected extrahepatic biliary tract cancer. *Int J Radiat Oncol* 2011;81(5):e853-9.

78. Kobayashi H, Murakami Y, Uemura K, et al. Human equilibrative nucleoside transporter 1 expression predicts survival of advanced cholangiocarcinoma patients treated with gemcitabine-based adjuvant chemotherapy after surgical resection. *Ann Surg* 2012;256(2):288-96.

79. Dover LL, Oster RA, McDonald AM, et al. Impact of adjuvant chemoradiation on survival in patients with resectable cholangiocarcinoma. *HPB* 2016;18(10):843-50.



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

80. Kim Y, Amini N, Wilson A, et al. Impact of Chemotherapy and External-Beam Radiation Therapy on Outcomes among Patients with Resected Gallbladder Cancer: A Multi-institutional Analysis. *Ann Surg Oncol* 2016;23(9):2998-3008.

81. Morine Y, Shimada M, Ikemoto T, et al. Effect of adjuvant gemcitabine combined with low-dose 5-fluorouracil and cisplatin chemotherapy for advanced biliary carcinoma. *Anticancer Res* 2017;37(11):6421-28.

82. Akahoshi K, Ban D, Kuboki R, et al. Orotate phosphoribosyltransferase as a predictor of benefit from S-1 adjuvant chemotherapy for cholangiocarcinoma patients. *J Gastroen Hepatol* 2018;34(6):1108-15.

83. Bergeat D, Turrini O, Courtin-Tanguy L, et al. Impact of adjuvant chemotherapy after pancreaticoduodenectomy for distal cholangiocarcinoma: a propensity score analysis from a French multicentric cohort. *Langenbeck Arch Surg* 2018;403(6):701-09.

84. Hester C, Nassour I, Adams-Huet B, et al. Improved Survival in Surgically Resected Distal Cholangiocarcinoma Treated with Adjuvant Therapy: a Propensity Score Matched Analysis. *J Gastrointest Surg* 2018;22(12):2080-87.

85. Zheng X, Chen B, Wu JX, et al. Benefit of adjuvant radiotherapy following narrow-margin hepatectomy in patients with intrahepatic cholangiocarcinoma that adhere to major vessels. *Cancer Manag Res* 2018;10:3973-81.

86. Jiang W, Zeng ZC, Tang ZY, et al. Benefit of radiotherapy for 90 patients with resected intrahepatic cholangiocarcinoma and concurrent lymph node metastases. *J Cancer Res Clin* 2010;136(9):1323-31.

87. Gu B, Qian L, Yu H, et al. Concurrent Chemoradiotherapy in Curatively Resected



For peer review only - http://bmjopen.bmj.com/site/about/guidelines.xhtml

726 Gallbladder Carcinoma: A Propensity Score-Matched Analysis. *Int J Radiat Oncol*
727 2018;100(1):138-45.
728 88. Choudhary S, Gupta N, Verma CP, et al. Influence of adjuvant therapy on pattern of
729 failure and survival in curatively resected gallbladder carcinoma. *J Cancer Res Ther*
730 2021;17(4):1064-68.
731 89. Im JH, Choi GH, Lee WJ, et al. Adjuvant radiotherapy and chemotherapy offer a
732 recurrence and survival benefit in patients with resected perihilar cholangiocarcinoma. *J*
733 *Cancer Res Clin Oncol* 2021;147(8):2435-45.
734 90. Miyata Y, Kogure R, Nakazawa A, et al. The efficacy of s-1 as adjuvant chemotherapy
735 for resected biliary tract carcinoma: A propensity score-matching analysis. *J Clin Med*
736 2021;10(5):1-10.
737 91. Wan W, Zheng B, Sun W, et al. Adjuvant Therapy in Resected Nonmetastatic Stage II–IV
738 Gallbladder Cancer: A Generalized Propensity Score Analysis. *Oncology Research &*
739 *Treatment* 2021;44(7-8):390-99.
740 92. Tariq NU, McNamara MG, Valle JW. Biliary tract cancers: current knowledge, clinical
741 candidates and future challenges. *Cancer Manag Res* 2019;11:2623-42.
742 93. Khan SA, Davidson BR, Goldin RD, et al. Guidelines for the diagnosis and treatment of
743 cholangiocarcinoma: an update. *Gut* 2012;61(12):1657-69.
744 94. Takada T, Amano H, Yasuda H, et al. Is postoperative adjuvant chemotherapy useful for
745 gallbladder carcinoma? A phase III multicenter prospective randomized controlled trial in
746 patients with resected pancreaticobiliary carcinoma. *Cancer* 2002;95(8):1685-95.
747 95. Neoptolemos JP, Moore MJ, Cox TF, et al. Effect of adjuvant chemotherapy with



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748 fluorouracil plus folinic acid or gemcitabine vs observation on survival in patients with
749 resected periampullary adenocarcinoma: the ESPAC-3 periampullary cancer randomized trial.
750 *JAMA* 2012;308(2):147-56.
751 96. Song S, Yang W, Tian H, et al. The efficacy and safety of 5-fluorouracil based adjuvant
752 therapy in resected biliary tract cancer: A systematic review and meta-analysis. *Clin Res*
753 *Hepatol Gas* 2022;46(2).
754 97. Ben-Josef E, Guthrie KA, El-Khoueiry AB, et al. SWOG S0809: A Phase II Intergroup
755 Trial of Adjuvant Capecitabine and Gemcitabine Followed by Radiotherapy and Concurrent
756 Capecitabine in Extrahepatic Cholangiocarcinoma and Gallbladder Carcinoma. *J Clin Oncol*
757 2015;33(24):2617-22.
758 98. Edeline J, Hirano S, Bertaut A, et al. Individual patient data meta-analysis of adjuvant
759 gemcitabine-based chemotherapy for biliary tract cancer: combined analysis of the BCAT and
760 PRODIGE-12 studies. *European journal of cancer (Oxford, England : 1990)* 2022;164:80-87.
761

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## Figure Legends

**Figure 1. A PRISMA flow diagram of the study selection.** BTC, biliary tract cancer; OS, overall survival; DFS, disease-free survival.

**Figure 2. Network plot of comparisons on overall survival and disease-free survival of treatments in patients with biliary tract cancer.** Each circular node represents a type of treatment. The node size corresponds to the total number of participants assigned to each treatment. Each line represents a type of direct comparison, and its width corresponds to the number of studies evaluating the connected treatments. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy.

**Figure 3. Network meta-analysis results for efficacy outcomes. (A)** League table of the network meta-analysis. Pooled HRs and 95% CIs for the overall survival (OS) and disease-free survival (DFS) are listed in the upper and lower triangle, respectively. The estimate in each cell compares the row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment. Significant results are in bold. **(B)** Forest plot of the network meta-analysis for OS. **(C)** Forest plot of the network meta-analysis for DFS. HRs and 95%



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CIs are provided and visually represented by the squares and error bars. **(D)** Bayesian ranking curves of comparable treatments on efficacy for patients with biliary tract cancer. Ranking curves indicate the probability of each treatment ranked first to last on OS (solid lines) and DFS (dotted lines). Data of the curves are presented in **Supplementary Table 9**. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy; HRs, hazard ratios; 95% CIs, 95% confidence intervals; AT, adjuvant therapy.

**Figure 4. Forest plots of the network meta-analysis for overall survival (OS) and disease-free survival (DFS) in the subgroup analyses.** (A) Forest plots of OS in patients stratified into the R0 and R1 groups. (B) and (E) Forest plots of OS and DFS in patients stratified into the GBC and CCA groups. (C) and (F) Forest plots of OS and DFS in patients stratified into the Asian and Western groups. (D) Forest plots of OS and DFS in patients with non-distant metastasis. Effect sizes are presented as hazard ratios with 95% confidence intervals.

**Figure 5. Bayesian ranking plots of comparable treatments on efficacy in patients with biliary tract cancer in the sensitivity analyses.** Ranking curves indicate the probability of each comparable treatment being ranked from first to last on overall survival (solid lines) and disease-free survival (dotted lines). Data of the



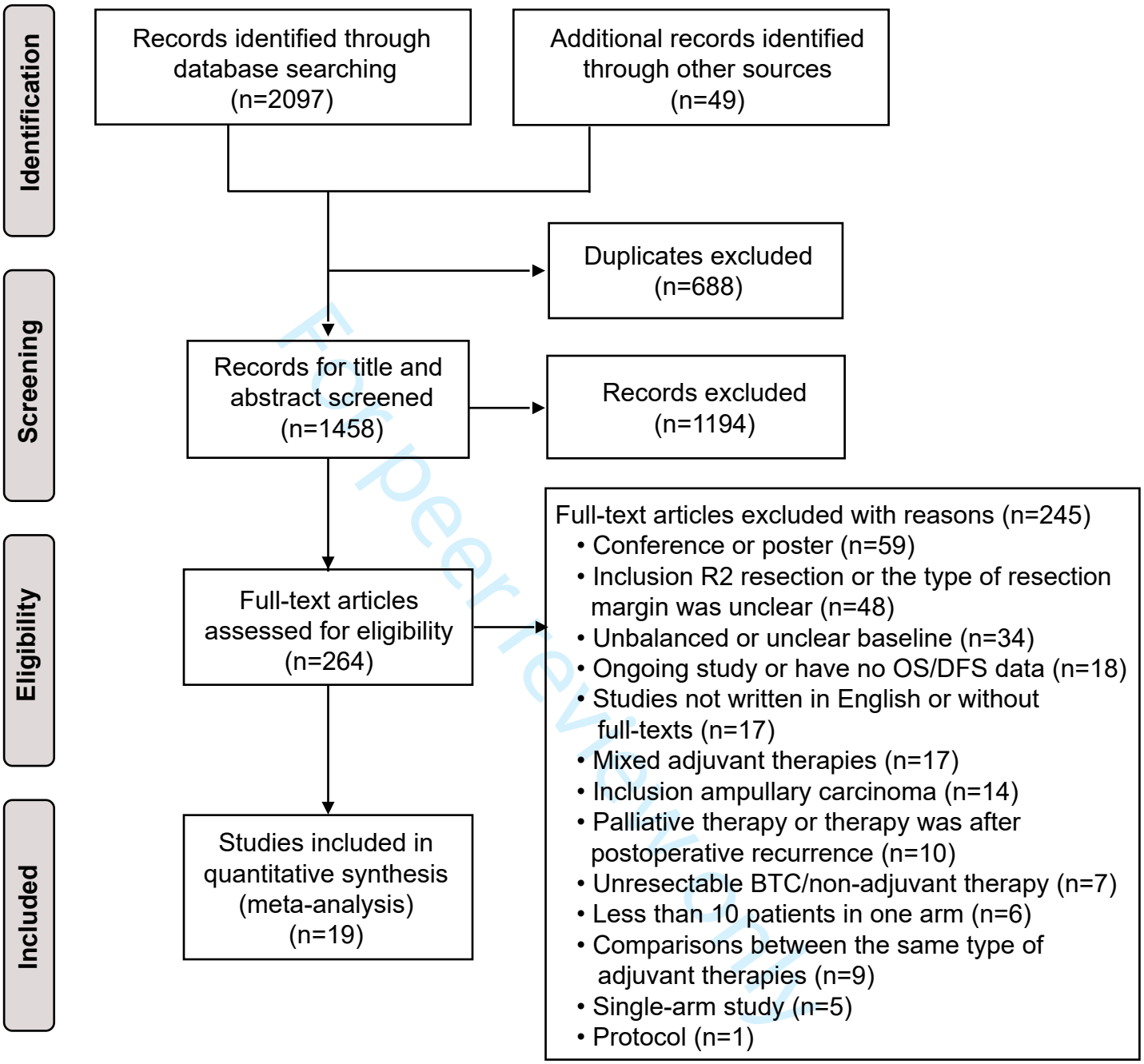
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curves are presented in **Supplementary Table 10**. OS, overall survival; DFS, disease-free survival; ACT, adjuvant chemotherapy; ART, adjuvant radiotherapy; ACRT, adjuvant chemoradiation therapy.

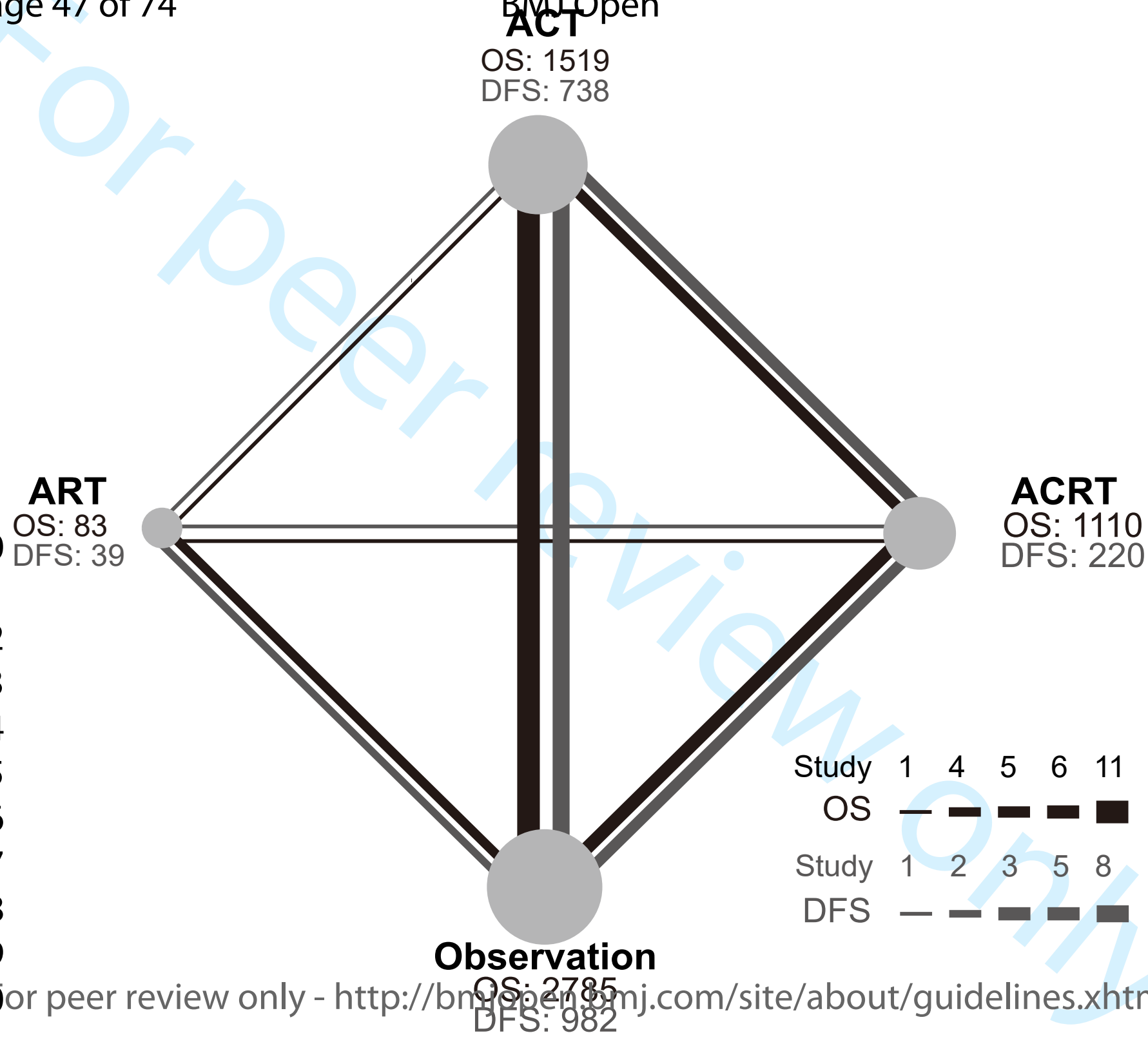
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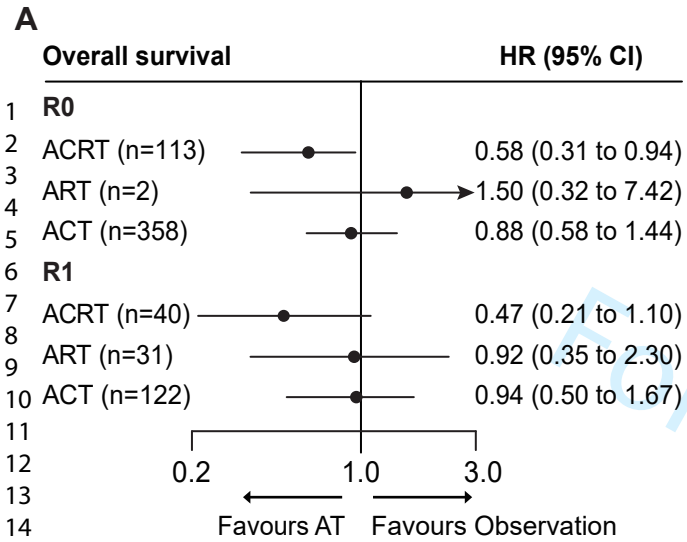
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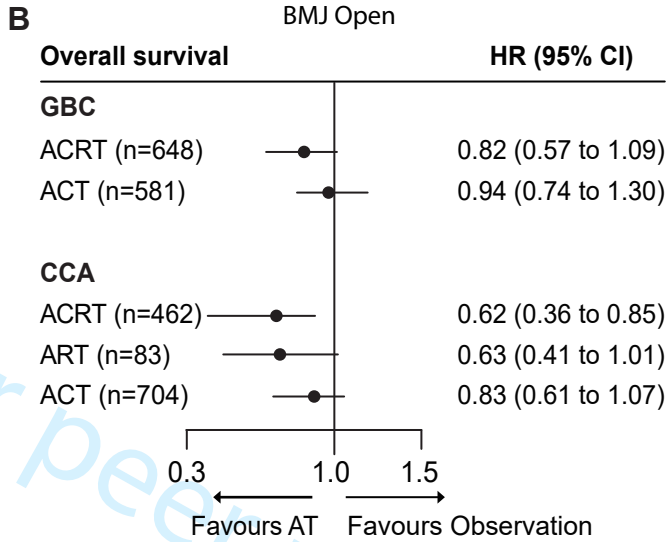


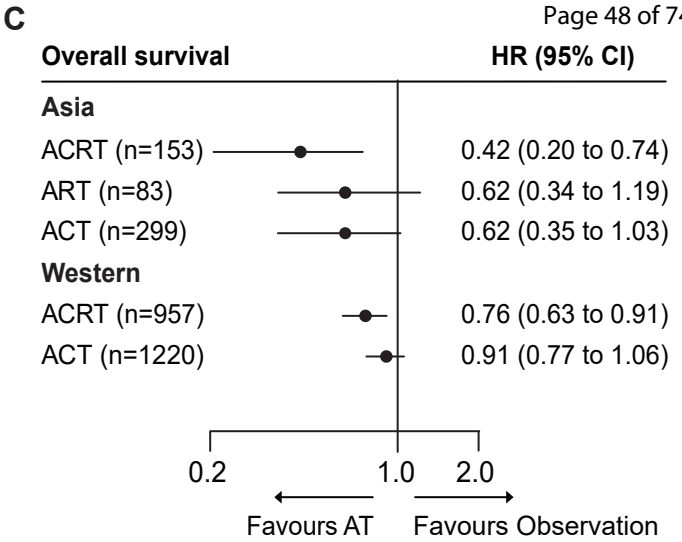

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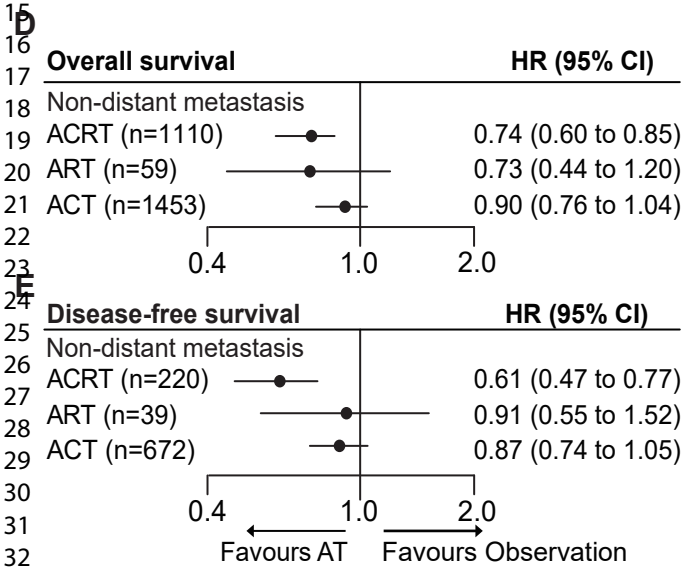


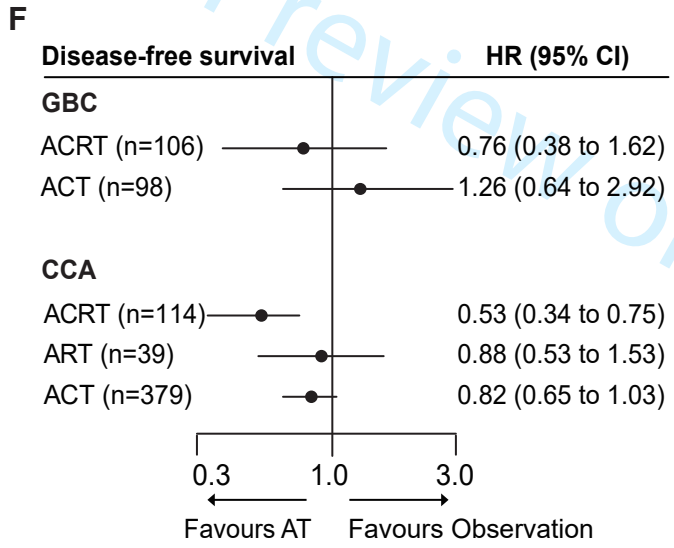

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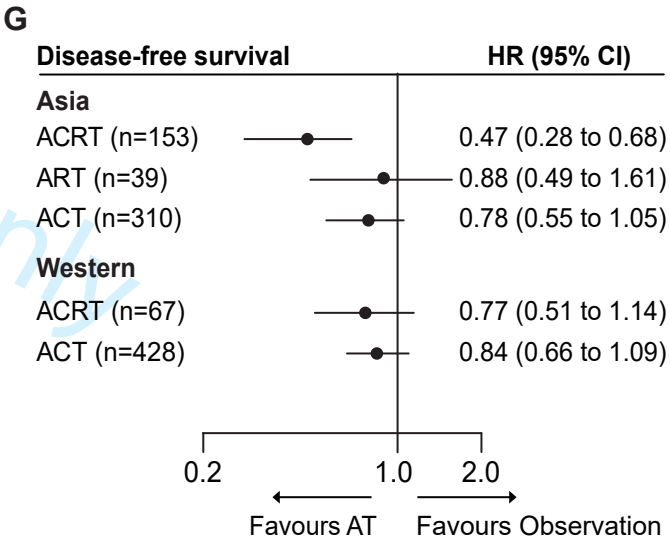


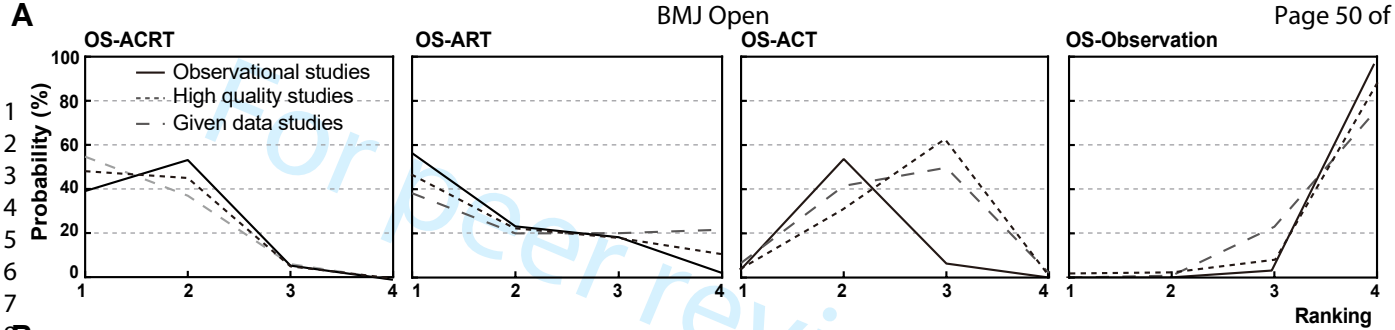



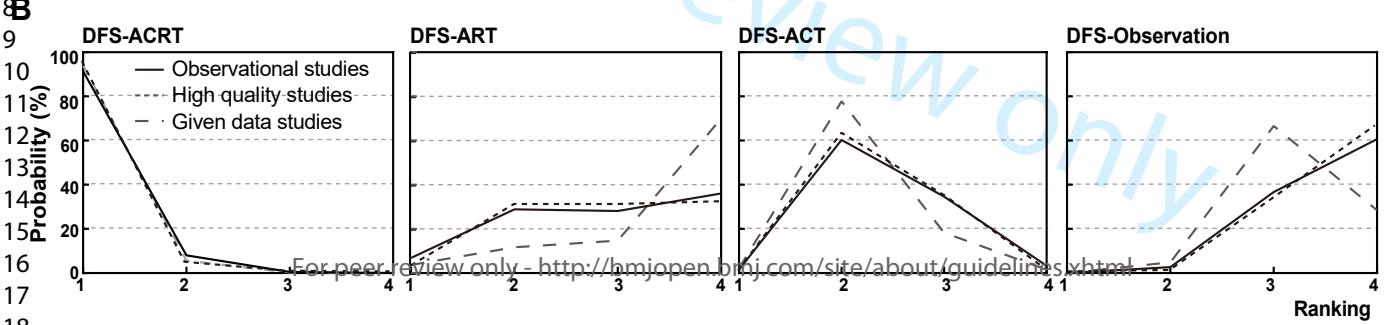




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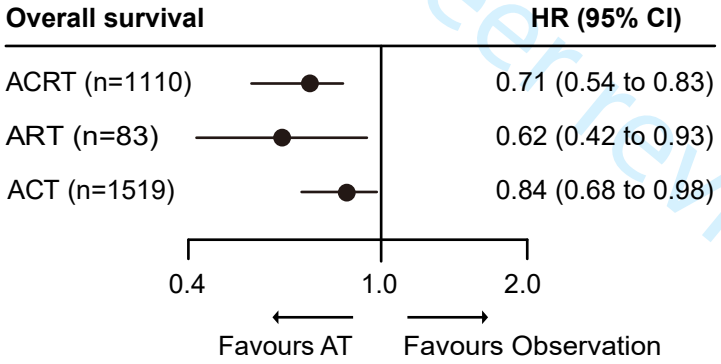




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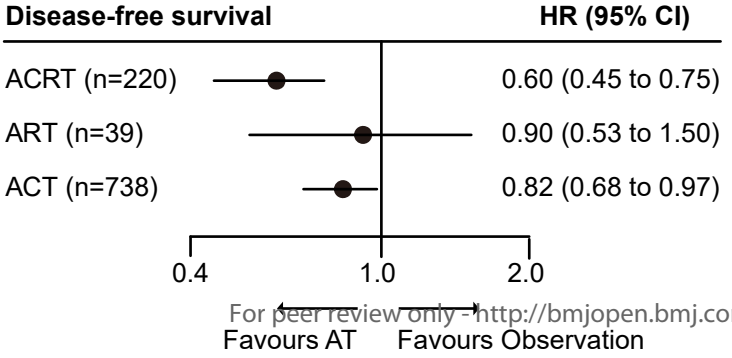
**A**

| Disease-free survival \ Overall survival | ACRT | ART | ACT | Observation |
|---|---|---|---|---|
| ACRT | | 1.14 (0.69 to 1.74) | 0.84 (0.67 to 1.01) | **0.71 (0.54 to 0.83)** |
| ART | 1.51 (0.86 to 2.73) | | 0.74 (0.49 to 1.17) | **0.62 (0.42 to 0.93)** |
| ACT | **1.37 (1.05 to 1.87)** | 0.91 (0.53 to 1.56) | | **0.84 (0.68 to 0.98)** |
| Observation | **1.68 (1.33 to 2.24)** | 1.11 (0.66 to 1.89) | **1.22 (1.03 to 1.46)** | |

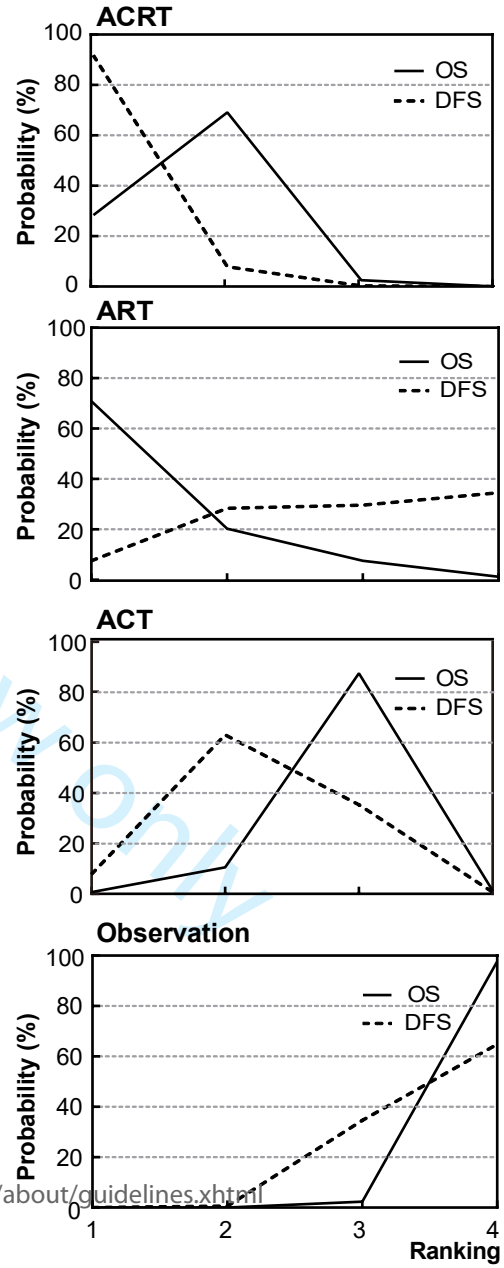
**B**




**C**




**D**




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# SUPPLEMENTAL MATERIAL

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**Supplementary Table 1. Checklist of the PRISMA extension for network meta-analysis.**

| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review *incorporating a network meta-analysis (or related form of meta-analysis).* | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable:<br>• Background: main objectives;<br>• Methods: data sources; study eligibility criteria, participants, and interventions; study appraisal and *synthesis methods, such as network meta-analysis.*<br>• Results: number of studies and participants identified; summary estimates with corresponding confidence/credible intervals; *treatment rankings may also be discussed. Authors may choose to summarize pairwise comparisons against a chosen treatment included in their analyses for brevity.*<br>• Discussion/Conclusions: limitations; conclusions and implications of findings.<br>• Other: primary source of funding; systematic review registration number with registry name. | 2-3 |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known, *including mention of why a network meta-analysis has been conducted.* | 4-7 |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 5-7 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | / |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. *Clearly describe eligible treatments included in the treatment network, and note whether any have been clustered or merged into the same node (with justification).* | 7-8 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 7 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | Supplemental Material page 6 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, | 9 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | included in the meta-analysis). | |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 8-9 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 8-9 |
| Geometry of the network | S1 | Describe methods used to explore the geometry of the treatment network under study and potential biases related to it. This should include how the evidence base has been graphically summarized for presentation, and what characteristics were compiled and used to describe the evidence base to readers. | 9 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 10-11 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). *Also describe the use of additional summary measures assessed, such as treatment rankings and surface under the cumulative ranking curve (SUCRA) values, as well as modified approaches used to present summary findings from meta-analyses.* | 9-10 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies for each network meta-analysis. This should include, but not be limited to:<br>• *Handling of multi-arm trials;*<br>• *Selection of variance structure;*<br>• *Selection of prior distributions in Bayesian analyses; and*<br>• *Assessment of model fit.* | 10 |
| Assessment of Inconsistency | S2 | Describe the statistical methods used to evaluate the agreement of direct and indirect evidence in the treatment network(s) studied. Describe efforts taken to address its presence when found. | 10 |
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 11 |
| Additional analyses | 16 | Describe methods of additional analyses, if done, indicating which were pre-specified. This may include, but not be limited to the following:<br>• Sensitivity or subgroup analyses;<br>• Meta-regression analyses;<br>• *Alternative formulations of the treatment network; and*<br>• *Use of alternative prior distributions for Bayesian analyses (if applicable).* | 11 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each | 11& Figure 1 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| | | stage, ideally with a flow diagram. | |
| Presentation of network structure | S3 | Provide a network graph of the included studies to enable visualization of the geometry of the treatment network | 18 & Figure 2 |
| Summary of network geometry | S4 | Provide a brief overview of characteristics of the treatment network. This may include commentary on the abundance of trials and randomized patients for the different interventions and pairwise comparisons in the network, gaps of evidence in the treatment network, and potential biases reflected by the network structure. | 18 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 13-15 & Table 1 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 16 & Supplemental Material page 7-8 and page18 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study:1) simple summary data for each intervention group, and 2) effect estimates and confidence/credible intervals. *Modified approaches may be needed to deal with information from larger networks.* | 16-17 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence/credible intervals. *In larger networks, authors may focus on comparisons versus a particular comparator (e.g. placebo or standard care), with full findings presented in an appendix. League tables and forest plots may be considered to summarize pairwise comparisons.* If additional summary measures were explored (such as treatment rankings), these should also be presented. | 17-18 |
| Exploration for inconsistency | S5 | Describe results from investigations of inconsistency. This may include such information as measures of model fit to compare consistency and inconsistency models, P values from statistical tests, or summary of inconsistency estimates from different parts of the treatment network. | 17 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 21 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression, *alternative network geometries studied, alternative choice of prior distributions for Bayesian analyses,* and so forth [see Item 16]). | 19-21 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 21-22 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). *Comment on the validity of the assumptions, such as transitivity and consistency. Comment on any concerns regarding network geometry (e.g., avoidance of certain comparisons).* | 24-25 |



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| Section/topic | Item # | Checklist item* | Reported on Page # |
|---|---|---|---|
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 22-24 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 27 |

PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analysis; PICOS = population, intervention, comparators, outcomes, study design.

*Text in italics indicates wording specific to reporting of network meta-analyses that has been added to guidance from the PRISMA statement.

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**Supplementary Table 2. Search strategies**

| Database | Query | Results |
|---|---|---|
| Pubmed | (((((((((""Biliary Tract Neoplasms""[Mesh]) OR ""Bile Duct Neoplasms""[Mesh]) OR ""Gallbladder Neoplasms""[Mesh]) OR ""Cholangiocarcinoma""[Mesh]) OR Biliary Tract Cancer) OR ""Biliary Tract Neoplasms"") OR ""Bile duct cancer"")) AND ((((((""Chemoradiotherapy, Adjuvant""[Mesh]) OR Radiotherapy, Adjuvant[Mesh]) OR ""Chemotherapy, Adjuvant""[Mesh]) OR ""Adjuvants, Immunologic""[Mesh]) OR ""adjuvant therapy"") OR ""adjuvant treatment"")) AND ((((((""Clinical Study"" [Publication Type]) OR ""Observational Study"" [Publication Type]) OR ""Clinical Trial"" [Publication Type]) OR case control study) OR ""Case-Control Studies""[Mesh]) OR (((random) OR control) OR randomized control trial)) | 509 |
| Embase | (('biliary tract cancer'/exp OR 'biliary tract tumor'/exp OR 'bile duct tumor'/exp OR 'gallbladder tumor'/exp OR 'bile duct carcinoma'/exp OR 'biliary tract cancer' OR 'biliary tract neoplasms' OR 'bile duct neoplasms' OR 'bile duct cancer' OR 'gallbladder cancer' OR cholangiocarcinoma) AND ('adjuvant therapy'/exp OR 'adjuvant chemotherapy'/exp OR 'adjuvant radiotherapy'/exp OR 'adjuvant chemoradiotherapy'/exp OR 'adjuvant treatment') AND ('clinical article'/de OR 'clinical study'/de OR 'clinical trial'/de OR 'clinical trial (topic)'/de OR 'cohort analysis'/de OR 'comparative effectiveness'/de OR 'comparative study'/de OR 'controlled clinical trial'/de OR 'controlled study'/de OR 'major clinical study'/de OR 'methodology'/de OR 'multicenter study'/de OR 'observational study'/de OR 'phase 2 clinical trial'/de OR 'phase 2 clinical trial (topic)'/de OR 'phase 3 clinical trial'/de OR 'phase 3 clinical trial (topic)'/de OR 'prospective study'/de OR 'randomized controlled trial'/de OR 'randomized controlled trial (topic)'/de OR 'retrospective study'/de) AND ([adult]/lim OR [aged]/lim OR [middle aged]/lim OR [very elderly]/lim OR [young adult]/lim)) | 892 |
| CINAHL | ((Biliary Tract Cancer OR "bile duct cancer" OR "Gallbladder" OR "biliary tract neoplasm" OR "Cholangiocarcinoma") AND ("adjuvant therapy" OR "adjuvant treatment" OR "adjuvant chemotherapy" OR "adjuvant radiotherapy" OR "adjuvant Chemoradiotherapy" OR "adjuvant radiochemotherapy")) | 274 |
| Cochrane | ("MeSH: [Biliary Tract Neoplasms]"/exp OR "MeSH: [cholangiocarcinoma]"/exp OR ("Biliary Tract Cancer"):ti,ab,kw) AND ("MeSH: [Radiotherapy, Adjuvant]"/exp OR "MeSH: [Chemotherapy, Adjuvant]"/exp) OR "MeSH: [Chemoradiotherapy, Adjuvant]"/exp OR "Adjuvant therapy" OR "Adjuvant treatment") | 52 |
| clinicaltrials.gov | (("Biliary Tract Cancer" OR "bile duct cancer" OR "Gallbladder" OR "Cholangiocarcinoma") AND "Adjuvant") | 46 |

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**Supplementary Table 3. Checklist for quality assessment and scoring using Newcastle-Ottawa Scale.**

| **Checklist items** |
|---|
| **Selection** |
| 1. How representative was the treatment group in comparison for BTC?<br>• If yes, one star;<br>• No star if the patients were selected or selection of group was not described. |
| 2. How representative was the observation group in comparison for BTC?<br>• If drawn from the same community as the treatment group, one star;<br>• No star if drawn from a different source or selection of group was not described. |
| 3. Ascertainment of treatment: secure record or structured interview?<br>• If the treatment were confirmed from surgical records or databases (eg NCDB or SEER), one star;<br>• No star if it was not described. |
| 4. Demonstration that outcome of interest was not present at start of study.<br>• Yes, one star;<br>• No, no star. |
| **Comparability** |
| 5. Group comparable for age, sex, primary tumour site, resection margin status, tumour stage, lymph node status, distant metastasis.<br>• If yes, two stars;<br>• One star was assigned if one of these eight characteristics was differed in groups or not reported.<br>• No star was assigned if there are at least 3 characteristics in the groups differed. |
| **Outcome** |
| 6. Assessment of outcome?<br>• One star if for information ascertained by record lincage or independent blind assessment;<br>• No star if this information was not reported. |
| 7. Was follow-up long enough for outcomes to occur?<br>• Yes, one star if the follow-up period was 5 years or more;<br>• No, no star if the follow-up period was less than 5 years or this information was not reported. |
| 8. Adequacy of follow-up.<br>• One star if follow-up 90%;<br>• No star if this information was not reported. |

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**Supplementary Table 4. Quality assessment of included retrospective studies.**

| Study | Selection | | | | Comparability | Outcome assessment | | | Score |
|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | |
| Todorokia 2000 | * | * | * | * | ** | * | * | * | 9 |
| Jiang 2010 | * | * | * | * | * | * | * | * | 8 |
| Kim 2011 | * | * | * | * | ** | * | * | - | 8 |
| Kobayashi 2012 | * | * | * | * | ** | * | * | * | 9 |
| Dover 2016 | * | * | * | * | * | * | * | - | 7 |
| Kim 2016 | * | * | * | * | * | * | * | - | 7 |
| Morine 2017 | * | * | * | * | ** | * | * | - | 8 |
| Gu 2017 | * | * | * | * | ** | * | * | - | 8 |
| Akahoshi 2018 | * | * | * | * | ** | * | * | - | 8 |
| Bergeat 2018 | * | * | * | * | ** | * | * | - | 8 |
| Hester 2018 | * | * | * | * | ** | * | * | - | 8 |
| Zheng 2018 | * | * | * | * | ** | * | * | * | 9 |
| Choudhary 2019 | * | * | * | * | * | * | * | - | 7 |
| Im 2021 | * | * | * | * | ** | * | * | * | 9 |
| Miyata 2021 | * | * | * | * | ** | * | * | - | 8 |
| Wan 2021 | * | * | * | * | ** | * | * | * | 9 |

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**Supplementary Table 5. Assessment of transitivity.**

| Items | Comparisons | Mean Diff. | Std.Error | Sig. | 95% CI |
|---|---|---|---|---|---|
| Age* | ART vs. Observation | -3.594 | 2.710 | 0.557 | -11.150 to 3.961 |
| | ACRT vs. Observation | -1.327 | 2.430 | 0.947 | -8.100 to 5.445 |
| | ACT vs. Observation | 1.416 | 2.012 | 0.895 | -4.192 to 7.024 |
| | ACRT vs. ART | 2.267 | 3.178 | 0.891 | -6.592 to 11.130 |
| | ACT vs. ART | 5.010 | 2.872 | 0.327 | -2.994 to 13.010 |
| | ACT vs. ACRT | 2.743 | 2.608 | 0.722 | -4.527 to 10.010 |
| Percentage female | ART vs. Observation | 0.81 | 9.782 | 0.9998 | -25.46 to 27.09 |
| | ACRT vs. Observation | 11.09 | 7.547 | 0.47 | -9.19 to 31.36 |
| | ACT vs. Observation | 6.10 | 6.485 | 0.78 | -11.33 to 23.52 |
| | ACRT vs. ART | 10.28 | 10.78 | 0.78 | -18.68 to 39.23 |
| | ACT vs. ART | 5.28 | 10.06 | 0.95 | -21.75 to 32.32 |
| | ACT vs. ACRT | -4.99 | 7.91 | 0.92 | -26.24 to 16.26 |
| Sample size | ART vs. Observation | -145.1 | 157.3 | 0.793 | -567.7 to 277.5 |
| | ACRT vs. Observation | -25.2 | 121.4 | 0.996 | -351.3 to 300.8 |
| | ACT vs. Observation | -44.7 | 104.3 | 0.973 | -324.9 to 235.5 |
| | ACRT vs. ART | 119.9 | 173.4 | 0.900 | -345.8 to 585.6 |
| | ACT vs. ART | 100.4 | 161.9 | 0.925 | -334.5 to 535.2 |
| | ACT vs. ACRT | -19.5 | 127.2 | 0.999 | -361.2 to 322.2 |
| Publication year | ART vs. Observation | -3.750 | 2.633 | 0.492 | -10.8 to 3.323 |
| | ACRT vs. Observation | 1.375 | 2.031 | 0.905 | -4.082 to 6.832 |
| | ACT vs. Observation | 2.231 | 1.746 | 0.582 | -2.459 to 6.920 |
| | ACRT vs. ART | 5.125 | 2.901 | 0.305 | -2.669 to 12.92 |
| | ACT vs. ART | 5.981 | 2.709 | 0.139 | -1.297 to 13.2 |
| | ACT vs. ACRT | 0.856 | 2.129 | 0.978 | -4.864 to 6.575 |
| Percentage R0 | ART vs. Observation | -12.66 | 17.17 | 0.881 | -58.75 to 33.42 |
| | ACRT vs. Observation | 7.61 | 13.97 | 0.947 | -29.87 to 45.09 |
| | ACT vs. Observation | 3.64 | 12.52 | 0.991 | -29.95 to 37.23 |
| | ACRT vs. ART | 20.27 | 18.70 | 0.701 | -29.90 to 70.44 |
| | ACT vs. ART | 16.30 | 17.64 | 0.792 | -31.03 to 63.64 |
| | ACT vs. ACRT | -3.97 | 14.54 | 0.993 | -42.98 to 35.03 |
| Percentage N0 | ART vs. Observation | 19.33 | 11.23 | 0.332 | -11.32 to 49.99 |
| | ACRT vs. Observation | -14.00 | 7.74 | 0.291 | -35.13 to 7.13 |
| | ACT vs. Observation | -10.90 | 5.99 | 0.285 | -27.24 to 5.44 |
| | ACRT vs. ART | -33.33 | 12.43 | 0.056 | -67.27 to 0.60 |
| | ACT vs. ART | -30.24 | 11.42 | 0.060 | -61.41 to 0.94 |
| | ACT vs. ACRT | 3.10 | 8.01 | 0.980 | -18.78 to 24.97 |

The characteristics included age, percentage female, sample size, publication year, percentage R0 and N0 have been evaluated in the network. All the comparisons had similar mean age and other main characteristics with P-value over 0.05.

* Median age was given instead of mean age in the Bergeat, Dover, Miyata, Gu, and Edeline studies. Information of age in the Jiang, Kobayashi, Akahoshi, Ebata, Choudhary, Im (2021), and Wan studies were presented as younger or older than a specific age that couldn't be integrated into the figure.



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**Supplementary Table 6. Results of global heterogeneity and local heterogeneity.**

| Comparisons | Local | | Global | | | Heterogeneity assessment |
|---|---|---|---|---|---|---|
| | Pair-wise ($I^2$) | Network ($I^2$) | Between study variance ($\tau^2$) | Pair-wise ($I^2$) | Consistency effect ($I^2$) | |
| **Overall survival for BTC patients** | | | | | | |
| ACT vs Observation | 20.3% | 20.2% | 0.022 | 0.0% | 19.4% | Low to high |
| ART vs Observation | 0.0% | **81.9%** | | | | |
| ACRT vs Observation | 33.4% | 47.5% | | | | |
| ACT vs ART | 0.0% | 49.4% | | | | |
| ACRT vs ART | 0.0% | 28.8% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients with R0 resection margin** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.069 | 0.96% | 0.0% | Low to moderate |
| ART vs Observation | NA | 0.0% | | | | |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACT vs ART | NA | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 33.5% | 21.6% | | | | |
| **Overall survival for BTC patients with R1 resection margin** | | | | | | |
| ACT vs Observation | 33.8% | 36.5% | 0.179 | 24.9% | 13.5% | Low to moderate |
| ART vs Observation | 0.0% | 34.5% | | | | |
| ACRT vs Observation | 28.1% | 28.5% | | | | |
| ACT vs ART | NA | 0.0% | | | | |
| ACRT vs ART | NA | 0.0% | | | | |
| ACRT vs ACT | NA | 0.0% | | | | |
| **Overall survival for GBC patients** | | | | | | |
| ACT vs Observation | 44.1% | 45.0% | 0.028 | 12.1% | 6.0% | Low to moderate |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.064 | 0.0% | 25.8% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | **66.7%** | **93.7%** | | | | |
| ACT vs ART | 0.0% | 47.7% | | | | |
| ACRT vs ART | 0.0% | 17.6% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | **56.5%** | **57.0%** | 0.271 | 27.5% | **64.9%** | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | **71.6%** | **91.7%** | | | | |
| ACT vs ART | 0.0% | 9.8% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.007 | 0.0% | 0.0% | Low |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Overall survival for BTC patients without distant-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.011 | 0.0% | 1.7% | Low to high |
| ART vs Observation | 0.0% | **63.2%** | | | | |
| ACRT vs Observation | 33.1% | 41.0% | | | | |
| ACT vs ART | 0.0% | 42.7% | | | | |
| ACRT vs ART | 0.0% | 20.6% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients** | | | | | | |



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| | | | | | | |
|---|---|---|---|---|---|---|
| ACT vs Observation | 6.1% | 6.6% | 0.017 | 0.0% | 0.0% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 20.0% | **82.2%** | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for GBC patients** | | | | | | |
| ACT vs Observation | **75.0%** | 48.1% | 0.169 | **51.2%** | 34.1% | Low to high |
| ACRT vs Observation | 9.7% | 0.0% | | | | |
| ACRT vs ACT | 3.7% | 0.0% | | | | |
| **Disease-free survival for CCA patients** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.029 | 0.0% | 0.0% | Low to moderate |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 35.0% | 47.0% | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients in Asia** | | | | | | |
| ACT vs Observation | 32.8% | 32.4% | 0.067 | 0.0% | 0.0% | Low to high |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 35.8% | **67.6%** | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |
| **Disease-free survival for BTC patients in Western** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.023 | 0.0% | 0.0% | Low |
| ACRT vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs ACT | NA | NA | | | | |
| **Disease-free survival for BTC patients for patients without distant-metastasis** | | | | | | |
| ACT vs Observation | 0.0% | 0.0% | 0.012 | 0.0% | 0.0% | Low to moderate |
| ART vs Observation | 0.0% | 0.0% | | | | |
| ACRT vs Observation | 20.0% | 38.7% | | | | |
| ACT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ART | 0.0% | 0.0% | | | | |
| ACRT vs ACT | 0.0% | 0.0% | | | | |

The numbers with high heterogeneity are in bold (We inferred the magnitude of heterogeneity by comparing the estimated $\tau 2$ to empirical distributions of heterogeneity typically found in meta-analyses. Low heterogeneity could be considered when the estimated $\tau^2$ is less than the 25% quantile of the empirical distribution, moderate heterogeneity for $\tau^2$ between 25% and 50% quantile, and high heterogeneity for $\tau^2$ larger than the 50% quantile.). NA=Not available, because only one study was included in this type of comparison; ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma.



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**Supplementary Table 7. Node-splitting analysis of inconsistency.**

| Comparison | Direct effect | Indirect effect | Overall | *P* |
|---|---|---|---|---|
| | lnHR (95% Cl) | | | |
| **Overall survival for BTC patients** | | | | |
| ACRT, ACT | 0.24 (0.02, 0.51) | 0.10 (-0.30, 0.58) | 0.18 (-0.01, 0.40) | 0.52 |
| ACRT, ART | 1.3 (0.63, 1.90) | -0.56 (-1.10, -0.08) | -0.13 (-0.53, 0.37) | **0.0000** |
| ACT, ART | 0.60 (0.05, 1.10) | -0.66 (-1.10, -0.18) | -0.31 (-0.71, 0.16) | **0.001** |
| ACRT, Observation | 0.35 (0.07, 0.73) | 0.83 (0.25, 1.60) | 0.35 (0.18, 0.61) | 0.15 |
| ACT, Observation | 0.24 (-0.03, 0.54) | 0.19 (-0.52, 0.94) | 0.17 (0.02, 0.38) | 0.89 |
| ART, Observation | 0.70 (0.29, 1.10) | -0.92 (-2.90, 1.10) | 0.48 (0.07, 0.86) | 0.13 |
| **Overall survival for BTC patients with R0 resection margin** | | | | |
| ACRT, ACT | 1.2 (-0.82, 3.3) | 0.35 (-0.31, 1.2) | 0.41 (-0.16, 1.20) | 0.43 |
| ACRT, ART | NA | NA | NA | NA |
| ACT, ART | 0.92 (-1.30, 3.20) | 0.05 (-2.20, 2.30) | 0.53 (-1.00, 2.10) | 0.58 |
| ACRT, Observation | 0.49 (-0.03, 1.20) | 1.4 (-0.69, 3.40) | 0.53 (0.07, 1.20) | 0.41 |
| ACT, Observation | 0.14 (-0.40, 0.65) | -0.24 (-2.00, 1.60) | 0.12 (-0.35, 0.54) | 0.69 |
| ART, Observation | 0.37 (-1.80, 2.40) | -0.90 (-3.00, 1.20) | -0.41 (-2.00, 1.10) | 0.41 |
| **Overall survival for BTC patients with R1 resection margin** | | | | |
| ACRT, ACT | 0.60 (-0.69, 1.90) | 0.57 (-0.80, 1.80) | 0.68 (-0.32, 1.60) | 0.97 |
| ACRT, ART | 1.20 (-0.17, 2.5) | -0.31 (-2.40, 1.60) | 0.66 (-0.52, 1.80) | 0.18 |
| ACT, ART | 0.57 (-0.67, 1.80) | -0.88 (-2.60, 0.86) | -0.017 (-1.10, 1.00) | 0.15 |
| **Overall survival for GBC patients** | | | | |
| ACRT, ACT | 0.17 (-0.37, 0.82) | 0.81 (-0.63, 2.50) | 0.20 (-0.09, 0.55) | 0.35 |
| ACRT, Observation | 0.25 (-0.20, 0.76) | 0.55 (-0.98, 1.90) | 0.25 (-0.02, 0.54) | 0.67 |
| ACT, Observation | 0.061 (-0.51, 0.45) | -0.35 (-2.00, 1.30) | 0.06 (-0.26, 0.30) | 0.58 |
| **Overall survival for CCA patients** | | | | |
| ACRT, ACT | 0.35 (0.03, 0.81) | 0.15 (-0.29, 0.72) | 0.30 (-0.03, 0.81) | 0.45 |
| ACRT, ART | 1.3 (0.60, 2.00) | -0.57 (-1.10, 0.01) | 0.04 (-0.51, 0.77) | **0.00035** |
| ACT, ART | 0.61 (-0.02, 1.20) | -0.70 (-1.20, -0.13) | -0.27 (-0.77, 0.29) | **0.0056** |
| ACRT, Observation | 0.56 (0.03, 1.30) | 0.69 (-0.13, 1.70) | 0.49 (0.16, 1.00) | 0.77 |
| ART, Observation | 0.69 (0.27, 1.10) | -0.98 (-3.0, 1.10) | 0.46 (-0.02, 0.91) | 0.12 |
| ACT, Observation | 0.19 (-0.20, 0.59) | 0.51 (-0.41, 1.60) | 0.18 (-0.07, 0.50) | 0.50 |
| **Overall survival for BTC patients in Asia** | | | | |
| ACRT, ACT | 0.71 (-0.18, 1.70) | -0.02 (-0.74, 0.95) | 0.33 (-0.37, 1.20) | 0.23 |
| ACRT, ART | 1.3 (0.53, 2.20) | -0.75 (-1.50, 0.05) | 0.18 (-0.56, 1.20) | **0.0018** |
| ACT, ART | 0.11 (-0.78, 1.00) | -0.27 (-1.30, 0.85) | -0.15 (-0.78, 0.56) | 0.56 |
| ACRT, Observation | 0.63 (-0.040, 1.50) | 2.00 (-0.17, 4.30) | 0.77 (0.17, 1.60) | 0.24 |
| ACT, Observation | 0.55 (-0.11, 1.20) | -0.057 (-1.4, 1.4) | 0.44 (-0.06, 0.99) | 0.41 |
| ART, Observation | 0.67 (0.07, 1.30) | 0.71 (-0.54, 1.80) | 0.59 (-0.01, 1.10) | 0.96 |
| **Overall survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.16 (-0.11, 0.39) | 0.49 (-0.28, 1.30) | 0.18 (-0.02, 0.37) | 0.40 |
| ACRT, Observation | 0.25 (-0.01, 0.55) | 0.35 (-0.26, 0.89) | 0.28 (0.09, 0.46) | 0.70 |
| ACT, Observation | 0.12 (-0.09, 0.31) | 0.32 (-0.48, 1.1) | 0.10 (-0.06, 0.26) | 0.61 |
| **Overall survival for BTC patients without distant-metastasis** | | | | |
| ACRT, ACT | 0.24 (0.03, 0.48) | 0.20 (-0.17, 0.69) | 0.20 (0.039, 0.40) | 0.86 |
| ACRT, ART | 1.30 (0.66, 1.90) | -0.74 (-1.40, -0.04) | -0.01 (-0.52, 0.53) | **0.0000** |
| ACT, ART | 0.59 (0.06, 1.10) | -0.82 (-1.50, -0.15) | -0.21 (-0.72, 0.31) | **0.001** |
| ART, Observation | 0.66 (0.13, 1.20) | -1.00 (-3.00, 0.98) | 0.32 (-0.18, 0.81) | 0.12 |
| ACRT, Observation | 0.31 (0.08, 0.69) | 0.61 (0.11, 1.40) | 0.31 (0.15, 0.52) | 0.28 |
| ACT, Observation | 0.14 (-0.12, 0.42) | 0.20 (-0.43, 0.94) | 0.10 (-0.04, 0.27) | 0.85 |
| **Disease-free survival for BTC patients** | | | | |
| ACRT, ACT | 0.49 (0.10, 0.93) | 0.26 (-0.14, 0.70) | 0.32 (0.05, 0.63) | 0.41 |
| ACRT, ART | 1.30 (0.60, 2.10) | -0.10 (-0.95, 0.72) | 0.41 (-0.14, 1.00) | **0.009** |
| ACT, ART | 0.33 (-0.27, 0.94) | -0.20 (-0.92, 0.54) | 0.09 (-0.44, 0.63) | 0.27 |
| ACRT, Observation | 0.43 (0.17, 0.75) | 1.10 (0.34, 1.80) | 0.52 (0.28, 0.81) | 0.11 |
| ACT, Observation | 0.22 (0.04, 0.41) | -0.14 (-0.87, 0.60) | 0.20 (0.03, 0.38) | 0.35 |



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| | | | | |
|---|---|---|---|---|
| ART, Observation | 0.29 (-0.30, 0.86) | -0.30 (-1.90, 1.30) | 0.11 (-0.41, 0.63) | 0.50 |
| **Disease-free survival for GBC patients** | | | | |
| ACRT, ACT | 0.30 (-0.52, 1.20) | 1.50 (-0.38, 3.30) | 0.50 (-0.20, 1.30) | 0.22 |
| ACRT, Observation | 0.30 (-0.47, 1.10) | -0.25 (-2.10, 1.60) | 0.27 (-0.48, 0.97) | 0.52 |
| ACT, Observation | -0.28 (-1.30, 0.60) | -0.15 (-2.10, 1.80) | -0.24 (-1.10, 0.44) | 0.89 |
| **Disease-free survival for CCA patients** | | | | |
| ACRT, ACT | 1.00 (0.33, 1.70) | 0.27 (-0.18, 0.78) | 0.45 (0.05, 0.95) | 0.09 |
| ACRT, ART | 1.30 (0.62, 2.10) | -0.14 (-1.10, 0.79) | 0.53 (-0.08, 1.30) | **0.017** |
| ACT, ART | 0.33 (-0.32, 0.98) | -0.19 (-0.98, 0.63) | 0.09 (-0.49, 0.67) | 0.32 |
| ACRT, Observation | 0.58 (0.20, 1.10) | 1.40 (0.02, 2.70) | 0.64 (0.29, 1.10) | 0.28 |
| ACT, Observation | 0.21 (-0.04, 0.49) | -0.15 (-1.40, 1.10) | 0.20 (-0.03, 0.44) | 0.55 |
| ART, Observation | 0.28 (-0.33, 0.89) | -0.33 (-2.00, 1.30) | 0.11 (-0.45, 0.66) | 0.50 |
| **Disease-free survival for BTC patients in Asia** | | | | |
| ACRT, ACT | 0.88 (0.29, 1.50) | 0.25 (-0.32, 0.92) | 0.50 (0.09, 1.00) | 0.14 |
| ACRT, ART | 1.30 (0.56, 2.20) | 0.02 (-0.99, 1.10) | 0.64 (-0.01, 1.40) | 0.052 |
| ACT, ART | 0.33 (-0.43, 1.10) | -0.13 (-1.00, 0.85) | 0.13 (-0.49, 0.78) | 0.44 |
| ACRT, Observation | 0.65 (0.21, 1.30) | 1.2 (0.30, 2.10) | 0.76 (0.39, 1.30) | 0.31 |
| ACT, Observation | 0.30 (-0.06, 0.70) | -0.01 (-0.90, 0.99) | 0.25 (-0.05, 0.59) | 0.52 |
| ART, Observation | 0.28 (-0.39, 0.95) | -0.30 (-2.00, 1.40) | 0.12 (-0.48, 0.72) | 0.54 |
| **Disease-free survival for BTC patients in Western** | | | | |
| ACRT, ACT | 0.09 (-0.51 to 0.69) | 0.23 (-0.51 to 0.99) | 0.09 (-0.33 to 0.55) | 0.76 |
| **Disease-free survival for BTC patients without distant-metastasis** | | | | |
| ACRT, ACT | 0.48 (0.12, 0.89) | 0.34 (-0.04, 0.77) | 0.35 (0.09, 0.65) | 0.61 |
| ACRT, ART | 1.30 (0.62, 2.00) | -0.12 (-0.94, 0.69) | 0.39 (-0.17, 0.96) | **0.01** |
| ACT, ART | 0.33 (-0.26, 0.92) | -0.26 (-0.98, 0.48) | 0.05 (-0.46, 0.57) | 0.19 |
| ACRT, Observation | 0.43 (0.18, 0.72) | 0.98 (0.27, 1.70) | 0.49 (0.25, 0.75) | 0.16 |
| ACT, Observation | 0.15 (-0.05, 0.33) | -0.13 (-0.85, 0.62) | 0.14 (-0.04, 0.30) | 0.46 |
| ART, Observation | 0.29 (-0.29, 0.86) | -0.33 (-2.00, 1.30) | 0.10 (-0.42, 0.62) | 0.46 |

Nonsignificant values ($P > 0.05$) indicate no inconsistency between direct and indirect effects. NA=Not available; CI=confidence interval. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 8. Comparisons of the fitness of consistency and inconsistency models using deviance information criteria**

| Item | Model | Overall | Resection margin status | | Tumor Site | | The regions | | Non-distant metastasis |
|---|---|---|---|---|---|---|---|---|---|
| | | | R0 | R1 | GBC | CCA | Asia | Western | |
| OS | Consistency | 46.17 | 15.37 | 16.03 | 16.95 | 31.39 | 226.5 | 11.88 | 37.79 |
| | Inconsistency | 46.05 | 17.96 | 15.99 | 16.70 | 32.54 | 228.1 | 12.93 | 38.31 |
| DFS | Consistency | 29.93 | NA | NA | 9.50 | 22.21 | 22.76 | 7.88 | 23.61 |
| | Inconsistency | 31.15 | NA | NA | 10.26 | 24.96 | 25.12 | 7.89 | 25.28 |

The fitness of the Bayesian model was evaluated by deviance information criteria (DIC), which is adjusted with the complexity of the model.

NA=Not applicable; OS=overall survival; DFS=disease-free survival; R0=negative resection margins; R1=microscopic positive resection margins; GBC=gallbladder cancer; CCA= cholangiocarcinoma.

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**Supplementary Table 9. Ranking results of network meta-analysis for overall survival (OS) and disease-free survival (DFS).**

| Treatment | Rank of probability for OS (%) | | | | | Rank of probability for DFS (%) | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| ACT | 0.7 | 10.5 | **87.5** | 1.3 | 36.9 | 7.9 | 63.1 | 35.4 | 0.7 | 54.7 |
| ART | **70.9** | 20.3 | 7.5 | 1.2 | **87.0** | 7.5 | 28.4 | 29.6 | 34.5 | 36.3 |
| ACRT | 28.4 | **69.1** | 2.5 | 0.0 | 75.3 | **91.7** | 7.9 | 0.3 | 0.0 | **97.1** |
| Observation | 0.0 | 0.0 | 2.4 | 97.5 | 0.8 | 0.0 | 0.6 | 34.6 | 64.8 | 11.9 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

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**Supplementary Table 10. Rank results for OS and DFS in the sensitivity analysis.**

| Treatment | Rank of probability (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| | OS for studies including observational studies | | | | | DFS for studies including observational studies | | | | |
| ACT | 3.8 | 53.7 | 6.4 | 0.0 | 43.2 | 1.6 | 60.3 | 34.6 | 3.4 | 53.4 |
| ART | **56.4** | 23.2 | 18.3 | 2.1 | **77.9** | 6.9 | 28.9 | 28.1 | 36.0 | 35.6 |
| ACRT | 39.8 | 53.7 | 6.4 | 0.0 | **77.8** | **91.4** | 8.0 | 0.5 | 0.0 | **97.0** |
| Observation | 0.0 | 0.0 | 3.2 | 96.8 | 1.1 | 0.0 | 2.7 | 36.7 | 60.6 | 14.1 |
| **Treatment** | **OS for studies with high quality** | | | | | **DFS for studies with high quality** | | | | |
| ACT | 4.0 | 31.0 | 63.0 | 2.0 | 45.7 | 0.9 | 63.3 | 34.9 | 0.9 | 54.7 |
| ART | 47.3 | 22.9 | 18.5 | 11.2 | 68.8 | 3.8 | 31.6 | 31.6 | 32.9 | 35.4 |
| ACRT | **48.7** | 45.6 | 5.6 | 0.1 | **90.0** | **95.3** | 4.4 | 0.3 | 0.0 | **98.3** |
| Observation | 0.0 | 0.5 | 6.3 | 86.7 | 1.6 | 0.0 | 0.7 | 33.2 | 66.1 | 11.5 |
| **Treatment** | **OS for original reported HRs (95%CI) in the studies** | | | | | **DFS for original reported HRs (95%CI) in the studies** | | | | |
| ACT | 6.6 | 41.5 | 49.8 | 2.1 | 50.8 | 2.2 | 77.8 | 18.3 | 1.6 | 60.2 |
| ART | 38.2 | 20.0 | 20.1 | 21.7 | 58.2 | 3.7 | 11.7 | 14.7 | 69.9 | 16.4 |
| ACRT | **55.2** | 37.7 | 7.0 | 0.1 | **82.7** | **94.0** | 5.6 | 0.3 | 0.0 | **97.9** |
| Observation | 0.0 | 0.9 | 23.0 | 76.1 | 8.3 | 0.0 | 4.9 | 66.6 | 28.5 | 25.5 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.

CI=confidence interval; OS=overall survival; DFS=disease-free survival.

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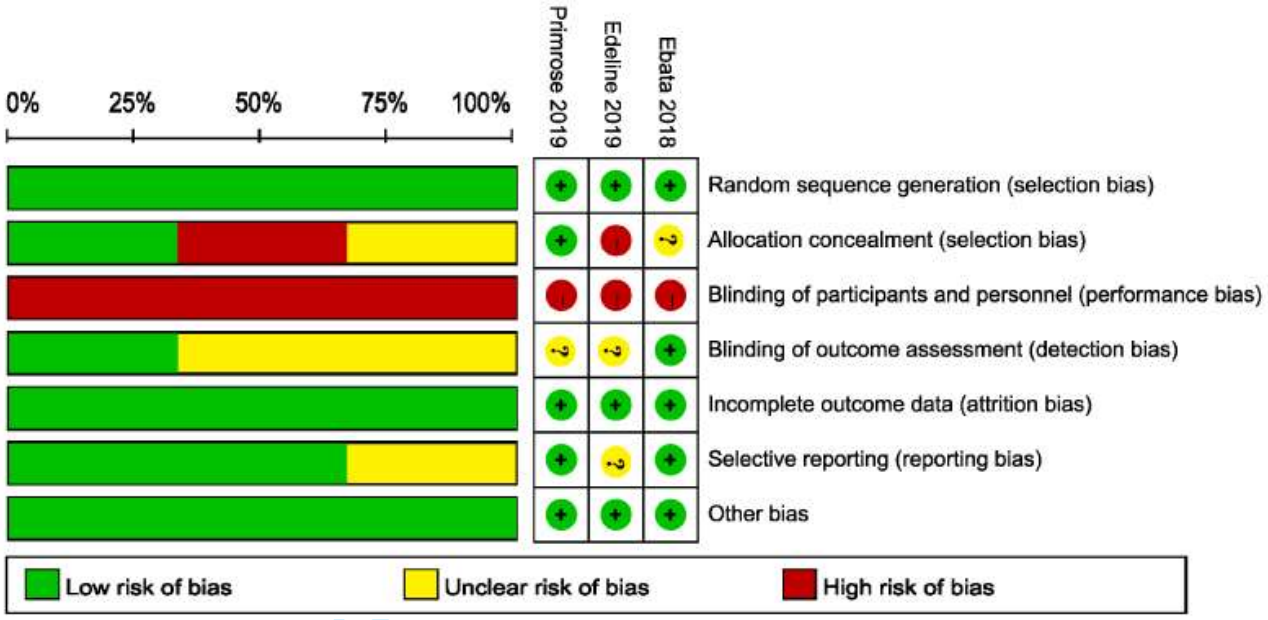
**Supplementary Table 11. Ranking results for overall survival (OS) and disease-free survival (DFS) in the subgroup analysis.**

| Treatment | Rank of probability (%) | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | SUCRA | 1 | 2 | 3 | 4 | SUCRA |
| | OS for BTC patients with R0 resection margin | | | | | OS for BTC patients with R1 resection margin | | | | |
| ACT | 6.0 | 54.0 | 32.3 | 7.6 | 52.8 | 5.0 | 31.9 | 36.3 | 26.5 | 38.5 |
| ART | 11.6 | 12.0 | 7.6 | 68.8 | 22.1 | 9.9 | 36.8 | 17.5 | 35.8 | 40.3 |
| ACRT | **81.9** | 15.6 | 2.2 | 0.3 | **93.1** | **83.9** | 11.2 | 3.3 | 1.6 | **92.5** |
| Observation | 0.4 | 18.4 | 57.9 | 23.3 | 32.0 | 28.8 | 1.2 | 20.1 | 42.5 | 36.2 |
| | OS for GBC patients | | | | | OS for CCA patients | | | | |
| ACT | 7.4 | 61.3 | 31.3 | NA | 38.1 | 1.1 | 15.9 | 75.7 | 7.3 | 37.0 |
| ART | NA | NA | NA | NA | NA | 45.4 | 39.8 | 12.2 | 2.7 | 75.9 |
| ACRT | **90.4** | 8.3 | 1.3 | NA | **94.5** | 53.5 | 43.9 | 2.5 | 0.1 | **83.6** |
| Observation | 2.2 | 30.4 | 67.4 | NA | 17.4 | 0.0 | 0.4 | 9.7 | 90.0 | 3.5 |
| | OS for BTC patients in Asia | | | | | OS for BTC patients in Western | | | | |
| ACT | 8.5 | 45.1 | 43.6 | 2.8 | 53.1 | 3.2 | 87.2 | 9.6 | NA | 46.8 |
| ART | 13.5 | 37.1 | 42.7 | 6.7 | 52.5 | NA | NA | NA | NA | NA |
| ACRT | **78.0** | 17.2 | 4.7 | 0.1 | **91.0** | **96.6** | 3.1 | 0.2 | NA | **98.2** |
| Observation | 0.0 | 0.6 | 9.0 | 90.5 | 3.4 | 0.2 | 96.6 | 90.1 | NA | 5.0 |
| | OS for BTC patients without distant metastasis | | | | | DFS for BTC patients without distant metastasis | | | | |
| ACT | 0.4 | 19.9 | 73.1 | 6.7 | 38.0 | 0.3 | 53.7 | 42.0 | 4.0 | 50.1 |
| ART | **50.2** | 29.2 | 10.6 | 10.0 | 73.2 | 7.1 | 36.2 | 21.1 | 35.5 | 38.3 |
| ACRT | 49.4 | 50.0 | 0.6 | 0.0 | **82.9** | **92.6** | 7.2 | 0.2 | 0.0 | **97.5** |
| Observation | 0.0 | 0.9 | 15.7 | 83.3 | 5.8 | 0.0 | 2.9 | 36.6 | 60.5 | 14.1 |
| | DFS for GBC patients | | | | | DFS for CCA patients | | | | |
| ACT | 4.1 | 20.1 | 75.8 | NA | 14.1 | 1.2 | 57.9 | 38.2 | 2.7 | 52.6 |
| ART | NA | NA | NA | NA | NA | 4.4 | 35.0 | 28.1 | 32.5 | 37.1 |
| ACRT | **78.2** | 18.0 | 3.8 | NA | **87.2** | **94.4** | 5.2 | 0.4 | 0.0 | 37.1 |
| Observation | 17.7 | 62.0 | 20.4 | NA | 48.7 | 0.0 | 1.8 | 33.3 | 64.8 | 12.3 |
| | DFS for BTC patients in Asia | | | | | DFS for BTC patients in Western | | | | |
| ACT | 0.9 | 62.8 | 33.3 | 3.0 | 53.9 | 32.3 | 60.4 | 7.3 | NA | 62.5 |
| ART | 3.1 | 31.1 | 33.0 | 32.8 | 34.8 | NA | NA | NA | NA | NA |
| ACRT | **96.0** | 3.8 | 0.2 | 0.0 | **98.6** | **66.4** | 25.2 | 8.5 | NA | **78.9** |
| Observation | 0.0 | 2.3 | 33.4 | 64.2 | 12.7 | 1.3 | 14.5 | 84.2 | NA | 8.6 |

The number in each cell represents the posterior probability of the row-defining treatment being ranked at the column-defining position. The numbers with the biggest probability of ranking first and last are in bold and underscored. NA=Not available; ACT=Adjuvant Chemotherapy; ART=Adjuvant Radiotherapy; ACRT=Adjuvant Chemoradiation therapy.



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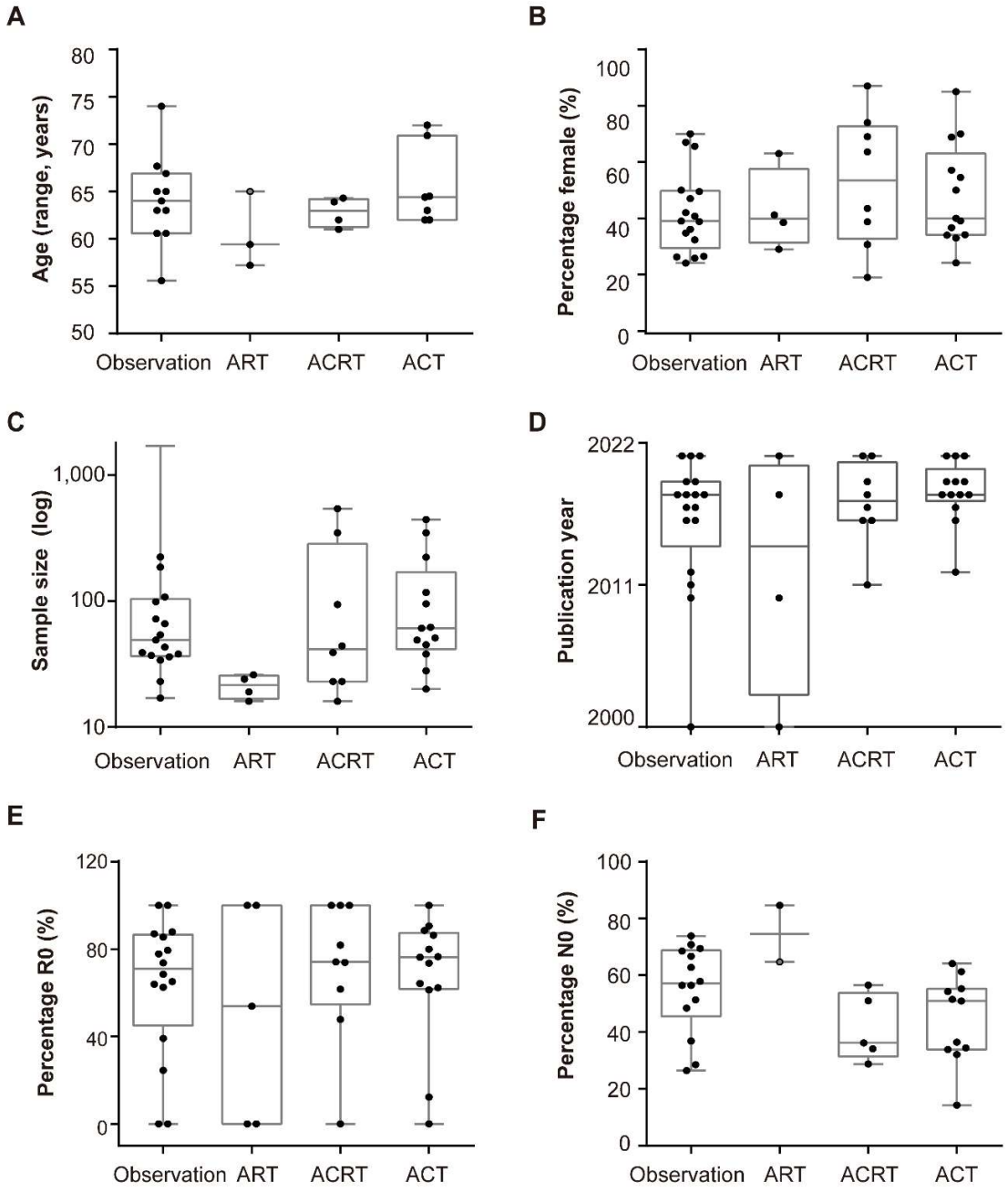



**Supplementary Figure 1. Summary of results from quality assessment of three randomized controlled studies using the Cochrane risk of bias tool.**

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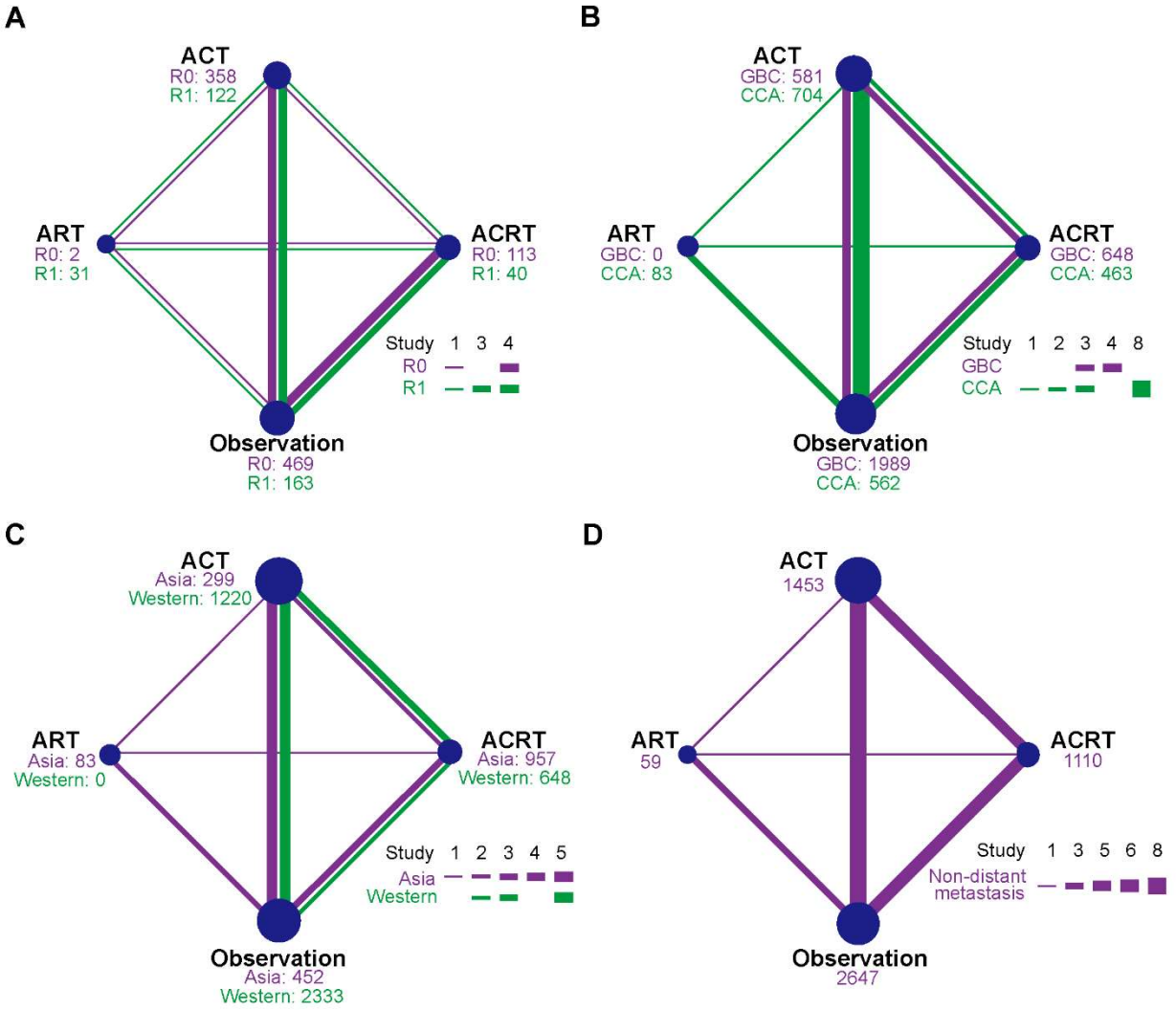
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**Supplementary Figure 2. Boxplots for distribution of mean age (A), percentage female (B), sample size (C), publication year (D), percentage R0 (E) and N0 (F) across comparisons.**



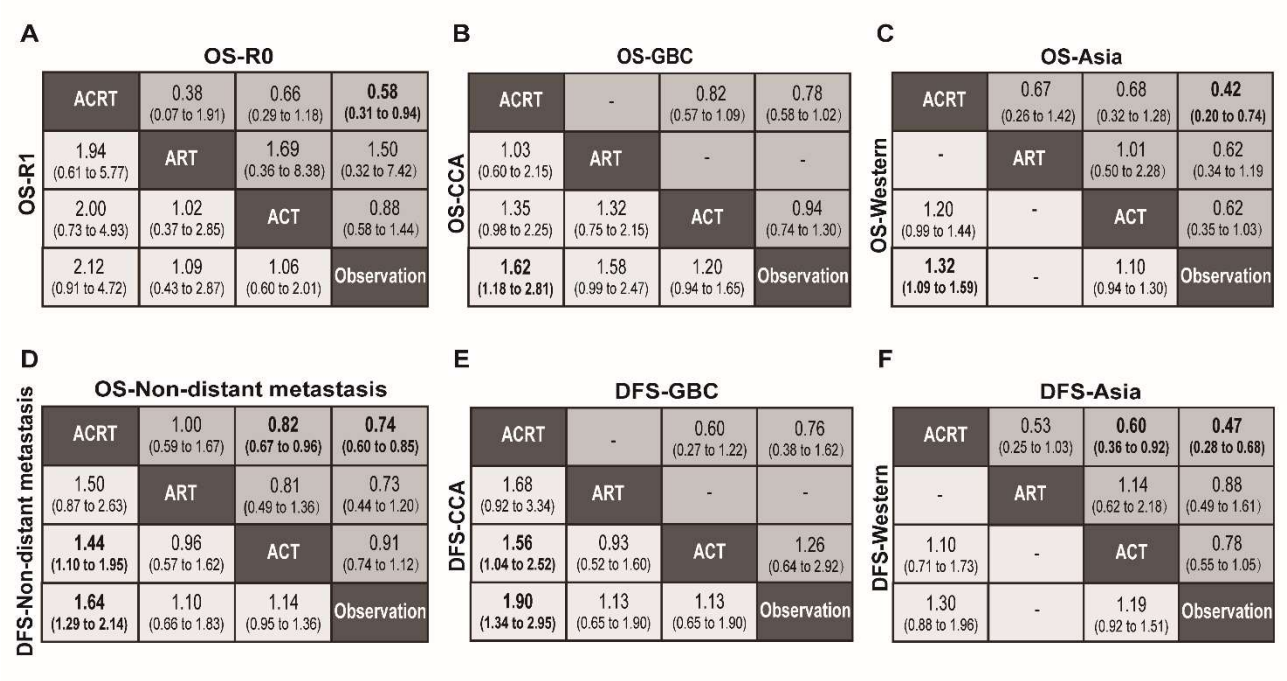
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**Supplementary Figure 3. Network plots of comparisons on overall survival of treatments in subgroups analyses (patients stratified into R0, R1, GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** Comparisons on overall survival in patients with GBC and CCA; **(C)** Comparisons on overall survival in subgroups of study patients in Asia and Western; **(D)** Comparisons on overall survival in patients without distal metastasis.



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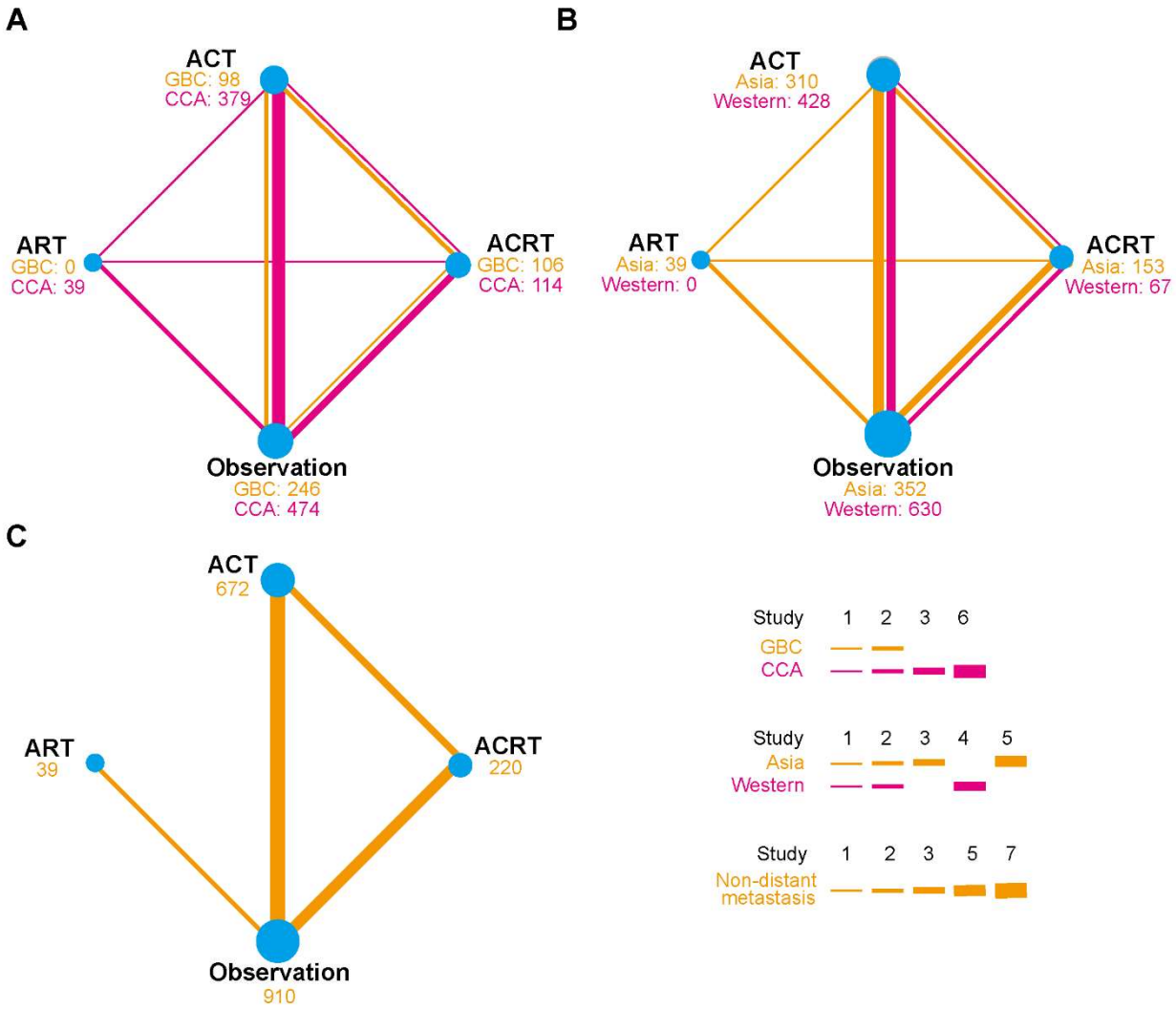



**Supplementary Figure 4. League tables of the network meta-analysis in subgroup analyses. (A)** League table of overall survival in subgroups of study patients with R0 and R1 resection margin; **(B)** and **(E)** League tables of overall survival and disease-free survival in subgroups of study patients with GBC and CCA; **(C)** and **(F)** League tables of overall survival and disease-free survival in subgroups of study patients in Asia and Western; **(D)** League table of overall survival and disease-free survival in subgroups of study patients without distal metastasis. Pooled hazard ratios (HRs) and 95% credible intervals (95% CIs) are listed in each cell. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment.

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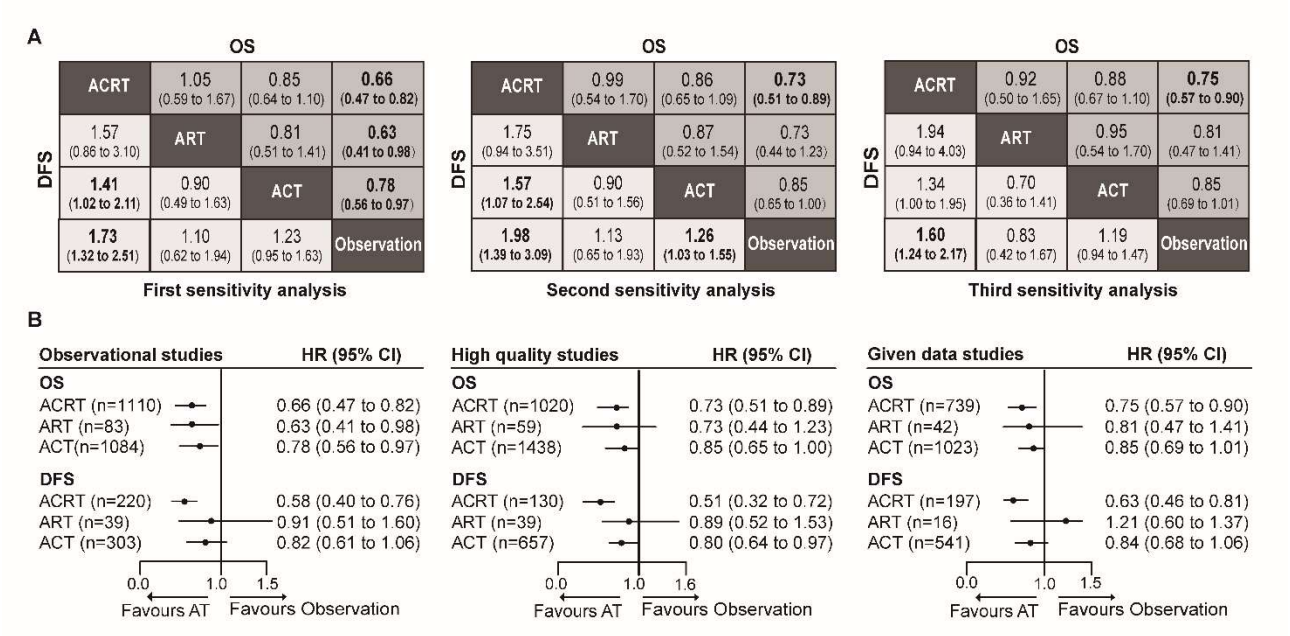
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**Supplementary Figure 5. Network plots of comparisons on disease-free survival of treatments in subgroups analyses (patients stratified into GBC, CCA, Asia, Western and non-distal metastasis). (A)** Comparisons on disease-free survival in subgroups of study patients with GBC and CCA; **(B)** Comparisons on disease-free survival in subgroups of study patients in Asia and Western; **(C)** Comparisons on disease-free survival in the subgroup of study patients without distal metastasis.



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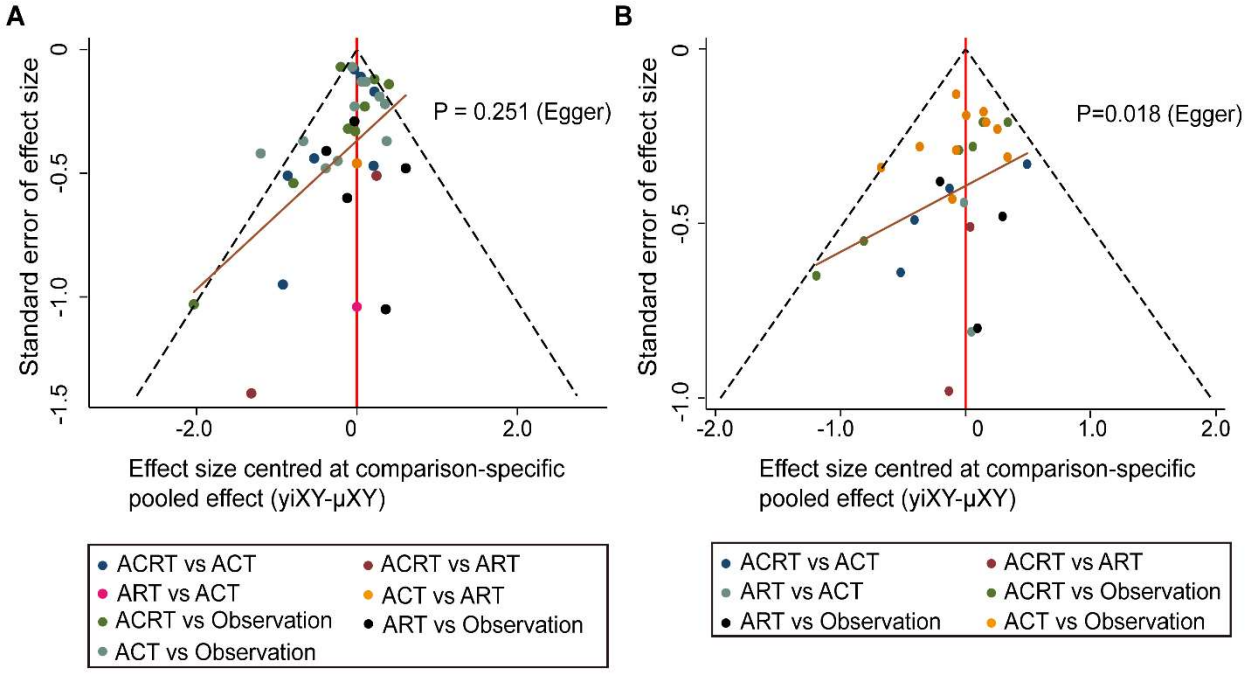



**Supplementary Figure 6. Pooled estimates of the sensitivity analysis. (A)** League tables of the network meta-analysis in sensitivity analyses. The estimate in each cell is for the comparison of row-defining treatment versus column-defining treatment. In the left lower half (DFS results), HR >1 favours the column-defining treatment, and in the upper right half (OS results), HR <1 favours the row-defining treatment. **(B)** Forest plots of the network meta-analysis in sensitivity analyses. Adjuvant treatments are ranked according to their surface under the cumulative ranking curve and compared with observation. Effect sizes are presented as hazard ratios (HRs) with 95% confidence intervals (95% CI). OS: overall survival; DFS: disease-free survival. First sensitivity analysis: excluding RCT studies; Second sensitivity analysis: removing low-and moderate-quality observational studies; Third sensitivity analysis: only including the studies for which HRs were reported in the original articles.

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**Supplementary Figure 7. The 'comparison-adjusted' funnel plot in network meta-analysis to assess funnel plot asymmetry on the efficacy outcomes. (A)** Funnel plot asymmetry on the overall survival; **(B)** Funnel plot asymmetry on the disease-free survival. Funnel-plot asymmetry was tested b Egger's regression tests. No publication bias is detected when the P-value is larger than 0.05.

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